

7 Series FPGAs SelectIO Resources

User Guide

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Revision History

The following table shows the revision history for this document.

Date	Version	Revision
03/01/2011	1.0	Initial Xilinx release.
04/06/2011	1.0.1	Updated disclaimer and copyright sections on page 2 .
05/31/2011	1.1	Added New Features . Updated the example device including Figure 1-15 and the <i>partgen</i> example on page 24 . Added VRN/VRP External Resistance Design Migration Guidelines . Updated the BITSLLIP Submodule section including Figure 3-12 . Removed Figure 3-13: <i>Bits from Data Input Stream (D) of Figure 3-12</i> .
07/20/2012	1.2	Updated paragraph before Table 1-1 . Added LVDS signaling to Table 1-1 . Updated V_{CCO} and V_{CCAUX_IO} . Updated Xilinx DCI . Removed V_{CCINT} . Added Match_cycle Configuration Option , DCIUpdateMode Configuration Option , DCIRESET Primitive , and Special DCI Requirements for Some Banks . Updated DCI Cascading . Updated DCI cascading guidelines after Figure 1-7 . Updated table note in Table 1-3 . Added Uncalibrated Split Termination in High-Range I/O Banks (IN_TERM) . Updated 7 Series FPGA SelectIO Primitives . Added DCI_CASCADE Constraint and VCCAUX_IO Constraint . Updated IBUF_LOW_PWR Attribute , Output Slew Rate Attributes , Output Drive Strength Attributes , PULLUP/PULLDOWN/KEEPER Attribute for IBUF , OBUFT , and IOBUF , and 7 Series FPGA I/O Resource VHDL/Verilog Examples . Put Internal V_{REF} inside Differential Termination Attribute , page 49 . Updated DRIVE attribute in Table 1-10 . Updated titles of Figure 1-41 through Figure 1-44 . Updated LVDS and LVDS_25 (Low Voltage Differential Signaling) , including adding Figure 1-72 . Added IN_TERM attribute to SSTL (Stub-Series Terminated Logic) . Added table note to Table 1-55 . Added Simultaneous Switching Outputs .

Date	Version	Revision
07/20/2012	1.2 (Cont'd)	Updated ILOGIC Resources. In Table 2-3, added $T_{\text{ICOCKD}}/T_{\text{IOCKDD}}$ and removed T_{ICEIQ}. Updated Input Delay Resources (IDELAY). Updated functional description of LD port in Table 2-4. In IDELAY Ports, updated Module Load - LD and Increment/Decrement Signals - CE, INC, and added Pipeline Register Load - LDPIPEEN and Pipeline Register Reset - REGRST. Removed Table 2-5: "Control Pin Descriptions." Updated descriptions of IDELAY_TYPE and IDELAY_VALUE in Table 2-5. Updated IDELAY_TYPE Attribute, IDELAY_VALUE Attribute, and HIGH_PERFORMANCE_MODE Attribute. Updated IDELAY Timing. Updated text before Figure 2-12. Updated Stability after an Increment/Decrement Operation. Updated IDELAYCTRL, including Figure 2-16. Added paragraph about OLOGICE2 and OLOGICE3 to OLOGIC Resources. Updated first paragraph of Output Delay Resources (ODELAY)—Not Available in HR Banks. Updated functions of REGRST, LD, CNTVALUEIN, LDPIPEEN, and CNTVALUEOUT in Table 2-13. Added description of VAR_LOAD_PIPE mode to Module Load - LD. Added Pipeline Register Load - LDPIPEEN and Pipeline Register Reset - REGRST. Updated Count Value In - CNTVALUEIN, Count Value Out - CNTVALUEOUT, and Increment/Decrement Signals - CE, INC. Removed Table 2-14: "Control Pin Descriptions." Updated descriptions of ODELAY_TYPE and ODELAY_VALUE in Table 2-14. Updated ODELAY Attributes. Added ODELAY Modes. Updated text before Figure 2-26. Updated Reset Input - RST, page 149. Added INIT_Q and SRVAL_Q attributes to Table 3-2. Updated bulleted list after Figure 3-6 and in MEMORY Interface Type. Updated Figure 3-7. Updated ISERDESE2 Width Expansion, BITSLLIP Submodule, and Data Parallel-to-Serial Converter. Deleted the OCBEXTEND pin in Figure 3-14. Updated descriptions of OFB and TFB in Table 3-6. Updated Output Feedback from OSERDESE2 - OFB, 3-state Control Output - TFB, and Reset Input - RST, page 164. Updated OSERDESE2 Clocking Methods and OSERDESE2 Width Expansion. Updated latencies in Table 3-11. Added IO_FIFO Overview. Updated Resetting the IO_FIFO. Added Appendix A, Termination Options for SSO Noise Analysis.
10/31/2012	1.3	Removed XC7V1500T from third bullet after Figure 1-7 .
05/13/2014	1.4	Updated V_{CCO}. Added item to bulleted list after Figure 1-7. Updated paragraph after Figure 1-10. In VRN/VRP External Resistance Design Migration Guidelines, updated first two paragraphs and added description of power rating. Updated title of Figure 1-11 and Figure 1-12. Updated step 4 of DCI in 7 Series FPGAs I/O Standards. Updated Figure 1-14. Updated first paragraph of Uncalibrated Split Termination in High-Range I/O Banks (IN_TERM). Added IOBUF_DCIEN, IOBUF_INTERMDISABLE, IOBUFDS_DIFF_OUT_DCIEN, IOBUFDS_DIFF_OUT_INTERMDISABLE, and IOBUFDS_INTERMDISABLE to 7 Series FPGA SelectIO Primitives. Removed O output from Figure 1-22 and following description. Updated HSTL_II_T_DCI and HSTL_II_T_DCI_18. Added IBUFDS_DIFF_OUT_INTERMDISABLE, IOBUF_DCIEN, and IOBUF_INTERMDISABLE. Updated connections in Figure 1-28, Figure 1-30, Figure 1-31, and Figure 1-32. Reversed R_{VRN} and R_{VRP} in Figure 1-46, Figure 1-48, Figure 1-49, Figure 1-50, Figure 1-52, Figure 1-54, Figure 1-55, Figure 1-56, Figure 1-57, Figure 1-58, Figure 1-60, Figure 1-62, and Figure 1-63. Added note to SSTL18_II, SSTL15, SSTL135, DIFF_SSTL18_II, DIFF_SSTL15, DIFF_SSTL135. Updated fifth paragraph of SSTL (Stub-Series Terminated Logic). Removed Thevenin equivalent of $R/2$ and description of source termination series resistors from SSTL18_I_DCI, DIFF_SSTL18_I_DCI, SSTL18_II, SSTL15, SSTL135, DIFF_SSTL18_II, DIFF_SSTL15, DIFF_SSTL135, SSTL18_II_DCI, SSTL15_DCI, SSTL135_DCI, DIFF_SSTL18_II_DCI, DIFF_SSTL15_DCI, DIFF_SSTL135_DCI, SSTL18_II_T_DCI, SSTL15_T_DCI, SSTL135_T_DCI, DIFF_SSTL18_II_T_DCI, DIFF_SSTL15_T_DCI, DIFF_SSTL135_T_DCI, and SSTL12, SSTL12_DCI, SSTL12_T_DCI, DIFF_SSTL12, DIFF_SSTL12_DCI, DIFF_SSTL12_T_DCI. Updated Figure 1-57 and Figure 1-59.

Date	Version	Revision
05/13/2014	1.4 (Cont'd)	Added to list of criteria after Table 1-44. Added note to Table 1-48. Updated description after Table 1-51. Updated V_{CCO} Input column in Table 1-55. Added note 3 to Table 1-56. Updated DLYIN connection in Figure 2-4. Updated Clock Input - C, page 117. Updated description of PIPE_SEL in Table 2-5 and Table 2-14. Added VAR_LOAD description to first paragraph of Stability after an Increment/Decrement Operation, page 123. Removed center I/Os from Figure 2-16. Updated Data Output - DATAOUT, page 135. In ODELAY Modes, replaced ODELAYCTRL with IDELAYCTRL. In Table 3-1, added CLKDIVP and updated descriptions of OCLK and OCLKB. Updated High-Speed Clock for Strobe-Based Memory Interfaces and Oversampling Mode - OCLK and Reset Input - RST. Added IOBDELAY to Table 3-2. Updated bullets in MEMORY Interface Type. Updated bullets in OVERSAMPLE Interface Type. Updated Figure 3-7. Added sentence about ISERDESE2 being reset to Guidelines for Using the Bitslip Submodule. Removed Bitslip submodule from description of CLKDIV in Table 3-6. Added TBYTE_CTL and TBYTE_SRC to Table 3-7. In Figure 3-18, shifted OQ, TQ, and OBUFT.O by one CLK edge.
05/15/2015	1.5	Added paragraph about overvoltage protection mode to V_{CCO}. Added State of I/Os During and After Configuration. Updated Special DCI Requirements for Some Banks. In IOSTANDARD Attribute, replaced DIFF_HSTL18_II with DIFF_HSTL_II_18. Reversed R_{VRN} and R_{VRP} resistors in left side IOB of DCI terminations in Figure 1-49, Figure 1-50, Figure 1-52, Figure 1-54, Figure 1-57, Figure 1-58, Figure 1-60, and Figure 1-62. Added note 2 to Table 1-55. Added Vivado Design Suite to Pin Planning to Mitigate SSO Sensitivity. Updated description of clock input C in IDELAY Ports and ODELAY Ports. Replaced SR with S/R in Figure 2-17, Figure 2-20, and Table 2-10.
09/18/2015	1.6	Replaced SR with S/R throughout. Added note about set and reset pins to Table 2-1 and Table 2-10. In RDY - Ready, updated sentence about RDY signal being deasserted if REFCLK is held High or Low for more than one clock period. In Table 3-6, changed TBYTEOUT port type from input to output.
09/15/2016	1.7	Updated first paragraph of V_{CCO}. Added reference to UG912: <i>Vivado Design Suite Properties Reference Guide</i> in Uncalibrated Split Termination in High-Range I/O Banks (IN_TERM) and 7 Series FPGA SelectIO Attributes/Constraints. Added description of termination for unused I/Os to PULLUP/PULLDOWN/KEEPER Attribute for IBUF, OBUFT, and IOBUF. Updated third bullet in OSERDESE2 Clocking Methods. In Table 3-11, updated DDR 14:1 latency to 5 CLK cycles.
09/27/2016	1.8	Added Spartan-7 family to Preface. Updated Reset Input - RST. Removed bullet describing CLK driven by BUFG and CLKDIV driven by a different BUFG from NETWORKING Interface Type and OSERDESE2 Clocking Methods.
08/22/2017	1.9	Replaced HR with HP in second paragraph of V_{CCO}. Expanded description of ZHOLD in ILOGIC Resources. Updated REFCLK_FREQUENCY value and description in Table 2-5 and Table 2-14. Updated description of Clock Input from Clock Buffer - CLKIN. Updated descriptions of SHIFTOUT1, SHIFTOUT2, SHIFTIN1, and SHIFTIN2 in Table 3-6. Updated input span to D1–D8 in Timing Characteristics of 8:1 DDR Serialization.
05/08/2018	1.10	Expanded descriptions in Parallel 3-state Inputs - T1 to T4 and DATA_RATE_TQ Attribute. Added note after Table 3-11.

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Appendix A: Termination Options for SSO Noise Analysis

About This Guide

Xilinx® 7 series FPGAs include four FPGA families that are all designed for lowest power to enable a common design to scale across families for optimal power, performance, and cost. The Spartan®-7 family is the lowest density with the lowest cost entry point into the 7 series portfolio. The Artix®-7 family is optimized for highest performance-per-watt and bandwidth-per-watt for cost-sensitive, high volume applications. The Kintex®-7 family is an innovative class of FPGAs optimized for the best price-performance. The Virtex®-7 family is optimized for highest system performance and capacity.

This guide serves as a technical reference describing the 7 series FPGAs SelectIO™ resources. This 7 series FPGAs SelectIO resources user guide, part of an overall set of documentation on the 7 series FPGAs, is available on the Xilinx website at www.xilinx.com/documentation.

Guide Contents

This manual contains the following chapters:

- [Chapter 1, SelectIO Resources](#)
- [Chapter 2, SelectIO Logic Resources](#)
- [Chapter 3, Advanced SelectIO Logic Resources](#)

Additional Resources

To find additional documentation, see the Xilinx website at:

<http://www.xilinx.com/support/documentation/index.htm>.

To search the Answer Database of silicon, software, and IP questions and answers, or to create a technical support WebCase, see the Xilinx website at:

<http://www.xilinx.com/support>.

SelectIO Resources

I/O Tile Overview

Input/output characteristics and logic resources are covered in three consecutive chapters.

[Chapter 1, SelectIO Resources](#) describes the electrical behavior of the output drivers and input receivers, and gives detailed examples of many standard interfaces. [Chapter 2, SelectIO Logic Resources](#) describes the input and output data registers and their double-data rate (DDR) operation, and the programmable input delay (IDELAY) and programmable output delay (ODELAY). [Chapter 3, Advanced SelectIO Logic Resources](#) describes the data serializer/deserializer (SERDES).

The 7 series FPGAs offer both high-performance (HP) and high-range (HR) I/O banks. The HP I/O banks are designed to meet the performance requirements of high-speed memory and other chip-to-chip interfaces with voltages up to 1.8V. The HR I/O banks are designed to support a wider range of I/O standards with voltages up to 3.3V. [Table 1-1](#) highlights the features supported in the HP and HR I/O banks. Refer to [Table 1-1](#) for help when making initial decisions on I/O banks for a particular design's requirements. See the specific device family data sheet for details on the performance and other electrical requirements of the HP and HR I/O banks.

The 7 series FPGAs contain different combinations of HR and HP I/O banks. The *7 Series FPGAs Overview* documents the available number of each type of bank for all devices.

Table 1-1: Supported Features in the HR and HP I/O Banks

Feature	HP I/O Banks	HR I/O Banks
3.3V I/O standards ⁽¹⁾	N/A	Supported
2.5V I/O standards ⁽¹⁾	N/A	Supported
1.8V I/O standards ⁽¹⁾	Supported	Supported
1.5V I/O standards ⁽¹⁾	Supported	Supported
1.35V I/O standards ⁽¹⁾	Supported	Supported
1.2V I/O standards ⁽¹⁾	Supported	Supported
LVDS signaling	Supported ⁽²⁾	Supported
24 mA drive option for LVCMOS18 and LVTTTL outputs	N/A	Supported
V _{CCAUX_IO} supply rail	Supported	N/A
Digitally-controlled impedance (DCI) and DCI cascading	Supported	N/A
Internal V _{REF}	Supported	Supported

Table 1-1: Supported Features in the HR and HP I/O Banks (Cont'd)

Feature	HP I/O Banks	HR I/O Banks
Internal differential termination (DIFF_TERM)	Supported	Supported
IDELAY	Supported	Supported
ODELAY	Supported	N/A
IDELAYCTRL	Supported	Supported
ISERDES	Supported	Supported
OSERDES	Supported	Supported
ZHOLD_DELAY	N/A	Supported

Notes:

1. Not all I/O standards and drive strengths are supported in both the HP and HR I/O banks. The [I/O Bank Availability](#) column in [Table 1-55](#) shows the specific I/O standards that are available in the HP and HR I/O banks.
2. Although LVDS is generally considered a 2.5V I/O standard, it is supported in both the HR and HP I/O banks.

New Features

The 7 series devices support many of the same features supported in the Virtex®-6 and Spartan®-6 FPGAs, however, some of these features are changed in form or functionality. These changes include:

- There are now two distinctly different types of I/O banks, HR and HP, and each type supports some unique I/O standards and features.
- The memory interface related I/O standards such as SSTL and HSTL now support the SLEW attribute, and are selectable between both FAST and SLOW edge rates. The default SLEW for all I/O standards is SLOW, which has been the case for all I/O standards that supported the SLEW attribute in all previous FPGA families (namely LVCMOS and LVTTTL). However, because this attribute is a new addition to the memory interface standards, if left unchanged (not specified in the RTL, UCF file, or I/O planning software), the default slew rates for these for these standards will result in much slower slew rates than in previous families. To achieve similar slew rates as in previous families, new designs now require the SLEW attribute to be specified and set to FAST. [Table 1-56](#) shows (among other features) which I/O standards support the SLEW attribute.
- The 7 series FPGA DCI calibration circuit has improved the accuracy of the internal termination resistance. As a result, the selection of values for the external precision resistors is different for the split-termination DCI standards. Specifically, the external resistors are now chosen to be double the target Thevenin-equivalent resistance, whereas in Virtex-6 FPGAs and earlier families they were chosen to be equal to the target Thevenin-equivalent resistance. See the [Xilinx DCI](#) section for more details.
- There are additional I/O Logic design primitives with new features and functions. See [Chapter 2, SelectIO Logic Resources](#) for more details on these primitives.

SelectIO Resources Introduction

All 7 series FPGAs have configurable SelectIO drivers and receivers, supporting a wide variety of standard interfaces. The robust feature set includes programmable control of output strength and slew rate, on-chip termination using digitally-controlled impedance (DCI), and the ability to internally generate a reference voltage (INTERNAL_VREF).

Note: HR banks do not have DCI. Therefore, any reference to DCI in this user guide does not apply to the HR banks.

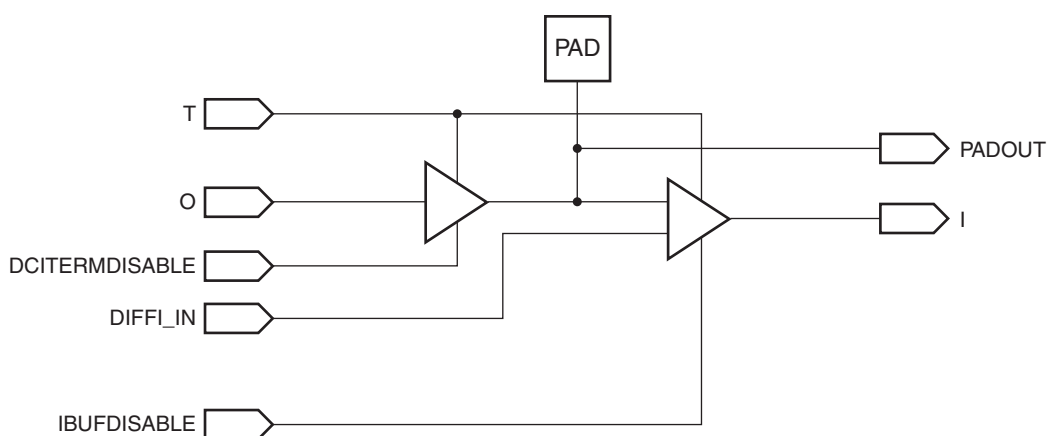
With some exceptions, each I/O bank contains 50 SelectIO pins. The two pins at the very ends of each bank can only be used with single-ended I/O standards. The remaining 48 pins can be used with either single-ended or differential standards using two SelectIO pins grouped together as positive/negative (P/N) pairs. Every SelectIO resource contains input, output, and 3-state drivers.

The SelectIO pins can be configured to various I/O standards, both single-ended and differential.

- Single-ended I/O standards (e.g., LVCMOS, LVTTL, HSTL, PCI, and SSTL)
- Differential I/O standards (e.g., LVDS, Mini_LVDS, RSDS, PPDS, BLVDS, and differential HSTL and SSTL)

Figure 1-1 shows the single-ended (only) HP I/O block (IOB) and its connections to the internal logic and the device pad. Figure 1-2 shows the regular HP IOB. Figure 1-3 shows the single-ended (only) HR IOB. Figure 1-4 shows the regular HR IOB. In both the HP and HR I/O banks, the single-ended (only) and regular IOBs are essentially equivalent, except the single-ended (only) IOBs do not have the connections to make the differential output signals. In most devices, the single-ended (only) IOBs are the two pins located at the ends of each I/O bank. The regular IOB that make up the other 48 pins in each bank can implement both single-ended and differential I/O standards.

Each IOB has a direct connection to an ILOGIC/OLOGIC pair containing the input and output logic resources for data and 3-state control for the IOB. Both ILOGIC and OLOGIC can be configured as ISERDES and OSERDES, respectively, as described in [Chapter 3, Advanced SelectIO Logic Resources](#).



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Figure 1-1: Single-Ended (Only) HP IOB Diagram

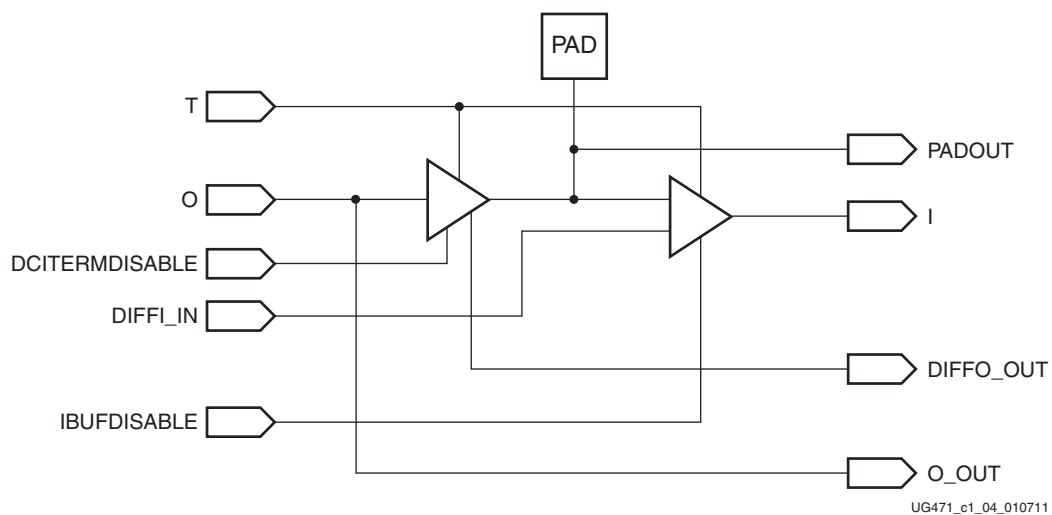


Figure 1-2: Regular HP IOB Diagram

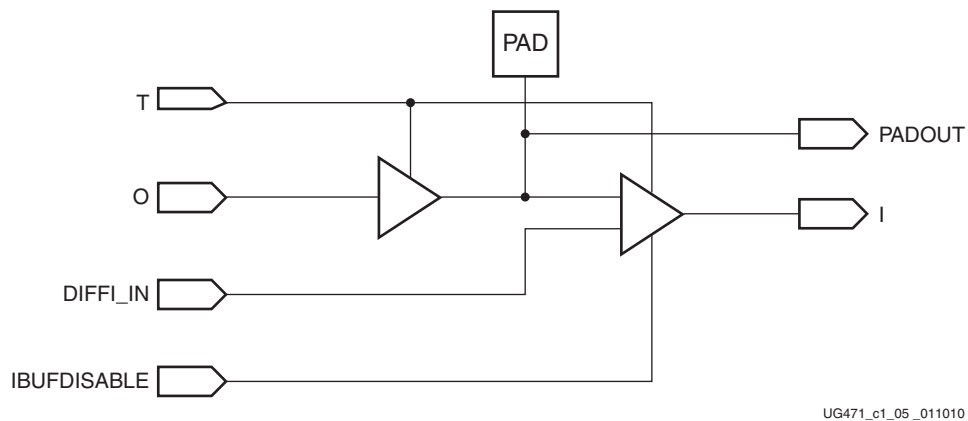


Figure 1-3: Single-Ended (Only) HR IOB Diagram

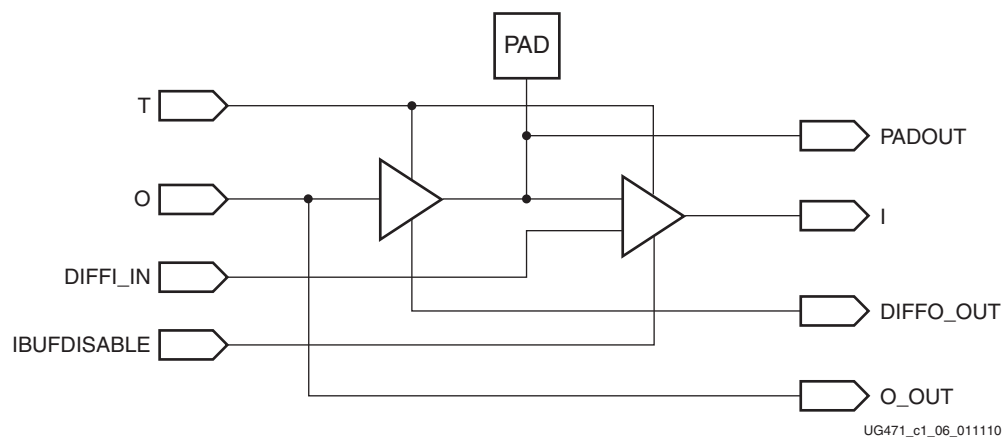


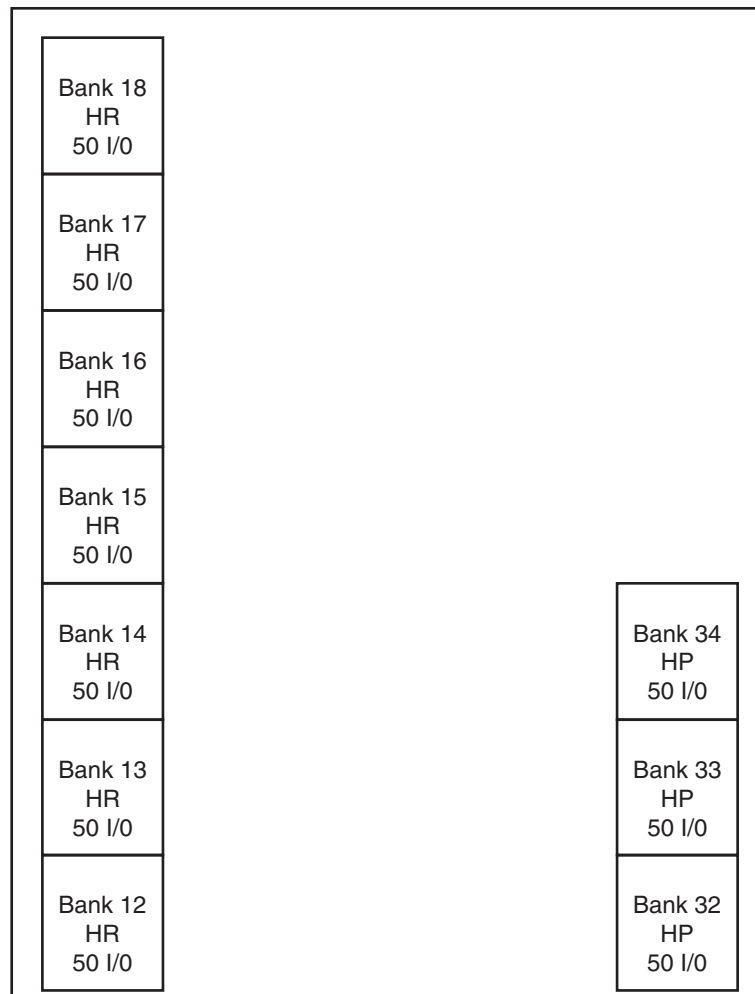
Figure 1-4: Regular HR IOB Diagram

SelectIO Resources General Guidelines

This section summarizes the general guidelines to be considered when designing with the SelectIO resources in 7 series FPGAs.

7 Series FPGA I/O Bank Rules

In 7 series devices, an I/O bank consists of 50 IOBs. The number of banks depends upon the device size and the package pinout. In the *7 Series FPGAs Overview* the total number of available I/O banks is listed by device type. For example, the XC7K325T has 10 usable I/O banks. [Figure 1-5](#) is an example of a columnar floorplan showing the XC7K325T I/O banks. [UG475: 7 Series FPGAs Packaging and Pinout Specifications](#) includes information on the I/O banks for each device/package combination.



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Figure 1-5: 7 Series FPGA XC7K325T I/O Banks

Supply Voltages for the SelectIO Pins

V_{CCO}

The V_{CCO} supply is the primary power supply of the 7 series I/O circuitry. The V_{CCO} (V) columns in [Table 1-55](#) provide the V_{CCO} requirements for each of the supported I/O standards, and illustrate the V_{CCO} requirements for both inputs and outputs as well as the optional internal differential termination circuit. All V_{CCO} pins for a given I/O bank must be connected to the same external voltage supply on the board, and as a result all of the I/O within a given I/O bank must share the same V_{CCO} level. The V_{CCO} voltage must match the requirements for the I/O standards that have been assigned to the I/O bank. An incorrect V_{CCO} voltage can result in loss of functionality or damage to the device. The 7 series power supply requirements, including power-on and power-off sequencing, are described in the 7 series FPGA data sheets.

In HP I/O banks, if the I/O standard voltage requirement is $\leq 1.8V$, but a $V_{CCO} \geq 2.5V$ is applied, the device automatically enters an overvoltage protection mode. Reconfiguring the device with the correct V_{CCO} level restores normal operation.

V_{REF}

Single-ended I/O standards with a differential input buffer require an input reference voltage (V_{REF}). When V_{REF} is required within an I/O bank, the two multi-function V_{REF} pins for the bank must be used as V_{REF} supply inputs. 7 series FPGAs can optionally use an internally generated reference voltage by enabling the INTERNAL_VREF constraint. For more information on this constraint, see [7 Series FPGA SelectIO Attributes/Constraints](#), [page 46](#).

V_{CCAUX}

The global auxiliary (V_{CCAUX}) supply rail is primarily used for providing power to the various block feature's interconnect logic inside the 7 series FPGAs. In the I/O banks, V_{CCAUX} is also used to power input buffer circuits for some of the I/O standards. These include all of the single-ended I/O standards at or below 1.8V, and also some of the 2.5V standards (HR I/O banks only). Additionally, the V_{CCAUX} rail provides power to the bank's differential input buffer circuits used for the differential and V_{REF} I/O standards.

The 7 series power supply requirements, including power-on and power-off sequencing, are described in the 7 series FPGA data sheets.

V_{CCAUX_IO}

The auxiliary I/O (V_{CCAUX_IO}) supply rail is only present in HP I/O banks and provides power to the I/O circuitry. The Kintex-7 and Virtex-7 FPGAs data sheets contain a table titled Maximum Physical Interface (PHY) Rate for Memory Interfaces that references V_{CCAUX_IO} . This table indicates how the V_{CCAUX_IO} pins can be powered at either 1.8V (default), or optionally at 2.0V to achieve higher frequency performance for certain types of memory interfaces. Although this table is designed for memory interfaces, it can also provide guidance on powering V_{CCAUX_IO} for other high-speed single-ended interfaces based on the target bit rates. The table does not apply to LVDS, which uses a different type of driver circuit than the single-ended drivers that are more affected by the V_{CCAUX_IO} level. Thus, for LVDS interfaces, it does not matter which voltage level the V_{CCAUX_IO} rail is powered at. The default value of 1.8V affords a lower-power consumption and provides very close to the same performance in the I/Os. The 2.0V option is available when the slightly-increased performance is required for the very fastest bit rates supported for the single-ended drivers.

There is a design constraint for I/O nets and primitives called `VCCAUX_IO`, which should be specified in the design if the `VCCAUX_IO` pins for any banks are to be set at 2.0V. See [7 Series FPGA SelectIO Attributes/Constraints, page 46](#) for information on this constraint.

The `VCCAUX_IO` pins are connected together internally inside Kintex-7 and Virtex-7 device packages in groups of three or four HP I/O banks. The package files chapter of [UG475: 7 Series FPGA Packaging and Pinout Specification](#) contains links to the ASCII package files, and the figures in the device diagrams chapter indicate which device/package combinations contain HP I/O banks with `VCCAUX_IO` pins. The ASCII package files indicate which bank's `VCCAUX_IO` pins are grouped together inside the package. The `VCCAUX_IO` package pin names have the syntax `VCCAUX_IO_G#`, where the # is the internal group number. The package files contain a column called "VCCAUX Group" that shows for every I/O pin which `VCCAUX` group that I/O bank is associated with. All I/O pins that are in the same `VCCAUX_IO` group must have `VCCAUX_IO` constraints on their nets or primitives that are compatible. All `VCCAUX_IO` pins that are grouped together should be tied to the same voltage rail on the board. FBG packages for Kintex-7 devices contain `VCCAUX_IO` pins but are no connects internally. All HP I/O banks in those packages are powered from the main `VCCAUX` rail instead.

State of I/Os During and After Configuration

7 series FPGAs have pins dedicated to configuration functions contained in I/O bank 0. Banks 14 and 15 also contain I/O pins known as multi-function or multi-purpose pins that can also be used for configuration, but then convert to normal I/O pins after configuration is complete. Additionally in SSI devices, pins in banks 11, 12, 17, 18, 20, and 21 have restrictions during configuration similar to multi-function pins. However, pins in these banks do not have any configuration functions.

In devices where bank 14 and/or bank 15 are HR banks and configured with a `VCCO` requirement $\leq 1.8V$, inputs might have a 0-1-0 transition to the interconnect logic during configuration if the input is tied to 0 or floating and the configuration voltage is $\geq 2.5V$. For further details, refer to [UG470: 7 Series FPGAs Configuration User Guide](#).

7 Series FPGA DCI—Only available in the HP I/O banks

Introduction

As FPGAs get bigger and system clock speeds get faster, PC board design and manufacturing becomes more difficult. With ever faster edge rates, maintaining signal integrity becomes a critical issue. PC board traces must be properly terminated to avoid reflections or ringing.

To terminate a trace, resistors are traditionally added to make the output and/or input match the impedance of the receiver or driver to the impedance of the trace. However, due to increased device I/Os, adding resistors close to the device pins increases the board area and component count, and can in some cases be physically impossible. To address these issues and to achieve better signal integrity, Xilinx developed the digitally controlled impedance (DCI) technology.

Depending on the I/O standard, DCI can either control the output impedance of a driver, or add a parallel termination present at the driver and/or receiver, with the goal of accurately matching the characteristic impedance of the transmission line. DCI actively adjusts these impedances inside the I/O to calibrate to external precision reference resistors placed on the VRN and VRP pins. This compensates for changes in I/O

impedance due to process variation. It also continuously adjusts the impedances to compensate for variations of temperature and supply voltage fluctuations.

For the I/O standards with controlled impedance drivers, DCI controls the driver impedance to either match the two reference resistors, or for some standards, to match half the value of these reference resistors.

For the I/O standards with controlled parallel termination, DCI provides the parallel termination for both transmitters and receivers. This eliminates the need for termination resistors on the board, reduces board routing difficulties and component count, and improves signal integrity by eliminating stub reflection. Stub reflection occurs when termination resistors are located too far from the end of the transmission line. With DCI, the termination resistors are as close as possible to the output driver or the input buffer, thus, eliminating stub reflections. DCI is only available in 7 series FPGAs HP I/O banks. DCI is not available in HR I/O banks.

Xilinx DCI

DCI uses two multi-purpose reference pins in each I/O bank to control the impedance of the driver or the parallel-termination value for all of the I/Os of that bank. The N reference pin (VRN) must be pulled up to V_{CCO} by a reference resistor, and the P reference pin (VRP) must be pulled down to ground by another reference resistor. The value of each reference resistor should be either equal to the characteristic impedance of the PC board traces or twice that value.

To implement DCI in a design:

1. Assign one of the DCI I/O standards in an HP I/O bank (see [Table 1-2](#) and [Table 1-3](#)).
2. Connect the VRN multi-function pin to a precision resistor tied to the V_{CCO} rail for the same bank.
3. Connect the VRP multi-function pin to a precision resistor tied to ground.

The following sections discuss how to determine the precision resistor values for VRN and VRP for the different I/O standards. Only one set of VRN and VRP resistors is used for each bank, so all DCI standards within each bank must be able to share the same external resistance values. If several I/O banks in the same I/O bank column are using DCI, and all of those I/O banks use the same VRN/VRP resistor values, the internal VRN and VRP nodes can be cascaded so that only one pair of pins for all of the I/O banks in the entire I/O column is required to be connected to precision resistors. This option is called DCI cascading and is detailed in [DCI Cascading, page 22](#). This section also describes how to determine if I/O banks share the same I/O bank column. If DCI I/O standards are not used in the bank, these pins are available as regular I/O pins. [UG475: 7 Series FPGAs Packaging and Pinout Specifications](#) gives detailed pin descriptions.

DCI adjusts the impedance of the I/O by selectively turning transistors in the I/Os on or off. The impedance is adjusted to match the external reference resistors. The adjustment starts during the device startup sequence. By default, the DONE pin does not go High until the first part of the impedance adjustment process is completed.

The DCI calibration can be reset by instantiating the DCIRESET primitive. Toggling the RST input to the DCIRESET primitive while the device is operating, resets the DCI state machine and restarts the calibration process. All I/Os using DCI will be unavailable until the LOCKED output from the DCIRESET block is asserted. This functionality is useful in applications where the temperature and/or supply voltage changes significantly from device power-up to the nominal operating condition.

For controlled impedance output drivers, the impedance can be adjusted either to match the reference resistors or half the resistance of the reference resistors. For on-chip termination, the termination is always adjusted to match the reference resistors.

For the I/O standards that support the DCI controlled impedance driver, DCI can configure output drivers to be the following types:

- [Controlled Impedance Driver \(Source Termination\)](#)
- [Controlled Impedance Driver with Half Impedance \(Source Termination\)](#)

For the I/O standards that support parallel termination, DCI creates a Thevenin equivalent, or split-termination resistance, to the $V_{CCO}/2$ voltage level. The I/O standards' naming convention adds:

- DCI in the name of the I/O standard if split-termination resistors are always present in the I/O, independent of whether the standard is used on an input, output, or bidirectional pin.
- T_DCI in the name of the I/O standard if split-termination resistors are only present when the output buffer is 3-stated.

Match_cycle Configuration Option

Match_cycle is a configuration option that optionally halts the startup sequence at the end of the FPGA configuration sequence until the DCI logic has performed the first match (calibration) to the external reference resistors. This option is also sometimes referred to as DCI match. For more information about the Match_cycle option, refer to the “Configuration Details” chapter in [UG470: 7 Series FPGAs Configuration User Guide](#). For information on how to invoke the option in a design and to set it to a specific startup cycle, refer to the Match_cycle option in [UG628: Command Line Tools User Guide](#).

DCIUpdateMode Configuration Option

DCIUpdateMode is a configuration option that can override control of how often the DCI circuit updates the impedance matching to the VRN and VRP reference resistors. This option defaults to AsRequired but also has an optional value of Quiet in the Xilinx implementation software. The settings for the DCIUpdateMode configuration option are:

- **AsRequired:** Initial impedance calibration is made at device initialization, and dynamic impedance adjustments are made as needed throughout device operation (default).
- **Continuous:** For 7 series FPGAs, this value has no effect (defaults back to AsRequired).
- **Quiet:** Impedance calibration is done only once at device initialization, or each time the RST pin is asserted on the DCIRESET primitive for designs that include this primitive.

It is strongly recommended that the DCIUpdateMode option be kept with the default value of AsRequired so that the DCI circuitry is allowed to operate normally. See [UG628: Command Line Tools User Guide](#) for more details if there is a special need to set this option to Quiet.

DCIRESET Primitive

DCIRESET is a Xilinx design primitive that provides the capability to perform a reset of the DCI controller state machine during normal operation of the design. Unless DCIUpdateMode is set to Quiet (see [DCIUpdateMode Configuration Option](#)) or for the case outlined below related to the use of multi-function pins set to use DCI, for most situations this primitive should not be required in a design. See [UG768: Xilinx 7 Series FPGA Libraries Guide for HDL Designs](#) for more details on the DCIRESET primitive.

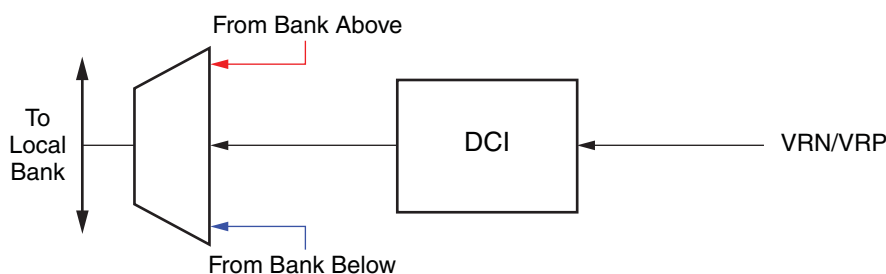
Special DCI Requirements for Some Banks

If any of the multi-function pins in I/O banks 14 or 15 (any device), or banks 11, 12, 17, 18, 20, and 21 (SSI devices only) are assigned DCI I/O standards in the user design, the DCIRESET primitive should also be included and used in the design. In that case, the design should pulse the RST input of DCIRESET and then wait for the LOCKED signal to be asserted prior to using any user input or outputs on these pins with DCI standards. This is required because these I/O pins ignore the initial DCI calibration that happens during the normal device initialization.

As a result, if the DCIRESET primitive had not been used and DCIUpdateMode was set to AsRequired, after those pins become normal I/O pins there would be an indeterministic delay between the end of configuration and when the DCI calibration algorithm updated those pins DCI settings. If DCIRESET was not used and DCIUpdateMode was set to Quiet, these pins would never have their DCI values set. In that case, the Controlled Impedance DCI I/O standards (such as LVDCI_18) would behave as if in the high-Z state all the time, and Split Termination DCI I/O standards (such as SSTL15_DCI) would behave as if there was no internal termination. Including and using the DCIRESET primitive in the design allows these pins to have DCI I/O standards and to perform without issue.

DCI Cascading

The 7 series FPGA HP I/O banks using DCI I/O standards have the option of deriving the DCI impedance values from another HP I/O bank. As shown in [Figure 1-6](#), a digital control bus is internally distributed throughout the bank to control the impedance of each I/O.

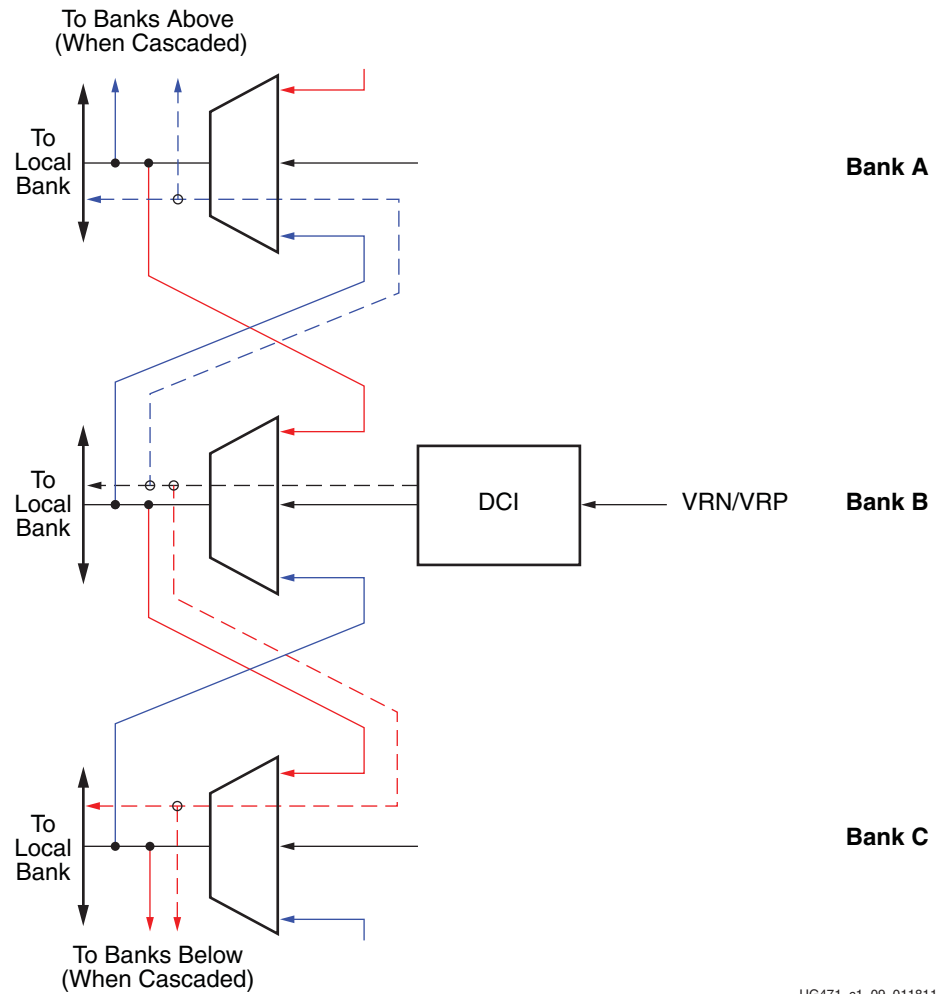


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Figure 1-6: DCI Use within a Bank

With DCI cascading, one I/O bank (the master bank) must have its VRN/VRP pins connected to external reference resistors. Other I/O banks in the same HP I/O bank column (slave banks) can use DCI standards with the same impedance as the master bank, without connecting the VRN/VRP pins on these slave banks to external resistors. DCI impedance control in cascaded banks is received from the I/O master bank.

Figure 1-7 shows DCI cascading support over multiple I/O banks. Bank B is the master I/O bank, and Banks A and C are considered slave I/O banks.



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Figure 1-7: DCI Cascading Supported Over Multiple I/O Banks

The guidelines when using DCI cascading are as follows:

- DCI cascading is only available through a column of HP I/O banks
- The master and slave SelectIO banks must all reside on the same HP I/O column on the device and can span the entire column unless there is an interposer boundary.
- DCI cascading cannot pass through the interposer boundaries of the larger Virtex-7 devices with stacked silicon interconnect (SSI) technology. This includes the XC7V2000T and XC7VX1140T devices. For these devices, the I/O banks that are separated by these interposer boundaries are shown in the figures of the “Die Level Bank Numbering Overview” section of [UG475: 7 Series FPGAs Packaging and Pinout Specifications](#).
- Master and slave I/O banks must have the same V_{CCO} and V_{REF} (if applicable) voltage.
- I/O banks in the same HP I/O column that are not using DCI (pass-through banks) do not have to comply with the V_{CCO} and V_{REF} voltage rules for combining DCI settings.
- DCI I/O banking compatibility rules must be satisfied across all master and slave banks (for example, only one DCI I/O standard using single termination type is allowed across all master and slave banks).
- To locate I/O banks that reside in the same I/O column, refer to the figures of the “Die Level Bank Numbering Overview” section of [UG475: 7 Series FPGAs Packaging and Pinout Specifications](#).
- For specific information on implementing DC cascading in a design, see [DCI_CASCADE Constraint](#), page 46.
- Xilinx recommends that unused banks be powered up because leaving the V_{CCO} pins of unused I/O banks floating reduces the level of ESD protection on these pins and the I/O pins in the bank. If the bank is unpowered, DCI can still be cascaded through the unpowered bank.

Controlled Impedance Driver (Source Termination)

To optimize signal integrity for high-speed or high-performance applications, extra measures are required to match the output impedance of drivers to the impedance of the transmission lines and receivers. Optimally, drivers must have an output impedance matching the characteristic impedance of the driven line, otherwise reflections can occur due to discontinuities. To solve this issue, designers sometimes use external-source series-termination resistors placed close to the pins of high-strength, low-impedance drivers. The resistance values are chosen such that the sum of the driver's output impedance plus the resistance of the source series-termination resistor roughly equals the impedance of the transmission line.

DCI can provide controlled impedance output drivers to eliminate reflections without requiring the use of an external source-termination resistor. The impedance is set by the external reference resistors with resistance equal to the trace impedance.

The DCI I/O standards supporting the controlled impedance driver are: LVDCI_15, LVDCI_18, HSLVDCI_15, HSLVDCI_18, HSUL_12_DCI, and DIFF_HSUL_12_DCI.

Figure 1-8 illustrates a controlled impedance driver in a 7 series device.

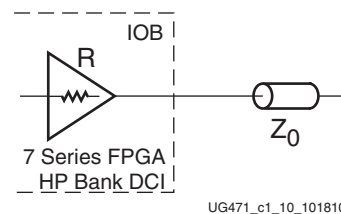


Figure 1-8: Controlled Impedance Driver

Controlled Impedance Driver with Half Impedance (Source Termination)

DCI also provides drivers with one half of the impedance of the reference resistors. This doubling of the reference resistor value reduces the static power consumption through these resistors by a factor of half. The DCI I/O standards supporting controlled impedance drivers with half-impedance are LVDCI_DV2_15 and LVDCI_DV2_18.

Figure 1-9 illustrates a controlled driver with half impedance inside a 7 series device. The reference resistors R must be $2 \times Z_0$ in order to match the impedance of Z_0 .

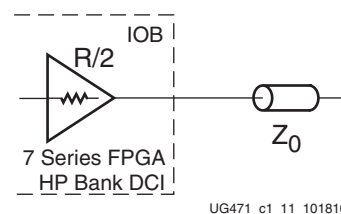


Figure 1-9: Controlled Impedance Driver with Half Impedance

Split-Termination DCI (Thevenin Equivalent Termination to $V_{CCO}/2$)

Some I/O standards (e.g., HSTL and SSTL) require an input termination resistance (R) to a V_{TT} voltage of $V_{CCO}/2$ (see [Figure 1-10](#)).

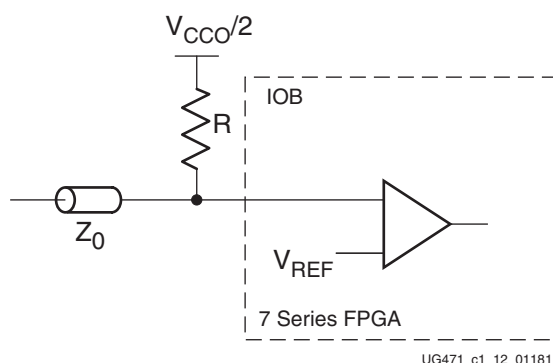


Figure 1-10: Input Termination to $V_{CCO}/2$ without DCI

Split-termination DCI creates a Thevenin equivalent circuit using two resistors of twice the resistance value ($2R$). One terminates to V_{CCO} , the other to ground. Split-termination DCI provides an equivalent termination to $V_{CCO}/2$ using this method. The $2R$ termination resistance is set by the external reference resistors. For example, to achieve the Thevenin equivalent parallel-termination circuit of 50Ω to $V_{CCO}/2$, would require 100Ω external precision resistors at the VRN and VRP pins. The DCI input standards supporting split termination are shown in [Table 1-2](#).

Table 1-2: All DCI I/O Standards Supporting Split-Termination DCI

HSTL_I_DCI	DIFF_HSTL_I_DCI	SSTL18_I_DCI	DIFF_SSTL18_I_DCI
HSTL_I_DCI_18	DIFF_HSTL_I_DCI_18	SSTL18_II_DCI	DIFF_SSTL18_II_DCI
HSTL_II_DCI	DIFF_HSTL_II_DCI	SSTL18_II_T_DCI	DIFF_SSTL18_II_T_DCI
HSTL_II_DCI_18	DIFF_HSTL_II_DCI_18	SSTL15_DCI	DIFF_SSTL15_DCI
HSTL_II_T_DCI	DIFF_HSTL_II_T_DCI	SSTL15_T_DCI	DIFF_SSTL15_T_DCI
HSTL_II_T_DCI_18	DIFF_HSTL_II_T_DCI_18	SSTL135_DCI	DIFF_SSTL135_DCI
		SSTL135_T_DCI	DIFF_SSTL135_T_DCI
		SSTL12_DCI	DIFF_SSTL12_DCI
		SSTL12_T_DCI	DIFF_SSTL12_T_DCI

Figure 1-11 illustrates split-termination DCI inside a 7 series device.

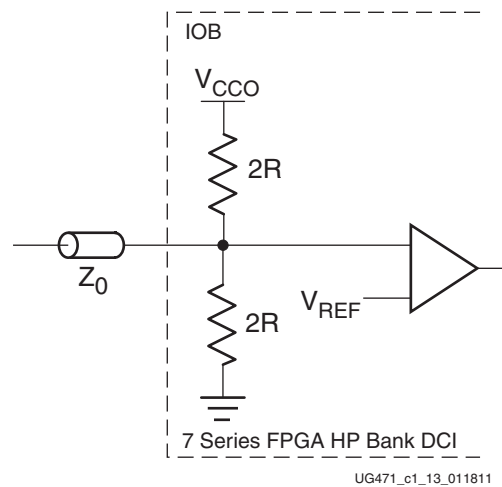


Figure 1-11: Input Termination to $V_{CCO}/2$ Using Split-Termination DCI (External Resistors on VRN, VRP = 2R)

VRN/VRP External Resistance Design Migration Guidelines

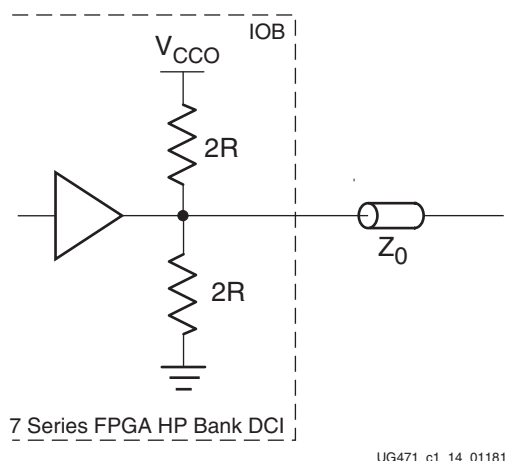
Previous Xilinx FPGA families featuring DCI used a slightly different circuit for calibrating the split-termination impedance from the external reference resistors placed on the VRN and VRP pins. The Virtex-6 FPGA DCI calibrates each leg of the split-termination circuit to be double the external resistor values. For example, in a Virtex-6 device with a target parallel termination of 50Ω to $V_{CCO}/2$ requires 50Ω external resistors on the VRN and VRP pins.

The 7 series FPGAs DCI calibrates each leg of the split termination circuit to be directly equal to the external resistor values. For example, in a 7 series device with a target parallel termination of 50Ω to $V_{CCO}/2$ requires 100Ω external resistors on the VRN and VRP pins. This is particularly important to consider when choosing the VRN and VRP values to be used in the same I/O bank (or multiple cascaded DCI banks) for both controlled-impedance DCI and split-termination DCI standards.

In a Virtex-6 FPGA design with a 50Ω target controlled impedance driver for an LVDCI_18 output and a 50Ω target split-termination receiver for an HSTL_I_DCI_18 input, can be implemented using 50Ω external resistors on the VRN and VRP pins. To migrate this same design to a 7 series FPGA would not change the HSTL_I_DCI_18 I/O standard; however, the external resistors must change to 100Ω and the controlled impedance driver changes to an LVDCI_DIV2_18 output. This example outcome is equivalent; however both resistor values and I/O standard changes are required. A power rating of 0.05W or higher can be safely used for the VRN and VRP external resistors.

DCI and 3-state DCI (T_DCI)

The class-I driver versions of the SSTL and HSTL I/O standards are only supported for unidirectional signaling; they can only be assigned to input-only or output-only pins in a design, not bidirectional pins. The DCI versions of class-I SSTL and HSTL I/O standards only have internal split-termination resistors present on inputs (not outputs). The class-II driver versions of SSTL and HSTL I/O standards are supported for bidirectional and unidirectional signaling; they can be assigned to input, output, or bidirectional pins in a design. The DCI versions of class-II SSTL and HSTL I/O standards always have internal split-termination resistors present on input, outputs, or bidirectional pins. Figure 1-12 illustrates a driver with split termination inside a 7 series device.



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Figure 1-12: Driver with Termination to $V_{CCO}/2$ Using DCI Split Termination (External Resistors on VRN, VRP = $2R$)

When the split-termination is present while driving, DCI only controls the impedance of the termination, but not the driver. However, many applications can benefit from having the split-termination resistors turned off whenever the pin is driving. The 3-state DCI (T_DCI) standards were designed to meet this requirement by turning off the split-termination resistors whenever the output buffer is driving, and turning on the split-termination resistors when the output is in 3-state (such as when receiving or in an idle state). The T_DCI standards can only be assigned to bidirectional pins. For unidirectional input pins, the DCI version of the standard can be assigned. For unidirectional output pins, either the non-DCI or the DCI version can be assigned.

The I/O standards with split-termination DCI resistors that are always present are shown in [Table 1-3](#).

Table 1-3: I/O Standards with Split-Termination DCI Always Present

HSTL_I_DCI ⁽¹⁾	DIFF_HSTL_I_DCI ⁽¹⁾	SSTL18_I_DCI ⁽¹⁾	DIFF_SSTL18_I_DCI ⁽¹⁾
HSTL_I_DCI_18 ⁽¹⁾	DIFF_HSTL_I_DCI_18 ⁽¹⁾	SSTL18_II_DCI	DIFF_SSTL18_II_DCI
HSTL_II_DCI	DIFF_HSTL_II_DCI	SSTL15_DCI ⁽¹⁾	DIFF_SSTL15_DCI ⁽¹⁾
HSTL_II_DCI_18	DIFF_HSTL_II_DCI_18	SSTL135_DCI ⁽¹⁾	DIFF_SSTL135_DCI ⁽¹⁾
		SSTL12_DCI ⁽¹⁾	DIFF_SSTL12_DCI ⁽¹⁾

Notes:

1. The non-class II versions of the HSTL and SSTL I/O standards only have the split-termination DCI resistors present on inputs, not outputs. Bidirectional pin assignments are not allowed for these standards.

The I/O standards with split-termination DCI (T_DCI) that are only enabled when 3-stated are shown in [Table 1-4](#).

Note: The T_DCI standards can only be assigned to bidirectional pins.

Table 1-4: I/O Standards with Split-Termination DCI Only When 3-Stated

HSTL_II_T_DCI	SSTL18_II_T_DCI	DIFF_SSTL18_II_T_DCI
HSTL_II_T_DCI_18	SSTL15_T_DCI	DIFF_SSTL15_T_DCI
DIFF_HSTL_II_T_DCI	SSTL135_T_DCI	DIFF_SSTL135_T_DCI
DIFF_HSTL_II_T_DCI_18	SSTL12_T_DCI	DIFF_SSTL12_T_DCI

DCI in 7 Series FPGAs I/O Standards

DCI supports the standards shown in [Table 1-5](#).

Table 1-5: All 7 Series Device DCI I/O Standards

LVDCI_18	HSTL_I_DCI	DIFF_HSTL_I_DCI	SSTL18_I_DCI	DIFF_SSTL18_I_DCI
LVDCI_15	HSTL_I_DCI_18	DIFF_HSTL_I_DCI_18	SSTL18_II_DCI	DIFF_SSTL18_II_DCI
LVDCI_DV2_18	HSTL_II_DCI	DIFF_HSTL_II_DCI	SSTL18_II_T_DCI	DIFF_SSTL18_II_T_DCI
LVDCI_DV2_15	HSTL_II_DCI_18	DIFF_HSTL_II_DCI_18	SSTL15_DCI	DIFF_SSTL15_DCI
HSLVDCI_18	HSTL_II_T_DCI	DIFF_HSTL_II_T_DCI	SSTL15_T_DCI	DIFF_SSTL15_T_DCI
HSLVDCI_15	HSTL_II_T_DCI_18	DIFF_HSTL_II_T_DCI_18	SSTL135_DCI	DIFF_SSTL135_DCI
			SSTL135_T_DCI	DIFF_SSTL135_T_DCI
			SSTL12_DCI	DIFF_SSTL12_DCI
			SSTL12_T_DCI	DIFF_SSTL12_T_DCI
			HSUL_12_DCI	DIFF_HSUL_12_DCI

To correctly use DCI in 7 series devices:

1. V_{CCO} pins must be connected to the appropriate V_{CCO} voltage based on the IOSTANDARDS in that I/O bank.
2. Correct DCI I/O buffers must be used in the software either by using IOSTANDARD attributes or instantiations in the HDL code.
3. DCI standards require connecting external reference resistors to the multipurpose pins (VRN and VRP). When this is required, these two multipurpose pins cannot be used as general-purpose I/O in the I/O bank using DCI or in the master I/O bank when cascading DCI. Refer to the 7 series FPGA pinout tables for the specific pin locations. Pin VRN must be pulled up to V_{CCO} by its reference resistor. Pin VRP must be pulled down to ground by its reference resistor. An exception to this requirement comes when cascading DCI in slave I/O banks since the VRN and VRP pins can be used as general-purpose I/O.

DCI standards with the controlled impedance driver can be used on input-only signals. For this case, if these pins are the only pins using DCI standards in a given I/O bank, that bank does not require connecting the external reference resistors to the VRP/VRN pins. When these DCI-based I/O standards are the only ones in a bank, the VRP and VRN pins in that bank *can be used* as general-purpose I/O.

- DCI inputs that do not require reference resistors on VRP/VRN are shown in [Table 1-6](#).

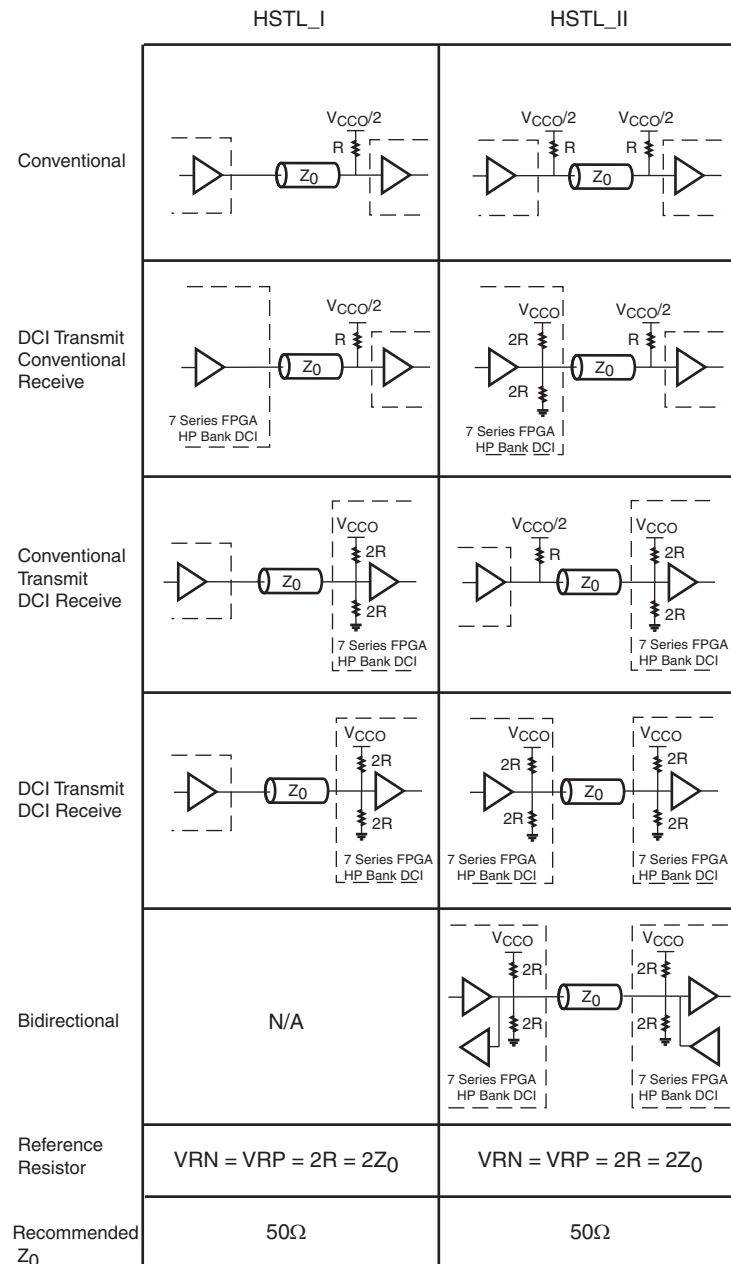
Table 1-6: I/O Standards with DCI Inputs that Do Not Require Reference Resistors

LVDCI_18	LVDCI_DV2_18	HSLVDCI_18	HSUL_12_DCI
LVDCI_15	LVDCI_DV2_15	HSLVDCI_15	DIFF_HSUL_12_DCI

4. The value of the external reference resistors should be selected to give the desired output driver impedance or split-termination impedance. For example, when using LVDCI_15, to achieve a 50Ω output driver impedance, the external reference resistors used on the VRN and VRP pins should each be 50Ω . When using SSTL15_T_DCI, to achieve a 50Ω Thevenin equivalent termination (R) to $V_{CCO}/2$, the external reference resistors should each be 100Ω , which is $(2R)$. Xilinx requires that the exact same value of the resistance be used on the VRP and VRN pins in order to achieve the expected DCI behavior.
5. Follow the DCI I/O banking rules:
 - a. V_{REF} must be compatible for all of the inputs in the same I/O bank or in a group of I/O banks when using DCI cascade.
 - b. V_{CCO} must be compatible for all of the inputs and outputs in the same I/O bank.
 - c. Split termination, controlled impedance driver, and controlled impedance driver with half impedance can co-exist in the same bank.

DCI Usage Examples

- Figure 1-13 provides examples illustrating the use of the HSTL_I_DCI and HSTL_II_DCI I/O standards.
- Figure 1-14 provides examples illustrating the use of the SSTL18_I_DCI and SSTL18_II_DCI I/O standards.

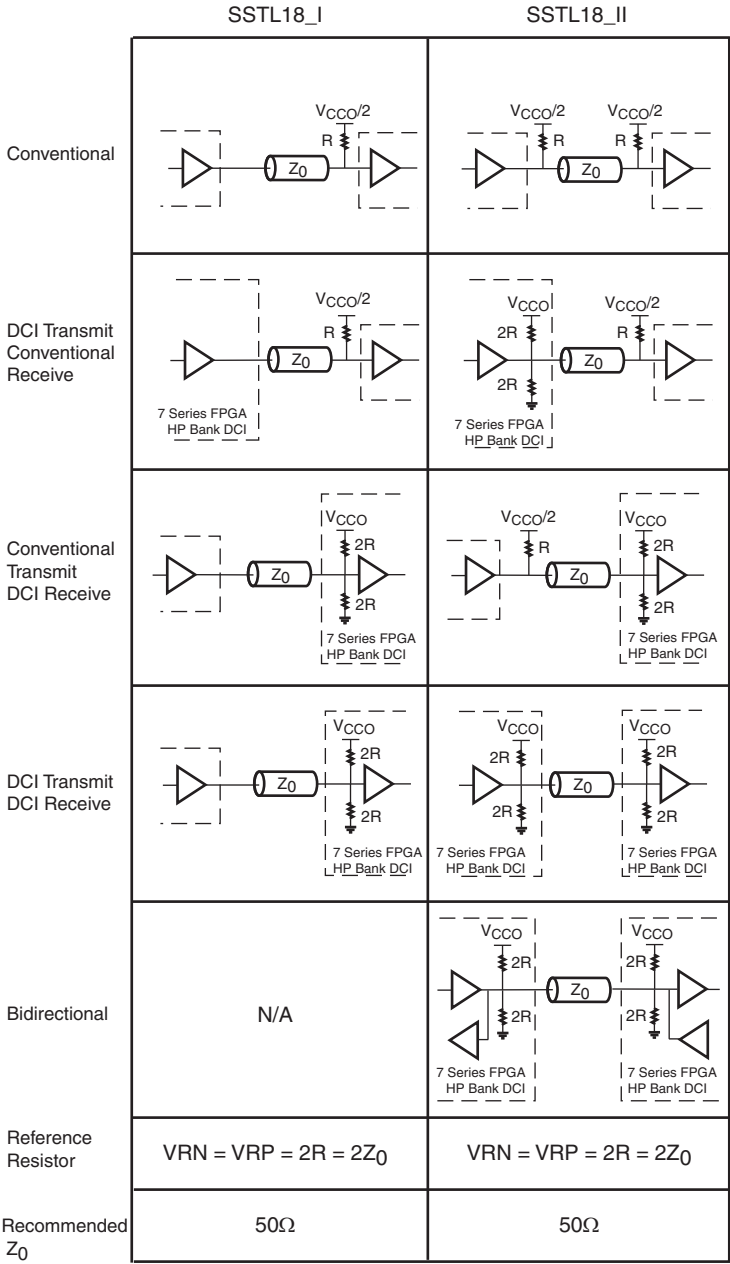


Notes:

- Z_0 is the recommended PCB trace impedance.

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Figure 1-13: HSTL DCI Usage Examples



Notes:
1. Z_0 is the recommended PCB trace impedance.

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Figure 1-14: SSTL DCI Usage Examples

Uncalibrated Split Termination in High-Range I/O Banks (IN_TERM)

The HR I/O banks have an optional on-chip split-termination feature very similar to the 3-state split-termination DCI feature available in the HP I/O banks. Similar to the 3-state split-termination DCI in the HP banks, the option in the HR banks creates a Thevenin equivalent circuit using two internal resistors of twice the target resistance value. One resistor terminates to V_{CCO} and the other to ground, providing a Thevenin equivalent termination circuit to the mid-point $V_{CCO}/2$. The termination is present constantly on inputs, and on bidirectional pins whenever the output buffer is 3-stated. However, an important difference between this uncalibrated split-termination option and 3-state split-termination DCI is that instead of calibrating to external reference resistors on the VRN and VRP pins when using DCI, this feature invokes internal resistors that have no calibration routine to compensate for temperature, process, or voltage variations. This option has target Thevenin equivalent resistance values of 40 Ω , 50 Ω , and 60 Ω .

Another difference from the DCI termination is how this uncalibrated termination is invoked in a design. While the 3-state split-termination DCI option is invoked by assigning the T_DCI I/O standards to I/O pins in HP I/O banks, the uncalibrated split-termination option is invoked by assigning IN_TERM constraints to the I/O pin nets in HR I/O banks. This can be done in several ways, including in the source HDL design, in the UCF, NCF, or XCF files, or in the PlanAhead™ software. For more details, see IN_TERM in [UG625: Constraints Guide](#) (for the ISE® tools) or [UG912: Vivado Design Suite Properties Reference Guide](#).

In HR banks, the IN_TERM constraint can be set to NONE (default), UNTUNED_SPLIT_40, UNTUNED_SPLIT_50, or UNTUNED_SPLIT_60. An example of the UCF syntax is:

```
NET "pad_net_name" IN_TERM = "UNTUNED_SPLIT_50";
```

[Table 1-7](#) shows a list of I/O standards that support the IN_TERM constraint in the HR I/O banks. IN_TERM is not supported in HP I/O banks.

Table 1-7: I/O Standards that Support IN_TERM

HSTL_I	DIFF_HSTL_I	SSTL15_R	DIFF_SSTL15_R
HSTL_II	DIFF_HSTL_II	SSTL15	DIFF_SSTL15
HSTL_I_18	DIFF_HSTL_I_18	SSTL135_R	DIFF_SSTL135_R
HSTL_II_18	DIFF_HSTL_II_18	SSTL135	DIFF_SSTL135
SSTL18_I	DIFF_SSTL18_I		
SSTL18_II	DIFF_SSTL18_II		

7 Series FPGA SelectIO Primitives

The Xilinx software library includes an extensive list of primitives to support a variety of I/O standards available in the 7 series FPGA I/O primitives. The following generic primitives can each support most of the available single-ended I/O standards.

- IBUF (input buffer)
- IBUF_IBUFDISABLE (input buffer with buffer disable control)
- IBUF_INTERMDISABLE (input buffer with buffer disable and IN_TERM disable controls)
- IBUFG (clock input buffer)
- IOBUF (bidirectional buffer)
- IOBUF_DCIEN (bidirectional buffer with DCI disable and input buffer disable)
- IOBUF_INTERMDISABLE (bidirectional buffer with IN_TERM disable and input buffer disable)
- OBUF (output buffer)
- OBUFT (3-state output buffer)

These eight generic primitives can each support most of the available differential I/O standards:

- IBUFDS (differential input buffer)
- IBUFDS_DIFF_OUT (differential input buffer with complementary outputs)
- IBUFDS_DIFF_OUT_IBUFDISABLE (differential input buffer with complementary outputs and buffer disable)
- IBUFDS_DIFF_OUT_INTERMDISABLE (differential input buffer with complementary outputs, buffer disable, and IN_TERM disable)
- IBUFDS_IBUFDISABLE (differential input buffer with buffer disable control)
- IBUFDS_INTERMDISABLE (differential input buffer with buffer disable, and IN_TERM disable)
- IBUFGDS (differential clock input buffer)
- IBUFGDS_DIFF_OUT (differential clock input buffer with complementary outputs)
- IOBUFDS (differential bidirectional buffer)
- IOBUFDS_DCIEN (differential bidirectional buffer with DCI disable and input buffer disable)
- IOBUFDS_DIFF_OUT (differential bidirectional buffer with complementary outputs from the input buffer)
- IOBUFDS_DIFF_OUT_DCIEN (differential bidirectional buffer with complementary outputs from the input buffer, with DCI disable and input buffer disable)
- IOBUFDS_DIFF_OUT_INTERMDISABLE (differential bidirectional buffer with complementary outputs from the input buffer with IN_TERM disable and buffer disable)
- IOBUFDS_INTERMDISABLE (differential bidirectional buffer with buffer disable and IN_TERM disable)
- OBUFDS (differential output buffer)
- OBUFTDS (differential 3-state output buffer)

More information including instantiation techniques and available attributes for these and all other design primitives is available in [UG768: Xilinx 7 Series FPGA Libraries Guide for HDL Designs](#).

IBUF and IBUFG

Signals used as inputs to 7 series devices must use an input buffer (IBUF). The generic 7 series FPGA IBUF primitive is shown in [Figure 1-15](#).

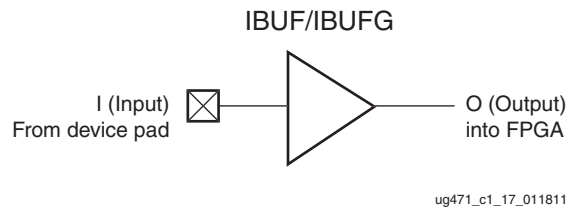


Figure 1-15: Input Buffer Primitives (IBUF/IBUFG)

The IBUF and IBUFG primitives are the same. IBUFGs are used when an input buffer is used as a clock input. In the Xilinx software tools, an IBUFG is automatically placed at clock input sites.

IBUF_IBUFDISABLE

The IBUF_IBUFDISABLE primitive shown in [Figure 1-16](#) is an input buffer with a disable port that can be used as an additional power saving feature for periods when the input is not used.

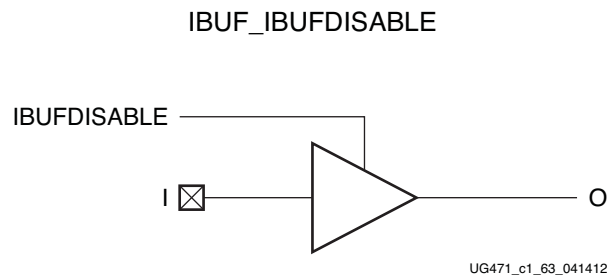


Figure 1-16: Input Buffer With Input Buffer Disable (IBUF_IBUFDISABLE)

The IBUF_IBUFDISABLE primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. This feature can be used to reduce power at times when the I/O is idle. Input buffers that use the V_{REF} power rail (such as SSTL and HSTL) benefit the most from the IBUFDISABLE being set to TRUE because they tend to have higher static power consumption than the non-VREF standards such as LVCMOS and LVTTTL.

IBUF_INTERMDISABLE

The IBUF_INTERMDISABLE primitive shown in [Figure 1-17](#) is available in the HR I/O banks and is similar to the IBUF_IBUFDISABLE primitive in that it has a IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IBUF_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details about this feature.

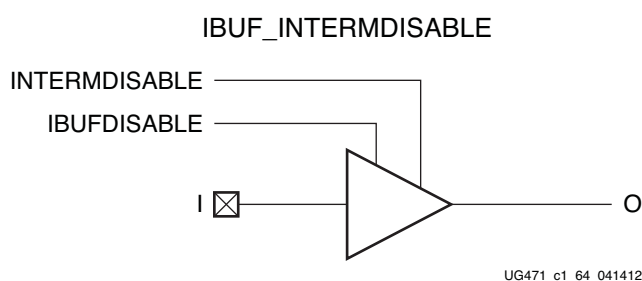


Figure 1-17: Input Buffer With Input Buffer Disable and IN_TERM Disable (IBUF_INTERMDISABLE)

The IBUF_INTERMDISABLE primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), those termination legs are disabled whenever the driver is active (T is low). The IBUF_INTERMDISABLE primitive further allows the termination legs to be disabled whenever the INTERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle. Input buffers that use the VREF power rail (such as SSTL and HSTL) benefit the most from the IBUFDISABLE signal being set to TRUE because they tend to have higher static power consumption than the non-VREF standards such as LVCMOS and LVTTL.

IBUFDS and IBUFGDS

The usage and rules corresponding to the differential primitives are similar to the single-ended SelectIO primitives. Differential SelectIO primitives have two pins to and from the device pads to show the P and N channel pins in a differential pair. N channel pins have a B suffix. The IBUFDS and IBUFGDS primitives are the same, IBUFGDS is used when an differential input buffer is used as a clock input.

[Figure 1-18](#) shows the differential input buffer primitive.

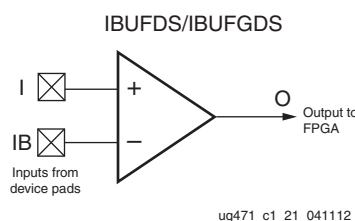


Figure 1-18: Differential Input Buffer Primitives (IBUFDS/IBUFGDS)

IBUFDS_DIFF_OUT and IBUFGDS_DIFF_OUT

Figure 1-19 shows the differential input buffer primitives with complementary outputs (O and OB). IBUFDS_DIFF_OUT and IBUFGDS_DIFF_OUT primitives are the same, IBUFGDS_DIFF_OUT is used for clock inputs. These primitives are only recommended for use by experienced Xilinx designers.

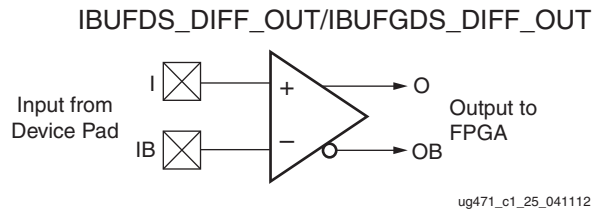


Figure 1-19: **Differential Input Buffer Primitives With Complementary Outputs (IBUFDS_DIFF_OUT/IBUFGDS_DIFF_OUT)**

IBUFDS_DIFF_OUT_IBUFDISABLE

The IBUFDS_DIFF_OUT_IBUFDISABLE primitive shown in Figure 1-20 is a differential input buffer with complementary differential outputs and a disable port that can be used as an additional power saving feature for periods when the input is not used.

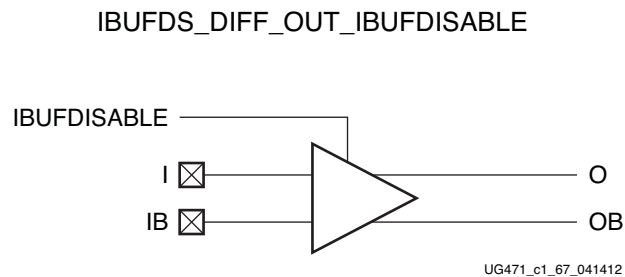


Figure 1-20: **Differential Input Buffer With Complementary Outputs and Input Buffer Disable (IBUFDS_DIFF_OUT_IBUFDISABLE)**

The IBUFDS_DIFF_OUT_IBUFDISABLE primitive can disable the input buffer and force both the O and OB outputs to the fabric high when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. This feature can be used to reduce power whenever the I/O is idle.

IBUFDS_IBUFDISABLE

The IBUFDS_IBUFDISABLE primitive shown in Figure 1-21 is a differential input buffer with a disable port that can be used as an additional power saving feature for periods when the input is not used.

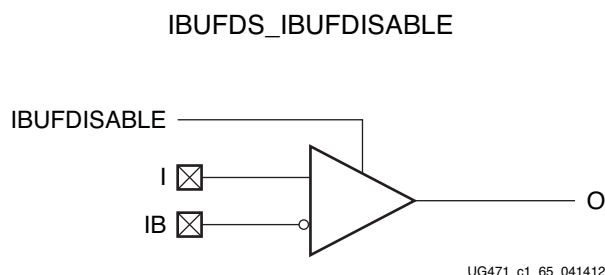


Figure 1-21: Differential Input Buffer With Input Buffer Disable (IBUFDS_IBUFDISABLE)

The IBUFDS_IBUFDISABLE primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. This feature can be used to reduce power whenever the I/O is idle.

IBUFDS_INTERMDISABLE

The IBUFDS_INTERMDISABLE primitive shown in Figure 1-22 is available in the HR I/O banks and is similar to the IBUFDS_IBUFDISABLE primitive in that it has an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IBUFDS_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details on this feature.

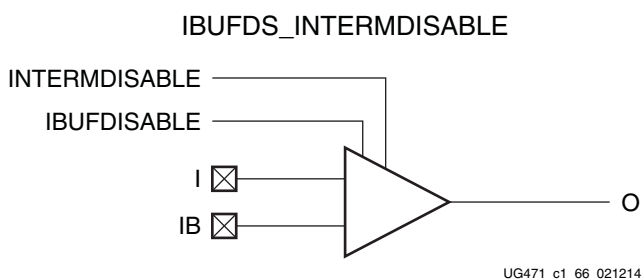


Figure 1-22: Differential Input Buffer With Input Buffer Disable and IN_TERM Disable (IBUFDS_INTERMDISABLE)

The IBUFDS_INTERMDISABLE primitive can disable the input buffer and force the O output to the fabric High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, the IBUFDISABLE input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), this primitive disables the termination legs whenever the INTERMDISABLE signal is asserted High. These features can both be combined to reduce power whenever the input is idle.

IBUFDS_DIFF_OUT_INTERMDISABLE

The IBUFDS_DIFF_OUT_INTERMDISABLE primitive shown in Figure 1-23 is available in the HR I/O banks and is similar to the IBUFDS_IBUFDISABLE primitive in that it has an

IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IBUFDS_DIFF_OUT_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details on this feature.

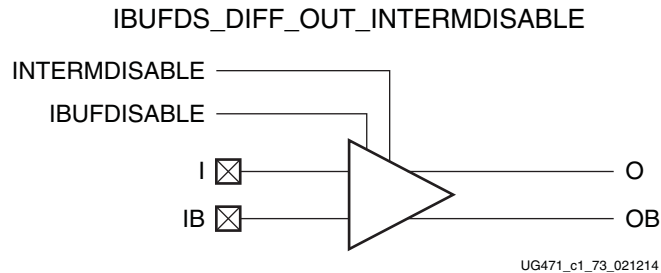


Figure 1-23: Differential Input Buffer With Input Buffer Disable and IN_TERM Disable (IBUFDS_DIFF_OUT_INTERMDISABLE)

The IBUFDS_DIFF_OUT_INTERMDISABLE primitive can disable the input buffer and force both the O and OB outputs to the fabric High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, the IBUFDISABLE input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), this primitive disables the termination legs whenever the INTERMDISABLE signal is asserted High. These features can both be combined to reduce power whenever the input is idle.

IOBUF

The IOBUF primitive is needed when bidirectional signals require both an input buffer and a 3-state output buffer with an active High 3-state T pin. [Figure 1-24](#) shows a generic 7 series FPGA IOBUF. A logic High on the T pin disables the output buffer.

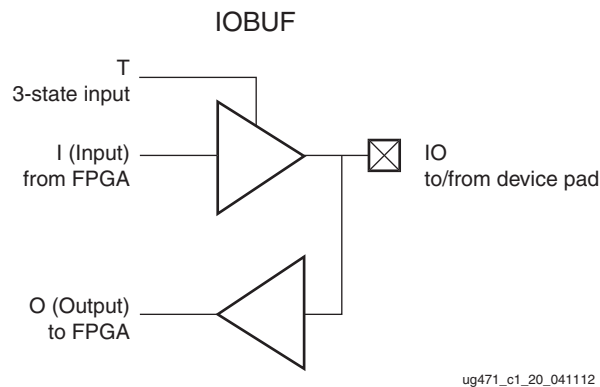


Figure 1-24: Input/Output Buffer Primitive (IOBUF)

IOBUF_DCEN

The IOBUF_DCEN primitive shown in [Figure 1-25](#) is available in the HP I/O banks. It has an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IOBUF_DCEN primitive also has a DCITERMDISABLE port that can be used to manually disable the optional DCI split-termination feature. See

Split-Termination DCI (Thevenin Equivalent Termination to $V_{CCO}/2$) and DCI and 3-state DCI (T_DCI) for more details.

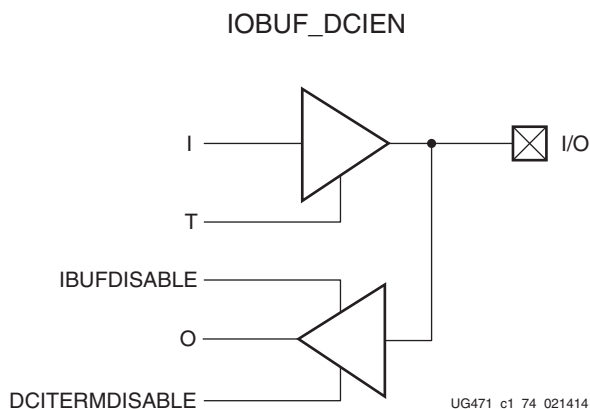


Figure 1-25: Bidirectional Buffer With Input Path Disable and DCI Disable (IOBUF_DCEN)

The IOBUF_DCEN primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the split-termination DCI feature, this primitive disables the termination legs whenever the DCITERMDISABLE signal is asserted High. Only the 3-state DCI I/O standards can be used on bidirectional signals. With 3-state DCI I/O standards, the DCI termination legs turn off whenever the driver is active. The IOBUF_DCEN primitive further allows the termination legs to be disabled whenever the DCITERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

IOBUF_INTERMDISABLE

The IOBUF_INTERMDISABLE primitive shown in Figure 1-26 is available in the HR I/O banks. It has an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IOBUF_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details on this feature.

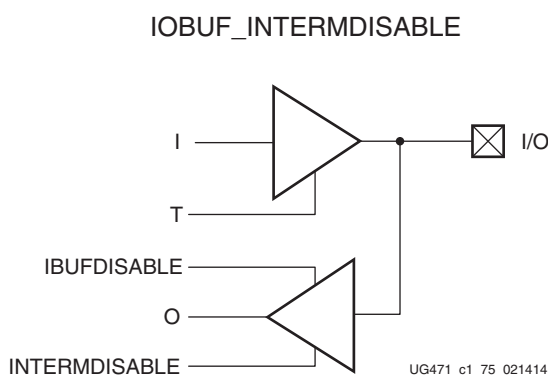


Figure 1-26: Bidirectional Buffer With Input Path Disable and IN_TERM Disable (IOBUF_INTERMDISABLE)

The IOBUF_INTERMDISABLE primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), those termination legs are disabled whenever the driver is active (T is low). This primitive further allows the termination legs to be disabled whenever the INTERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

IOBUFDS

Figure 1-27 shows the differential input/output buffer primitive. A logic High on the T pin disables the output buffer.

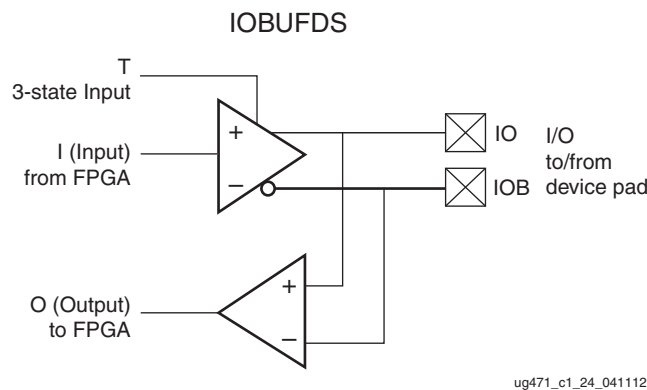


Figure 1-27: Differential Input/Output Buffer Primitive (IOBUFDS)

IOBUFDS_DCEN

The IOBUFDS_DCEN primitive shown in Figure 1-28 is available in the HP I/O banks. It has a IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IOBUFDS_DCEN primitive also has a DCITERMDISABLE port that can be used to manually disable the optional DCI split-termination feature. See [Split-Termination DCI \(Thevenin Equivalent Termination to VCCO/2\)](#) and [DCI and 3-state DCI \(T_DCI\)](#) for more details.

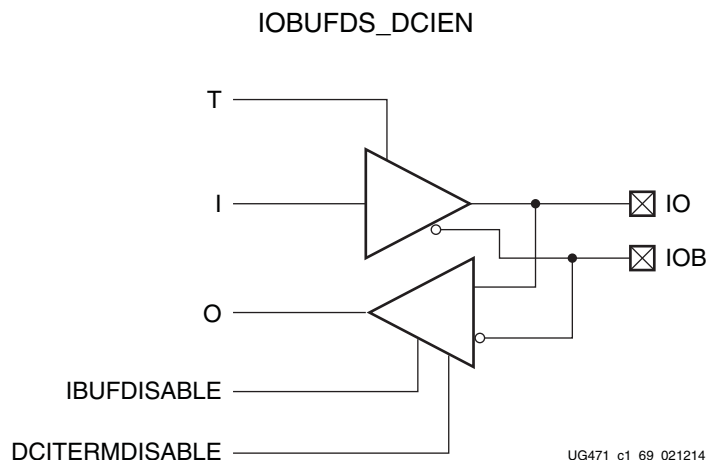


Figure 1-28: Differential Bidirectional Buffer With Input Path Disable and DCI Disable (IOBUFDS_DCIEN)

The IOBUFDS_DCIEN primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the split-termination DCI feature, this primitive disables the termination legs whenever the DCITERMDISABLE signal is asserted High. Only the 3-state DCI I/O standards can be used on bidirectional signals. With 3-state DCI I/O standards, the DCI termination legs turn off whenever the driver is active. The IOBUFDS_DCIEN primitive further allows the termination legs to be disabled whenever the DCITERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

IOBUFDS_DIFF_OUT

[Figure 1-29](#) shows the differential input/output buffer primitive with complementary outputs (O and OB). This primitive is only recommended for use by experienced Xilinx designers with memory interface applications. A logic High on the T pin disables the output buffer.

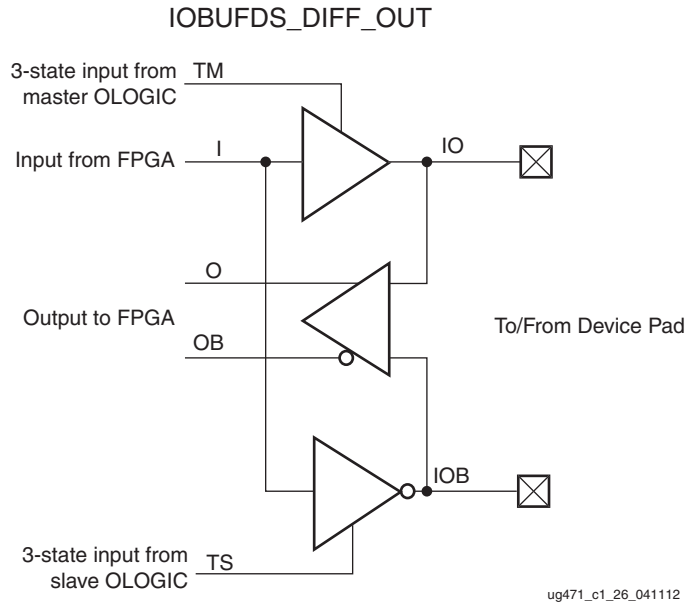


Figure 1-29: Differential Input/Output Buffer Primitive With Complementary Outputs for the Input Buffer (IOBUFDS_DIFF_OUT)

IOBUFDS_DIFF_OUT_DCIEEN

The IOBUFDS_DIFF_OUT_DCIEEN primitive shown in [Figure 1-30](#) is available in the HP I/O banks. It has complementary differential outputs, an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used, and a DCITERMDISABLE port that can be used to manually disable the optional DCI split-termination feature. See [Split-Termination DCI \(Thevenin Equivalent Termination to VCCO/2\)](#) and [DCI and 3-state DCI \(T_DCI\)](#) for more details.

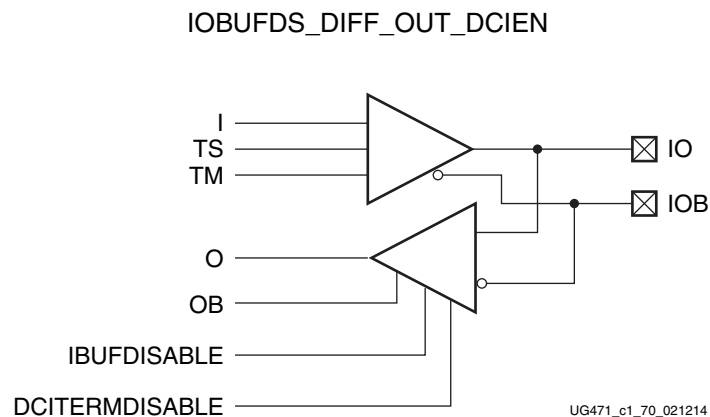


Figure 1-30: Differential Bidirectional Buffer with Complementary Outputs, Input Path Disable, and DCI Disable (IOBUFDS_DCIEEN)

The IOBUFDS_DIFF_OUT_DCIEEN primitive can disable the input buffer and force both the O and OB outputs to the fabric High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the

split-termination DCI feature, this primitive disables the termination legs whenever the DCITERMDISABLE signal is asserted high. Only the 3-state DCI I/O standards can be used on bidirectional signals. With 3-state DCI I/O standards, the DCI termination legs turn off whenever the driver is active (TS is low for the IO output, TM is low for the IOB output). The IOBUFDS_DIFF_OUT_DCIEN primitive further allows the termination legs to be disabled whenever the DCITERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

IOBUFDS_DIFF_OUT_INTERMDISABLE

The IOBUFDS_DIFF_OUT_INTERMDISABLE primitive shown in [Figure 1-31](#) is available in the HR I/O banks. It has an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used.

The IOBUFDS_DIFF_OUT_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details on this feature.

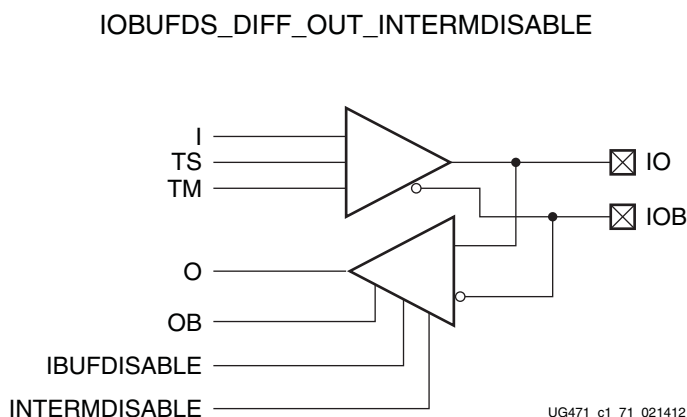


Figure 1-31: Differential Bidirectional Buffer with Complementary Outputs, Input Buffer Disable, and IN_TERM Disable (IOBUFDS_DIFF_OUT_INTERMDISABLE)

The IOBUFDS_DIFF_OUT_INTERMDISABLE primitive can disable the input buffer and force both the O and OB outputs to the fabric High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), those termination legs are also disabled whenever the driver is active (TS is low for the IO output, TM is low for the IOB output). The IOBUFDS_DIFF_OUT_INTERMDISABLE primitive further allows the termination legs to be disabled whenever the INTERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

IOBUFDS_INTERMDISABLE

The IOBUFDS_INTERMDISABLE primitive shown in [Figure 1-32](#) is available in the HR I/O banks. It has an IBUFDISABLE port that can be used to disable the input buffer during periods that the buffer is not being used. The IOBUFDS_INTERMDISABLE primitive also has an INTERMDISABLE port that can be used to disable the optional uncalibrated

split-termination feature. See [Uncalibrated Split Termination in High-Range I/O Banks \(IN_TERM\)](#) for more details on this feature.

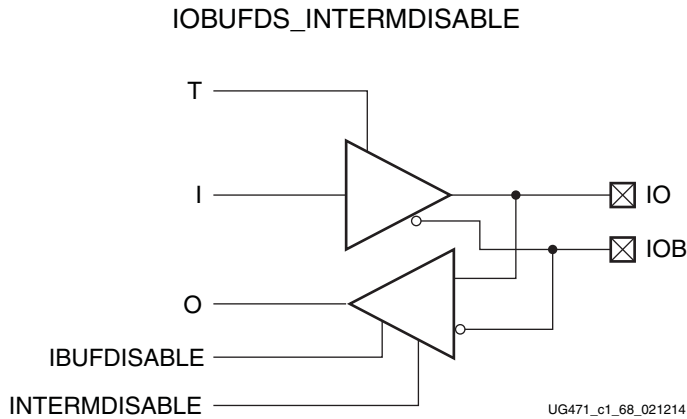


Figure 1-32: Differential Bidirectional Buffer With Input Buffer Disable and IN_TERM Disable (IOBUFDS_INTERMDISABLE)

The IOBUFDS_INTERMDISABLE primitive can disable the input buffer and force the O output to the fabric to a logic High when the USE_IBUFDISABLE attribute is set to TRUE and the IBUFDISABLE signal is asserted High. If USE_IBUFDISABLE is set to FALSE, this input is ignored and should be tied to ground. If the I/O is using the optional uncalibrated split-termination feature (IN_TERM), those termination legs are disabled whenever the driver is active (T is low). This primitive further allows the termination legs to be disabled whenever the INTERMDISABLE signal is asserted High. These features can be combined to reduce power whenever the input is idle for a period of time.

OBUF

An output buffer (OBUF) must be used to drive signals from 7 series devices to external output pads. A generic 7 series FPGA OBUF primitive is shown in [Figure 1-33](#).

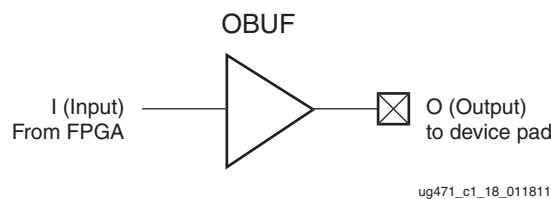


Figure 1-33: Output Buffer Primitive (OBUF)

OBUFDS

[Figure 1-34](#) shows the differential output buffer primitive.

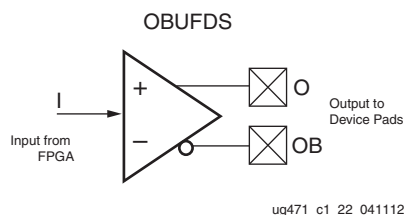


Figure 1-34: Differential Output Buffer Primitive (OBUFDS)

OBUFT

The generic 3-state output buffer OBUFT, shown in Figure 1-35, typically implements 3-state outputs or bidirectional I/O.

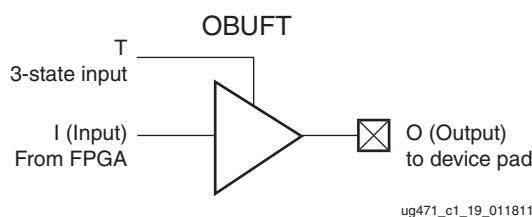


Figure 1-35: 3-State Output Buffer Primitive (OBUFT)

OBUFTDS

Figure 1-36 shows the differential 3-state output buffer primitive.

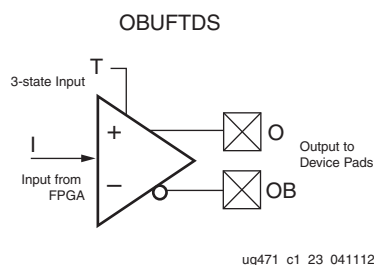


Figure 1-36: Differential 3-state Output Buffer Primitive (OBUFTDS)

7 Series FPGA SelectIO Attributes/Constraints

Access to some 7 series FPGA I/O resource features (e.g., location constraints, input delay, output drive strength, and slew rate) is available through the attributes/constraints associated with these features. For more information about implementing these constraints and attributes as well as others, see [UG625: Constraints Guide](#) (for the ISE tools) or [UG912: Vivado Design Suite Properties Reference Guide](#).

DCI_CASCADE Constraint

The DCI_CASCADE constraint identifies a DCI master bank and its corresponding slave banks. See [DCI Cascading](#), page 22 for more information.

The DCI_CASCADE attribute uses this syntax in the UCF file:

```
CONFIG DCI_CASCADE = "<master> <slave1> <slave2> ...";
```

For example:

```
CONFIG DCI_CASCADE = "11 13 15 17";
```

Location Constraints

The location constraint (LOC) must be used to specify the I/O location of an instantiated I/O primitive. The possible values for the location constraint are all the external port identifiers (e.g., A8, M5, AM6, etc.). These values are device and package size dependent.

The LOC attribute uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> LOC =  
"<EXTERNAL_PORT_IDENTIFIER>";
```

Example:

```
INST MY_IO LOC=R7;
```

IOSTANDARD Attribute

The IOSTANDARD attribute is available to choose the values for an I/O standard for all I/O buffers. The supported I/O standards are listed in the specific 7 series FPGAs data sheets, however, [Table 1-56](#) lists the IOSTANDARD support by bank type (HR, HP, or both). The IOSTANDARD attribute uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> IOSTANDARD="<IOSTANDARD VALUE>";
```

The IOSTANDARD default for single-ended I/O is LVCMOS18, for differential I/Os the default is DIFF_HSTL_II_18.

IBUF_LOW_PWR Attribute

The IBUF_LOW_PWR attribute is available for the following inputs:

- All I/O standards with differential inputs, including:
 - LVDS
 - LVDS_25
 - PPDS_25
 - RSDS_25
 - MINI_LVDS_25
 - BLVDS_25
 - DIFF_HSTL (all variations)
 - DIFF_SSTL (all variations)
 - DIFF_MOBILE_DDR
 - DIFF_HSUL (all variations)

- All V_{REF} -based inputs such as HSLVDCI, SSTL, HSTL, and HSUL
- All input and bidirectional primitives

The IBUF_LOW_PWR attribute allows an optional trade-off between performance and power. The change in the performance is reflected in the delay through the input buffer and can be measured in the static timing report for the design. The change in power can be estimated using the XPower Estimator (XPE) or XPower Analyzer (XPA) tools.

This attribute is set to TRUE by default, which implements the input buffer in the lower-power mode rather than the higher-performance mode. The IBUF_LOW_PWR attribute is applied to the I/O buffer instance and uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> IBUF_LOW_PWR=[TRUE|FALSE];
```

Output Slew Rate Attributes

A variety of attribute values provide the option of choosing the desired slew rate for I/O output buffers. For LVCMOS, LVTTTL, SSTL, HSTL, MOBILE_DDR, and HSUL output buffers, including the differential versions, the desired slew rate can be specified with the SLEW attribute.

It might be important to specify FAST slew rate for high-performance applications such as high-frequency memory interfaces. However, faster slew rates can also lead to reflections or increased noise issues if not properly designed (such as with terminations, transmission line impedance continuity, and cross-coupling).

The allowed values for the SLEW attribute are:

- SLEW = SLOW (Default)
- SLEW = FAST

The SLEW attribute uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> SLEW = "<SLEW_VALUE>";
```

By default, the slew rate for each output buffer is set to SLOW. This is the default used to minimize the power bus transients when switching non-critical signals.

Output Drive Strength Attributes

For LVCMOS and LVTTTL output buffers (OBUF, OBUFT, and IOBUF), the desired drive strength (in mA) can be specified with the DRIVE attribute.

The allowed values for the DRIVE attribute are shown in [Table 1-8](#). The default DRIVE value is 12.

Table 1-8: Allowed Values for the DRIVE Attribute

Standard	HR Bank Current Drive (mA)	HP Bank Current Drive (mA)
LVCMOS12	4, 8, or 12	2, 4, 6, or 8
LVCMOS15	4, 8, 12, or 16	2, 4, 6, 8, 12, or 16
LVCMOS18	4, 8, 12, 16, or 24	2, 4, 6, 8, 12, or 16
LVCMOS25	4, 8, 12, or 16	N/A
LVCMOS33	4, 8, 12, or 16	N/A
LVTTTL	4, 8, 12, 16, or 24	N/A

The DRIVE attribute uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> DRIVE = "<DRIVE_VALUE>";
```

PULLUP/PULLDOWN/KEEPER Attribute for IBUF, OBUFT, and IOBUF

Input buffers (e.g., IBUF), 3-state output (e.g., OBUFT), and bidirectional (e.g., IOBUF) buffers can have a weak pull-up resistor, a weak pull-down resistor, or a weak “keeper” circuit. This feature can be invoked by adding the following possible constraint values to the relevant net of the buffers:

- PULLUP
- PULLDOWN
- KEEPER

The logic level can be set globally for unused I/O using the `BITSTREAM.CONFIG.UNUSEDPIN` property. For more information on implementing these attributes on either individual I/Os or globally for all I/Os, see the pull-up, pull-down, and keeper descriptions in [UG625: Constraints Guide](#) (for the ISE tools) or [UG912: Vivado Design Suite Properties Reference Guide](#). For more information on implementing device configuration bitstream settings, refer to [UG628: Command Line Tools User Guide](#) (for the ISE tools) or [UG908: Vivado Design Suite User Guide: Programming and Debugging](#).

Differential Termination Attribute

The differential termination (`DIFF_TERM`) attribute supports the differential I/O standards when used as inputs. It is used to turn the built-in, 100Ω, differential termination on or off. The on-chip input differential termination in 7 series devices provides major advantages over using a discrete resistor by removing the stub at the receiver completely and therefore greatly improving signal integrity. Additionally it:

- Consumes less power than DCI termination
- Does not use VRP/VRN pins (DCI)

This attribute can be applied to input pins for the following I/O standards:

- LVDS
- LVDS_25
- MINI_LVDS_25
- PPDS_25
- RSDS_25

The V_{CCO} of the I/O bank must be connected to 1.8V for LVDS, and 2.5V for the other differential I/O standards to provide 100Ω of effective differential termination. `DIFF_TERM` is only available for inputs and can *only* be used the appropriate V_{CCO} voltage.

The `DIFF_TERM` attribute can be specified in the UCF constraints file or by setting the appropriate value in the generic map (VHDL) or in-line parameter (Verilog) of the instantiated `IBUFDS`, `IBUFGDS`, `IBUFDS_DIFF_OUT`, or `IOBUFDS_DIFF_OUT` primitives. Refer to the ISE tools language templates or the 7 series FPGA HDL Libraries Guide for the proper syntax for instantiating these primitives and setting the `DIFF_TERM` attribute.

The allowed values for the `DIFF_TERM` attribute are:

- `DIFF_TERM = TRUE`

- `DIFF_TERM = FALSE` (Default)

The `DIFF_TERM` attribute uses the following syntax in the UCF file:

```
INST <I/O_BUFFER_INSTANTIATION_NAME> DIFF_TERM = " [TRUE|FALSE] " ;
```

Internal V_{REF}

The V_{REF} for an I/O bank can be (optionally) generated inside the 7 series FPGA. Internal generation removes the need to provide for a particular V_{REF} supply rail on the printed circuit board (PCB) and frees the multi-purpose V_{REF} pins in a given I/O bank to be used as normal I/O pins. Consider this alternative when the 7 series FPGA is the only device on the board/system requiring a particular V_{REF} voltage supply level, or if there is a shortage of I/O pins in a given I/O bank. The internally generated V_{REF} (`INTERNAL_VREF`) is sourced from the V_{CCAUX} . Each bank has a single V_{REF} plane and each I/O bank can therefore only have the optional `INTERNAL_VREF` set to a single voltage level for the entire bank.

The constraint `INTERNAL_VREF` is assigned to one bank at time.

Example 1: `INTERNAL_VREF` for Bank 14 using `HSTL_II` (1.5V), which requires a 0.75V reference voltage, uses the following constraint:

```
INTERNAL_VREF_BANK14 = 0.75;
```

Example 2: `INTERNAL_VREF` for Bank 15 using `HSTL_II_18` (1.8V), which requires a 0.9V reference voltage, uses the following constraint.

```
INTERNAL_VREF_BANK15 = 0.90;
```

The rules for using `INTERNAL_VREF` are:

- One value of V_{REF} can be set for the bank.
- `INTERNAL_VREF` can only be set to the nominal reference voltage value of a given I/O standard.
- Valid settings of `INTERNAL_VREF` are:
 - 0.60
 - 0.675
 - 0.75
 - 0.90
- When using `INTERNAL_VREF` in a bank, the multi-purpose V_{REF} pins in that bank can be used as normal I/O.

The rules for combining I/O standards in the same bank also apply for `INTERNAL_VREF`.

V_{CCAUX_IO} Constraint

`VCCAUX_IO` is a constraint available for I/O nets and primitives that should be specified in the design if the V_{CCAUX_IO} pins for any HP banks are going to be set to 2.0V. `VCCAUX_IO` defaults to a value of `DONTCARE` but can be set to `NORMAL` (1.8V) or `HIGH` (2.0V). If the V_{CCAUX_IO} pins in a given bank are to be powered at 2.0V, at least one I/O net or primitive in that bank should have its `VCCAUX_IO` constraint set to `HIGH`, and all other I/O nets and primitives that must either be set to `HIGH` or `DONTCARE`. If the V_{CCAUX_IO} pins in a bank are to be powered at 1.8V, at least one I/O net or primitive in that bank should have this constraint set to `NORMAL`, and all other I/O nets or primitives should be set to either `NORMAL` or `DONTCARE`.

In VHDL, the VHDL constraint associated with the IOB primitive instantiation is declared as follows:

```
attribute VCCAUX_IO of {component_name | label_name}:
{component | label} is "{NORMAL | HIGH | DONTCARE}";
```

In Verilog, the Verilog constraint is placed immediately before the module or instantiation of the IOB primitive. The Verilog constraint is specified as follows:

```
(* VCCAUX_IO = {NORMAL | HIGH | DONTCARE} *)
UCF and NCF Syntax
NET "net_name" VCCAUX_IO=(0 | NORMAL | HIGH | DONTCARE);
INST "instance_name " VCCAUX_IO=(NORMAL | HIGH | DONTCARE);
```

7 Series FPGA I/O Resource VHDL/Verilog Examples

The VHDL and Verilog example syntaxes for instantiating 7 series FPGA I/O resources are found in [UG768: Xilinx 7 Series FPGA Libraries Guide for HDL Designs](#).

Supported I/O Standards and Terminations

The following sections provide an overview of the I/O standards and options supported by all 7 series devices.

While most 7 series FPGA I/O supported standards specify a range of allowed voltages, this chapter records typical voltage values only. Detailed information on each specification can be found on the Electronic Industry Alliance JEDEC web site at <http://www.jedec.org>.

LVTTL (Low Voltage TTL)

Table 1-9: Available I/O Bank Type

HR	HP
Available	N/A

LVTTL is a general-purpose EIA/JESD standard for 3.3V applications that uses a single-ended CMOS input buffer and a push-pull output buffer. This standard requires a 3.3V output source voltage (V_{CCO}), but does not require the use of a reference voltage (V_{REF}) or a termination voltage (V_{TT}). This standard is defined by JEDEC (JESD 8C.01).

Sample circuits illustrating both unidirectional and bidirectional LVTTL termination techniques are shown in [Figure 1-37](#) and [Figure 1-38](#). These two diagrams show examples of source-series and parallel terminated topologies.

Figure 1-37 shows unidirectional terminated topologies.

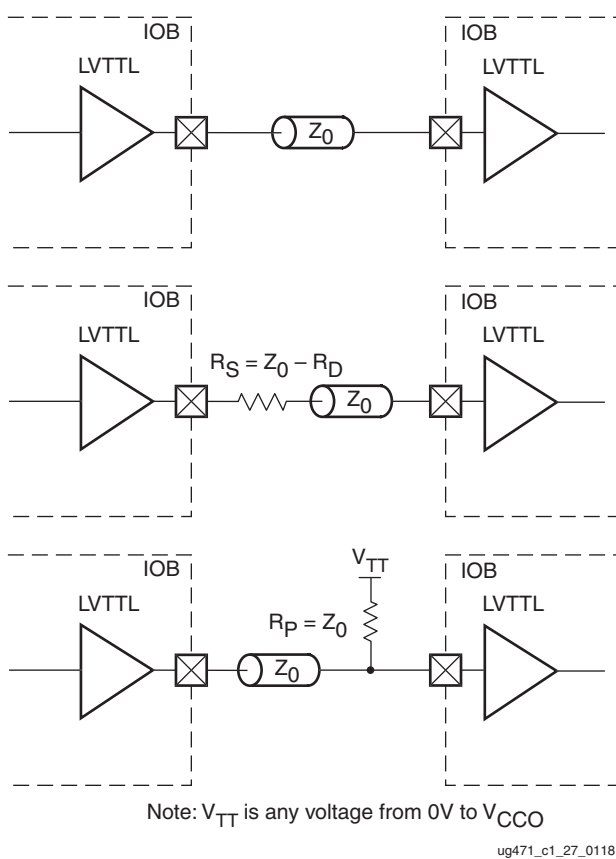
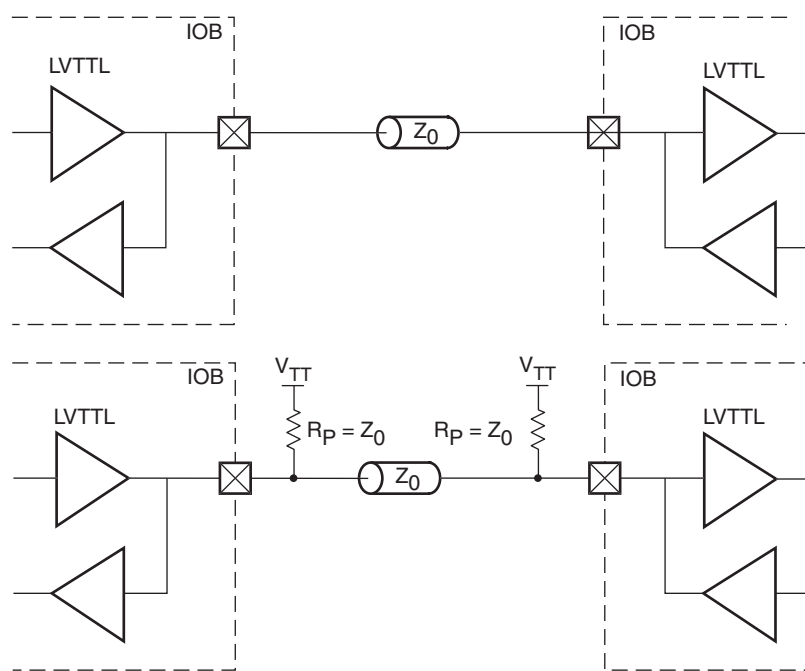


Figure 1-37: LVTTL Unidirectional Termination

Figure 1-38 shows a bidirectional, parallel-terminated topology.



Note: V_{TT} is any voltage from 0V to V_{CCO}

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Figure 1-38: LVTTL Bidirectional Termination

Table 1-10 details the allowed attributes that can be applied to the LVTTL I/O standard. This standard is only available in the HR I/O banks.

Table 1-10: Allowed Attributes for the LVTTL I/O Standards

Attributes	Primitives	
	IBUF/IBUFG	OBUF/OBUFT/ IOBUF
IOSTANDARD	LVTTL	LVTTL
DRIVE	N/A	4, 8, 12 (default), 16, 24
SLEW	N/A	{FAST, SLOW}

LVC MOS (Low Voltage CMOS)

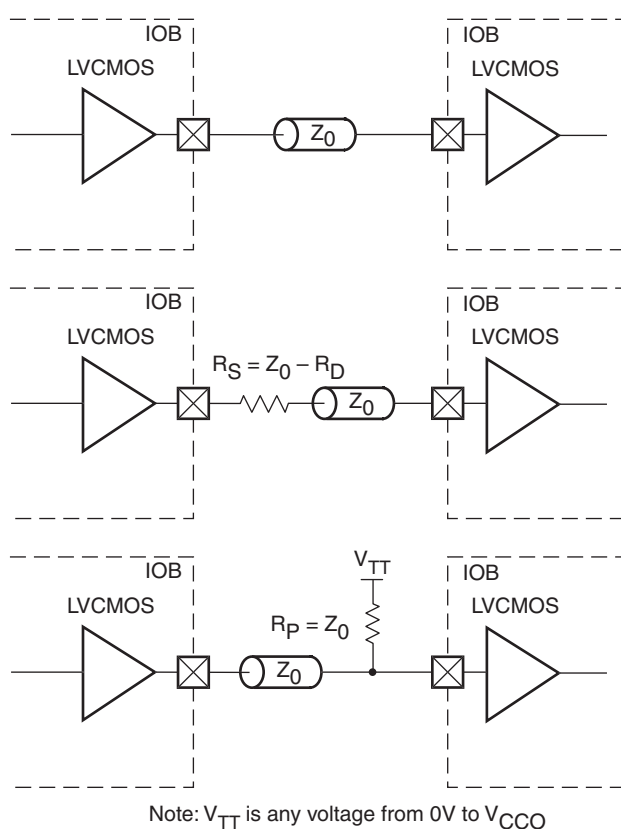
Table 1-11: Available I/O Bank Type

HR	HP
Available	Available

LVC MOS is a widely used switching standard implemented in CMOS transistors. This standard is defined by JEDEC (JESD 8C.01). The LVC MOS standards supported in 7 series FPGAs are: LVC MOS12, LVC MOS15, LVC MOS18, LVC MOS25, and LVC MOS33.

Sample circuits illustrating both unidirectional and bidirectional LVC MOS termination techniques are shown in Figure 1-39 and Figure 1-40. These two diagrams show examples of source-series and parallel terminated topologies.

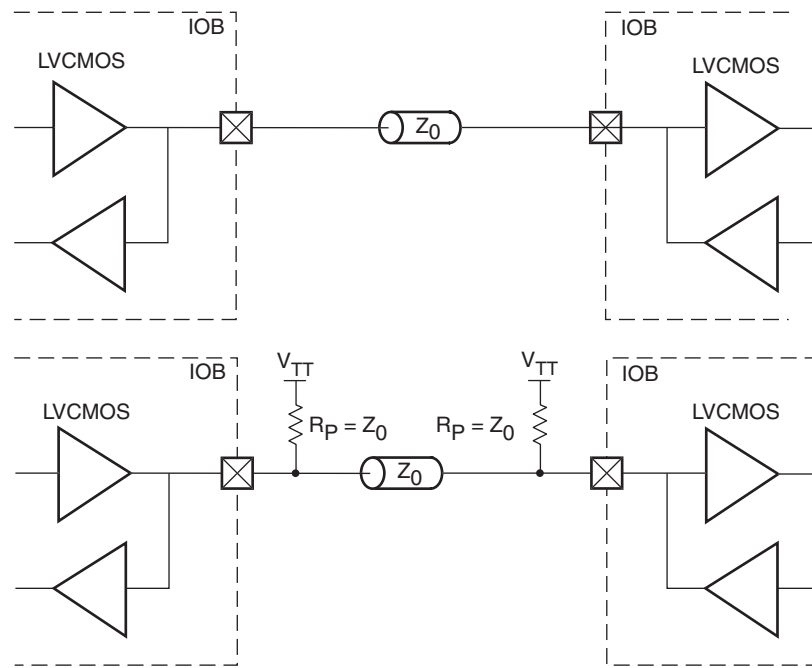
Figure 1-39 shows unidirectional terminated topologies.



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Figure 1-39: LVC MOS Unidirectional Termination

Figure 1-40 shows a bidirectional, parallel-terminated topology.



Note: V_{TT} is any voltage from 0V to V_{CC0}

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Figure 1-40: LVC MOS Bidirectional Termination

Table 1-12 details the allowed attributes that can be applied to the LVC MOS33 and LVC MOS25 I/O standards. These standards are only available in the HR I/O banks.

Table 1-12: Allowed Attributes for the LVC MOS33 and LVC MOS25 I/O Standards

Attributes	Primitives	
	IBUF/IBUFG	OBUF/OBUFT/ IOBUF
IOSTANDARD	LVC MOS33, LVC MOS25	LVC MOS33, LVC MOS25
DRIVE	N/A	4, 8, 12, 16
SLEW	N/A	{FAST, SLOW}

Table 1-13 details the allowed attributes that can be applied to the LVC MOS18 I/O standard. This standard is available in both the HR and HP I/O banks.

Table 1-13: Allowed Attributes for the LVC MOS18 I/O Standard

Attributes	Primitives		
	IBUF/IBUFG	OBUF/OBUFT/IOBUF	
		HP I/O Banks	HR I/O Banks
IOSTANDARD	LVC MOS18	LVC MOS18	LVC MOS18
DRIVE	N/A	2, 4, 6, 8, 12, 16	4, 8, 12, 16, 24
SLEW	N/A	{FAST, SLOW}	{FAST, SLOW}

Table 1-14 details the allowed attributes that can be applied to the LVCMOS15 I/O standard. This standard is available in both the HR and HP I/O banks.

Table 1-14: Allowed Attributes for the LVCMOS15 I/O Standard

Attributes	Primitives		
	IBUF/IBUFG	OBUF/OBUFT/IOBUF	
		HP I/O Banks	HR I/O Banks
IOSTANDARD	LVCMOS15	LVCMOS15	LVCMOS15
DRIVE	N/A	2, 4, 6, 8, 12, 16	4, 8, 12, 16
SLEW	N/A	{FAST, SLOW}	{FAST, SLOW}

Table 1-15 details the allowed attributes that can be applied to the LVCMOS12 I/O standard. This standard is available in both the HR and HP I/O banks.

Table 1-15: Allowed Attributes for the LVCMOS12 I/O Standard

Attributes	Primitives		
	IBUF/IBUFG	OBUF/OBUFT/IOBUF	
		HP I/O Banks	HR I/O Banks
IOSTANDARD	LVCMOS12	LVCMOS12	LVCMOS12
DRIVE	N/A	2, 4, 6, 8	4, 8, 12
SLEW	N/A	{FAST, SLOW}	{FAST, SLOW}

LVDCI (Low-Voltage Digitally Controlled Impedance)

Table 1-16: Available I/O Bank Type

HR	HP
N/A	Available

Using these I/O buffers configures the outputs as controlled impedance drivers. The receiver of LVDCI is identical to a LVCMOS receiver. Some I/O standards, such as LVCMOS, must have a drive impedance that matches the characteristic impedance of the driven line. The HP I/O banks in the 7 series devices provide a controlled impedance output driver to provide series termination without external-source termination resistors. The impedance is set by the common external reference resistors, with resistance equal to the trace characteristic impedance, Z_0 .

Sample circuits illustrating both unidirectional and bidirectional topologies for a controlled impedance driver are shown in Figure 1-41 and Figure 1-42. The DCI I/O standards supporting a controlled impedance driver are: LVDCI_15 and LVDCI_18.

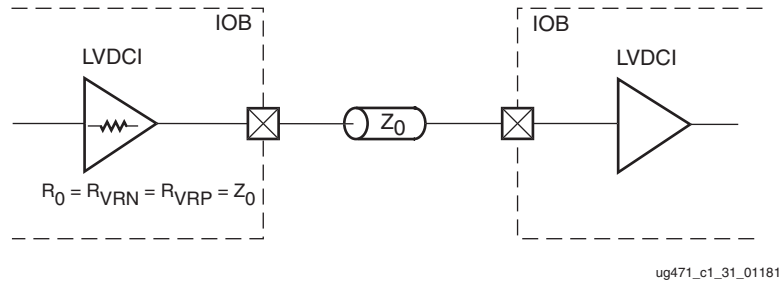


Figure 1-41: Unidirectional Controlled Impedance Driver Topology

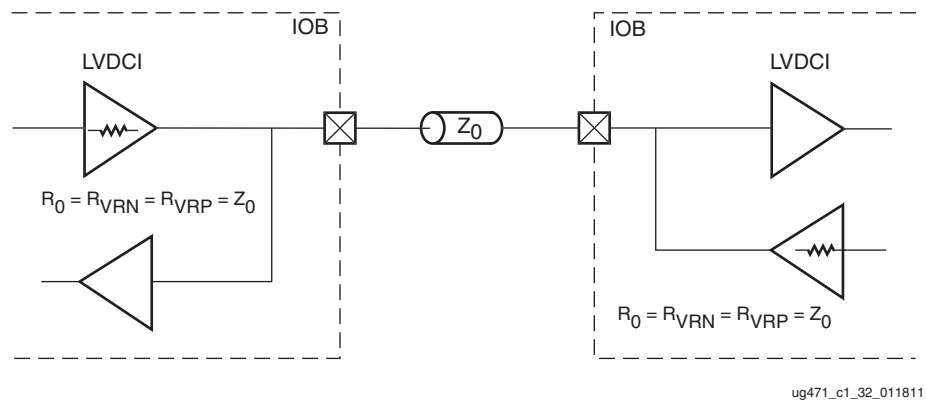


Figure 1-42: Bidirectional Controlled Impedance Driver Topology

LVDCI_DV2

Table 1-17: Available I/O Bank Type

HR	HP
N/A	Available

A controlled impedance driver with half impedance (source termination) can also provide drivers with one half of the impedance of the reference resistors. This allows reference resistors to be twice as large, thus reducing static power consumption through VRN/VRP. The I/O standards supporting a controlled impedance driver with half impedance are: LVDCI_DV2_15 and LVDCI_DV2_18. [Figure 1-43](#) and [Figure 1-44](#) illustrate a controlled driver with half impedance unidirectional topologies.

To match the drive impedance to Z_0 when using a driver with half impedance, the reference resistor R must be twice Z_0 .

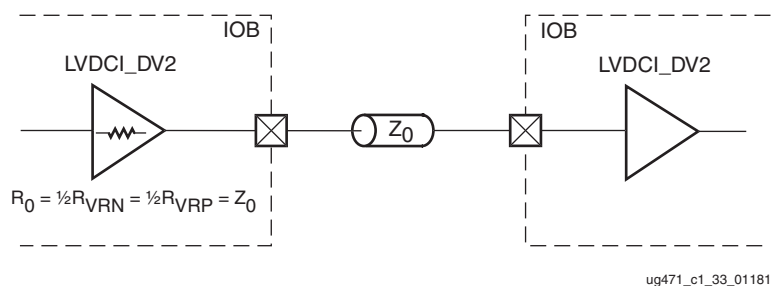


Figure 1-43: Unidirectional Controlled Impedance Driver with Half Impedance Topology

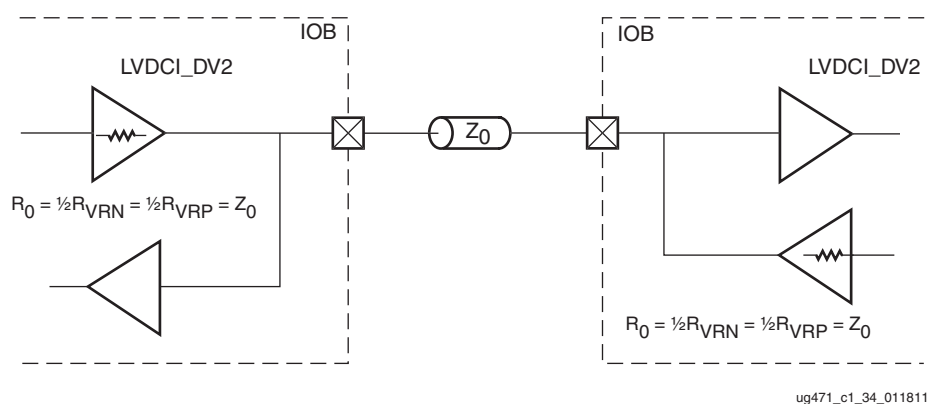


Figure 1-44: Bidirectional Controlled Impedance Driver with Half Impedance Topology

There are no optional current drive strength settings for LVDCI drivers. When the driver impedance is one-half of the VRN/VRP reference resistors, it is indicated by the addition of DV2 to the attribute name.

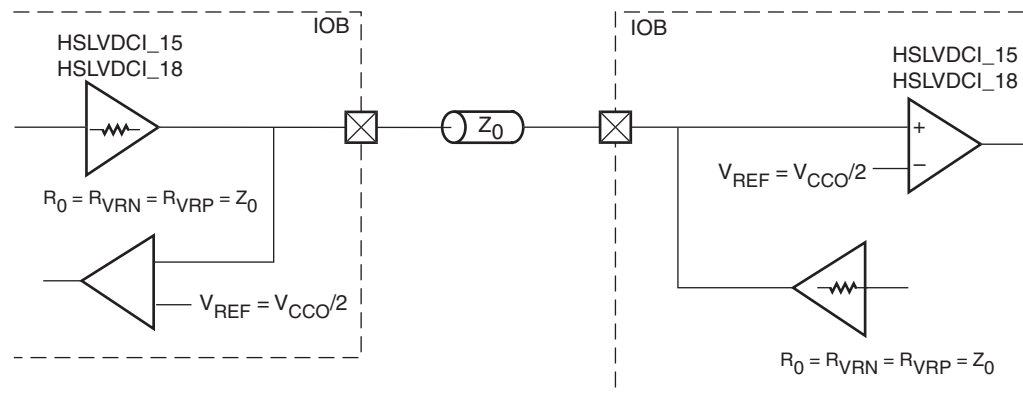
HSLVDCI (High-Speed LVDCI)

Table 1-18: Available I/O Bank Type

HR	HP
N/A	Available

The HSLVDCI standard is intended for bidirectional use. The driver is identical to LVDCI, while the input is identical to HSTL and SSTL. By using a V_{REF} -referenced input, HSLVDCI allows greater input sensitivity at the receiver than when using a single-ended LVCMOS-type receiver.

A sample circuit illustrating bidirectional termination techniques for an HSLVDCI controlled impedance driver is shown in Figure 1-45. The DCI I/O standards supporting a controlled impedance driver with a V_{REF} referenced input are: HSLVDCI_15 and HSLVDCI_18.



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Figure 1-45: HSLVDCI Controlled Impedance Driver with Bidirectional Termination

For electrical specifications, refer to the LVDCI V_{OH} and V_{OL} entries in the 7 series data sheets.

HSTL (High-Speed Transceiver Logic)

The high-speed transceiver logic (HSTL) standard is a general purpose high-speed bus standard defined by JEDEC (JESD8-6). The HSTL standards have four variations (classes). To support clocking high-speed memory interfaces, differential versions are also available. 7 series FPGA I/O supports class-I for the 1.2V version of HSTL (in HP banks), and class-I and II for the 1.5V and 1.8V versions, including the differential versions. The differential versions of the standard require a differential amplifier input buffer and a push-pull output buffer. The HP I/O banks also support DCI versions.

HSTL_I and HSTL_I_18

Table 1-19: Available I/O Bank Type

HR	HP
Available	Available

HSTL_I and HSTL_I_18 use $V_{CCO}/2$ as a parallel-termination voltage (V_{TT}) and are intended for use in unidirectional links.

HSTL_I_12

Table 1-20: Available I/O Bank Type

HR	HP
N/A	Available

HSTL_I_12 uses $V_{CCO}/2$ as a parallel-termination voltage (V_{TT}) and is intended for use in unidirectional links.

HSTL_I_DCI and HSTL_I_DCI_18

Table 1-21: Available I/O Bank Type

HR	HP
N/A	Available

HSTL_I_DCI and HSTL_I_DCI_18 provide on-chip split thevenin termination powered from V_{CCO} , creating an equivalent parallel-termination voltage (V_{TT}) of $V_{CCO}/2$, and are intended for use in unidirectional links.

HSTL_II and HSTL_II_18

Table 1-22: Available I/O Bank Type

HR	HP
Available	Available

HSTL_II and HSTL_II_18 use $V_{CCO}/2$ as a parallel-termination voltage (V_{TT}) and are intended for use in bidirectional links.

HSTL_II_DCI and HSTL_II_DCI_18

Table 1-23: Available I/O Bank Type

HR	HP
N/A	Available

HSTL_II_DCI and HSTL_II_DCI_18 provide on-chip split thevenin termination powered from V_{CCO} , creating an equivalent termination voltage of $V_{CCO}/2$, and are intended for use in bidirectional links.

HSTL_II_T_DCI and HSTL_II_T_DCI_18

Table 1-24: Available I/O Bank Type

HR	HP
N/A	Available

HSTL_II_T_DCI and HSTL_II_T_DCI_18 provide on-chip split-thevenin termination powered from V_{CCO} that creates an equivalent termination voltage of $V_{CCO}/2$ at the receiver when the driver is 3-stated. When the driver is not 3-stated, these two standards do not have termination.

DIFF_HSTL_I and DIFF_HSTL_I_18

Table 1-25: Available I/O Bank Type

HR	HP
Available	Available

Differential HSTL class-I pairs complementary single-ended HSTL_I type drivers with a differential receiver, and are intended to be used in unidirectional links.

DIFF_HSTL_I_DCI and DIFF_HSTL_I_DCI_18

Table 1-26: Available I/O Bank Type

HR	HP
N/A	Available

Differential HSTL class-I pairs complementary single-ended HSTL_I type drivers with a differential receiver, including on-chip split-thevenin termination, and are intended to be used in unidirectional links.

DIFF_HSTL_II and DIFF_HSTL_II_18

Table 1-27: Available I/O Bank Type

HR	HP
Available	Available

Differential HSTL class-II pairs complementary single-ended HSTL_II type drivers with a differential receiver. Differential HSTL class-II is intended to be used in bidirectional links.

Differential HSTL can also be used for differential clock and DQS signals in memory interface designs.

DIFF_HSTL_II_DCI and DIFF_HSTL_II_DCI_18

Table 1-28: Available I/O Bank Type

HR	HP
N/A	Available

Differential HSTL class-II pairs complementary single-ended HSTL_II type drivers with a differential receiver, including on-chip split-thevenin termination. Differential HSTL class-II is intended to be used in bidirectional links. Differential HSTL can also be used for differential clock and DQS signals in memory interface designs.

DIFF_HSTL_II_T_DCI and DIFF_HSTL_II_T_DCI_18

Table 1-29: Available I/O Bank Type

HR	HP
N/A	Available

These standards are almost the same as the DIFF_HSTL_II_DCI and DIFF_HSTL_II_DCI_18 standards except that the termination is only present when the driver is 3-stated.

HSTL Class I (1.2V, 1.5V, or 1.8V)

[Figure 1-46](#) shows a sample circuit illustrating a termination technique for HSTL class-I for the 1.2V, 1.5V, or 1.8V versions. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.2V, 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the DCI standards.

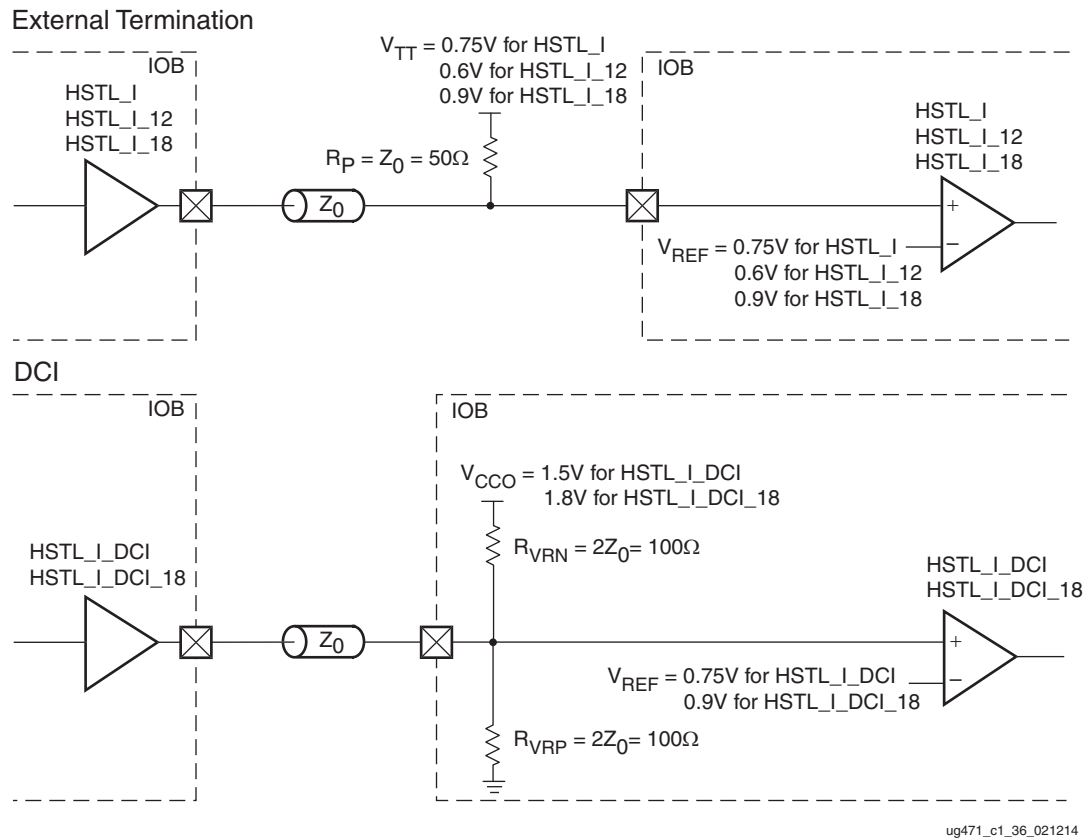


Figure 1-46: HSTL Class I (1.2V, 1.5V, or 1.8V) Termination

Differential HSTL Class I

Figure 1-47 shows a sample circuit illustrating a termination technique for differential HSTL class-I (1.5V or 1.8V) with unidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable.

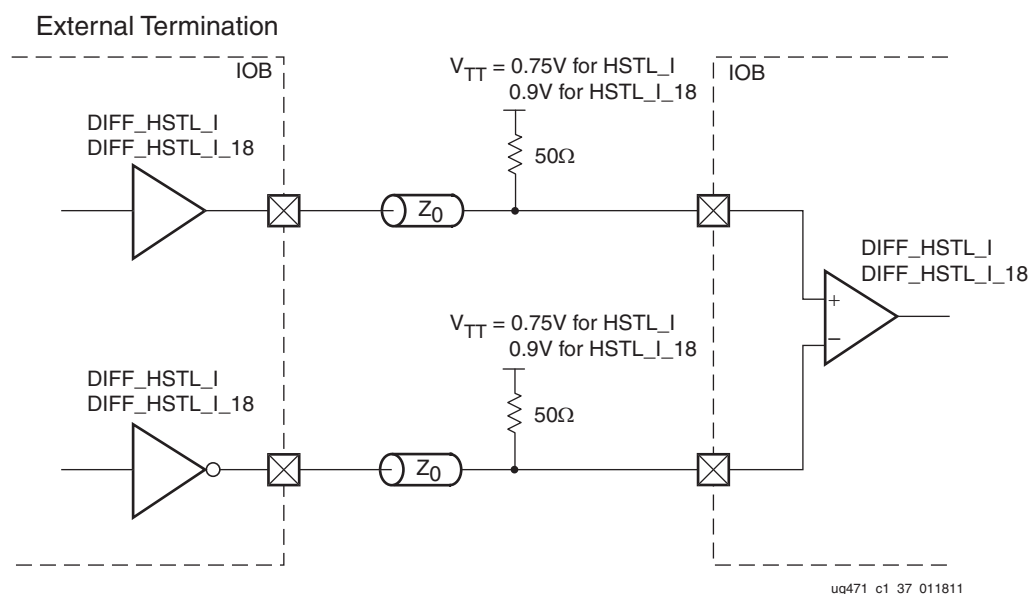


Figure 1-47: Differential HSTL Class I (1.5V or 1.8V) Unidirectional Termination

Figure 1-48 shows a sample circuit illustrating a termination technique for differential HSTL class-I (1.5V or 1.8V) with unidirectional DCI termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support these DCI standards.

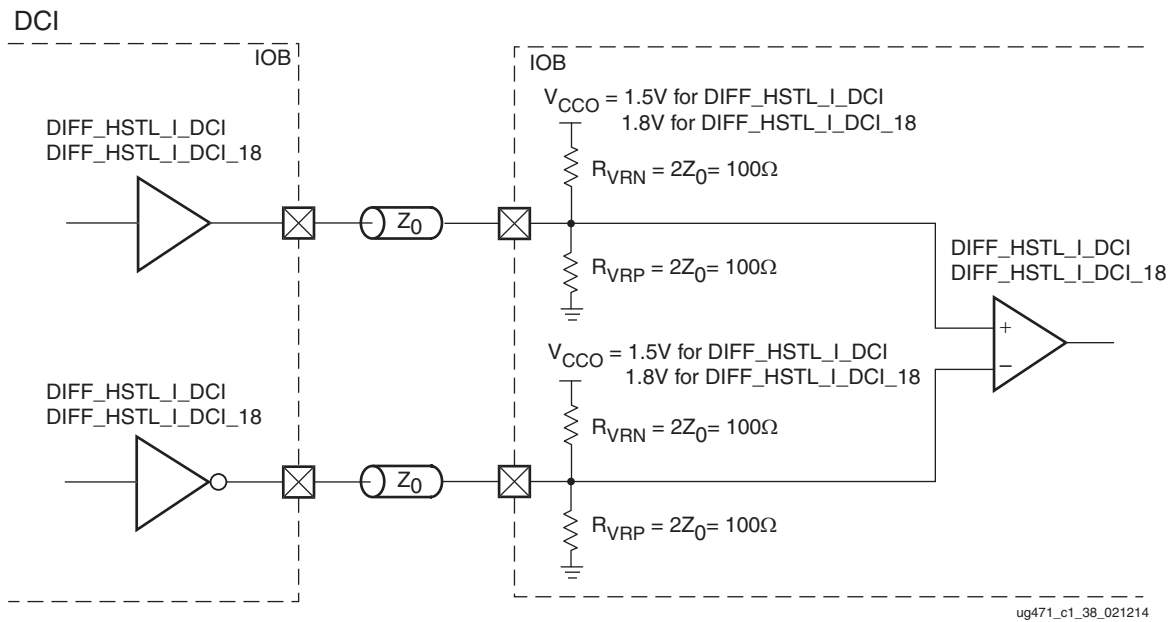
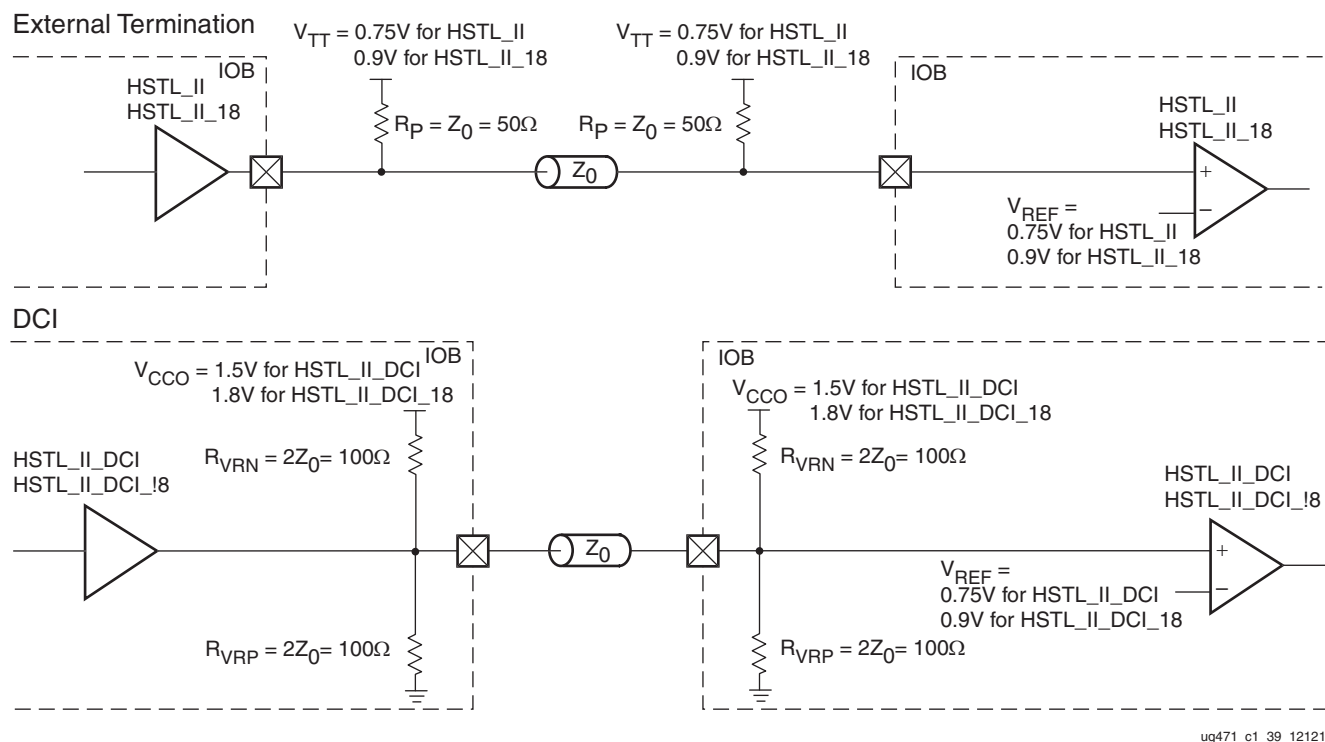


Figure 1-48: Differential HSTL Class I (1.5V or 1.8V) DCI Unidirectional Termination

HSTL Class II

Figure 1-49 shows a sample circuit illustrating a termination technique for HSTL class-II (1.5V or 1.8V) with unidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the DCI standards. The internal split-termination resistors are always present, independent of whether the drivers are 3-stated.



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Figure 1-49: HSTL Class II (1.5V or 1.8V) Unidirectional Termination

Figure 1-50 shows a sample circuit illustrating a termination technique for HSTL class-II (1.5V or 1.8V) with bidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the DCI standards. The internal split-termination resistors are always present, independent of whether the drivers are 3-stated.

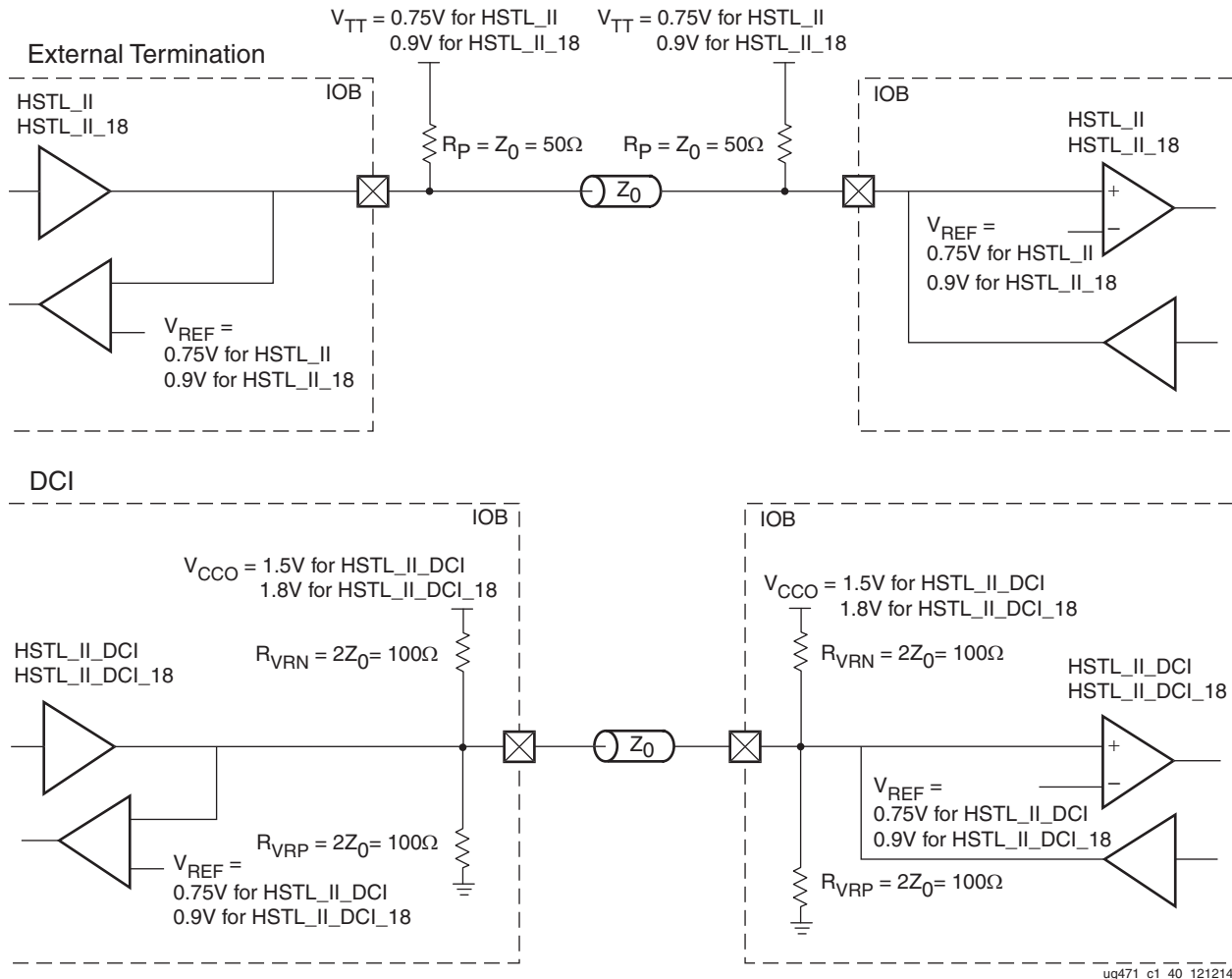


Figure 1-50: HSTL Class II (1.5V or 1.8V) Bidirectional Termination

Differential HSTL Class II

Figure 1-51 shows a sample circuit illustrating a termination technique for differential HSTL class-II (1.5V or 1.8V) with unidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable.

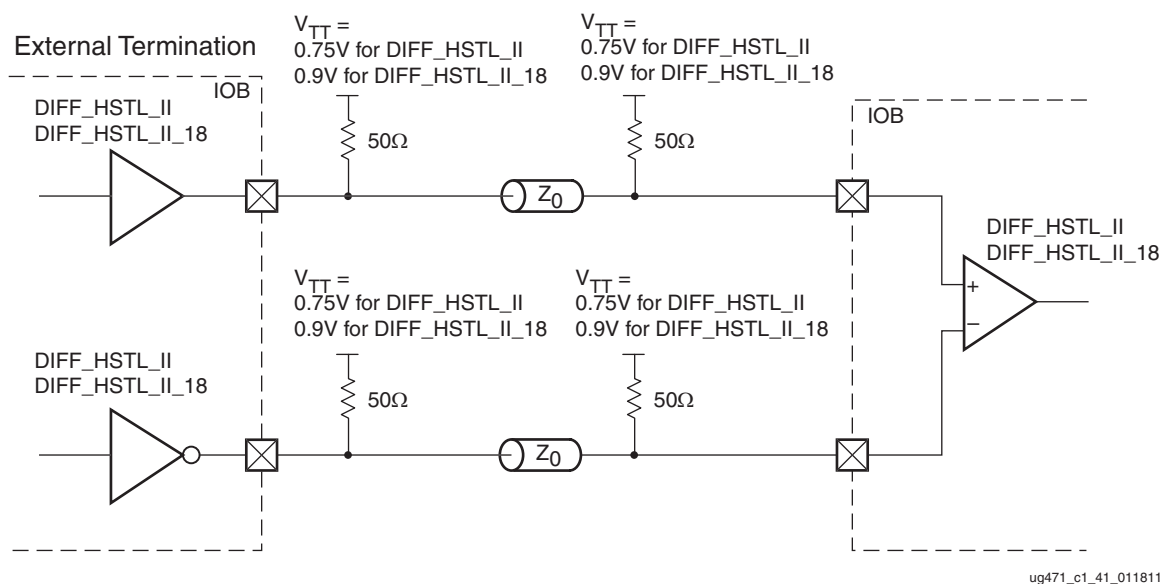


Figure 1-51: Differential HSTL Class II (1.5V or 1.8V) Unidirectional Termination

Figure 1-52 shows a sample circuit illustrating a termination technique for differential HSTL class-II (1.5V or 1.8V) with unidirectional DCI termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the DCI standards. The internal split-termination resistors are always present, independent of whether the drivers are 3-stated.

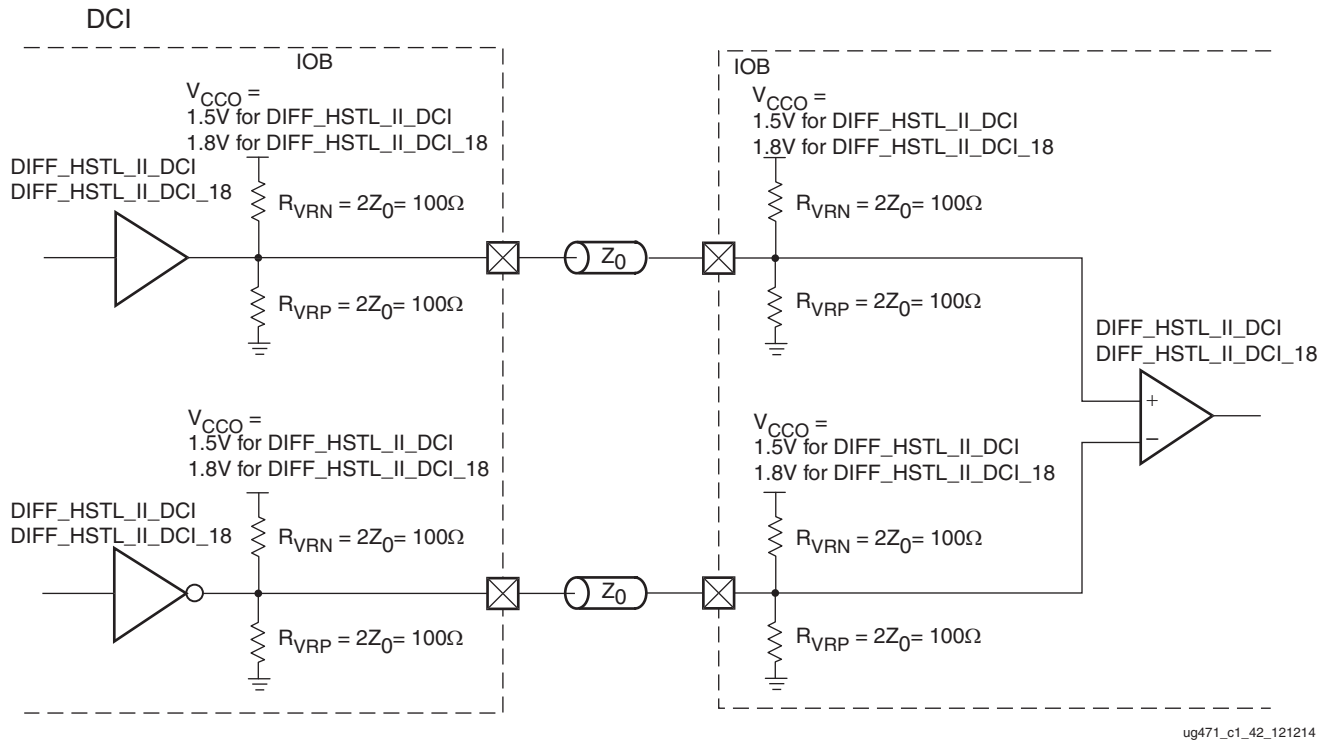


Figure 1-52: Differential HSTL Class II (1.5V or 1.8V) DCI Unidirectional Termination

Figure 1-53 shows a sample circuit illustrating a termination technique for differential HSTL class-II (1.5V or 1.8V) with bidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable.

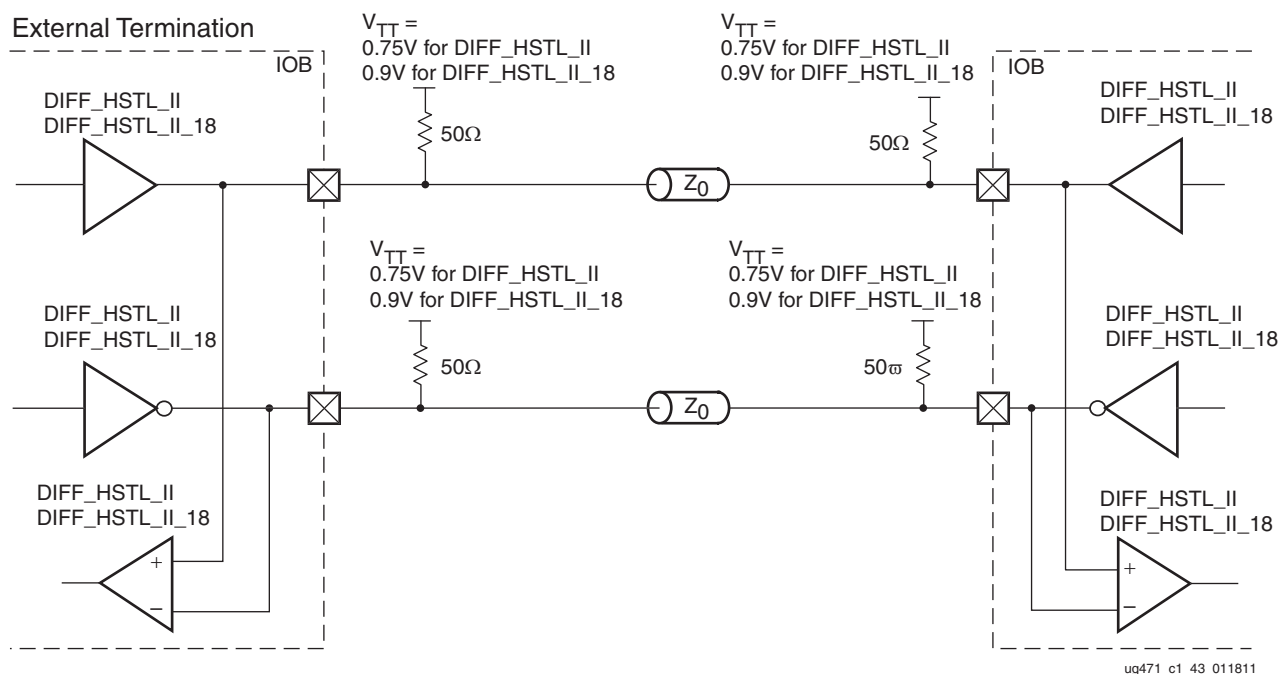


Figure 1-53: Differential HSTL Class II (1.5V or 1.8V) Bidirectional Termination

Figure 1-54: Differential HSTL Class II (1.5V or 1.8V) DCI Bidirectional Termination

HSTL_II_T_DCI (1.5V or 1.8V) Split-Thevenin Termination (3-state)

Figure 1-55 shows a sample circuit illustrating a termination technique for HSTL_II_T_DCI (1.5V) and HSTL_II_T_DCI_18 (1.8V) with on-chip split-thevenin termination. In this bidirectional case, when 3-stated, the termination is invoked on the receiver and not on the driver. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the T_DCI standards. The internal split-termination resistors are only present when the output buffers are 3-stated.

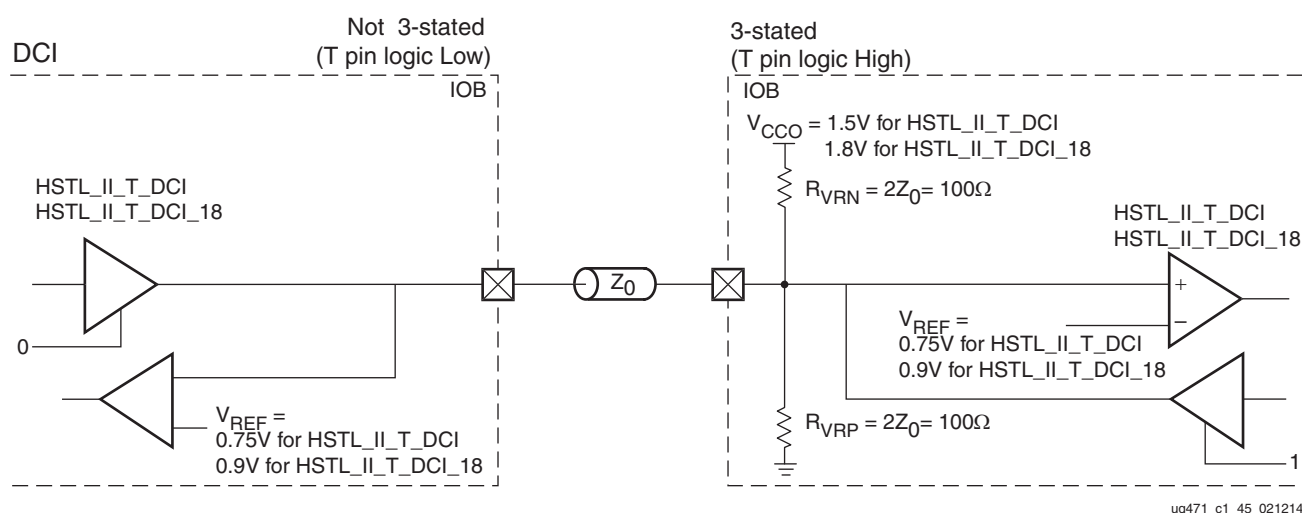
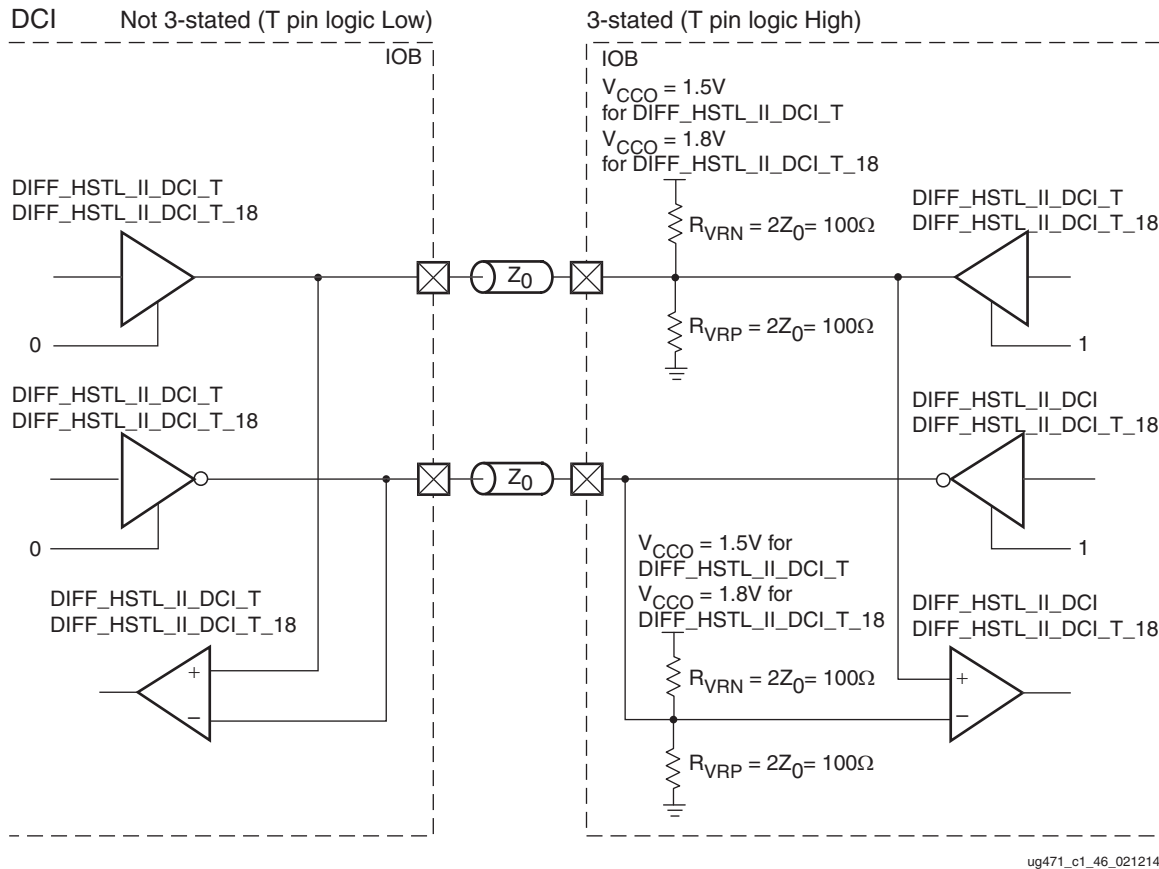


Figure 1-55: HSTL_II_T_DCI (1.5V) and HSTL_II_T_DCI_18 (1.8V) Split-Thevenin Termination (3-state)

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Figure 1-56 shows a sample circuit illustrating a termination technique for differential HSTL class-II (1.5V or 1.8V) with on-chip split-thevenin termination. In a specific circuit, all drivers and receivers must be at the same voltage level (either 1.5V or 1.8V); they are not interchangeable. Only HP I/O banks support the T_DCI standards. The internal split-termination resistors are only present when the output buffers are 3-stated.



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Figure 1-56: Differential HSTL Class II (1.5V or 1.8V) DCI with Split-Thevenin Termination (3-state)

SSTL (Stub-Series Terminated Logic)

The Stub-Series Terminated Logic (SSTL) for 1.8V (SSTL18), 1.5V (SSTL15), and 1.35V (SSTL135) are I/O standards used for general purpose memory buses.

While example termination techniques are discussed in this section, the optimal termination schemes for a given memory interface are determined using signal-integrity analysis of the actual PCB topology including the memory devices used, the board layout, and transmission line impedances. Xilinx provides both IBIS model files and encrypted HSPICE model files for all of the I/O standards. 7 series FPGAs support these SSTL standards for both single-ended signaling and differential signaling. The differential versions use a true differential amplifier input buffer and complementary push-pull output buffers. The 3-state DCI (T_DCI) versions of these standards are the preferred I/O standards to use for memory interfaces implemented in the HP I/O banks. The IN_TERM (untuned internal termination) attribute is recommended for interfaces implemented in HR I/O banks.

New to the 7 series FPGAs is the option to specify the slew rate of the output buffer for all of the memory interface related I/O standards: HSTL, SSTL, HSUL, and MOBILE_DDR. This is similar to the Xilinx LVCMOS and LVTTL I/O standards, where both slow and fast slew options are available. Although slow is the default setting, for most fast interface frequencies the fast slew option is preferable. However, the optimal selection is determined through signal-integrity analysis.

SSTL18 is defined by the JEDEC standard JESD8-15, and is used for DDR2 SDRAM memory interfaces. The class-I driver can only be used in unidirectional topologies (no bidirectional support). The class-II driver can be used for both for bidirectional and unidirectional signaling. For some topologies (such as short, point-to-point interfaces), the class-I driver can result in reduced overshoot and better signal integrity.

SSTL18 class-I and class-II are available in both the HP and HR I/O banks, with the HP banks providing DCI and T_DCI options for tuned internal parallel split-termination resistors. The T_DCI option is only available for bidirectional signals (no input-only or output-only). HR banks provide IN_TERM options for untuned internal parallel split-termination resistors. Although the optimal drive and termination scheme for any new design is determined through careful signal-integrity analysis, initial considerations include:

- HP I/O banks: SSTL18_II_T_DCI at the 7 series FPGAs bidirectional pins (DQ and DQS), and SSTL18_II at the unidirectional pins (all other pins). Use on-die termination (ODT) at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.
- HR I/O banks: SSTL18_II at the 7 series FPGAs pins for both bidirectional (DQ/DQS) and unidirectional (all other pins) signals, combined with the IN_TERM (internal termination) attribute on the bidirectional pins. Use ODT at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.

SSTL15 is used for DDR3 SDRAM memory interfaces and is roughly defined (not by name) in the JEDEC standard JESD79-3E. For this standard, the full-strength driver (SSTL15) is available in both the HP and HR I/O banks. A weaker, reduced-strength driver, designated by an R in the standard name (SSTL15_R), is available in the HR I/O banks. Both drivers support bidirectional and unidirectional signaling. For some topologies (such as short point-to-point interfaces), the reduced-strength driver can result in reduced overshoot and better signal integrity. The HP I/O banks provide DCI and T_DCI options for tuned internal parallel split-termination resistors. While the DCI option is not available for bidirectional signals (input or output only), the T_DCI option is only available for

bidirectional signals (no input-only or output-only). HR banks provide IN_TERM options for untuned internal parallel split-termination resistors. Although the optimal drive and termination scheme for any new design is determined through careful signal-integrity analysis, initial considerations include:

- HP I/O banks: SSTL15_T_DCI at the 7 series FPGAs bidirectional pins (DQ and DQS), and SSTL15 at the unidirectional pins (all other pins). ODT used at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.
- HR I/O banks: SSTL15 at the 7 series FPGAs pins for both bidirectional (DQ/DQS) and unidirectional (all other pins) signals, combined with the IN_TERM (internal termination) attribute on the bidirectional pins. Use ODT at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.

SSTL135 is used for DDR3L SDRAM memory interfaces and is roughly defined (not by name) in the JEDEC standard JESD79-3-1. For this standard, the full-strength driver (SSTL135) is available in both the HP and HR I/O banks. A weaker, reduced-strength driver, designated by an R in the standard name (SSTL135_R), is available in the HR I/O banks. Both drivers support bidirectional and unidirectional signaling. For some topologies (such as short point-to-point interfaces), the reduced-strength driver can result in reduced overshoot and better signal integrity.

The HP I/O banks also provide DCI and T_DCI options for tuned internal parallel split-termination resistors. While the DCI option is not available for bidirectional signals (input or output only), the T_DCI option is only available for bidirectional signals (no input-only or output-only). HR banks provide IN_TERM options for untuned internal parallel split-termination resistors. Although the optimal drive and termination scheme for any new design is determined through careful signal-integrity analysis, initial considerations include:

- HP I/O banks: SSTL135_T_DCI at the 7 series FPGAs bidirectional pins (DQ and DQS), and SSTL135 at the unidirectional pins (all other pins). ODT used at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.
- HR I/O banks: SSTL135 at the 7 series FPGAs pins for both bidirectional (DQ/DQS) and unidirectional (all other pins) signals, combined with the IN_TERM (internal termination) attribute on the bidirectional pins. Use ODT at the memory device on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ for the unidirectional signals.

SSTL12 supports Micron's next-generation RDRAM3 memory. This standard is only available in the HP I/O banks. Both DCI and T_DCI options are available to improve the signal integrity through the use of tuned internal split-termination resistors. While the DCI option is not available for bidirectional signals (input or output only), the T_DCI option is only available for bidirectional signals (no input-only or output-only). Although the optimal drive and termination scheme for any new design is determined through careful signal-integrity analysis, initial considerations include:

- SSTL12_T_DCI at the 7 series FPGAs bidirectional pins (DQ and DQS)
- SSTL12 at the unidirectional pins (all other pins).
- ODT used at the memory device, if available on the bidirectional signals, and external parallel-termination resistors to $V_{TT} = V_{CCO}/2$ where ODT is not available.

SSTL15_R, SSTL135_R, DIFF_SSTL15_R, DIFF_SSTL135_R

Table 1-30: Available I/O Bank Type

HR	HP
Available	N/A

The reduced drive-strength R standards are versions of the standard drivers, which can be preferred for short, point-to-point board topologies. Parallel end-termination resistors (commonly 50Ω) to $V_{TT} = (V_{CCO}/2)$ are typically placed on the board close to any receiver. The differential (DIFF_) versions use complementary single-ended drivers for outputs and differential receivers for inputs.

SSTL18_I, DIFF_SSTL18_I

Table 1-31: Available I/O Bank Type

HR	HP
Available	Available

These standards are only available for unidirectional (input or output) signals. Class-I drivers can be preferred for short, point-to-point board topologies. Parallel end-termination resistors (commonly 50Ω) to $V_{TT} = (V_{CCO}/2)$ are typically placed on the board close to any receiver. The differential (DIFF_) version uses complementary single-ended drivers for outputs and differential receivers for inputs.

SSTL18_I_DCI, DIFF_SSTL18_I_DCI

Table 1-32: Available I/O Bank Type

HR	HP
N/A	Available

These standards are only available for unidirectional (input or output) signals. Class-I drivers can be preferred for short, point-to-point board topologies. DCI provides tuned internal parallel split-termination resistors that are always present (for receivers). The value of both the pull-up and pull-down resistors mirror the resistance measured on the VRN/VRP pins, creating the Thevenin equivalent resistance to the $V_{CCO}/2$ mid-point level. The differential (DIFF_) version uses complementary single-ended drivers for outputs and differential receivers for inputs.

SSTL18_II, SSTL15, SSTL135, DIFF_SSTL18_II, DIFF_SSTL15, DIFF_SSTL135

Table 1-33: Available I/O Bank Type

HR	HP
Available	Available

Parallel end-termination resistors (commonly 50Ω) to $V_{TT} = (V_{CCO}/2)$ are typically placed on the board close to any receiver. The differential (DIFF_) versions use complementary single-ended drivers for outputs and differential receivers for inputs.

Note: A lower resistance value can be used for the parallel end-termination resistors in some DDR3 applications. Refer to [UG586: Zynq-7000 SoC and 7 Series Devices Memory Interface Solutions v2.0 User Guide](#) for details.

SSTL18_II_DCI, SSTL_15_DCI, SSTL135_DCI, DIFF_SSTL18_II_DCI, DIFF_SSTL_15_DCI, DIFF_SSTL135_DCI

Table 1-34: Available I/O Bank Type

HR	HP
N/A	Available

The DCI standards provide tuned internal parallel split-termination resistors that are always present (for receivers). The value of both the pull-up and pull-down resistors mirror the resistance measured on the VRN/VRP pins, creating the Thevenin equivalent resistance to the $V_{CCO}/2$ mid-point level. The differential (DIFF_) versions use complementary single-ended drivers for outputs and differential receivers for inputs.

SSTL18_II_T_DCI, SSTL15_T_DCI, SSTL135_T_DCI, DIFF_SSTL18_II_T_DCI, DIFF_SSTL15_T_DCI, DIFF_SSTL135_T_DCI

Table 1-35: Available I/O Bank Type

HR	HP
N/A	Available

These standards are only available for bidirectional (input and output) signals. The T_DCI standards provide tuned internal parallel split-termination resistors that are only present when the 3-state control is enabled on the output buffer. The termination is disabled whenever the output buffer is driving. The value of both the pull-up and pull-down resistors mirror the resistance measured on the VRN/VRP pins, creating the Thevenin equivalent resistance to the $V_{CCO}/2$ mid-point level. The differential (DIFF_) versions use complementary single-ended drivers for outputs and differential receivers for inputs.

SSTL12, SSTL12_DCI, SSTL12_T_DCI, DIFF_SSTL12, DIFF_SSTL12_DCI, DIFF_SSTL12_T_DCI

Table 1-36: Available I/O Bank Type

HR	HP
N/A	Available

DCI provides tuned internal parallel split-termination resistors that are always present for receivers. DCI versions are only available for unidirectional (input or output) signals. The T_DCI versions limit the resistors to only be present when the 3-state control is enabled on the output buffer (only when receiving). With T_DCI, the termination is disabled whenever the output buffer is driving. The T_DCI versions are only available for bidirectional signals (input and output). The value of both the pull-up and pull-down resistors mirror the resistance measured on the VRN/VRP pins, creating the Thevenin equivalent resistance to the $V_{CCO}/2$ mid-point level. The differential (DIFF_) versions use complementary single-ended drivers for outputs, and differential receivers for inputs.

SSTL18, SSTL15, SSTL135, SSTL12

Figure 1-57 shows a sample circuit illustrating a unidirectional termination technique for SSTL18, SSTL15, SSTL135, or SSTL12. In a specific circuit, all drivers and receivers must be at the same voltage level (1.8V, 1.5V, 1.35V, or 1.2V); they are not interchangeable. Also shown in Figure 1-57, only SSTL18_II_DCI has internal split-termination resistors present in an output pin.

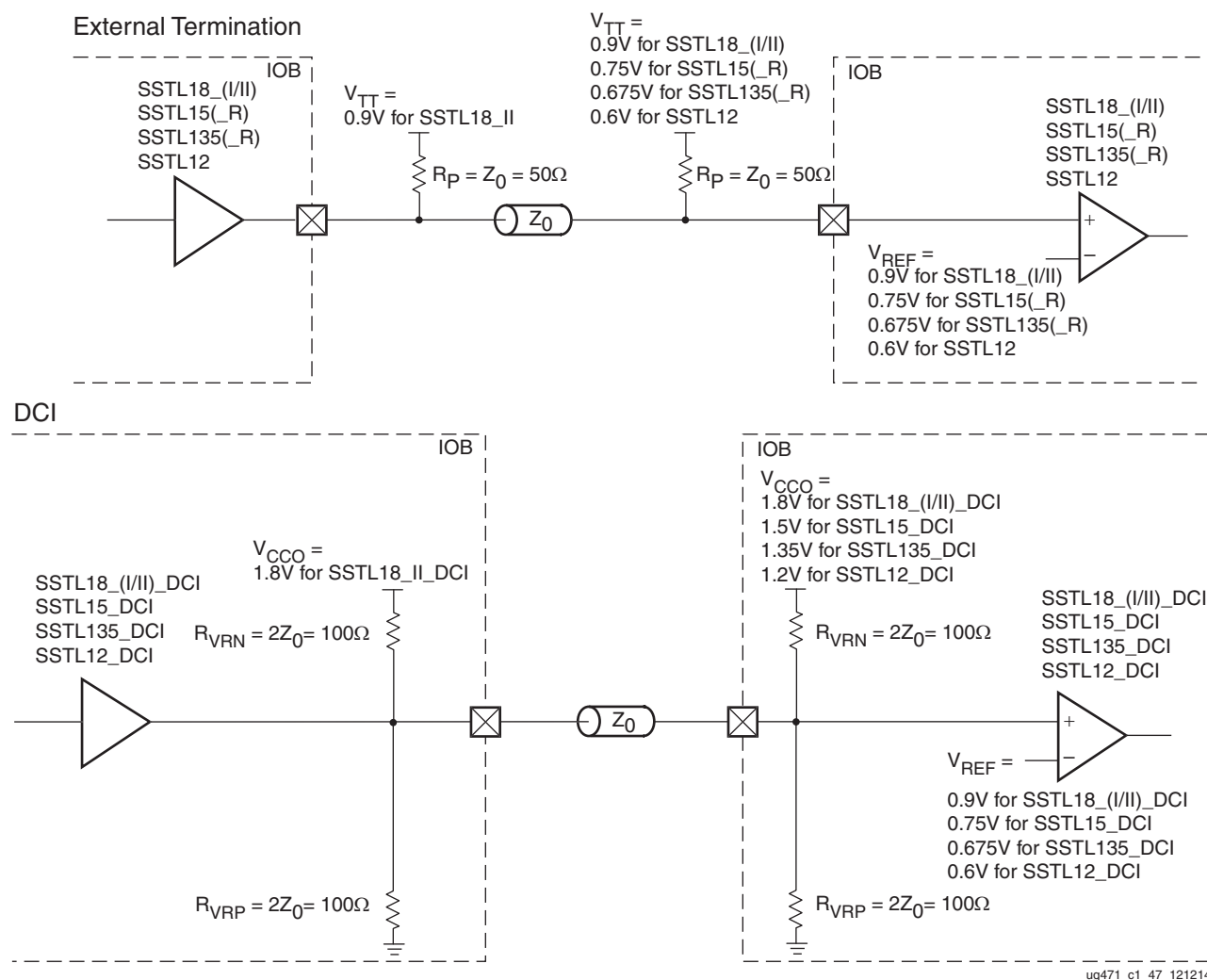
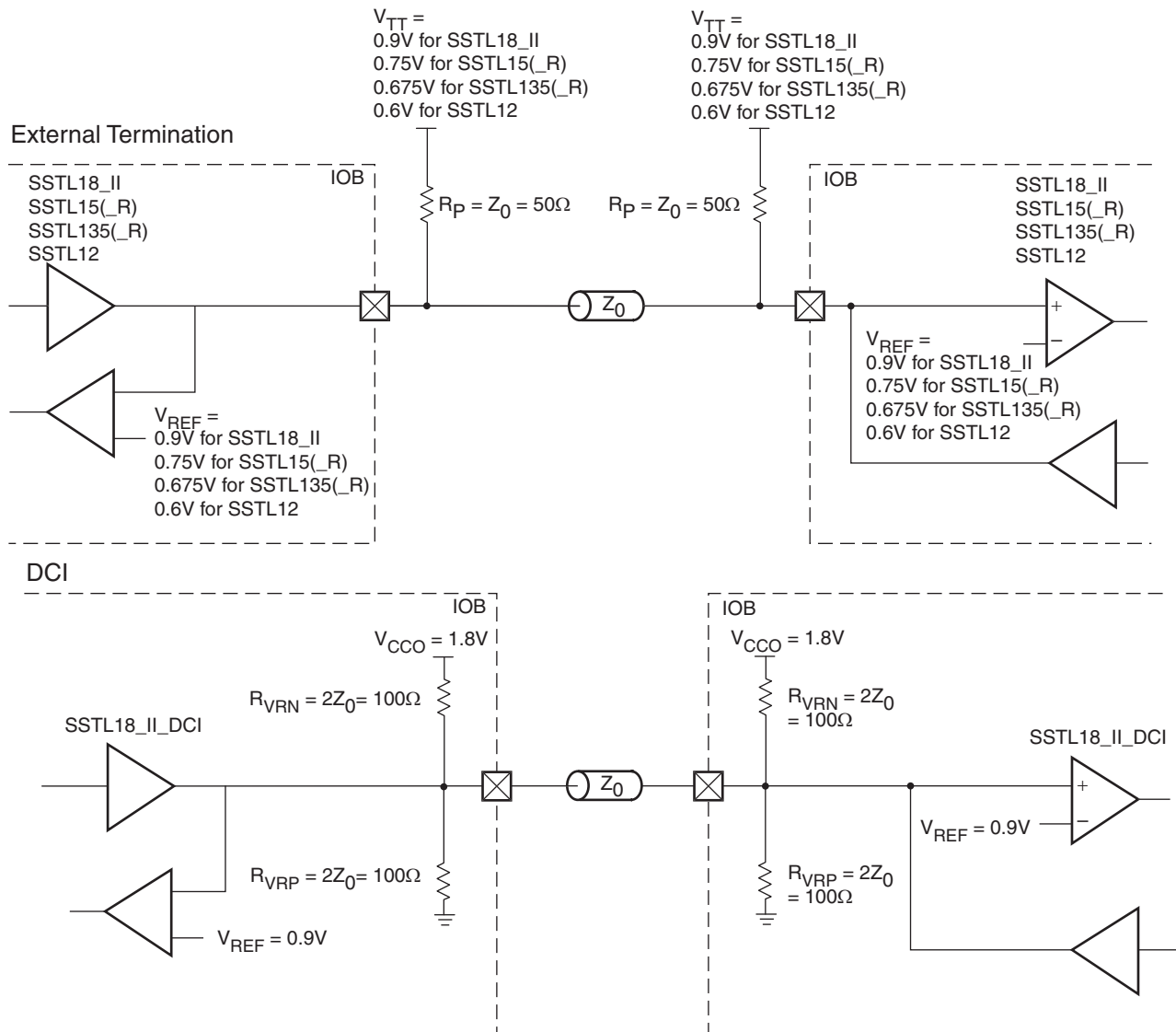


Figure 1-57: SSTL18, SSTL15, SSTL135, or SSTL12 Unidirectional Termination

Figure 1-58 shows a sample circuit illustrating a bidirectional termination technique for SSTL18, SSTL15, SSTL135, or SSTL12. In a specific circuit, all drivers and receivers must be at the same voltage level (1.8V, 1.5V, 1.35V, or 1.2V); they are not interchangeable. SSTL18 class-I is not available for bidirectional signaling. Also, SSTL18_II_DCI is the only available DCI standard available for bidirectional signaling. The DCI versions of SSTL18_I, SSTL15, SSTL135, and SSTL12 are only available for unidirectional signaling. Use the T_DCI standards for bidirectional signaling of SSTL15, SSTL135, and SSTL12 with DCI termination. The internal split-termination resistors are always present on SSTL18_II_DCI, independent of whether the drivers are 3-stated.



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Figure 1-58: SSTL18, SSTL15, SSTL135, or SSTL12 Bidirectional Termination

Differential SSTL18, SSTL15, SSTL135, SSTL12

Figure 1-59 shows a sample circuit illustrating a termination technique for differential SSTL18, SSTL15, SSTL135, or SSTL12 with unidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (1.8V, 1.5V, 1.35V, or 1.2V); they are not interchangeable.

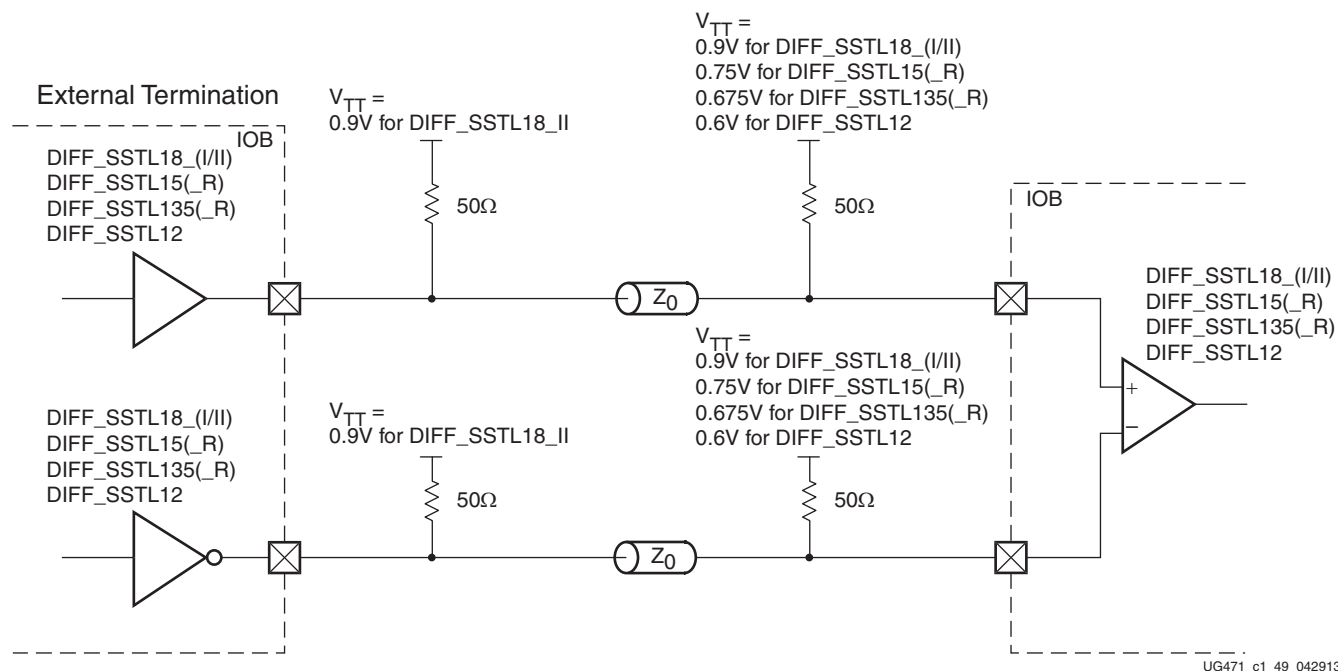


Figure 1-59: Differential SSTL18, SSTL15, SSTL135, or SSTL12 Unidirectional Termination

Figure 1-60 shows a sample circuit illustrating a termination technique for differential SSTL18, SSTL15, SSTL135, or SSTL12 with unidirectional DCI termination. In a specific circuit, all drivers and receivers must be at the same voltage level (1.8V, 1.5V, 1.35V, or 1.2V); they are not interchangeable. Also shown in Figure 1-60, only SSTL18_II_DCI has internal split-termination resistors present in an output pin.

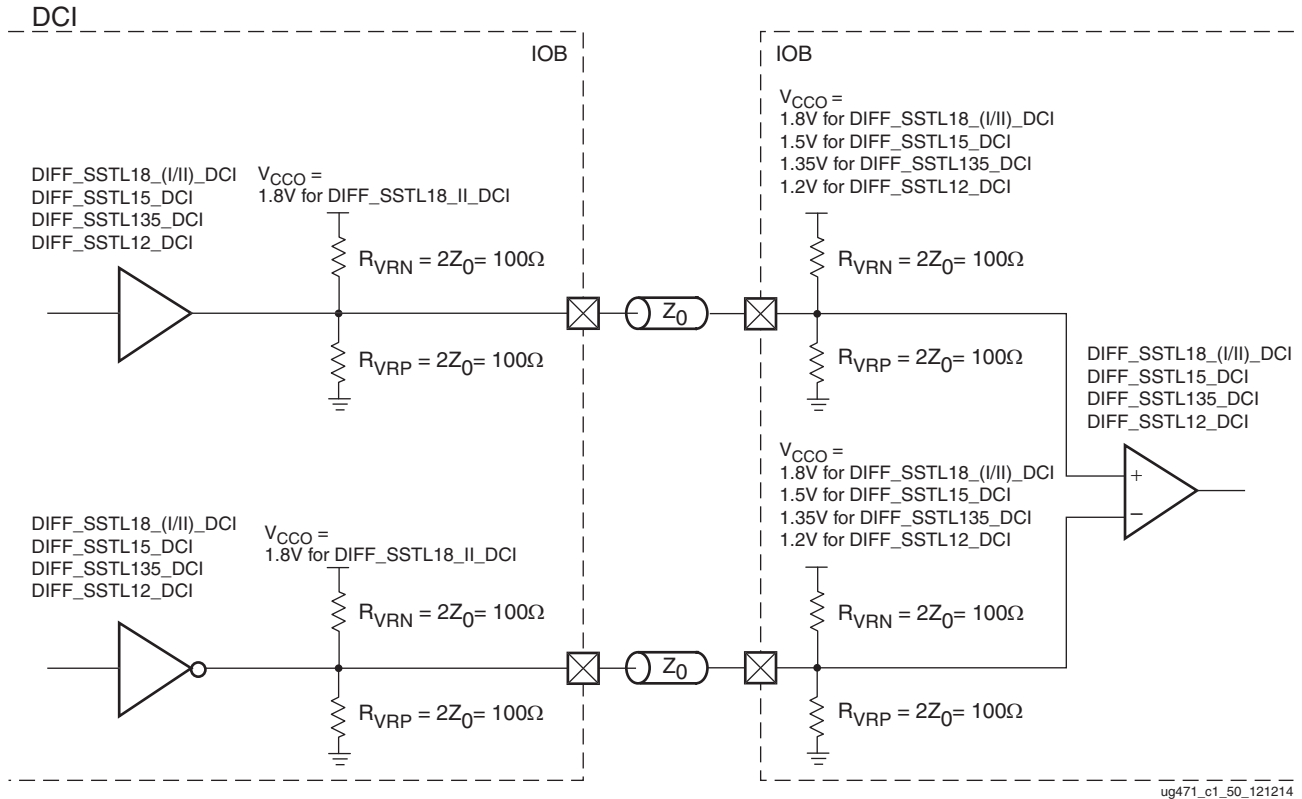


Figure 1-60: Differential SSTL18, SSTL15, SSTL135, or SSTL12 Unidirectional DCI Termination

Figure 1-61 shows a sample circuit illustrating a termination technique for differential SSTL18, SSTL15, SSTL135, or SSTL12 with bidirectional termination. In a specific circuit, all drivers and receivers must be at the same voltage level (1.8V, 1.5V, 1.35V, or 1.2V); they are not interchangeable. Differential SSTL18 class-I is not available for bidirectional use.

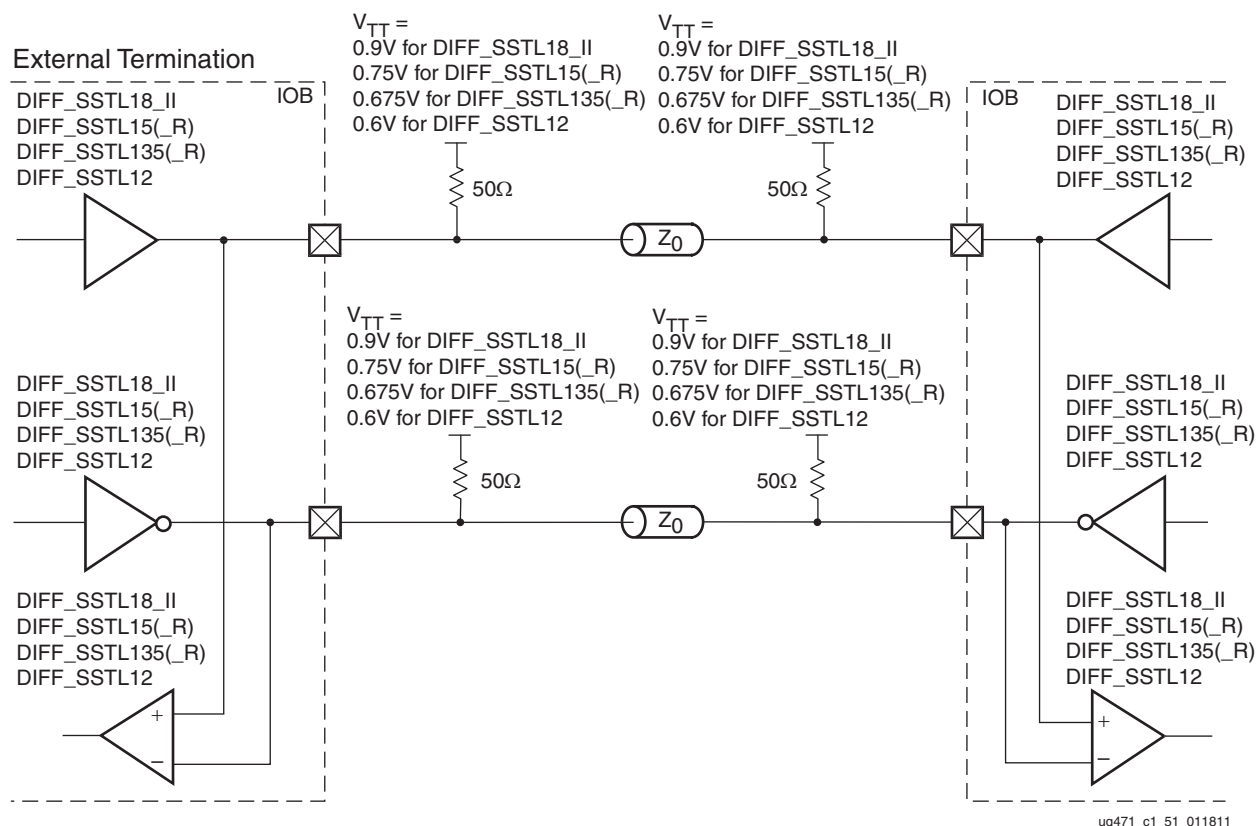


Figure 1-61: Differential SSTL18, SSTL15, SSTL135, or SSTL12 with Bidirectional Termination

Figure 1-62 shows a sample circuit illustrating a termination technique for differential SSTL18 with bidirectional DCI termination. DIFF_SSTL18_II_DCI is the only available DCI standard for bidirectional use signaling. The DCI versions of DIFF_SSTL18_I, DIFF_SSTL15, DIFF_SSTL135, and DIFF_SSTL12 are only available for unidirectional signaling. Use the T_DCI standards for bidirectional signaling of DIFF_SSTL15, DIFF_SSTL135, and DIFF_SSTL12 with DCI termination.

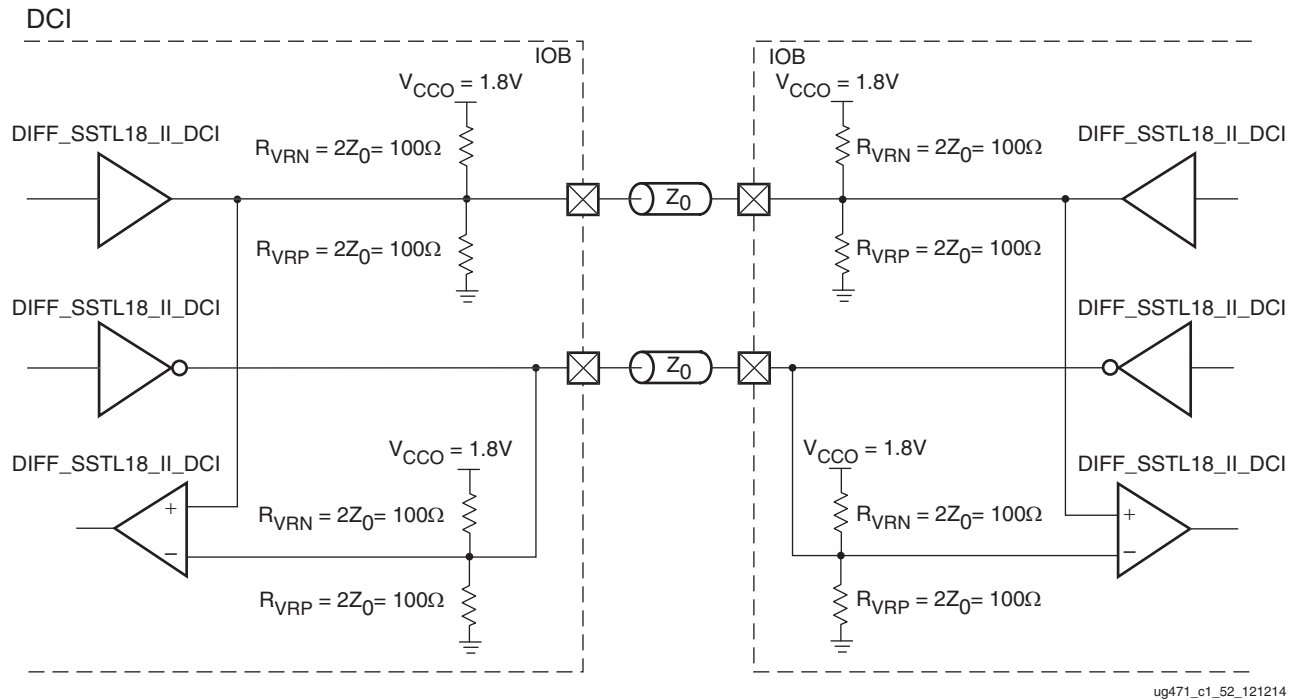


Figure 1-62: Differential SSTL18 Class II with DCI Bidirectional Termination

SSTL18, SSTL15, SSTL135, or SSTL12 (T_DCI) Termination

Figure 1-63 shows a sample circuit illustrating a termination technique for SSTL18, SSTL15, SSTL135, or SSTL12 (T_DCI) with on-chip split-thevenin termination. In this bidirectional I/O standard, when 3-stated, the internal split-termination is invoked on the receiver and not on the driver.

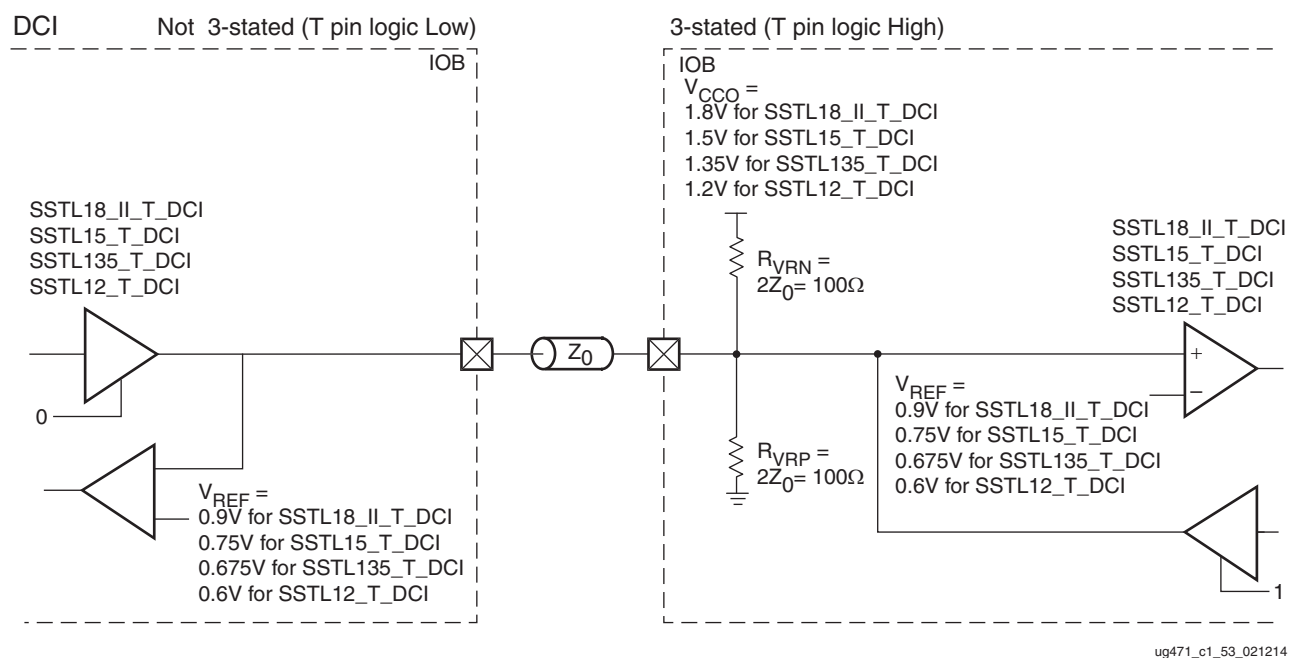


Figure 1-63: SSTL18, SSTL15, SSTL135, or SSTL12 (T_DCI) Termination

HSUL_12 (High Speed Unterminated Logic)

The HSUL_12 standard is for LPDDR2 memory buses. HSUL_12 is defined by the JEDEC standard JESD8-22. 7 series FPGAs support this standard for single-ended signaling and differential signaling. Similar to SSTL, this standard also requires a differential amplifier input buffer and a push-pull output buffer.

HSUL_12 and DIFF_HSUL_12

Table 1-37: Available I/O Bank Type

HR	HP
Available	Available

The differential (DIFF_) version uses complementary single-ended drivers for outputs and differential receivers for inputs.

HSUL_DCI_12 and DIFF_HSUL_12_DCI

Table 1-38: Available I/O Bank Type

HR	HP
N/A	Available

DCI provides a tuned output impedance driver that matches the output impedance to the reference resistors on the VRP and VRN pins. No split termination resistors are present for either drivers or receivers. The differential (DIFF_) versions use complementary single-ended drivers for outputs and differential receivers for inputs.

HSUL_12

Figure 1-64 shows a sample circuit illustrating a unidirectional board topology (with no termination) for HSUL_12. Only HP I/O banks support the DCI version.

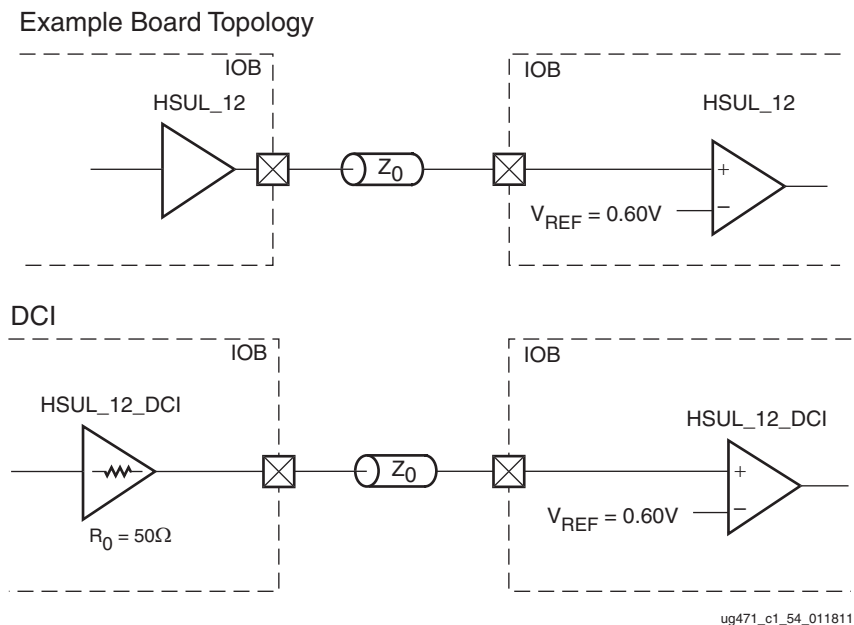


Figure 1-64: HSUL_12 with Unidirectional Signalling

Figure 1-65 shows a sample circuit illustrating a bidirectional board topology (with no termination) for HSUL_12. Only HP I/O banks support the DCI version.

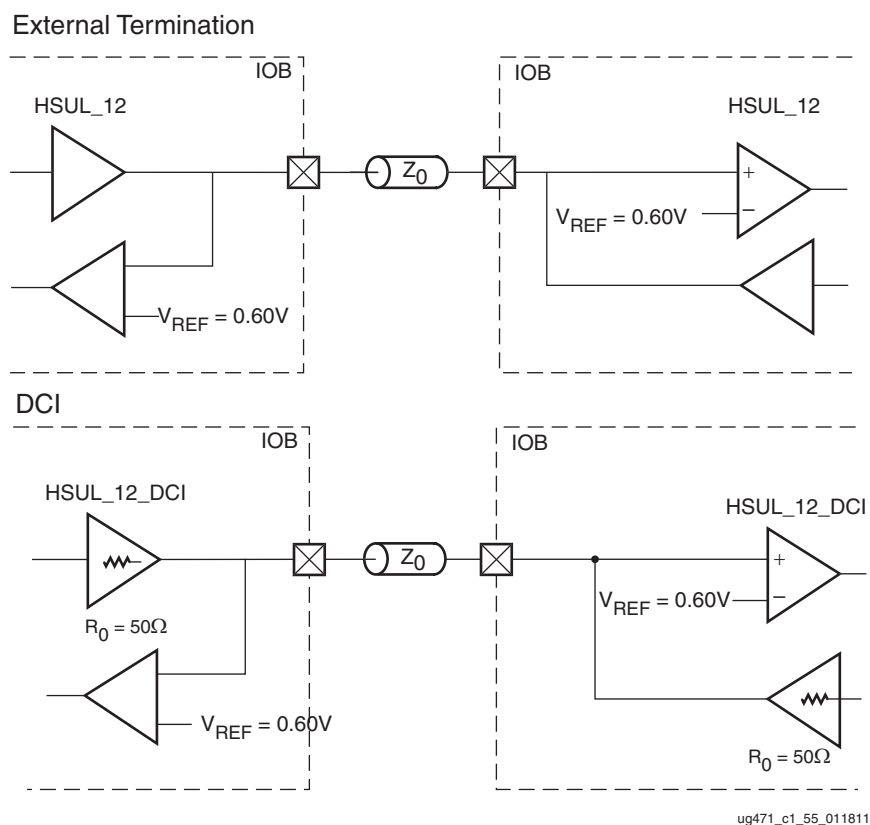


Figure 1-65: HSUL_12 with Bidirectional Signalling

Differential HSUL_12

Figure 1-66 shows a sample circuit illustrating a board topology (with no termination) for differential HSUL_12 with unidirectional signalling.

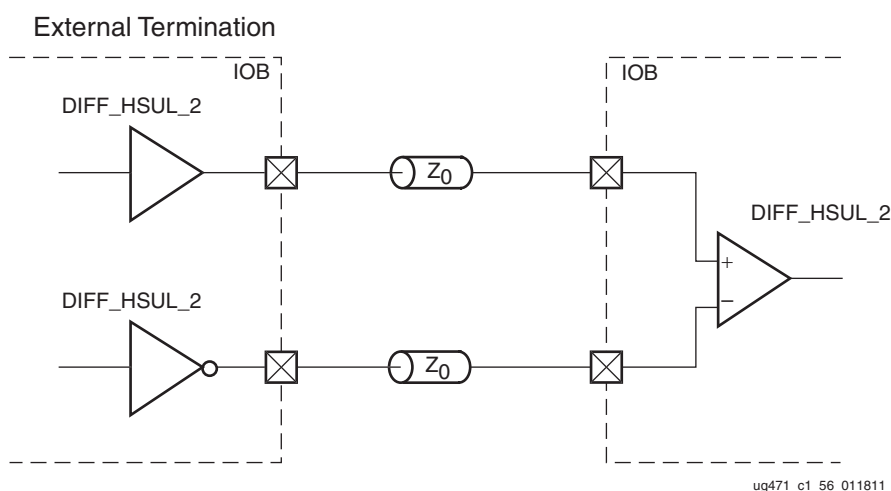


Figure 1-66: Differential HSUL_12 with Unidirectional Signalling

Figure 1-67 shows a sample circuit illustrating a board topology (with no termination) for differential HSUL_12 with unidirectional DCI signalling.

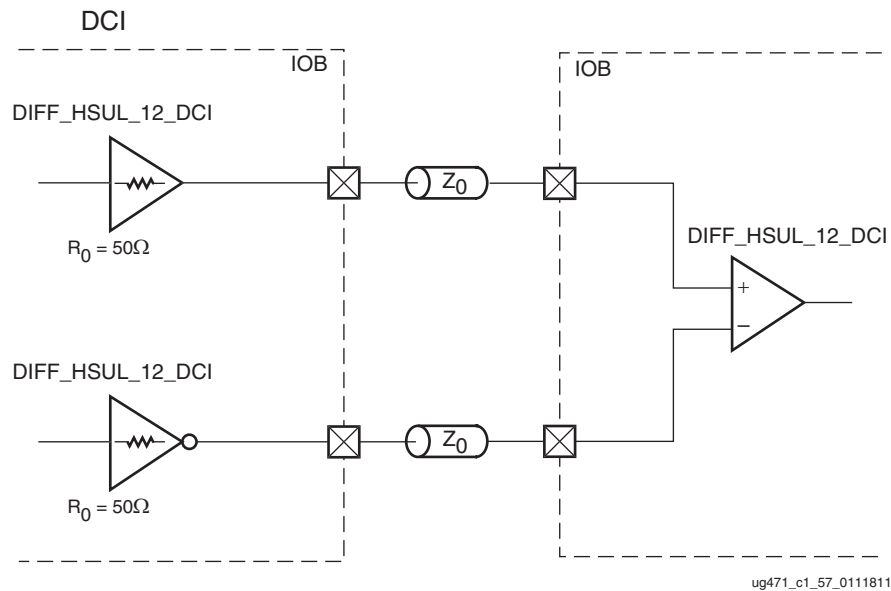


Figure 1-67: Differential HSUL_12 with Unidirectional DCI Signalling

Figure 1-68 shows a sample circuit illustrating a board topology (with no termination) for differential HSUL_12 with bidirectional signalling.

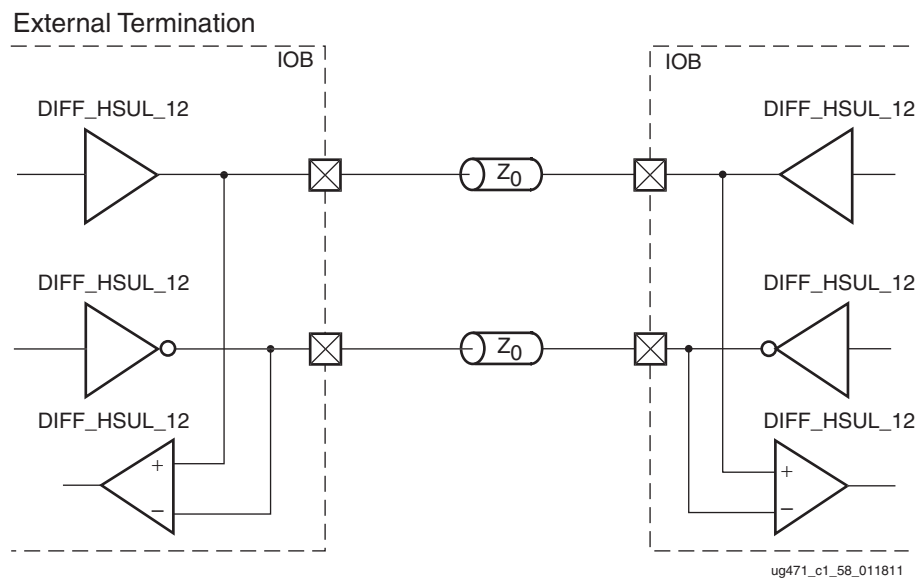


Figure 1-68: Differential HSUL_12 with Bidirectional Signalling

Figure 1-69 shows a sample circuit illustrating a board topology (with no termination) for differential HSUL_12 with bidirectional DCI signalling.

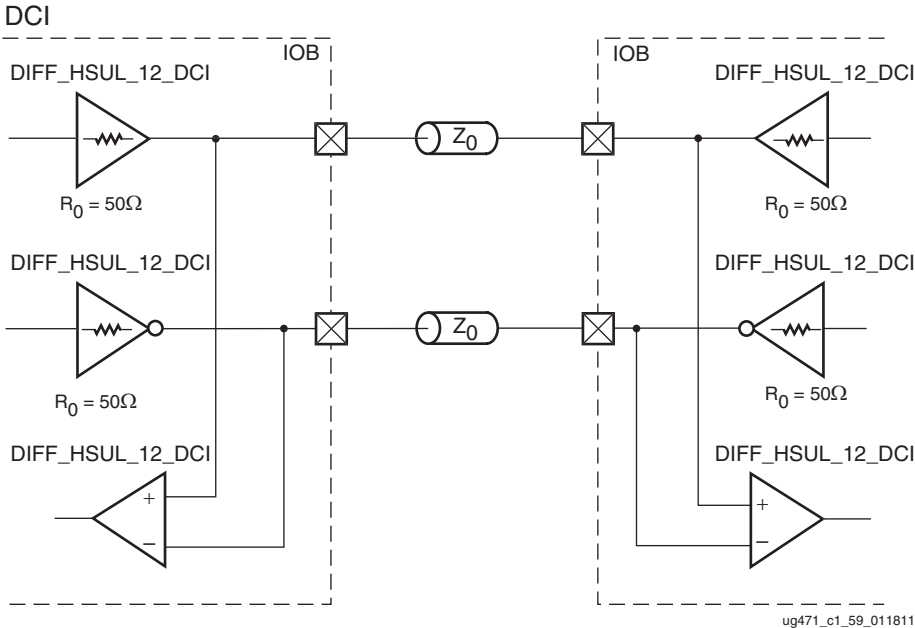


Figure 1-69: Differential HSUL_12 with DCI Bidirectional Signalling

MOBILE_DDR (Low Power DDR)

Table 1-39: Available I/O Bank Type

HR	HP
Available	N/A

The MOBILE_DDR standard is for LPDDR and Mobile DDR memory buses. MOBILE_DDR is defined by the JEDEC I/O standard JESD209A. It is a 1.8V single-ended I/O standard that eliminates the need for V_{REF} and V_{TT} voltage supplies. 7 series FPGAs support this standard for single-ended signaling and differential outputs. The differential outputs drive the CK/CK# pins.

The differential (DIFF_) version uses complementary single-ended drivers for outputs, and differential receivers for inputs.

Summary of Memory Interface IOSTANDARDS and Attributes Supported

Table 1-40 lists the available 7 series FPGA single-ended HSTL, SSTL, HSUL, and MOBILE_DDR I/O standards and attributes supported.

Table 1-41 lists the available 7 series FPGA differential HSTL, SSTL, HSUL, and MOBILE_DDR I/O standards and attributes supported.

Table 1-42 lists the SLEW attribute for the 7 series FPGA single-ended and differential HSTL, SSTL, HSUL, and MOBILE_DDR I/O standards.

Table 1-40: IOSTANDARD Attributes for Single-Ended HSTL, SSTL, HSUL, or MOBILE_DDR I/O Standards

Attributes	Primitives			
	IBUF, IBUFG, OBUF, or OBUFT		IOBUF	
	HP I/O Banks	HR I/O Banks	HP I/O Banks	HR I/O Banks
IOSTANDARD	HSTL_I	HSTL_I	N/A	N/A
	HSTL_I_12	N/A	N/A	N/A
	HSTL_I_18	HSTL_I_18	N/A	N/A
	HSTL_I_DCI	N/A	N/A	N/A
	HSTL_I_DCI_18	N/A	N/A	N/A
	HSTL_II	HSTL_II	HSTL_II	HSTL_II
	HSTL_II_18	HSTL_II_18	HSTL_II_18	HSTL_II_18
	HSTL_II_DCI	N/A	HSTL_II_DCI	N/A
	HSTL_II_DCI_18	N/A	HSTL_II_DCI_18	N/A
	N/A	N/A	HSTL_II_T_DCI	N/A
	N/A	N/A	HSTL_II_T_DCI_18	N/A
	SSTL12	N/A	SSTL12	N/A
	SSTL12_DCI	N/A	N/A	N/A
	N/A	N/A	SSTL12_T_DCI	N/A
	N/A	SSTL135_R	N/A	SSTL135_R
	SSTL135	SSTL135	SSTL135	SSTL135
	SSTL135_DCI	N/A	N/A	N/A
	N/A	N/A	SSTL135_T_DCI	N/A
	N/A	SSTL15_R	N/A	SSTL15_R
	SSTL15	SSTL15	SSTL15	SSTL15
	SSTL15_DCI	N/A	N/A	N/A
	N/A	N/A	SSTL15_T_DCI	N/A
	SSTL18_I	SSTL18_I	N/A	N/A
	SSTL18_I_DCI	N/A	N/A	N/A
	SSTL18_II	SSTL18_II	SSTL18_II	SSTL18_II
	SSTL18_II_DCI	N/A	SSTL18_II_DCI	N/A
	N/A	N/A	SSTL18_II_T_DCI	N/A
	HSUL_12	HSUL_12	HSUL_12	HSUL_12
	HSUL_12_DCI	N/A	HSUL_12_DCI	N/A
	N/A	MOBILE_DDR	N/A	MOBILE_DDR

Table 1-41: IOSTANDARD Attributes for Differential HSTL, SSTL, HSUL, or MOBILE_DDR I/O Standards

Attributes	Primitives			
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, IBUFGDS_DIFF_OUT, OBUFDS, or OBUFTDS		IOBUFDS or IOBUFDS_DIFF_OUT	
	HP I/O Banks	HR I/O Banks	HP I/O Banks	HR I/O Banks
IOSTANDARD	DIFF_HSTL_I	DIFF_HSTL_I	N/A	N/A
	DIFF_HSTL_I_18	DIFF_HSTL_I_18	N/A	N/A
	DIFF_HSTL_I_DCI	N/A	N/A	N/A
	DIFF_HSTL_I_DCI_18	N/A	N/A	N/A
	DIFF_HSTL_II	DIFF_HSTL_II	DIFF_HSTL_II	DIFF_HSTL_II
	DIFF_HSTL_II_18	DIFF_HSTL_II_18	DIFF_HSTL_II_18	DIFF_HSTL_II_18
	DIFF_HSTL_II_DCI	N/A	DIFF_HSTL_II_DCI	N/A
	DIFF_HSTL_II_DCI_18	N/A	DIFF_HSTL_II_DCI_18	N/A
	DIFF_HSTL_II_T_DCI	N/A	DIFF_HSTL_II_T_DCI	N/A
	DIFF_HSTL_II_T_DCI_18	N/A	DIFF_HSTL_II_T_DCI_18	N/A
	DIFF_SSTL12	N/A	DIFF_SSTL12	N/A
	DIFF_SSTL12_DCI	N/A	N/A	N/A
	N/A	N/A	DIFF_SSTL12_T_DCI	N/A
	N/A	DIFF_SSTL135_R	N/A	DIFF_SSTL135_R
	DIFF_SSTL135	DIFF_SSTL135	DIFF_SSTL135	DIFF_SSTL135
	DIFF_SSTL135_DCI	N/A	N/A	N/A
	N/A	N/A	DIFF_SSTL135_T_DCI	N/A
	N/A	DIFF_SSTL15_R	N/A	DIFF_SSTL15_R
	DIFF_SSTL15	DIFF_SSTL15	DIFF_SSTL15	DIFF_SSTL15
	DIFF_SSTL15_DCI	N/A	N/A	N/A
	N/A	N/A	DIFF_SSTL15_T_DCI	N/A
	DIFF_SSTL18_I	DIFF_SSTL18_I	N/A	N/A
	DIFF_SSTL18_I_DCI	N/A	N/A	N/A
	DIFF_SSTL18_II	DIFF_SSTL18_II	DIFF_SSTL18_II	DIFF_SSTL18_II
	DIFF_SSTL18_II_DCI	N/A	DIFF_SSTL18_II_DCI	N/A
	N/A	N/A	DIFF_SSTL18_II_T_DCI	N/A
	DIFF_HSUL_12	DIFF_HSUL_12	DIFF_HSUL_12	DIFF_HSUL_12
	DIFF_HSUL_12_DCI	N/A	DIFF_HSUL_12_DCI	N/A
	N/A	DIFF_MOBILE_DDR	N/A	DIFF_MOBILE_DDR

Table 1-42: SLEW Attribute for All Single-Ended and Differential HSTL, SSTL, HSUL, and Mobile_DDR IOSTANDARDS

Attribute	Primitives			
	IBUF, IBUFG, IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT		OBUF, OBUFT, OBUFDS, or OBUFTDS, IOBUF, IOBUFDS or IOBUFDS_DIFF_OUT	
	HP I/O Banks	HR I/O Banks	HP I/O Banks	HR I/O Banks
SLEW	N/A	N/A	{FAST, SLOW}	{FAST, SLOW}

LVDS and LVDS_25 (Low Voltage Differential Signaling)

Low-voltage differential signaling (LVDS) is a powerful high-speed interface in many system applications. 7 series FPGA I/Os are designed to comply with the EIA/TIA electrical specifications for LVDS to make system and board design easier. With the use of an LVDS current-mode driver in the IOBs and the optional internal differential termination feature, the need for external source termination in point-to-point applications is eliminated. 7 series devices provide the most flexible solution for doing an LVDS design in an FPGA.

The LVDS I/O standard is only available in the HP I/O banks. It requires a V_{CCO} to be powered at 1.8V for outputs and for inputs when the optional internal differential termination is implemented (DIFF_TERM = TRUE).

The LVDS_25 I/O standard is only available in the HR I/O banks. It requires a V_{CCO} to be powered at 2.5V for outputs and for inputs when the optional internal differential termination is implemented (DIFF_TERM = TRUE).

Table 1-43: Available I/O Bank Type

HR	HP
Available for LVDS_25 only	Available for LVDS only

Transmitter Termination

The 7 series FPGA LVDS transmitter does not require any external termination. [Table 1-44](#) lists the allowed attributes corresponding to the 7 series FPGA LVDS current-mode drivers. 7 series FPGA LVDS current-mode drivers are a true current source and produce the proper (EIA/TIA compliant) LVDS signal.

Receiver Termination

[Figure 1-70](#) is an example of differential termination for an LVDS or LVDS_25 receiver on a board with 50Ω transmission lines.

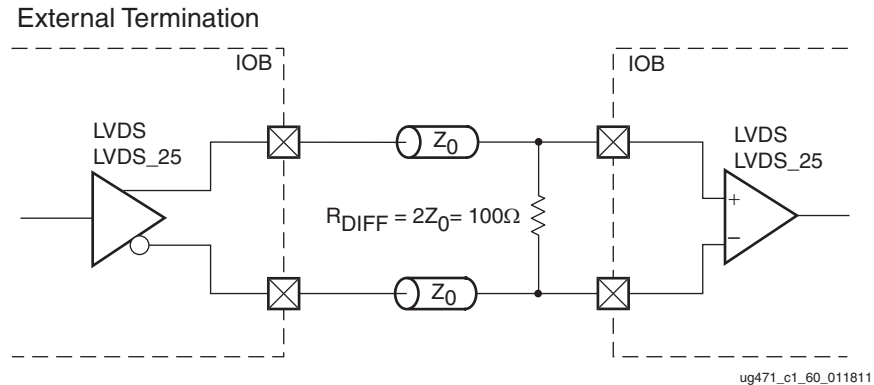


Figure 1-70: LVDS or LVDS_25 Receiver Termination

Figure 1-71 is an example of a differential termination for an LVDS or LVDS_25 receiver on a board with 50Ω transmission lines.

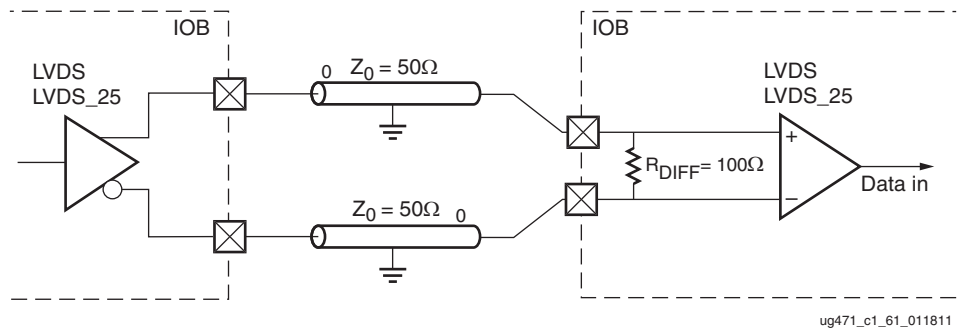


Figure 1-71: LVDS or LVDS_25 With DIFF_TERM Receiver Termination

Table 1-44 lists the available 7 series FPGA LVDS I/O standards and attributes supported.

Table 1-44: Allowed Attributes of the LVDS I/O Standards

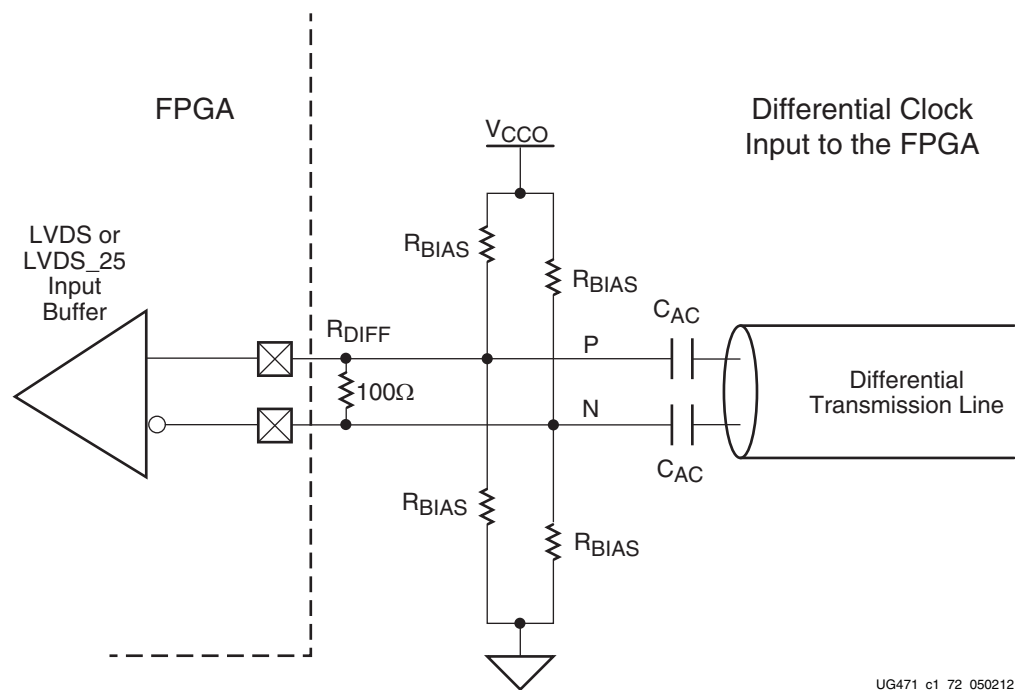
Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS or OBUFTDS
IOSTANDARD	LVDS (HP I/O Banks) or LVDS_25 (HR I/O Banks)	
DIFF_TERM	TRUE, FALSE	N/A

It is acceptable to have differential inputs such as LVDS and LVDS_25 in I/O banks that are powered at voltage levels other than the nominal voltages required for the outputs of those standards (1.8V for LVDS outputs, and 2.5V for LVDS_25 outputs). However, these criteria must be met:

- The optional internal differential termination is not used (DIFF_TERM = FALSE, which is the default value).
- The differential signals at the input pins meet the V_{IN} requirements in the Recommended Operating Conditions table of the specific device family data sheet.

- The differential signals at the input pins meet the V_{IDIFF} (min) requirements in the corresponding LVDS or LVDS_25 DC specifications tables of the specific device family data sheet.
- For HR I/O banks in bidirectional configuration, internal differential termination is always used.

One way to accomplish the above criteria is to use an external circuit that both AC-couples and DC-biases the input signals. Figure 1-72 shows an example circuit for providing an AC-coupled and DC-biased circuit for a differential clock input. R_{DIFF} provides the 100Ω differential receiver termination because the internal DIFF_TERM is set to FALSE. To maximize the input noise margin, all R_{BIAS} resistors should be the same value, essentially creating a V_{ICM} level of $V_{CCO}/2$. Resistors in the $10k$ – $100k\Omega$ range are recommended. The typical values for the AC coupling capacitors C_{AC} are in the range of 100 nF. All components should be placed physically close to the FPGA inputs.



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Figure 1-72: Example Circuit for AC-Coupled and DC-Biased Differential Clock Input

RSDS (Reduced Swing Differential Signaling)

Table 1-45: Available I/O Bank Type

HR	HP
Available	N/A

Reduced-swing differential signaling (RSDS) is similar to an LVDS high-speed interface using differential signaling. RSDS has a similar implementation to LVDS_25 in 7 series FPGAs and is only intended for point-to-point applications. RSDS is only available in HR I/O banks and requires a V_{CCO} voltage level of 2.5V. The IOSTANDARD is called RSDS_25. Table 1-46 summarizes all the possible RSDS I/O standards and attributes supported.

Table 1-46: Allowed Attributes of the RSDS I/O Standard

Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS or OBUFTDS
IOSTANDARD	RSDS_25	
DIFF_TERM	TRUE, FALSE	N/A

Mini-LVDS (Mini Low Voltage Differential Signaling)

Table 1-47: Available I/O Bank Type

HR	HP
Available	N/A

Mini-LVDS is a serial, intra-flat panel differential I/O standard that serves as an interface between the timing control function and an LCD source driver. Mini-LVDS inputs require a parallel-termination resistor, either by using a discrete resistor on the PCB, or by using the 7 series FPGAs DIFF_TERM attribute to enable internal termination. Mini-LVDS is only available in HR I/O banks and requires a V_{CCO} voltage level of 2.5V. The IOSTANDARD is called MINI_LVDS_25. Table 1-48 summarizes all the possible Mini-LVDS I/O standards and attributes supported.

Table 1-48: Allowed Attributes of the Mini-LVDS I/O Standard⁽¹⁾

Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS or OBUFTDS
IOSTANDARD	MINI_LVDS_25	
DIFF_TERM	TRUE, FALSE	N/A

Notes:

1. When in bidirectional configuration, internal differential termination is always used for this standard.

PPDS (Point-to-Point Differential Signaling)

Table 1-49: Available I/O Bank Type

HR	HP
Available	N/A

PPDS is a differential I/O standard for next-generation LCD interface row and column drivers. PPDS inputs require a parallel-termination resistor, either through the use of a discrete resistor on the PCB, or by using the 7 series FPGAs DIFF_TERM attribute to enable internal termination. PPDS is only available in HR I/O banks and requires a V_{CCO} voltage level of 2.5V. The IOSTANDARD is called PPDS_25. [Table 1-50](#) summarizes all the possible PPDS I/O standards and attributes supported.

Table 1-50: Allowed Attributes of the PPDS I/O Standard

Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS or OBUFTDS
IOSTANDARD	PPDS_25	
DIFF_TERM	TRUE, FALSE	N/A

TMDS (Transition Minimized Differential Signaling)

Table 1-51: Available I/O Bank Type

HR	HP
Available	N/A

TMDS is a differential I/O standard for transmitting high-speed serial data used by the DVI and HDMI video interfaces. The TMDS standard requires external 50Ω pull-up resistors to 3.3V on the inputs. TMDS inputs do not require differential input termination resistors. TMDS is only available in HR I/O banks and requires a V_{CCO} voltage level of 3.3V. The IOSTANDARD is called TMDS_33. [Table 1-52](#) summarizes all the possible TMDS I/O standards.

Table 1-52: Allowed Attributes of the TMDS I/O Standard

Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS or OBUFTDS
IOSTANDARD	TMDS_33	

BLVDS (Bus LVDS)

Table 1-53: Available I/O Bank Type

HR	HP
Available	N/A

Since LVDS is intended for point-to-point applications, BLVDS is not an EIA/TIA standard implementation and requires careful adaptation of I/O and PCB layout design rules. The primitive supplied in the software library for bidirectional LVDS does not use the 7 series FPGA LVDS current-mode driver, instead, it uses complementary single-ended differential drivers. Therefore, source termination is required. BLVDS is only available in HR I/O banks and requires a V_{CCO} voltage level of 2.5V. The IOSTANDARD is called BLVDS_25. Table 1-54 summarizes all the possible BLVDS I/O standards.

Table 1-54: Allowed Attributes of the BLVDS I/O Standard

Attributes	Primitives	
	IBUFDS, IBUFGDS, IBUFDS_DIFF_OUT, or IBUFGDS_DIFF_OUT	OBUFDS, OBUFTDS, IOBUFDS, or IOBUFDS_DIFF_OUT
IOSTANDARD	BLVDS_25	

Figure 1-73 shows the BLVDS transmitter termination.

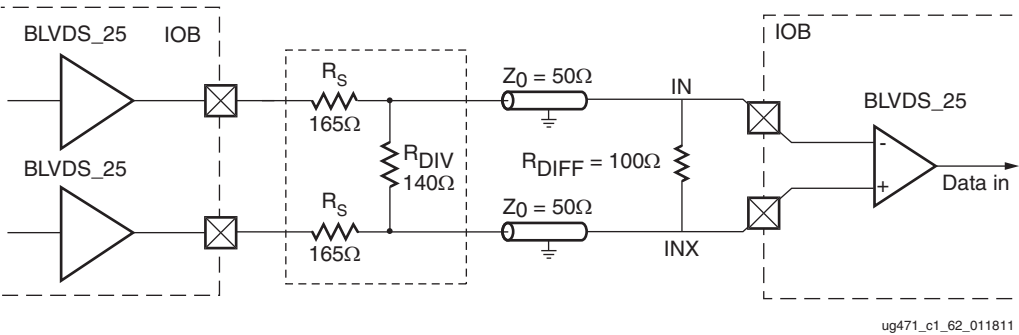


Figure 1-73: BLVDS Transmitter Termination

Rules for Combining I/O Standards in the Same Bank

The following rules must be obeyed to combine different input, output, and bidirectional standards in the same bank:

1. **Combining output standards only.** Output standards with the same output V_{CCO} requirement can be combined in the same bank.

Compatible example:

SSTL15_I and LVDCI_15 outputs

Incompatible example:

SSTL15 (output $V_{CCO} = 1.5V$) and
LVCMOS18 (output $V_{CCO} = 1.8V$) outputs

2. **Combining input standards only.** Input standards with the same V_{CCO} and V_{REF} requirements can be combined in the same bank.

Compatible example:

LVCMOS15 and HSTL_II inputs

Incompatible example:

LVCMOS15 (input $V_{CCO} = 1.5V$) and
LVCMOS18 (input $V_{CCO} = 1.8V$) inputs

Incompatible example:

HSTL_I_DCI_18 ($V_{REF} = 0.9V$) and
HSTL_I_DCI ($V_{REF} = 0.75V$) inputs

3. **Combining input standards and output standards.** Input standards and output standards with the same V_{CCO} requirement can be combined in the same bank.

Compatible example:

LVDS_25 output and LVCMOS25 input

Incompatible example:

LVDS_25 output (output $V_{CCO} = 2.5V$) and
HSTL_I_DCI_18 input (input $V_{CCO} = 1.8V$)

4. **Combining bidirectional standards with input or output standards.** When combining bidirectional I/O with other standards, make sure the bidirectional standard can meet the first three rules.

5. **Additional rules for combining DCI I/O standards.**

- a. Only one DCI target resistance value (controlled driver output impedance or split termination) can be used in any one HP I/O bank (or group of banks when using DCI chaining).

Incompatible example:

HSUL_12_DCI output with a 40Ω output impedance and SSTL12_T_DCI input with $100\Omega/100\Omega$ split termination

The implementation tools enforce these design rules.

Table 1-55, summarizes the V_{CCO} and V_{REF} requirements for each 7 series FPGA supported I/O standard. For more detailed DC specifications, including the recommended operating ranges of the supplies for each supported I/O standard, see the 7 series FPGA data sheets.

Table 1-55: V_{CCO} and V_{REF} Requirements for Each Supported I/O Standard

I/O Standard	I/O Bank Availability	V_{CCO} (V)			V_{REF} (V)
		Output	Input	Input with DIFF_TERM = TRUE	Input
BLVDS_25	HR	2.5	2.5 ⁽¹⁾	N/A	N/A
DIFF_HSTL_I	Both	1.5	Any	N/A	N/A
DIFF_HSTL_I_18	Both	1.8	Any	N/A	N/A
DIFF_HSTL_I_DCI	HP	1.5	1.5	N/A	N/A
DIFF_HSTL_I_DCI_18	HP	1.8	1.8	N/A	N/A
DIFF_HSTL_II	Both	1.5	Any	N/A	N/A
DIFF_HSTL_II_18	Both	1.8	Any	N/A	N/A
DIFF_HSTL_II_DCI	HP	1.5	1.5	N/A	N/A
DIFF_HSTL_II_DCI_18	HP	1.8	1.8	N/A	N/A
DIFF_HSTL_II_T_DCI	HP	1.5	1.5	N/A	N/A
DIFF_HSTL_II_T_DCI_18	HP	1.8	1.8	N/A	N/A
DIFF_HSUL_12	Both	1.2	Any	N/A	N/A
DIFF_HSUL_12_DCI	HP	1.2	1.2	N/A	N/A
DIFF_MOBILE_DDR	HR	1.8	1.8	N/A	N/A
DIFF_SSTL12	HP	1.2	Any	N/A	N/A
DIFF_SSTL12_DCI	HP	1.2	1.2	N/A	N/A
DIFF_SSTL12_T_DCI	HP	1.2	1.2	N/A	N/A
DIFF_SSTL135	Both	1.35	Any	N/A	N/A
DIFF_SSTL135_R	HR	1.35	Any	N/A	N/A
DIFF_SSTL135_DCI	HP	1.35	1.35	N/A	N/A
DIFF_SSTL135_T_DCI	HP	1.35	1.35	N/A	N/A
DIFF_SSTL15	Both	1.5	Any	N/A	N/A
DIFF_SSTL15_R	HR	1.5	Any	N/A	N/A
DIFF_SSTL15_DCI	HP	1.5	1.5	N/A	N/A
DIFF_SSTL15_T_DCI	HP	1.5	1.5	N/A	N/A
DIFF_SSTL18_I	Both	1.8	Any	N/A	N/A
DIFF_SSTL18_I_DCI	HP	1.8	1.8	N/A	N/A
DIFF_SSTL18_II	Both	1.8	Any	N/A	N/A

Table 1-55: V_{CCO} and V_{REF} Requirements for Each Supported I/O Standard (Cont'd)

I/O Standard	I/O Bank Availability	V_{CCO} (V)			V_{REF} (V)
		Output	Input	Input with DIFF_TERM = TRUE	Input
DIFF_SSTL18_I1_DCI	HP	1.8	1.8	N/A	N/A
DIFF_SSTL18_I1_T_DCI	HP	1.8	1.8	N/A	N/A
HSLVDCI_15	HP	1.5	Any	N/A	0.75
HSLVDCI_18	HP	1.8	Any	N/A	0.9
HSTL_I	Both	1.5	Any	N/A	0.75
HSTL_I_12	HP	1.2	Any	N/A	0.6
HSTL_I_18	Both	1.8	Any	N/A	0.9
HSTL_I_DCI	HP	1.5	1.5	N/A	0.75
HSTL_I_DCI_18	HP	1.8	1.8	N/A	0.9
HSTL_I1	Both	1.5	Any	N/A	0.75
HSTL_I1_18	Both	1.8	Any	N/A	0.9
HSTL_I1_DCI	HP	1.5	1.5	N/A	0.75
HSTL_I1_DCI_18	HP	1.8	1.8	N/A	0.9
HSTL_I1_T_DCI	HP	1.5	1.5	N/A	0.75
HSTL_I1_T_DCI_18	HP	1.8	1.8	N/A	0.9
HSUL_12	Both	1.2	Any	N/A	0.6
HSUL_12_DCI	HP	1.2	1.2	N/A	0.6
LVC MOS12	Both	1.2	1.2	N/A	N/A
LVC MOS15	Both	1.5	1.5	N/A	N/A
LVC MOS18	Both	1.8	1.8	N/A	N/A
LVC MOS25	HR	2.5	2.5	N/A	N/A
LVC MOS33	HR	3.3	3.3	N/A	N/A
LVDCI_15	HP	1.5	1.5	N/A	N/A
LVDCI_18	HP	1.8	1.8	N/A	N/A
LVDCI_DV2_15	HP	1.5	1.5	N/A	N/A
LVDCI_DV2_18	HP	1.8	1.8	N/A	N/A
LVDS	HP	1.8	1.8 ⁽¹⁾	1.8	N/A
LVDS_25	HR	2.5 ⁽²⁾	2.5 ⁽¹⁾	2.5	N/A
SSTL12	HP	1.2	Any	N/A	0.6
SSTL12_DCI	HP	1.2	1.2	N/A	0.6
SSTL12_T_DCI	HP	1.2	1.2	N/A	0.6

Table 1-55: V_{CCO} and V_{REF} Requirements for Each Supported I/O Standard (Cont'd)

I/O Standard	I/O Bank Availability	V_{CCO} (V)			V_{REF} (V)
		Output	Input	Input with DIFF_TERM = TRUE	Input
LVTTL	HR	3.3	3.3	N/A	N/A
MINI_LVDS_25	HR	2.5 ⁽²⁾	2.5 ⁽¹⁾	2.5	N/A
MOBILE_DDR	HR	1.8	1.8	N/A	N/A
PCI33_3	HR	3.3	3.3	N/A	N/A
PPDS_25	HR	2.5 ⁽²⁾	2.5 ⁽¹⁾	2.5	N/A
RSDS_25	HR	2.5 ⁽²⁾	2.5 ⁽¹⁾	2.5	N/A
SSTL135	Both	1.35	Any	N/A	0.675
SSTL135_R	HR	1.35	Any	N/A	0.675
SSTL135_DCI	HP	1.35	1.35	N/A	0.675
SSTL135_T_DCI	HP	1.35	1.35	N/A	0.675
SSTL15	Both	1.5	Any	N/A	0.75
SSTL15_R	HR	1.5	Any	N/A	0.75
SSTL15_DCI	HP	1.5	1.5	N/A	0.75
SSTL15_T_DCI	HP	1.5	1.5	N/A	0.75
SSTL18_I	Both	1.8	Any	N/A	0.9
SSTL18_I_DCI	HP	1.8	1.8	N/A	0.9
SSTL18_II	Both	1.8	Any	N/A	0.9
SSTL18_II_DCI	HP	1.8	1.8	N/A	0.9
SSTL18_II_T_DCI	HP	1.8	1.8	N/A	0.9
TMDS_33	HR	3.3	Any	N/A	N/A

Notes:

- Differential inputs for these standards can be placed in banks with V_{CCO} levels that are different from the required level for outputs. There are some important criteria that need to be considered:
 - The optional internal differential termination is not used (DIFF_TERM = FALSE, which is the default value) unless the V_{CCO} voltage is at the level required for outputs.
 - The differential signals at the input pins meet the V_{IN} requirements in the Recommended Operating Conditions table of the specific device family data sheet.
 - The differential signals at the input pins meet the V_{IDIFF} and V_{ICM} requirements in the corresponding LVDS or LVDS_25 DC Specifications tables in the specific device family data sheet. In some cases, to accomplish this it might be necessary to provide an external circuit to both AC-couple and DC-bias the pins.
- If the voltage exceeds 2.85V, the outputs will be in a high-Z state. The device should always be operated within the recommended operating range as specified in the 7 series FPGA data sheets.

Table 1-56, summarizes the DRIVE and SLEW attribute options, bidirectional buffer availability, and DCI termination type for each 7 series FPGA supported I/O standard.

Table 1-56: DRIVE and SLEW Attributes, Bidirectional Buffers, and DCI Termination Type

I/O Standard	I/O Bank Availability	DRIVE (mA)	SLEW	Bidirectional Buffers ⁽¹⁾	DCI Type ⁽²⁾	
		Outputs	Outputs		Outputs	Inputs
BLVDS_25	HR	N/A	N/A	Yes	None	None
DIFF_HSTL_I	Both	N/A	SLOW, FAST	No	None	None
DIFF_HSTL_I_18	Both	N/A	SLOW, FAST	No	None	None
DIFF_HSTL_I_DCI	HP	N/A	SLOW, FAST	No	None	Split
DIFF_HSTL_I_DCI_18	HP	N/A	SLOW, FAST	No	None	Split
DIFF_HSTL_II	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_HSTL_II_18	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_HSTL_II_DCI	HP	N/A	SLOW, FAST	Yes	Split	Split
DIFF_HSTL_II_DCI_18	HP	N/A	SLOW, FAST	Yes	Split	Split
DIFF_HSTL_II_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
DIFF_HSTL_II_T_DCI_18	HP	N/A	SLOW, FAST	Required	None	Split
DIFF_HSUL_12	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_HSUL_12_DCI	HP	N/A	SLOW, FAST	Yes	Driver	None
DIFF_MOBILE_DDR	HR	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL12	HP	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL12_DCI	HP	N/A	SLOW, FAST	No	None	Split
DIFF_SSTL12_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
DIFF_SSTL135	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL135_R	HR	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL135_DCI	HP	N/A	SLOW, FAST	No	None	Split
DIFF_SSTL135_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
DIFF_SSTL15	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL15_R	HR	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL15_DCI	HP	N/A	SLOW, FAST	No	None	Split
DIFF_SSTL15_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
DIFF_SSTL18_I	Both	N/A	SLOW, FAST	No	None	None
DIFF_SSTL18_I_DCI	HP	N/A	SLOW, FAST	No	None	Split
DIFF_SSTL18_II	Both	N/A	SLOW, FAST	Yes	None	None
DIFF_SSTL18_II_DCI	HP	N/A	SLOW, FAST	Yes	Split	Split
DIFF_SSTL18_II_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split

Table 1-56: DRIVE and SLEW Attributes, Bidirectional Buffers, and DCI Termination Type (Cont'd)

I/O Standard	I/O Bank Availability	DRIVE (mA)	SLEW	Bidirectional Buffers ⁽¹⁾	DCI Type ⁽²⁾	
		Outputs	Outputs		Outputs	Inputs
HSLVDCI_15	HP	N/A	N/A	Yes	Driver	None
HSLVDCI_18	HP	N/A	N/A	Yes	Driver	None
HSTL_I	Both	N/A	SLOW, FAST	No	None	None
HSTL_I_12	HP	N/A	SLOW, FAST	No	None	None
HSTL_I_18	Both	N/A	SLOW, FAST	No	None	None
HSTL_I_DCI	HP	N/A	SLOW, FAST	No	None	Split
HSTL_I_DCI_18	HP	N/A	SLOW, FAST	No	None	Split
HSTL_II	Both	N/A	SLOW, FAST	Yes	None	None
HSTL_II_18	Both	N/A	SLOW, FAST	Yes	None	None
HSTL_II_DCI	HP	N/A	SLOW, FAST	Yes	Split	Split
HSTL_II_DCI_18	HP	N/A	SLOW, FAST	Yes	Split	Split
HSTL_II_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
HSTL_II_T_DCI_18	HP	N/A	SLOW, FAST	Required	None	Split
HSUL_12	Both	N/A	SLOW, FAST	Yes	None	None
HSUL_12_DCI	HP	N/A	SLOW, FAST	Yes	Driver	None
LVC MOS12	Both	HP: 2, 4, 6, 8 HR: 4, 8, 12	SLOW, FAST	Yes	None	None
LVC MOS15	Both	HP: 2, 4, 6, 8, 12, 16 HR: 4, 8, 12, 16	SLOW, FAST	Yes	None	None
LVC MOS18	Both	HP: 2, 4, 6, 8, 12, 16 HR: 4, 8, 12, 16, 24	SLOW, FAST	Yes	None	None
LVC MOS25	HR	4, 8, 12, 16	SLOW, FAST	Yes	None	None
LVC MOS33	HR	4, 8, 12, 16	SLOW, FAST	Yes	None	None
LVDCI_15	HP	N/A	N/A	Yes	Driver	None
LVDCI_18	HP	N/A	N/A	Yes	Driver	None
LVDCI_DV2_15	HP	N/A	N/A	Yes	Driver	None
LVDCI_DV2_18	HP	N/A	N/A	Yes	Driver	None
LVDS	HP	N/A	N/A	Yes	None	None
LVDS_25	HR	N/A	N/A	Yes ⁽³⁾	None	None
SSTL12	HP	N/A	SLOW, FAST	Yes	None	None
SSTL12_DCI	HP	N/A	SLOW, FAST	No	None	Split
SSTL12_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
LVTTL	HR	4, 8, 12, 16, 24	SLOW, FAST	Yes	None	None

Table 1-56: DRIVE and SLEW Attributes, Bidirectional Buffers, and DCI Termination Type (Cont'd)

I/O Standard	I/O Bank Availability	DRIVE (mA)	SLEW	Bidirectional Buffers ⁽¹⁾	DCI Type ⁽²⁾	
		Outputs	Outputs		Outputs	Inputs
MINI_LVDS_25	HR	N/A	N/A	Yes ⁽³⁾	None	None
MOBILE_DDR	HR	N/A	SLOW, FAST	Yes	None	None
PCI33_3	HR	N/A	N/A	Yes	None	None
PPDS_25	HR	N/A	N/A	No	None	None
RSDS_25	HR	N/A	N/A	No	None	None
SSTL135	Both	N/A	SLOW, FAST	Yes	None	None
SSTL135_R	HR	N/A	SLOW, FAST	Yes	None	None
SSTL135_DCI	HP	N/A	SLOW, FAST	No	None	Split
SSTL135_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
SSTL15	Both	N/A	SLOW, FAST	Yes	None	None
SSTL15_R	HR	N/A	SLOW, FAST	Yes	None	None
SSTL15_DCI	HP	N/A	SLOW, FAST	No	None	Split
SSTL15_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
SSTL18_I	Both	N/A	SLOW, FAST	No	None	None
SSTL18_I_DCI	HP	N/A	SLOW, FAST	No	None	Split
SSTL18_II	Both	N/A	SLOW, FAST	Yes	None	None
SSTL18_II_DCI	HP	N/A	SLOW, FAST	Yes	Split	Split
SSTL18_II_T_DCI	HP	N/A	SLOW, FAST	Required	None	Split
TMDS_33	HR	N/A	N/A	No	None	None

Notes:

1. The bidirectional buffers column describes the I/O standards use of a bidirectional signal. The standards labeled as required can only be used with bidirectional signals and require primitives such as the IOBUF and IOBUFDS.
2. The DCI termination type column describes the type of termination available for the DCI I/O standards. Split refers to the split-termination resistors.
3. Internal differential termination is always used in bidirectional configuration.

Simultaneous Switching Outputs

Due to package inductance, each part/package supports a limited number of simultaneous switching outputs (SSOs), particularly when using fast, high-drive outputs. Fast, high-drive outputs should only be used when required by the application.

The SSN predictor tool within the PlanAhead software provides a way of analyzing the amount of noise margin on each I/O pin in a design based on information for the pin (the victim), as well as all other pins (aggressors) in the design. The tool takes into account I/O pin locations, I/O standards, slew rates, and terminations used, and provides a value for the noise margin for each pin based on these characteristics. The noise margin does not include any system-level characteristics such as board trace cross-talk or reflections due to board impedance discontinuities.

Ground or power bounce occurs when a large number of outputs simultaneously switch in the same direction. The output drive transistors all conduct current to a common rail. Low-to-High transitions connect to the V_{CCO} rail, while High-to-Low transitions connect to the ground rail. The resulting cumulative current transient induces a voltage difference across the inductance that exists between the internal and external ground levels, or internal and external V_{CCO} levels. The inductance is associated with bonding wires, package lead frame, die routing, package routing, and ball inductance. Any SSO-induced voltage consequently affects internal switching noise margins and ultimately signal quality.

The SSN predictor results assume that the FPGA is soldered on the PCB and that the board uses sound design practices. The noise margin values do not apply for FPGAs mounted in sockets due to the additional BGA ball inductance introduced by the socket.

Pin Planning to Mitigate SSO Sensitivity

When performing pin planning of a design, it is important to choose I/O pin placements that separate strong outputs and/or SSOs from sensitive inputs and outputs (particularly asynchronous inputs). Strong outputs tend to be the class-II versions of HSTL and SSTL drivers, PCI variants, and any LVCMOS or LVTTL with drive strengths over 8 mA. Sensitive inputs and outputs can have a low noise margin and tend to be high-speed signals or signals where the swing is reduced by parallel receiver termination. Because localized SSO noise in 7 series FPGAs is based on the proximity of signals to one another, it is important to try to separate signals based on the position of the package solder balls. To further reduce potential noise induced from SSOs, outputs should be distributed evenly rather than clustered in one area. SSOs within a bank should be spread across the bank as much as possible. Whenever possible, SSOs should be distributed into multiple banks.

The floorplanning capability in the Vivado® Design Suite and the PlanAhead tool in the ISE software can help accomplish pin planning to avoid SSO sensitivity issues. By clicking on a package pin in the Package window, a corresponding IOB is highlighted in the Device window. These IOB site types represent the die pads and show the relative physical location around the die edge. Through the use of the PlanAhead tool and the floorplanning capability in the Vivado Design Suite, intelligent pin placement can be used to separate the die pads of pins. This is implemented by separating the die pads of pins with strong outputs and SSOs from the die pads of pins with sensitive inputs and outputs. SSO effects can also be minimized by adding virtual ground pins and virtual V_{CCO} pins. A virtual ground is created by defining an output pin driven by a logic 0 at the highest drive strength available and connected to ground on the board. Similarly, a virtual V_{CCO} pin is created by defining an output pin driven by a logic 1 at the highest drive strength and connected to V_{CCO} on the board.

SelectIO Logic Resources

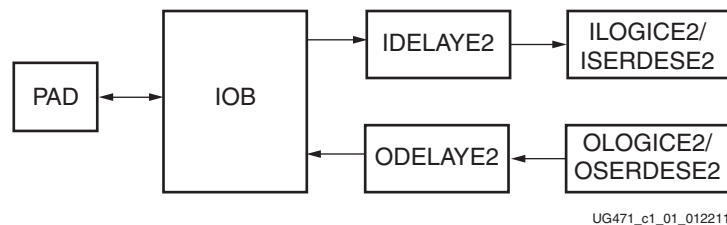
Introduction

This chapter describes the logic directly behind the I/O drivers and receivers covered in [Chapter 1, SelectIO Resources](#).

7 series FPGAs contain the basic I/O logic resources from previous Xilinx FPGAs. These resources include the following:

- Combinatorial input/output
- 3-state output control
- Registered input/output
- Registered 3-state output control
- Double-Data-Rate (DDR) input/output
- DDR output 3-state control
- IDELAY provides users control of an adjustable, fine-resolution delay taps
- ODELAY provides users control of an adjustable, fine-resolution delay taps
- SAME_EDGE output DDR mode
- SAME_EDGE and SAME_EDGE_PIPELINED input DDR mode

[Figure 2-1](#) shows a I/O tile for the 1.8V HP banks. [Figure 2-2](#) shows an I/O tile for a 3.3V HR bank. The SelectIO™ input, output, and 3-state drivers are in the input/output buffer (IOB). The HP banks have separate IDELAY and ODELAY blocks. The HR bank has the same logic elements as the HP banks except for the ODELAY block.



UG471_c1_01_012211

Figure 2-1: 7 Series FPGA HP Bank I/O Tile

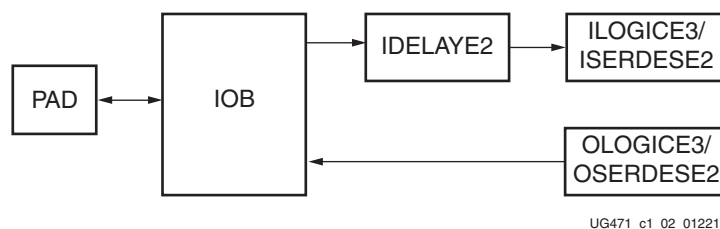


Figure 2-2: 7 Series FPGA HR Bank I/O Tile

ILOGIC Resources

The ILOGIC block is located next to the I/O block (IOB). The ILOGIC block contains the synchronous elements for capturing data as it comes into the FPGA through the IOB. The possibilities for ILOGIC configuration in 7 series devices are the ILOGICE2 (HP I/O banks) and ILOGICE3 (HR I/O banks). Although always described as ILOGIC in this guide, unless explicitly delineated, ILOGICE2 and ILOGICE3 are functionally identical and so are their ports. The only differences between ILOGICE2 and ILOGICE3 are:

- ILOGICE3 is located in the HR banks and has a zero hold delay element (ZHOLD).
- ILOGICE2 is located in the HP banks and does not have a ZHOLD element.

These differences are shown in Figure 2-3 and Figure 2-4. The ZHOLD delay at the D-input of the input/output interconnect (IOI) storage element eliminates any pad-to-pad hold time requirement. The ZHOLD delay is automatically matched to the internal clock-distribution delay, and when used, assures that the pad-to-pad hold time is zero. The ILOGIC block supports an optional static uncompensated zero hold (ZHOLD) delay line on inputs to compensate for clock insertion delay. The ZHOLD feature is optimized to compensate for clock insertion delays when the clocking path is directly sourced from a BUFG/BUFGCE that is sourced in the same bank or on an adjacent bank. ZHOLD is enabled by default unless the clock source is an MMCM or PLL, or unless the IOBDELAY attribute is set in the Xilinx design constraints (XDC).

Important: ZHOLD might not be appropriate for all applications, so consult the timing report to verify the impact to a specific clocking scheme.

ILOGICE2 and ILOGICE3 are not primitives in the sense that they cannot be instantiated. They contain user-instantiated elements such as an input flip-flop (IFD) or an input DDR element (IDDR) after place and route.

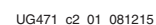
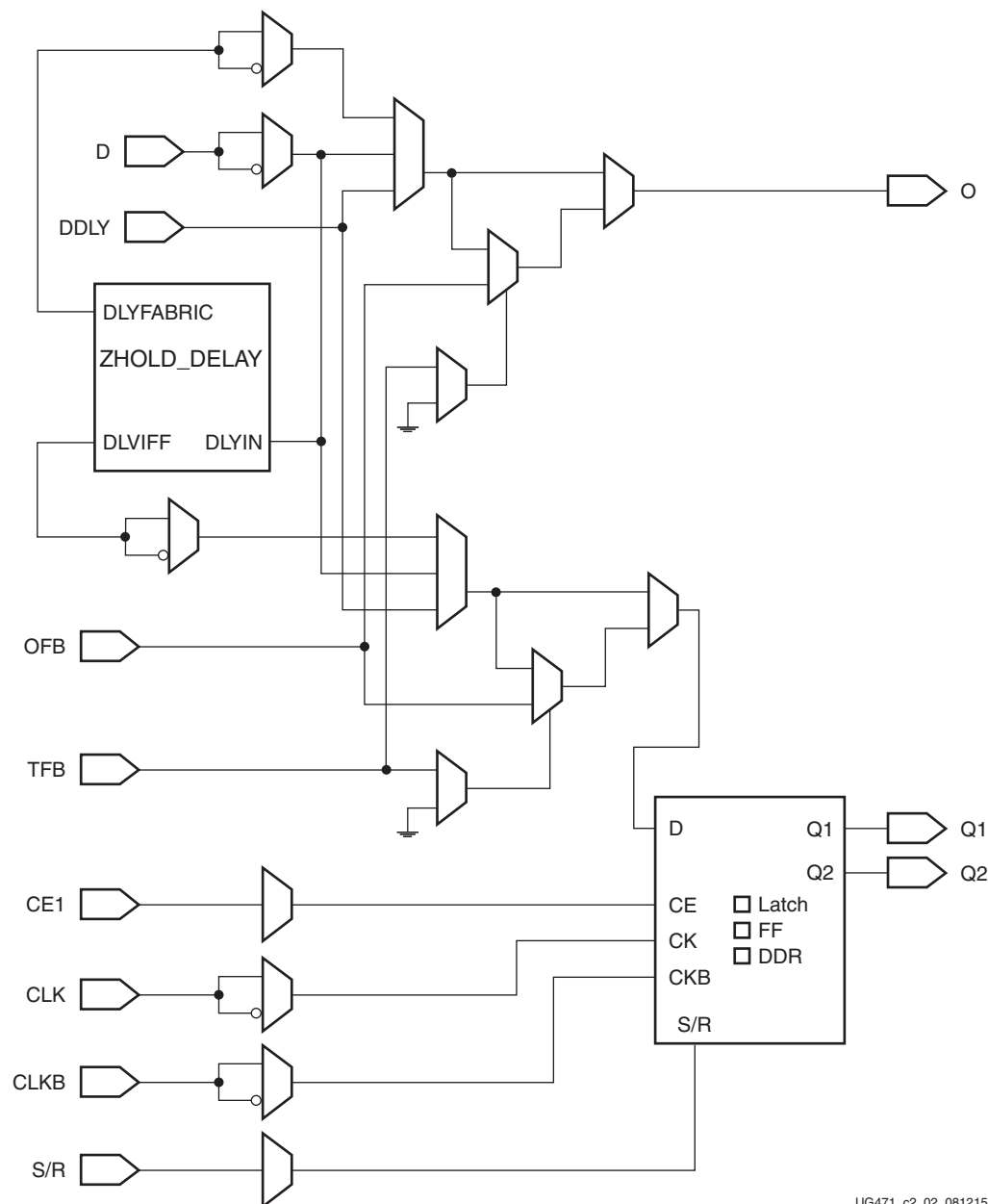


Figure 2-3: ILOGICE2 Block Diagram



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Figure 2-4: ILOGICE3 Block Diagram

ILOGIC can support the following operations:

- Edge-triggered D-type flip-flop
- IDDR mode (OPPOSITE_EDGE or SAME_EDGE or SAME_EDGE_PIPELINED). See [Input DDR Overview \(IDDR\)](#), page 109 for further discussion on input DDR.
- Level sensitive latch
- Asynchronous/combinatorial

The ILOGIC block registers have a common clock enable signal (CE1) that is active High by default. If left unconnected, the clock enable pin for any storage element defaults to the active state.

The ILOGIC block registers have a common synchronous or asynchronous set and reset (SR signal). The set/reset input pin, SR forces the storage element into the state specified by the SRVAL attributes. The reset condition predominates over the set condition.

The SRVAL attributes can be set individually for each storage element in the ILOGIC block, but the choice of synchronous or asynchronous set/reset (SRTYPE) can not be set individually for each storage element in the ILOGIC block.

The following sections discuss the various resources within the ILOGIC blocks.

Combinatorial Input Path

The combinatorial input path is used to create a direct connection from the input driver to the FPGA logic. This path is used by software automatically when:

1. There is a direct (unregistered) connection from input data to logic resources in the FPGA logic.
2. The **pack I/O register/latches into IOBs** software map directive is set to OFF.

Input DDR Overview (IDDR)

7 series devices have dedicated registers in the ILOGIC blocks to implement input double-data-rate (DDR) registers. This feature is used by instantiating the IDDR primitive.

All clocks feeding into the I/O tile are fully multiplexed, i.e., there is no clock sharing between ILOGIC and OLOGIC blocks. The IDDR primitive supports the following modes of operation:

- OPPOSITE_EDGE mode
- SAME_EDGE mode
- SAME_EDGE_PIPELINED mode

The SAME_EDGE and SAME_EDGE_PIPELINED modes are the same as for the Virtex-6 architecture. These modes allow designers to transfer falling edge data to the rising edge domain within the ILOGIC block, saving CLB and clock resources, and increasing performance. These modes are implemented using the DDR_CLK_EDGE attribute. The following sections describe each of the modes in detail.

OPPOSITE_EDGE Mode

A traditional input DDR solution, or OPPOSITE_EDGE mode, is accomplished via a single input in the ILOGIC block. The data is presented to the FPGA logic via the output Q1 on the rising edge of the clock and via the output Q2 on the falling edge of the clock. This structure is similar to the Virtex-6 FPGA implementation. [Figure 2-5](#) shows the timing diagram of the input DDR using the OPPOSITE_EDGE mode.

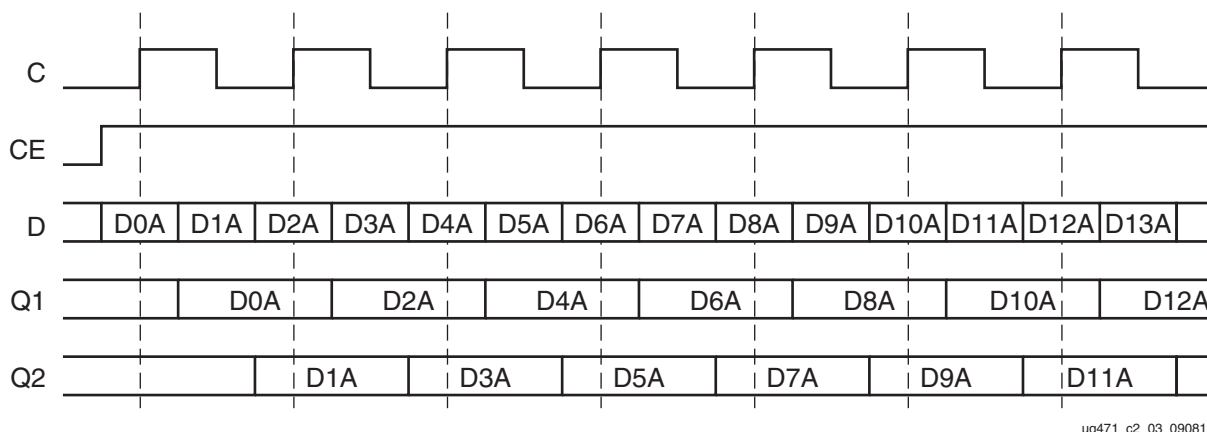


Figure 2-5: Input DDR Timing in OPPOSITE_EDGE Mode

SAME_EDGE Mode

In the SAME_EDGE mode, the data is presented into the FPGA logic on the same clock edge. This structure is similar to the Virtex-6 FPGA implementation.

Figure 2-6 shows the timing diagram of the input DDR using SAME_EDGE mode. In the timing diagram, the output pairs Q1 and Q2 are no longer (0) and (1). Instead, the first pair presented is pair Q1 (0) and Q2 (don't care), followed by pair (1) and (2) on the next clock cycle.

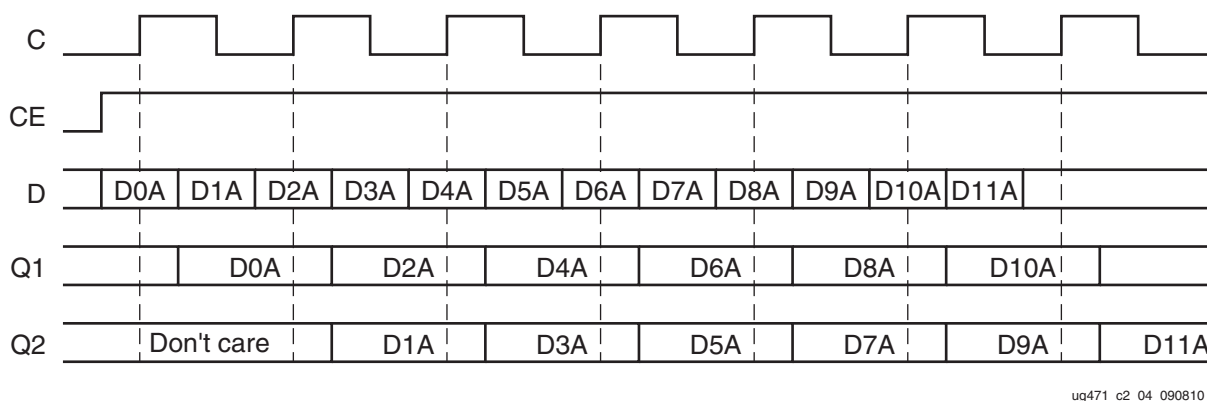
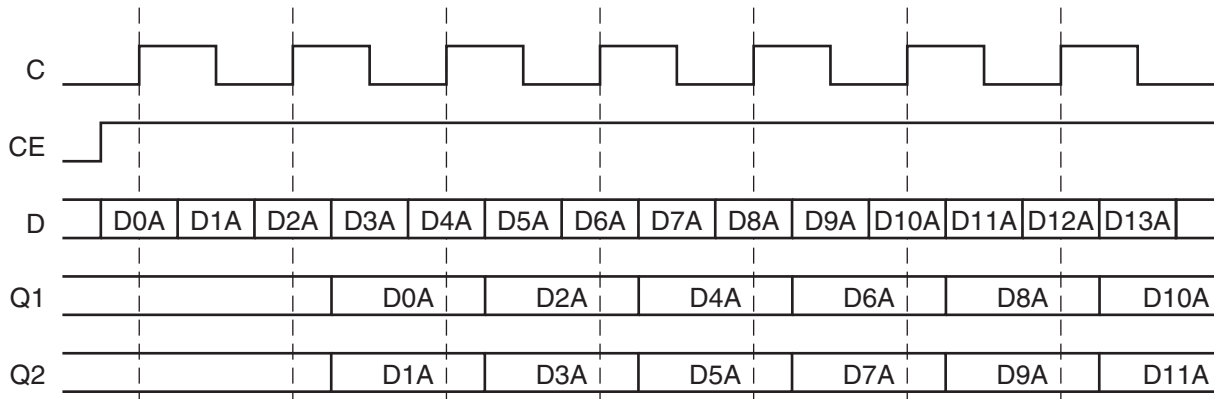


Figure 2-6: Input DDR Timing in SAME_EDGE Mode

SAME_EDGE_PIPELINED Mode

In the SAME_EDGE_PIPELINED mode, the data is presented into the FPGA logic on the same clock edge.

Unlike the SAME_EDGE mode, the data pair is not separated by one clock cycle. However, an additional clock latency is required to remove the separated effect of the SAME_EDGE mode. Figure 2-7 shows the timing diagram of the input DDR using the SAME_EDGE_PIPELINED mode. The output pairs Q1 and Q2 are presented to the FPGA logic at the same time.

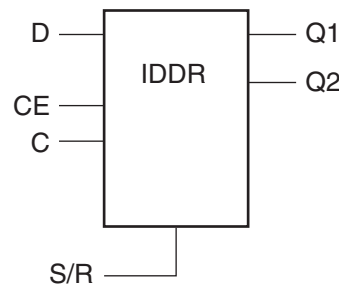


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Figure 2-7: Input DDR Timing in SAME_EDGE_PIPELINED Mode

Input DDR Resources (IDDR)

Figure 2-8 shows the block diagram of the IDDR primitive. Set and Reset are not supported at the same time. Table 2-1 lists the IDDR port signals. Table 2-2 describes the various attributes available and default values for the IDDR primitive.



ug471_c2_06_061215

Figure 2-8: IDDR Primitive Block Diagram

Table 2-1: IDDR Port Signals

Port Name	Function	Description
Q1 and Q2	Data outputs	IDDR register outputs.
C	Clock input port	The C pin represents the clock input pin.
CE	Clock enable port	The enable pin affects the loading of data into the DDR flip-flop. When Low, clock transitions are ignored and new data is not loaded into the DDR flip-flop. CE must be High to load new data into the DDR flip-flop.
D	Data input (DDR)	IDDR register input from IOB.

Table 2-1: IDDR Port Signals (*Cont'd*)

Port Name	Function	Description
S/R ⁽¹⁾	Set/Reset	Synchronous/Asynchronous Set/Reset pin. S/R is asserted High.

Notes:

1. The IDDR primitive contains both set and reset pins. However only one can be used per IDDR. As a result, S/R is described instead of separate set and reset pins.

Table 2-2: IDDR Attributes

Attribute Name	Description	Possible Values
DDR_CLK_EDGE	Sets the IDDR mode of operation with respect to clock edge	OPPOSITE_EDGE (default), SAME_EDGE, SAME_EDGE_PIPELINED
INIT_Q1	Sets the initial value for Q1 port	0 (default), 1
INIT_Q2	Sets the initial value for Q2 port	0 (default), 1
SRTYPE	Set/Reset type with respect to clock (C)	ASYNCR (default), SYNC

IDDR VHDL and Verilog Templates

The Libraries Guide includes templates for instantiation of the IDDR primitive in VHDL and Verilog.

ILOGIC Timing Models

This section describes the timing associated with the various resources within the ILOGIC block.

ILOGIC Timing Characteristics

Figure 2-9 illustrates ILOGIC register timing. When IDELAY is used, T_{IDOCK} is replaced by T_{IDOCKD} .

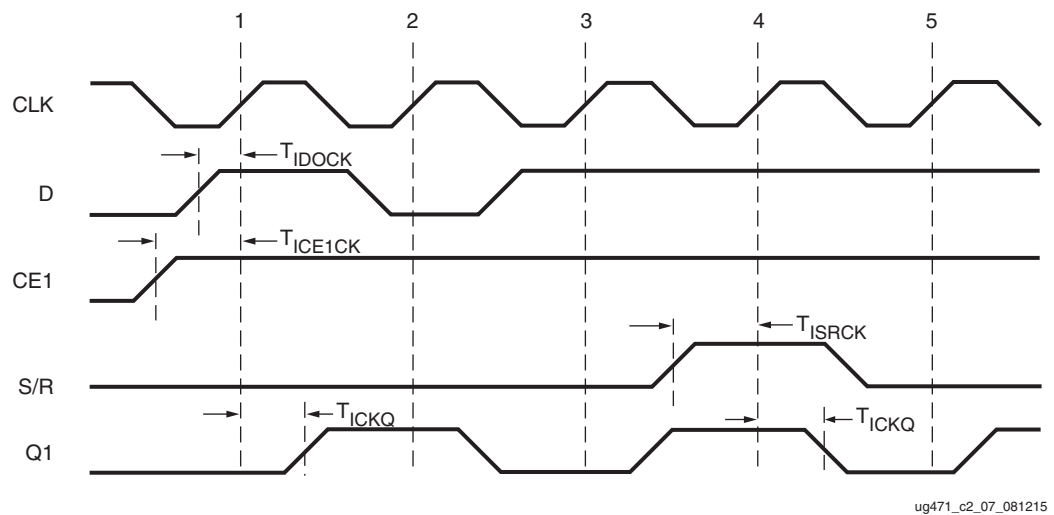


Figure 2-9: ILOGIC Input Register Timing Characteristics

Clock Event 1

- At time T_{ICE1CK} before Clock Event 1, the input clock enable signal becomes valid-high at the CE1 input of the input register, enabling the input register for incoming data.
- At time T_{IDOCK} before Clock Event 1, the input signal becomes valid-high at the D input of the input register and is reflected on the Q1 output of the input register at time T_{ICKQ} after Clock Event 1.

Clock Event 4

- At time T_{ISRCK} before Clock Event 4, the S/R signal (configured as synchronous reset in this case) becomes valid-high resetting the input register and reflected at the Q1 output of the IOB at time T_{ICKQ} after Clock Event 4.

ILOGIC Timing Characteristics, DDR

Figure 2-10 illustrates the ILOGIC in IDDR mode timing characteristics. When IDELAY is used, T_{IDOCK} is replaced by T_{IDOCKD} . The example shown uses IDDR in OPPOSITE_EDGE mode. For other modes, add the appropriate latencies as shown in Figure 2-7, page 111.

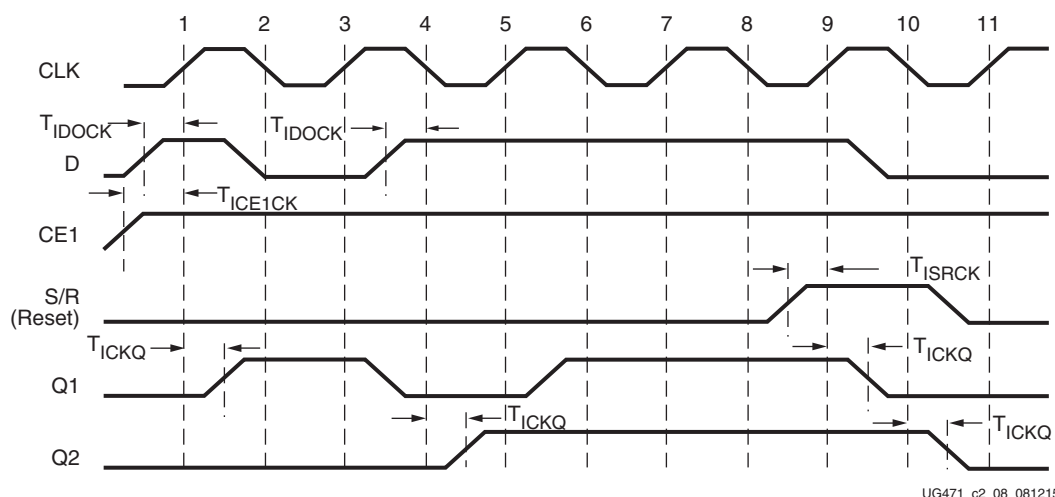


Figure 2-10: ILOGIC in IDDR Mode Timing Characteristics (OPPOSITE_EDGE Mode)

Clock Event 1

- At time T_{ICE1CK} before Clock Event 1, the input clock enable signal becomes valid-high at the CE1 input of both of the DDR input registers, enabling them for incoming data. Since the CE1 and D signals are common to both DDR registers, care must be taken to toggle these signals between the rising edges and falling edges of CLK as well as meeting the register setup-time relative to both edges.
- At time T_{IDOCK} before Clock Event 1 (rising edge of CLK), the input signal becomes valid-high at the D input of both registers and is reflected on the Q1 output of input-register 1 at time T_{ICKQ} after Clock Event 1.

Clock Event 4

- At time T_{IDOCK} before Clock Event 4 (falling edge of CLK), the input signal becomes valid-low at the D input of both registers and is reflected on the Q2 output of input-register 2 at time T_{ICKQ} after Clock Event 4 (no change in this case).

Clock Event 9

- At time T_{ISRCK} before Clock Event 9, the S/R signal (configured as synchronous reset in this case) becomes valid-high resetting Q1 at time T_{ICKQ} after Clock Event 9, and Q2 at time T_{ICKQ} after Clock Event 10.

Table 2-3 describes the timing parameters of the ILOGIC switching characteristics in the 7 series FPGA data sheets.

Table 2-3: ILOGIC Switching Characteristics

Symbol	Description
Setup/Hold	
T_{ICE1CK}/T_{ICKCE1}	CE1 pin Setup/Hold with respect to CLK
T_{ISRCK}/T_{ICKSR}	S/R pin Setup/Hold with respect to CLK
T_{IDOCK}/T_{IOCKD}	D pin Setup/Hold with respect to CLK

Table 2-3: ILOGIC Switching Characteristics (Cont'd)

Symbol	Description
$T_{\text{ICOCKD}}/T_{\text{IOCKDD}}$	DDLY pin Setup/Hold with respect to CLK
Combinatorial	
T_{IDI}	D pin to O pin propagation delay, no Delay
Sequential Delays	
T_{IDLO}	D pin to Q1 pin using flip-flop as a latch without Delay
T_{ICKQ}	CLK to Q outputs
T_{RQ}	S/R pin to OQ/TQ out

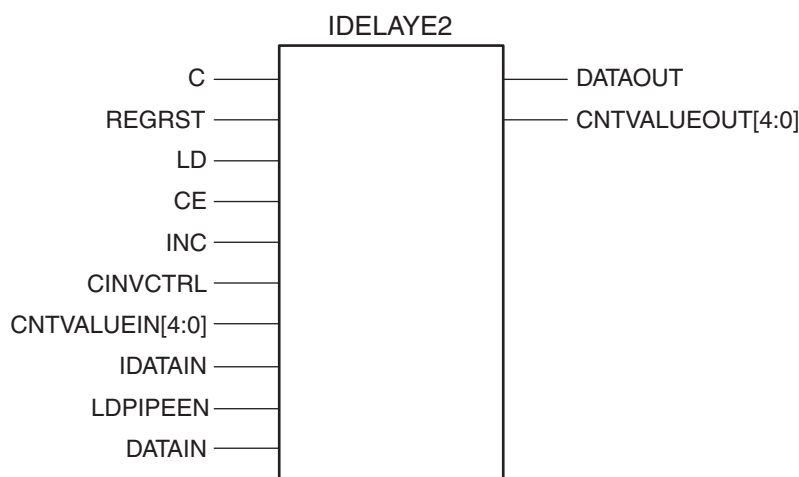
Note: The DDLY pin timing diagrams and parameters are identical to the D pin timing diagrams and parameters.

Input Delay Resources (IDELAY)

Every I/O block contains a programmable delay primitive called IDELAYE2. The IDELAY can be connected to an ILOGICE2/ISERDESE2 or ILOGICE3/ISERDESE2 block. IDELAYE2 is a 31-tap, wraparound, delay primitive with a calibrated tap resolution. Refer to the 7 series FPGA data sheets for delay values. It can be applied to the combinatorial input path, registered input path, or both. It can also be accessed directly from the FPGA logic. IDELAY allows incoming signals to be delayed on an individual input pin basis. The tap delay resolution is contiguously calibrated by the use of an IDELAYCTRL reference clock from the range specified in the 7 series FPGA data sheets.

IDELAYE2 Primitive

Figure 2-11 shows the IDELAYE2 primitive.



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Figure 2-11: IDELAYE2 Primitive

Table 2-4 lists the available ports in the IDELAYE2 primitive.

Table 2-4: IDELAYE2 Primitive Ports

Port Name	Direction	Width	Function
C	Input	1	Clock input used in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode.
REGRST	Input	1	Reset for the pipeline register. Only used in VAR_LOAD_PIPE mode.
LD	Input	1	Loads the IDELAYE2 primitive to the pre-programmed value in VARIABLE mode. In VAR_LOAD mode, it loads the value of CNTVALUEIN. In VAR_LOAD_PIPE mode it loads the value currently in the pipeline register.
CE	Input	1	Enable increment/decrement function.
INC	Input	1	Increment/decrement number of tap delays.
CINVCTRL	Input	1	Dynamically inverts the clock (C) polarity.
CNTVALUEIN	Input	5	Counter value from FPGA logic for dynamically loadable tap value.
IDATAIN	Input	1	Data input for IDELAY from the IBUF.
DATAIN	Input	1	Data input for IDELAY from the FPGA logic.
LDPIPEEN	Input	1	Enables the pipeline register to load data.
DATAOUT	Output	1	Delayed data from one of two data input ports (IDATAIN or DATAIN).
CNTVALUEOUT	Output	5	Counter value going to FPGA logic for monitoring tap value.

IDELAY Ports

Data Input from the IOB - IDATAIN

The IDATAIN input is driven by its associated IOB. IDELAY can drive data to either an ILOGICE2/ISERDESE2 or ILOGICE3/ISERDESE2 block, directly into the FPGA logic, or to both through the DATAOUT port with a delay set by the IDELAY_VALUE.

Data Input from the FPGA Logic - DATAIN

The DATAIN input is directly driven by the FPGA logic providing a logic accessible delay line. The data is driven back into the FPGA logic through the DATAOUT port with a delay set by the IDELAY_VALUE. DATAIN can be locally inverted. The data cannot be driven to an IOB.

Data Output - DATAOUT

Delayed data from the two data input ports. DATAOUT can drive to either an ILOGICE2/ISERDESE2 or ILOGICE3/ISERDESE2 block, directly into the FPGA logic, or to both.

Clock Input - C

All control inputs to IDELAYE2 primitive (REGRST, LD, CE, and INC) are synchronous to the clock input (C). A clock must be connected to this port when IDELAY is configured in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode. C can be locally inverted, and must be supplied by a global or regional clock buffer. If the ODELAYE2 primitive is used in the same I/O bank as the IDELAYE2 primitive, C must use the same clock net for both primitives.

Module Load - LD

When in VARIABLE mode, the IDELAY load port, LD, loads the value set by the IDELAY_VALUE attribute. The default value of the IDELAY_VALUE attribute is zero. When the default value is used, the LD port acts as an asynchronous reset for the IDELAY. The LD signal is an active-High signal and is synchronous to the input clock signal (C).

When in VAR_LOAD mode, the IDELAY load port, LD, loads the value set by the CNTVALUEIN. The value present at CNTVALUEIN[4:0] will be the new tap value. When in VAR_LOAD_PIPE mode, the IDELAY load port LD loads the value currently in the pipeline register. The value present in the pipeline register will be the new tap value.

C Pin Polarity Switch - CINVCTRL

The CINVCTRL pin is used for dynamically switching the polarity of the C pin. This is for use in applications when glitches are not an issue. When switching the polarity, do not use IDELAY control pins for two clock cycles.

Count Value In - CNTVALUEIN

The CNTVALUEIN pins are used for dynamically switching the loadable tap value.

Count Value Out - CNTVALUEOUT

The CNTVALUEOUT pins are used for reporting the loaded tap value.

Pipeline Register Load - LDPIPEEN

When High, this input loads the pipeline register with the value currently on the CNTVALUEIN pins.

Pipeline Register Reset - REGRST

When high, this input resets the pipeline register to all zeroes.

Increment/Decrement Signals - CE, INC

The increment/decrement is controlled by the enable signal (CE). This interface is only available when the IDELAY is in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode.

As long as CE remains High, IDELAY will increment or decrement by $T_{IDELAYRESOLUTION}$ every clock (C) cycle. The state of INC determines whether IDELAY will increment or decrement; INC = 1 increments, INC = 0 decrements, synchronously to the clock (C). If CE is Low the delay through IDELAY will not change regardless of the state of INC.

When CE goes High, the increment/decrement operation begins on the next positive clock edge. When CE goes Low, the increment/decrement operation ceases on the next positive clock edge.

The programmable delay taps in the IDELAYE2 primitive wrap-around. When the last tap delay is reached (tap 31) a subsequent increment function will return to tap 0. The same applies to the decrement function: decrementing from zero moves to tap 31.

The pipeline register functionality in VAR_LOAD_PIPE mode is extremely useful in bus structure designs. Individual delays can be (pipeline) loaded one at a time using LDPIPEEN and then all delays updated to their new values at the same time using the LD pin.

IDELAY Attributes

Table 2-5 summarizes the IDELAY attributes.

Table 2-5: IDELAY Attribute Summary

Attribute	Value	Default Value	Description
IDELAY_TYPE	String: FIXED, VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE	FIXED	Sets the type of tap delay line. FIXED delay sets a static delay value. VAR_LOAD dynamically loads tap values. VARIABLE delay dynamically adjusts the delay value. VAR_LOAD_PIPE is similar to VAR_LOAD mode with the ability to store the CNTVALUEIN value ready for a future update.
DELAY_SRC	String: IDATAIN, DATAIN	IDATAIN	IDATAIN: IDELAY chain input is IDATAIN DATAIN: IDELAY chain input is DATAIN
IDELAY_VALUE	Integer: 0 to 31	0	Specifies the fixed number of delay taps in fixed mode or the initial starting number of taps in VARIABLE mode (input path). When IDELAY_TYPE is set to VAR_LOAD, or VAR_LOAD_PIPE mode, this value is ignored and assumed to be zero.
HIGH_PERFORMANCE_MODE	Boolean: FALSE or TRUE	TRUE	When TRUE, this attribute reduces the output jitter. The difference in power consumption is quantified in the Xilinx Power Estimator tool.
SIGNAL_PATTERN	String: DATA, CLOCK	DATA	Causes the timing analyzer to account for the appropriate amount of delay-chain jitter in the data or clock path.
REFCLK_FREQUENCY	Real: 190 to 210, 290 to 310, or 390 to 410	200	Sets the tap value (in MHz) used by the timing analyzer for static timing analysis. The ranges of 290.0 to 310.0 and 390 to 410 are not available in all speed grades. See the 7 series FPGA data sheets.
CINVCTRL_SEL	Boolean: FALSE or TRUE	FALSE	Enables the CINVCTRL_SEL pin to dynamically switch the polarity of the C pin.
PIPE_SEL	Boolean: FALSE or TRUE	FALSE	Selects pipeline mode. This attribute should only be set to TRUE when using the VAR_LOAD_PIPE mode of operation.

IDELAY_TYPE Attribute

The IDELAY_TYPE attribute sets the type of delay used.

When the IDELAY_TYPE attribute is set to FIXED, the tap-delay value is fixed at the number of taps determined by the IDELAY_VALUE attribute setting. This value is preset and cannot be changed after configuration.

When the IDELAY_TYPE attribute is set to VARIABLE, the variable tap delay is selected. The tap delay can be incremented by setting CE = 1 and INC = 1, or decremented by CE = 1 and INC = 0. The increment/decrement operation is synchronous to C.

When the IDELAY_TYPE attribute is set to VAR_LOAD or VAR_LOAD_PIPE, the variable tap delay can be changed and dynamically loaded. The tap delay can be incremented by setting CE = 1 and INC = 1, or decremented by CE = 1 and INC = 0. The increment/decrement operation is synchronous to C. The LD pin in this mode loads the value presented on CNTVALUEIN in VAR_LOAD mode or the value previously written to the pipeline register in VAR_LOAD_PIPE mode. This allows the tap value to be dynamically set.

IDELAY_VALUE Attribute

The IDELAY_VALUE attribute specifies the initial number of tap delays. The possible values are any integer from 0 to 31. The default value is zero. The value of the tap delay reverts to IDELAY_VALUE when the tap delay is reset (by asserting the LD pin). In VARIABLE mode this attribute determines the initial setting of the delay line. In VAR_LOAD or VAR_LOAD_PIPE mode, this attribute is not used, and the initial value of the delay line is therefore always zero.

HIGH_PERFORMANCE_MODE Attribute

When TRUE, this attribute reduces the output jitter. This reduction in jitter results in a slight increase in power dissipation from the IDELAYE2 primitive.

SIGNAL_PATTERN Attribute

Clock and data signals have different electrical profiles and therefore accumulate different amounts of jitter in the IDELAY chain. By setting the SIGNAL_PATTERN attribute, the user enables timing analyzer to account for jitter appropriately when calculating timing. A clock signal is periodic in nature and does not have long sequences of consecutive ones or zeroes, while data is random in nature and can have long and short sequences of ones and zeroes.

IDELAY Modes

When used as IDELAY, the data input comes from either IBUF or the FPGA logic and the output goes to ILOGICE2/ISERDESE2 or ILOGICE3/ISERDESE2. There are four modes of operation available:

- Fixed delay mode (IDELAY_TYPE = FIXED)
In the fixed delay mode, the delay value is preset at configuration to the tap number determined by the attribute IDELAY_VALUE. Once configured, this value cannot be changed. When used in this mode, the IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details.
- Variable delay mode (IDELAY_TYPE = VARIABLE)
In the variable delay mode, the delay value can be changed after configuration by manipulating the control signals CE and INC. When used in this mode, the

IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details. The control pins being used in VARIABLE mode are described in [Table 2-6](#).

Table 2-6: Control Pin when IDELAY_TYPE = VARIABLE

C	LD	CE	INC	TAP Setting
0	x	x	x	No Change
1	1	x	x	IDELAY_VALUE
1	0	0	x	No Change
1	0	1	1	Current Value +1
1	0	1	0	Current Value -1
1	0	0	0	No Change

- Loadable variable delay mode (IDELAY_TYPE = VAR_LOAD)
In addition to having the same functionality of (IDELAY_TYPE = VARIABLE) in this mode the IDELAY tap can be loaded via the 5-input bits CNTVALUEIN<4:0> from the FPGA logic. When LD is pulsed the value present at CNTVALUEIN<4:0> will be the new tap value. As a results of this functionality the IDELAY_VALUE attribute is ignored. When used in this mode, the IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details. The control pins being used in VAR_LOAD mode are described in [Table 2-7](#).

Table 2-7: Control Pin when IDELAY_TYPE = VAR_LOAD

C	LD	CE	INC	CNTVALUEIN	CNTVALUEOUT	TAP Setting
0	x	x	x	x	No Change	No Change
1	1	x	x	CNTVALUEIN	CNTVALUEIN	CNTVALUEIN
1	0	0	x	x	No Change	No Change
1	0	1	1	x	Current Value +1	Current Value +1
1	0	1	0	x	Current Value -1	Current Value -1
1	0	0	0	0	No Change	No Change

IDELAY Timing

[Table 2-8](#) shows the IDELAY switching characteristics.

Table 2-8: IDELAY Switching Characteristics

Symbol	Description
$T_{IDELAYRESOLUTION}$	IDELAY tap resolution
T_{ICECK}/T_{ICKCE}	CE pin Setup/Hold with respect to C
T_{IINCCK}/T_{ICKINC}	INC pin Setup/Hold with respect to C
T_{IRSTCK}/T_{ICKRST}	LD pin Setup/Hold with respect to C

Figure 2-12 shows an IDELAY (IDELAY_TYPE = VARIABLE, IDELAY_VALUE = 0, and DELAY_SRC = IDATAIN) timing diagram.

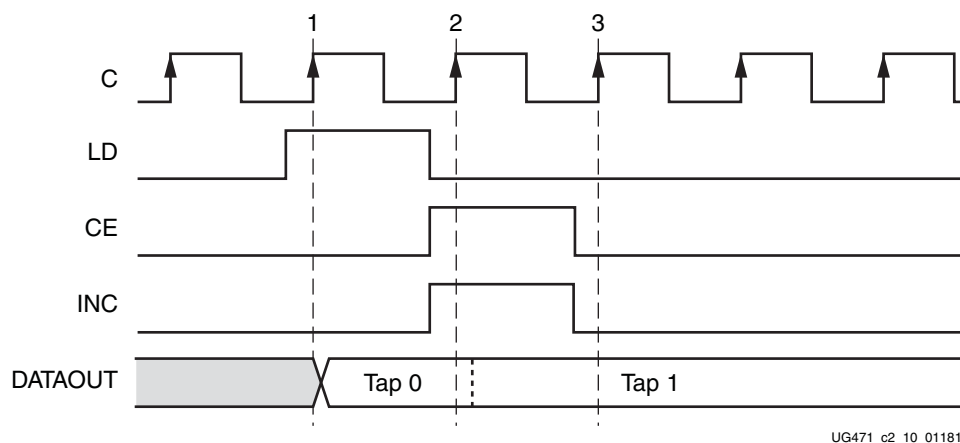


Figure 2-12: IDELAY Timing Diagram

Clock Event 1

On the rising edge of C, a reset is detected (LD is High), causing the output DATAOUT to select tap 0 as the output from the 31-tap chain.

Clock Event 2

A pulse on CE and INC is captured on the rising edge of C. This indicates an increment operation. The output changes without glitches from tap 0 to tap 1. See [Stability after an Increment/Decrement Operation](#).

Clock Event 3

CE and INC are no longer asserted, thus completing the increment operation. The output remains at tap 1 indefinitely until there is further activity on the LD, CE, or INC pins.

Figure 2-13 shows an IDELAY timing diagram in VAR_LOAD mode.

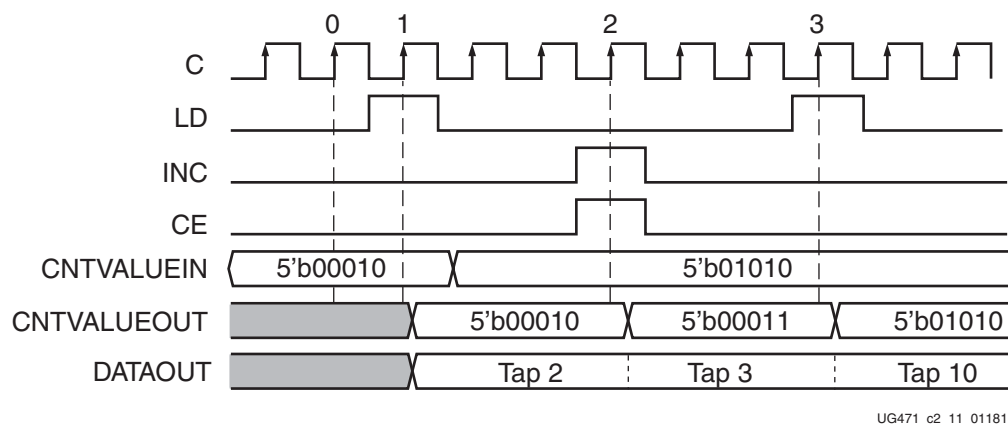


Figure 2-13: IDELAY in VAR_LOAD Timing Diagram

Clock Event 0

Before LD is pulsed the tap setting and therefore CNTVALUEOUT are at an unknown value.

Clock Event 1

On the rising edge of C, LD is detected as High causing the output DATAOUT to have a delay defined by the CNTINVALUE, and changing the tap setting to tap 2. The CNTVALUEOUT is updated to represent the new tap value.

Clock Event 2

A pulse on CE and INC are captured on the rising edge of C. This indicates an increment operation. The output changes without glitches from tap 2 to tap 3. The CNTVALUEOUT is updated to represent the new tap value.

Clock Event 3

On the rising edge of C, a LD is detected as High causing the output DATAOUT to be delayed by the CNTINVALUE. The CNTVALUEOUT shows the value of the tap setting. The output will remain at tap 10 indefinitely until there is further activity on the LD, CE, or INC pins.

Stability after an Increment/Decrement Operation

Figure 2-12 shows the delay line changing from tap 0 to tap 1 in response to an INC and CE command. Clearly, when the data value at tap 0 is different from the data value at tap 1, the output must change state. However, when the data values at tap 0 and tap 1 are the same (e.g., both 0 or both 1), then the transition from tap 0 to tap 1 causes no glitch or disruption on the output. This concept can be better comprehended by imagining the receiver data signal passing through the IDELAY tap chain. If tap 0 and tap 1 are both near the center of the receiver data eye, then the data sampled at tap 0 will be no different than the data sampled at tap 1. In this case, the transition from tap 0 to tap 1 causes no change to the output. To ensure that this is the case, the increment/decrement operation of IDELAY is designed to be glitchless. The same explanation also applies to the VAR_LOAD behavior shown in Figure 2-13. VAR_LOAD does, however, give the possibility of changing the delay by more than one tap, which could potentially result in a sample point that is well away from the current eye centre point.

The user can therefore dynamically adjust the IDELAY tap setting in real-time while live user data is passing through the IDELAYE2 primitive. The adjustments do not disrupt the live user data, provided that the current delay line value is near the middle of the received data eye.

The glitchless behavior also applies when an IDELAYE2 primitive is used in the path of a clock signal. Adjusting the tap setting does not cause a glitch or disruption on the output, provided that the delay line value is not near the edges seen in the received clock signal. In this case, the tap setting of the IDELAYE2 primitive in the clock path can be adjusted without disrupting any clock management elements or state machines that could be running on that clock.

IDELAY VHDL and Verilog Instantiation Template

VHDL and Verilog instantiation templates are available in the Libraries Guide for all primitives and submodules.

In VHDL, each template has a component declaration section and an architecture section. Each part of the template should be inserted within the VHDL design file. The port map of the architecture section should include the design signal names.

IDELAYCTRL

IDELAYCTRL Overview

If the IDELAYE2 or ODELAYE2 primitives are instantiated, the IDELAYCTRL module must also be instantiated. The IDELAYCTRL module continuously calibrates the individual delay taps (IDELAY/ODELAY) in its region (see [Figure 2-16, page 126](#)), to reduce the effects of process, voltage, and temperature variations. The IDELAYCTRL module calibrates IDELAY and ODELAY using the user supplied REFCLK.

IDELAYCTRL Primitive

[Figure 2-14](#) shows the IDELAYCTRL primitive.

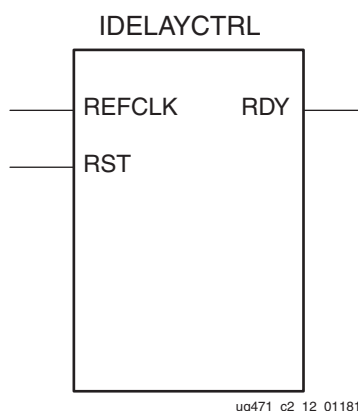


Figure 2-14: IDELAYCTRL Primitive

IDELAYCTRL Ports

RST - Reset

The reset input pin (RST) is an active-High asynchronous reset. To ensure proper IDELAY and ODELAY operation, IDELAYCTRL must be reset after configuration and the REFCLK signal is stable. A reset pulse width $T_{IDELAYCTRL_RPW}$ is required.

REFCLK - Reference Clock

The reference clock (REFCLK) provides a time reference to IDELAYCTRL to calibrate all IDELAY and ODELAY modules in the same region. This clock must be driven by a global or horizontal clock buffer (BUFG or BUFH). REFCLK must be $F_{IDELAYCTRL_REF} \pm$ the specified ppm tolerance (IDELAYCTRL_REF_PRECISION) to guarantee a specified IDELAY and ODELAY resolution ($T_{IDELAYRESOLUTION}$). REFCLK can be supplied directly from a user-supplied source or the MMCM and must be routed on a global clock buffer.

RDY - Ready

The ready (RDY) signal indicates when the IDELAY and ODELAY modules in the specific region are calibrated. The RDY signal is deasserted if REFCLK is held High or Low for more than one clock period. If RDY is deasserted Low, the IDELAYCTRL module must be reset. The implementation tools allow RDY to be unconnected/ignored. [Figure 2-15](#) illustrates the timing relationship between RDY and RST.

IDELAYCTRL Timing

[Table 2-9](#) shows the IDELAYCTRL switching characteristics.

Table 2-9: IDELAYCTRL Switching Characteristics

Symbol	Description
$F_{\text{IDELAYCTRL_REF}}$	REFCLK frequency
IDELAYCTRL_REF_PRECISION	REFCLK precision
$T_{\text{IDELAYCTRLCO_RDY}}$	Reset/Startup to Ready for IDELAYCTRL

As shown in [Figure 2-15](#), the 7 series FPGA IDELAYCTRL RST is an edge-triggered signal.

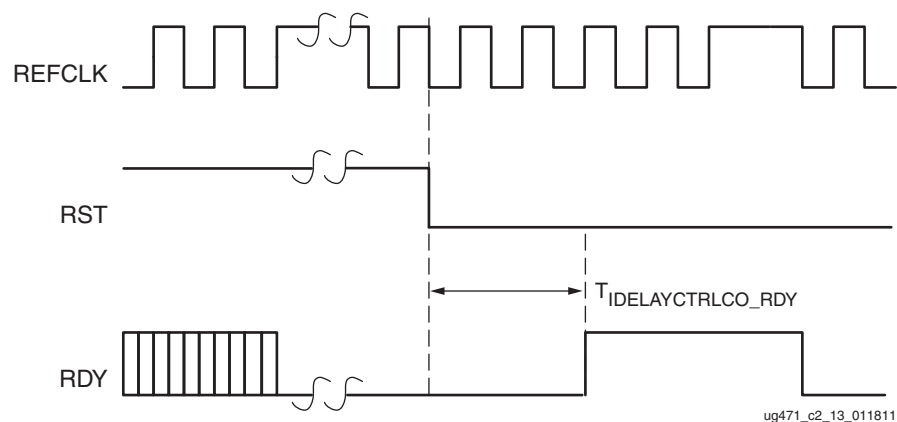


Figure 2-15: Timing Relationship Between RST and RDY

IDELAYCTRL Locations

IDELAYCTRL modules exist in every I/O column in every clock region. An IDELAYCTRL module calibrates all the IDELAYE2 and ODELAYE2 modules within its clock region. See the *7 Series FPGA Clocking User Guide* for the definition of a clock region.

[Figure 2-16](#) illustrates the relative locations of the IDELAYCTRL modules.

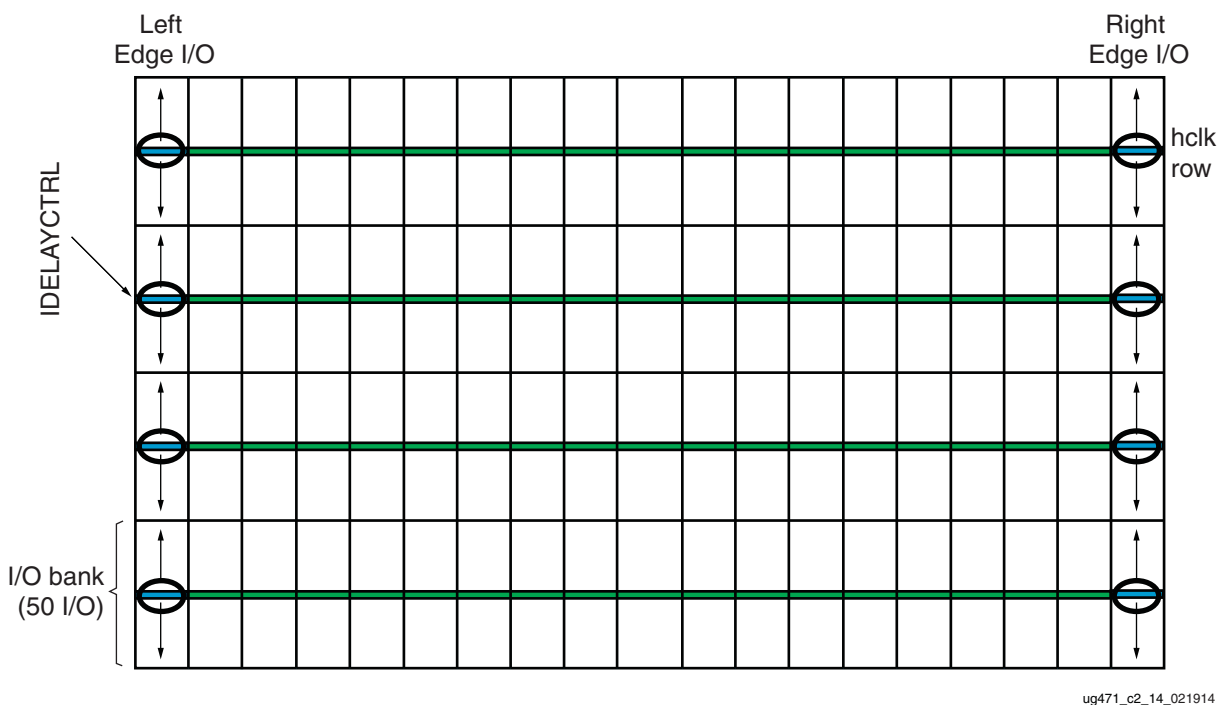


Figure 2-16: Relative Locations of IDELAYCTRL Modules

IDELAYCTRL Usage and Design Guidelines

For more information on placing and locking IDELAYCTRLs, see the constraints guide.

OLOGIC Resources

The OLOGIC block is located next to the I/O block (IOB). OLOGIC is a dedicated synchronous block sending data out of the FPGA through the IOB. The types of OLOGIC resources are OLOGICE2 (HP I/O banks) and OLOGICE3 (HR I/O banks). Although described as OLOGIC in this guide, unless explicitly delineated, OLOGICE2 and OLOGICE3 are functionally identical and so are their ports.

OLOGICE2 and OLOGICE3 are not primitives in the sense that they cannot be instantiated. They contain user-instantiated elements such as an output flip-flop (OFD) or an output DDR element (ODDR) after place and route.

OLOGIC consists of two major blocks, one to configure the output data path and the other to configure the 3-state control path. These two blocks have a common clock (CLK) but different enable signals, OCE and TCE. Both have asynchronous and synchronous set and reset (S/R signal) controlled by an independent SRVAL attribute.

The output and the 3-state paths can be independently configured in one of the following modes.

- Edge triggered D type flip-flop
- DDR mode (SAME_EDGE or OPPOSITE_EDGE)
- Level sensitive latch
- Asynchronous/combinatorial

Figure 2-17 illustrates the various logic resources in the OLOGIC block.

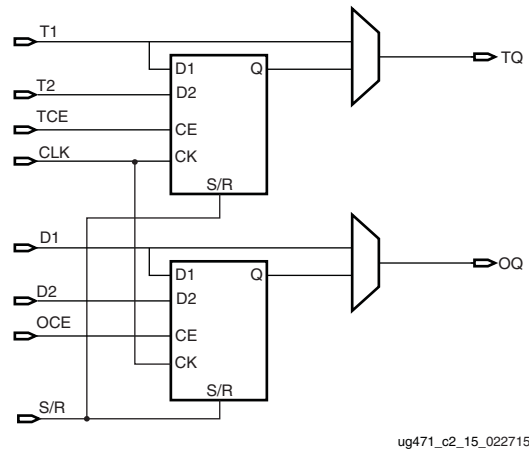


Figure 2-17: **OLOGIC Block Diagram**

This section of the documentation discusses the various features available using the OLOGIC resources.

Combinatorial Output Data and 3-State Control Path

The combinatorial output paths create a direct connection from the FPGA logic to the output driver or output driver control. These paths are used automatically by software when:

1. There is direct (unregistered) connection from logic resources in the FPGA logic to the output data or 3-state control.
2. The **pack I/O register/latches into IOBs** software map directive is set to OFF.

Output DDR Overview (ODDR)

7 series devices have dedicated registers in the OLOGIC to implement output DDR registers. This feature is accessed when instantiating the ODDR primitive. DDR multiplexing is automatic when using OLOGIC. No manual control of the mux-select is needed. This control is generated from the clock.

There is only one clock input to the ODDR primitive. Falling edge data is clocked by a locally inverted version of the input clock. All clocks feeding into the I/O tile are fully multiplexed, i.e., there is no clock sharing between the ILOGIC or the OLOGIC blocks. The ODDR primitive supports the following modes of operation:

- OPPOSITE_EDGE mode
- SAME_EDGE mode

The SAME_EDGE mode is the same as for the Virtex-6 architecture. This mode allows designers to present both data inputs to the ODDR primitive on the rising-edge of the ODDR clock, saving CLB and clock resources, and increasing performance. This mode is implemented using the DDR_CLK_EDGE attribute. It is supported for 3-state control as well. The following sections describe each of the modes in detail.

OPPOSITE_EDGE Mode

In OPPOSITE_EDGE mode, both the edges of the clock (CLK) are used to capture the data from the FPGA logic at twice the throughput. This structure is similar to the Virtex-6 FPGA implementation. Both outputs are presented to the data input or 3-state control input of the IOB. The timing diagram of the output DDR using the OPPOSITE_EDGE mode is shown in Figure 2-18.

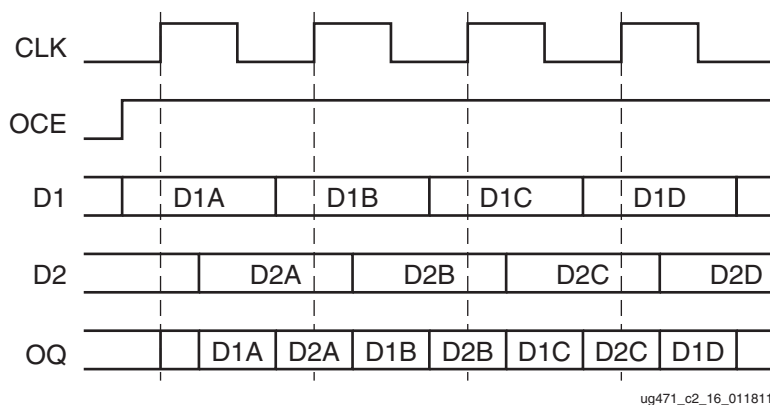


Figure 2-18: Output DDR Timing in OPPOSITE_EDGE Mode

SAME_EDGE Mode

In SAME_EDGE mode, data can be presented to the IOB on the same clock edge. Presenting the data to the IOB on the same clock edge avoids setup time violations and allows the user to perform higher DDR frequency with minimal register to register delay, as opposed to using the CLB registers. Figure 2-19 shows the timing diagram of the output DDR using the SAME_EDGE mode.

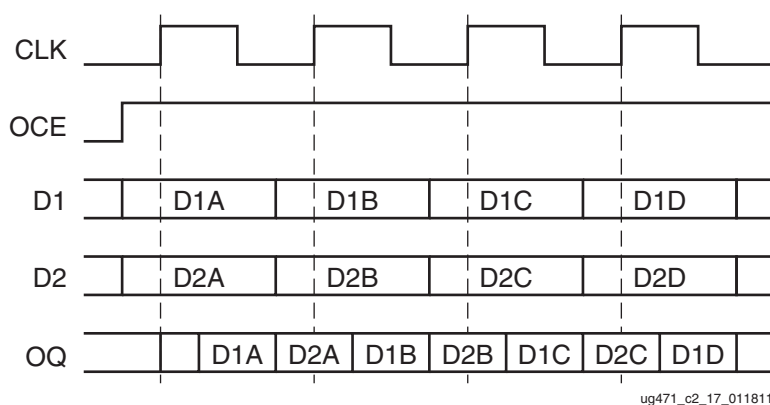


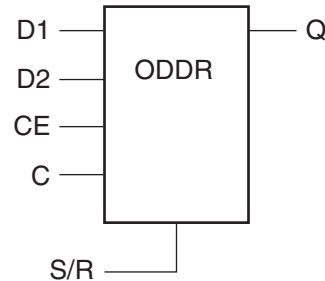
Figure 2-19: Output DDR Timing in SAME_EDGE Mode

Clock Forwarding

Output DDR can forward a copy of the clock to the output. This is useful for propagating a clock and DDR data with identical delays, and for multiple clock generation, where every clock load has a unique clock driver. This is accomplished by tying the D1 input of the ODDR primitive High, and the D2 input Low. Xilinx recommends using this scheme to forward clocks from the FPGA logic to the output pins.

Output DDR Primitive (ODDR)

Figure 2-20 shows the ODDR primitive block diagram. Set and Reset are not supported at the same time. Table 2-10 lists the ODDR port signals. Table 2-11 describes the various attributes available and default values for the ODDR primitive.



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Figure 2-20: ODDR Primitive Block Diagram

Table 2-10: ODDR Port Signals

Port Name	Function	Description
Q	Data output (DDR)	ODDR register output.
C	Clock input port	The CLK pin represents the clock input pin.
CE	Clock enable port	CE represents the clock enable pin. When asserted Low, this port disables the output clock on port Q.
D1 and D2	Data inputs	ODDR register inputs.
S/R ⁽¹⁾	Set/Reset	Synchronous/Asynchronous set/reset pin. Set/Reset is asserted High.

Notes:

1. The ODDR primitive contains both set and reset pins. However only one can be used per ODDR. As a result, S/R is described instead of separate set and reset pins.

Table 2-11: ODDR Attributes

Attribute Name	Description	Possible Values
DDR_CLK_EDGE	Sets the ODDR mode of operation with respect to clock edge	OPPOSITE_EDGE (default), SAME_EDGE
INIT	Sets the initial value for Q port	0 (default), 1
SRTYPE	Set/Reset type with respect to clock (C)	ASYNC, SYNC (default)

ODDR VHDL and Verilog Templates

The Libraries Guide includes templates for instantiation of the ODDR module in VHDL and Verilog.

OLOGIC Timing Models

This section discusses all timing models associated with the OLOGIC block. [Table 2-12](#) describes the function and control signals of the OLOGIC switching characteristics in the 7 series FPGA data sheets.

Table 2-12: OLOGIC Switching Characteristics

Symbol	Description
Setup/Hold	
T_{ODCK}/T_{OCKD}	D1/D2 pins Setup/Hold with respect to CLK
T_{OOCECK}/T_{OCKOCE}	OCE pin Setup/Hold with respect to CLK
T_{OSRCK}/T_{OCKSR}	S/R pin Setup/Hold with respect to CLK
T_{OTCK}/T_{OCKT}	T1/T2 pins Setup/Hold with respect to CLK
T_{OTCECK}/T_{OCKTCE}	TCE pin Setup/Hold with respect to CLK
Clock to Out	
T_{OCKQ}	CLK to OQ/TQ out
T_{RQ}	S/R pin to OQ/TQ out

Timing Characteristics

[Figure 2-21](#) illustrates the OLOGIC output register timing.

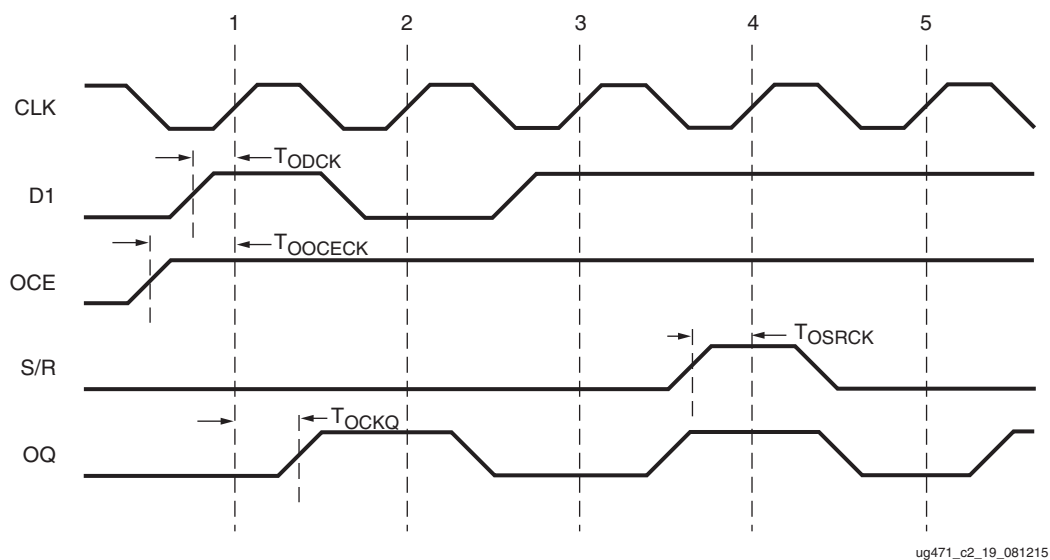


Figure 2-21: OLOGIC Output Register Timing Characteristics

Clock Event 1

- At time T_{OOCECK} before Clock Event 1, the output clock enable signal becomes valid-high at the OCE input of the output register, enabling the output register for incoming data.

- At time T_{ODCK} before Clock Event 1, the output signal becomes valid-high at the D1 input of the output register and is reflected at the OQ output at time T_{OCKQ} after Clock Event 1.

Clock Event 4

At time T_{OSRCK} before Clock Event 4, the S/R signal (configured as synchronous reset in this case) becomes valid-High, resetting the output register and reflected at the OQ output at time T_{RQ} after Clock Event 4.

Figure 2-22 illustrates the OLOGIC ODDR register timing.

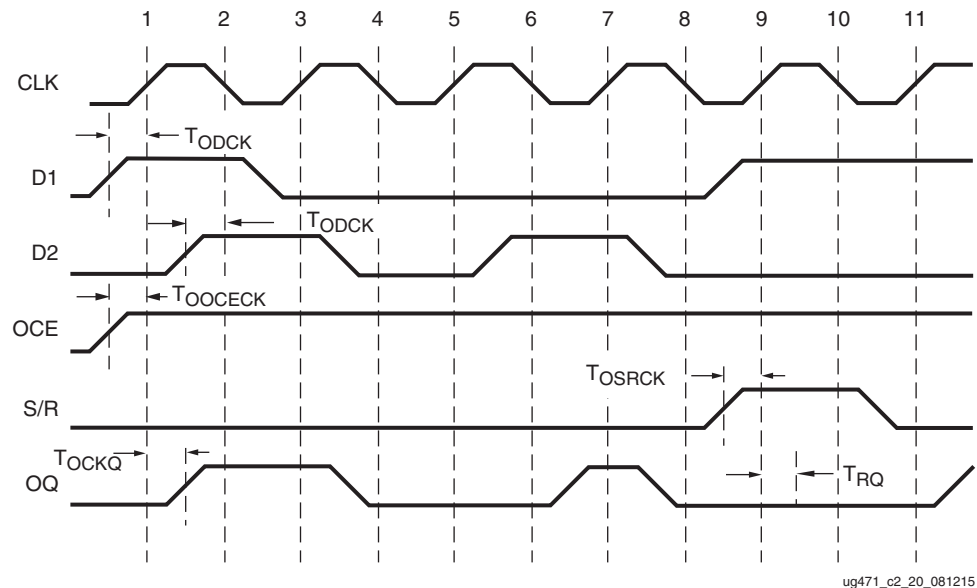


Figure 2-22: OLOGIC ODDR Register Timing Characteristics (OPPOSITE_EDGE Mode)

Clock Event 1

- At time T_{OOCECK} before Clock Event 1, the ODDR clock enable signal becomes valid-High at the OCE input of the ODDR, enabling ODDR for incoming data. Care must be taken to toggle the OCE signal of the ODDR register between the rising edges and falling edges of CLK as well as meeting the register setup-time relative to both clock edges.
- At time T_{ODCK} before Clock Event 1 (rising edge of CLK), the data signal D1 becomes valid-high at the D1 input of ODDR register and is reflected on the OQ output at time T_{OCKQ} after Clock Event 1.

Clock Event 2

- At time T_{ODCK} before Clock Event 2 (falling edge of CLK), the data signal D2 becomes valid-high at the D2 input of ODDR register and is reflected on the OQ output at time T_{OCKQ} after Clock Event 2 (no change at the OQ output in this case).

Clock Event 9

At time T_{OSRCK} before Clock Event 9 (rising edge of CLK), the S/R signal (configured as synchronous reset in this case) becomes valid-high resetting ODDR register, reflected at the OQ output at time T_{RQ} after Clock Event 9 (no change at the OQ output in this case) and

resetting ODDR register, reflected at the OQ output at time T_{RQ} after Clock Event 10 (no change at the OQ output in this case).

Figure 2-23 illustrates the OLOGIC 3-state register timing.

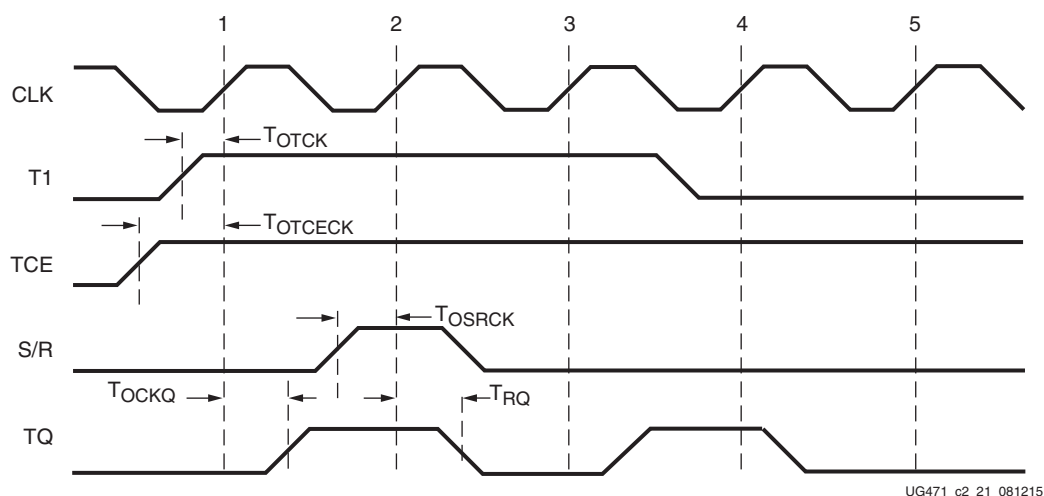


Figure 2-23: OLOGIC 3-State Register Timing Characteristics

Clock Event 1

- At time T_{OTCECK} before Clock Event 1, the 3-state clock enable signal becomes valid-high at the TCE input of the 3-state register, enabling the 3-state register for incoming data.
- At time T_{OTCK} before Clock Event 1 the 3-state signal becomes valid-high at the T input of the 3-state register, returning the pad to high-impedance at time T_{OCKQ} after Clock Event 1.

Clock Event 2

- At time T_{OSRCCK} before Clock Event 2, the S/R signal (configured as synchronous reset in this case) becomes valid-high, resetting the 3-state register at time T_{RQ} after Clock Event 2.

Figure 2-24 illustrates IOB DDR 3-state register timing. This example is shown using DDR in opposite edge mode. For other modes add the appropriate latencies as shown in Figure 2-7, page 111.

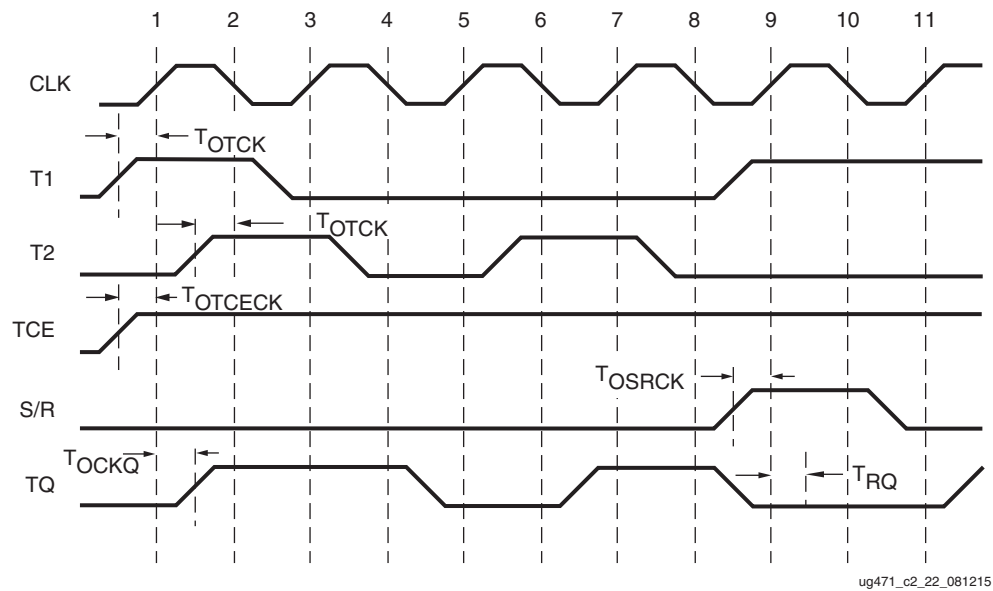


Figure 2-24: OLOGIC ODDR 3-State Register Timing Characteristics

Clock Event 1

- At time T_{OTCECK} before Clock Event 1, the 3-state clock enable signal becomes valid-High at the TCE input of the 3-state ODDR register, enabling them for incoming data. Care must be taken to toggle the TCE signal of the 3-state ODDR between the rising edges and falling edges of CLK as well as meeting the register setup-time relative to both clock edges.
- At time T_{OTCK} before Clock Event 1 (rising edge of CLK), the 3-state signal T1 becomes valid-high at the T1 input of 3-state register and is reflected on the TQ output at time T_{OCKQ} after Clock Event 1.

Clock Event 2

- At time T_{OTCK} before Clock Event 2 (falling edge of CLK), the 3-state signal T2 becomes valid-high at the T2 input of 3-state register and is reflected on the TQ output at time T_{OCKQ} after Clock Event 2 (no change at the TQ output in this case).

Clock Event 9

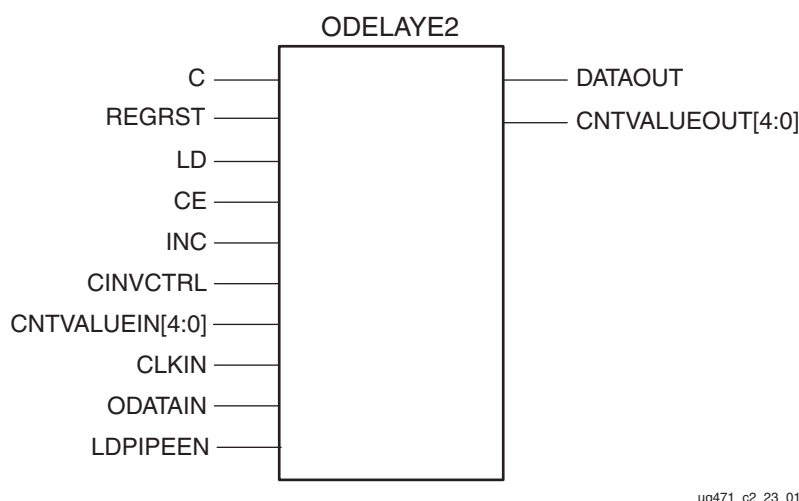
- At time T_{OSRCCK} before Clock Event 9 (rising edge of CLK), the S/R signal (configured as synchronous reset in this case) becomes valid-high resetting 3-state Register, reflected at the TQ output at time T_{RQ} after Clock Event 9 (no change at the TQ output in this case) and resetting 3-state Register, reflected at the TQ output at time T_{RQ} after Clock Event 10 (no change at the TQ output in this case).

Output Delay Resources (ODELAY)—Not Available in HR Banks

Every HP I/O block contains a programmable absolute delay primitive called ODELAYE2. The ODELAY can be connected to an OLOGICE2/OSERDESE2 block. ODELAY is a 31-tap, wraparound, delay primitive with a calibrated tap resolution. Refer to the 7 series FPGA data sheets for delay values. It can be applied to the combinatorial output path or registered output path. It can also be accessed directly from the FPGA logic. ODELAY allows outgoing signals to be delayed on an individual basis. The tap delay resolution is varied by selecting an IDELAYCTRL reference clock from the range specified in the 7 series FPGA data sheets.

ODELAYE2 Primitive

Figure 2-25 shows the ODELAYE2 primitive.



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Figure 2-25: ODELAYE2 Primitive

Table 2-13 lists the available ports in the ODELAYE2 primitive.

Table 2-13: ODELAYE2 Primitive Ports

Port Name	Direction	Width	Function
C	Input	1	Clock input used in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode.
REGRST	Input	1	Reset to all zeroes for the pipeline register.
LD	Input	1	Loads the ODELAY primitive to the pre-programmed value in VARIABLE mode. In VAR_LOAD mode, it loads the value of CNTVALUEIN. In VAR_LOAD_PIPE mode, it loads the value currently in the pipeline register.
CE	Input	1	Enable increment/decrement function.
INC	Input	1	Increment/decrement number of tap delays.
CINVCTRL	Input	1	Dynamically inverts the clock (C) polarity.
CNTVALUEIN	Input	5	Input value from FPGA logic for dynamically loadable tap value.
CLKIN	Input	1	Clock Access into the ODELAY (from the I/O CLKMUX).

Table 2-13: ODELAYE2 Primitive Ports (Cont'd)

Port Name	Direction	Width	Function
ODATAIN	Input	1	Data input for ODELAY from the OLOGICE2/OSERDESE2.
LDPIPEEN	Input	1	Enables the pipeline register to load data from CNTVALUEIN.
DATAOUT	Output	1	Delayed data from one of two data input ports (ODATAIN and CLKIN).
CNTVALUEOUT	Output	5	Current delay value going to FPGA logic for monitoring tap value.

ODELAY Ports

Data Input from the FPGA OLOGICE2/OSERDESE2 - ODATAIN

The ODATAIN input is driven by OLOGICE2/OSERDESE2. The ODATAIN drives the DATAOUT port which is connected to an IOB with a delay set by the ODELAY_VALUE.

Clock Input from Clock Buffer - CLKIN

The CLKIN input is driven from clock buffers (BUFIO, BUFG or BUFR). This clock is then delayed by a value set to ODELAY_VALUE and output through the DATAOUT and output buffer (OBUFT or OBUFTDS). When an IOBUF is used, the delayed clock can be routed back to the FPGA logic.

Caution! Be aware that a package pin of the FPGA is used when using an IOBUF to route a clock back into the FPGA.

Data Output - DATAOUT

Delayed data from one of the two data input ports. DATAOUT connects to the IOB.

Clock Input - C

All control inputs to ODELAYE2 primitive (LD, CE, and INC) are synchronous to the clock input (C). A clock must be connected to this port when ODELAY is configured in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode. C can be locally inverted, and must be supplied by a global or regional clock buffer. This clock must be connected to the same clock as used in the SelectIO logic resources. For example, when using the OSERDESE2, C is connected to the same clock as CLKDIV. If the IDELAYE2 primitive is used in the same I/O bank as the ODELAYE2 primitive, C must use the same clock net for both primitives.

Module Load - LD

When in VARIABLE mode, the ODELAY load port, LD, loads the delay primitive to a value set by the ODELAY_VALUE attribute. If these attributes are not specified, a value of zero is assumed. The LD signal is an active-High signal and is synchronous to the input clock signal (C).

When in VAR_LOAD mode, the ODELAY load port, LD, loads the delay primitive to a value set by the CNTVALUEIN. The value present at CNTVALUEIN[4:0] will be the new tap value. As a result of this functionality the ODELAY_VALUE attribute is ignored.

When in VAR_LOAD_PIPE mode, the IDELAY load port, LD, loads the value currently in the pipeline register. The value present in the pipeline register will be the new tap value.

Pipeline Register Load - LDPIPEEN

When High, this input loads the pipeline register with the value currently on the CNTVALUEIN pins.

Pipeline Register Reset - REGRST

When high, this input resets the pipeline register to all zeroes.

C Pin Polarity Switch - CINVCTRL

The CINVCTRL pin is used for dynamically switching the polarity of C pin. This is for use in applications when glitches are not an issue. When switching the polarity, do not use ODELAY control pins for two clock cycles.

Count Value In - CNTVALUEIN

The CNTVALUEIN pins are used together with the LD pin for dynamically switching the loadable tap value.

Count Value Out - CNTVALUEOUT

The CNTVALUEOUT pins are used for reporting the loaded tap value.

Increment/Decrement Signals - CE, INC

The increment/decrement is controlled by the enable signal (CE). This interface is only available when ODELAY is in VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE mode.

As long as CE remains High, ODELAY will increment or decrement by $T_{\text{IDELAYRESOLUTION}}$ every clock (C) cycle. The state of INC determines whether ODELAY will increment or decrement; INC = 1 increments, INC = 0 decrements, synchronously to the clock (C). If CE is Low the delay through ODELAY will not change regardless of the state of INC.

When CE goes High, the increment/decrement operation begins on the next positive clock cycle. When CE goes Low, the increment/decrement operation ceases on the next positive clock edge.

The programmable delay taps in the ODELAYE2 primitive wrap-around. When the end of the delay tap is reached (tap 31) a subsequent increment function will return to tap 0. The same applies to the decrement function: decrementing below zero moves to tap 31.

The pipeline register functionality in VAR_LOAD_PIPE mode is extremely useful in bus structure designs. Individual delays might be (pipeline) loaded one at a time using LDPIPEEN and then all delays updated to their new values at the same time using the LD pin.

ODELAY Attributes

Table 2-14 summarizes the ODELAY attributes.

Table 2-14: ODELAY Attribute Summary

Attribute	Value	Default Value	Description
ODELAY_TYPE	String: FIXED, VARIABLE, VAR_LOAD, or VAR_LOAD_PIPE	FIXED	Sets the type of tap delay line. FIXED delay sets a static delay value. VAR_LOAD dynamically loads tap values. VARIABLE delay dynamically adjusts the delay value. VAR_LOAD_PIPE is similar to VAR_LOAD mode with the ability to store the CNTVALUEIN value for future use.
ODELAY_VALUE	Integer: 0 to 31	0	Specifies the fixed number of delay taps in fixed mode or the initial starting number of taps in VARIABLE mode (output path). When ODELAY_TYPE is set to VAR_LOAD or VAR_LOAD_PIPE mode, this value is ignored and assumed to be all zeroes.
HIGH_PERFORMANCE_MODE	Boolean: FALSE or TRUE	FALSE	When TRUE, this attribute reduces the output jitter. The difference in power consumption is quantified in the Xilinx Power Estimator tool.
SIGNAL_PATTERN	String: DATA, CLOCK	DATA	Causes the timing analyzer to account for the appropriate amount of delay-chain jitter in the data or clock path.
REFCLK_FREQUENCY	Real: 190–210, 290 to 310, or 390 to 410	200	Sets the tap value (in MHz) used by the timing analyzer for static timing analysis. The ranges of 290.0 to 310.0 and 390 to 410 are not available in all speed grades. See the 7 series FPGA data sheets.
CINVCTRL_SEL	Boolean: FALSE or TRUE	FALSE	Enables the CINVCTRL_SEL pin to dynamically switch the polarity of the C pin.
PIPE_SEL	Boolean: FALSE or TRUE	FALSE	Selects pipeline mode. This attribute should only be set to TRUE when using the VAR_LOAD_PIPE mode of operation.
DELAY_SRC	String: ODATAIN, CLKIN	ODATAIN	Selects source for data input to ODELAY block.

ODELAY_TYPE Attribute

When set to FIXED, the tap-delay value is fixed at the number of taps determined by the ODELAY_VALUE attribute setting. This value is preset and cannot be changed after configuration.

When set to VARIABLE, the variable tap delay is selected. The tap delay can be incremented by setting CE = 1 and INC = 1, or decremented by CE = 1 and INC = 0. The increment/decrement operation is synchronous to C.

When set to VAR_LOAD or VAR_LOAD_PIPE, the variable tap delay can be changed and dynamically loaded. The tap delay can be incremented by setting CE = 1 and INC = 1, or decremented by CE = 1 and INC = 0. The increment/decrement operation is synchronous to C. The LD pin in VAR_LOAD mode loads the value presented on CNTVALUEIN. This

allows the tap value to be dynamically set. When in VAR_LOAD_PIPE mode, the LD pin enables the current value in the pipeline register to be loaded into the output delay.

ODELAY_VALUE Attribute

The ODELAY_VALUE attribute specifies tap delays. The possible values are any integer from 0 to 31. The default value is zero. The value of the tap delay reverts to ODELAY_VALUE when the tap delay is reset by asserting the LD signal. In VAR_LOAD or VAR_LOAD_PIPE mode, this attribute is assumed to be zero.

HIGH_PERFORMANCE_MODE Attribute

When TRUE, this attribute reduces the output jitter. This reduction in jitter results in a slight increase in power dissipation from the ODELAYE2 primitive.

SIGNAL_PATTERN Attribute

Clock and data signals have different electrical profiles and therefore accumulate different amounts of jitter in the ODELAY chain. By setting the SIGNAL_PATTERN attribute, the user enables timing analyzer to account for jitter appropriately when calculating timing. A clock signal is periodic in nature and does not have long sequences of consecutive ones or zeroes, while data is random in nature and can have long and short sequences of ones and zeroes.

ODELAY Modes

When used as ODELAY, the data input comes from either IBUF or the FPGA logic and the output goes to ILOGICE2/ISERDESE2 or ILOGICE3/ISERDESE2. There are four modes of operation available:

- Fixed delay mode (ODELAY_TYPE = FIXED)
In the fixed delay mode, the delay value is preset at configuration to the tap number determined by the attribute ODELAY_VALUE. Once configured, this value cannot be changed. When used in this mode, the IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details.
- Variable delay mode (ODELAY_TYPE = VARIABLE)
In the variable delay mode, the delay value can be changed after configuration by manipulating the control signals CE and INC. When used in this mode, the IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details. The control pins being used in VARIABLE mode are described in [Table 2-6](#).

Table 2-15: Control Pin when ODELAY_TYPE = VARIABLE

C	LD	CE	INC	TAP Setting
0	x	x	x	No Change
1	1	x	x	ODELAY_VALUE
1	0	0	x	No Change
1	0	1	1	Current Value +1
1	0	1	0	Current Value -1
1	0	0	0	No Change

- Loadable variable delay mode (ODELAY_TYPE = VAR_LOAD)

In addition to having the same functionality of (ODELAY_TYPE = VARIABLE) in this mode the ODELAY tap can be loaded via the 5-input bits CNTVALUEIN<4:0> from the FPGA logic. When LD is pulsed the value present at CNTVALUEIN<4:0> will be the new tap value. As a results of this functionality the ODELAY_VALUE attribute is ignored. When used in this mode, the IDELAYCTRL primitive must be instantiated. See [IDELAYCTRL Usage and Design Guidelines](#) for more details. The control pins being used in VAR_LOAD mode are described in [Table 2-7](#).

Table 2-16: Control Pin when ODELAY_TYPE = VAR_LOAD

C	LD	CE	INC	CNTVALUEIN	CNTVALUEOUT	TAP Setting
0	x	x	x	x	No Change	No Change
1	1	x	x	CNTVALUEIN	CNTVALUEIN	CNTVALUEIN
1	0	0	x	x	No Change	No Change
1	0	1	1	x	Current Value +1	Current Value +1
1	0	1	0	x	Current Value -1	Current Value -1
1	0	0	0	0	No Change	No Change

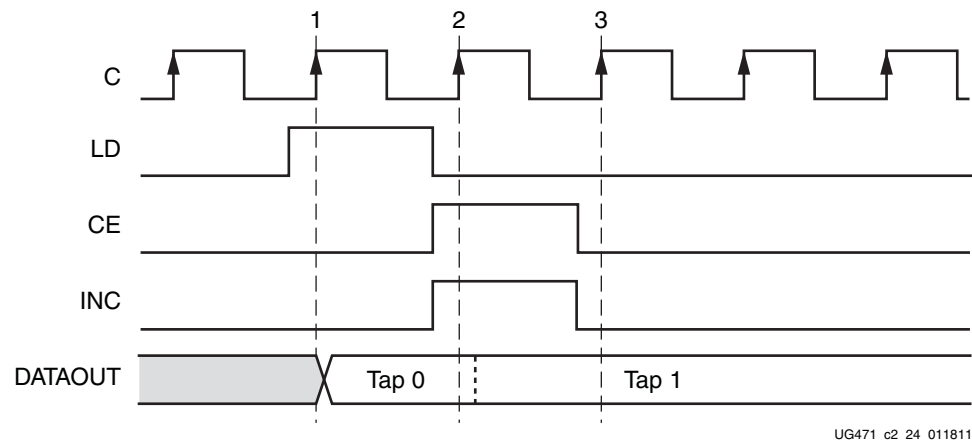
ODELAY Timing

[Table 2-17](#) shows the ODELAY switching characteristics.

Table 2-17: ODELAY Switching Characteristics

Symbol	Description
$T_{IDELAYRESOLUTION}$	IDELAY tap resolution
T_{ICECK}/T_{ICKCE}	CE pin Setup/Hold with respect to C
T_{IINCCK}/T_{ICKINC}	INC pin Setup/Hold with respect to C
T_{IRSTCK}/T_{ICKRST}	LD pin Setup/Hold with respect to C

[Figure 2-26](#) shows an ODELAYE2 (ODELAY_TYPE = VARIABLE, ODELAY_VALUE = 0, and DELAY_SRC = CLKIN/ODATAIN) timing diagram.



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Figure 2-26: ODELAY Timing Diagram (VARIABLE Mode)

Clock Event 1

On the rising edge of C, a reset is detected (LD is High), causing the output DATAOUT to select tap 0 as the output from the 31-tap chain.

Clock Event 2

A pulse on CE and INC is captured on the rising edge of C. This indicates an increment operation. The output changes without glitches from tap 0 to tap 1. See [Stability after an Increment/Decrement Operation](#).

Clock Event 3

CE and INC are no longer asserted, thus completing the increment operation. The output remains at tap 1 indefinitely until there is further activity on the LD, CE, or INC pins.

[Figure 2-27](#) shows an ODELAY timing diagram.

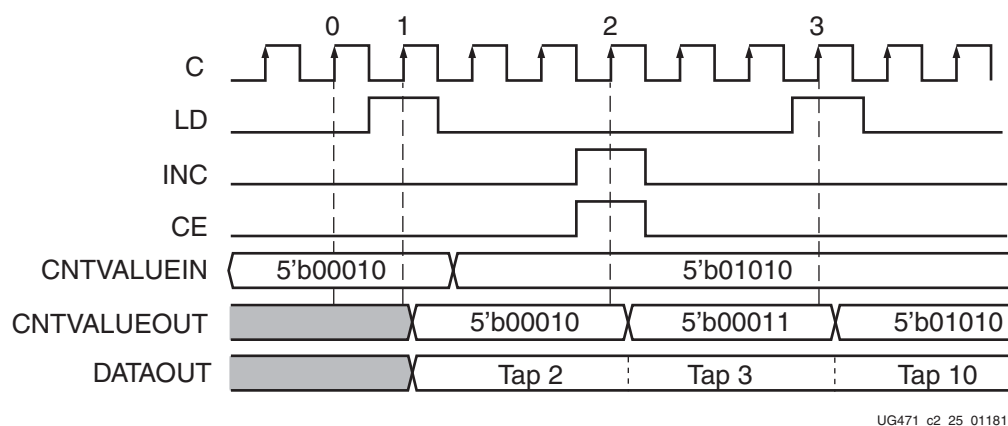


Figure 2-27: ODELAY in VAR_LOAD Timing Diagram

Clock Event 0

Before LD is pulsed the tap setting and CNTVALUEOUT are at an unknown value.

Clock Event 1

On the rising edge of C, LD is detected as High causing the output DATAOUT to be equal to the CNTINVALUE, and changing the tap setting to tap 2. The CNTVALUEOUT is updated to represent the new tap value.

Clock Event 2

A pulse on CE and INC are captured on the rising edge of C. This indicates an increment operation. The output changes without glitches from tap 2 to tap 3. The CNTVALUEOUT is updated to represent the new tap value.

Clock Event 3

On the rising edge of C, a LD is detected causing the output DATAOUT to be equal to the CNTINVALUE. The CNTVALUEOUT shows the value of the tap setting. The output will remain at tap 10 indefinitely until there is further activity on the LD, CE, or INC pins.

Stability after an Increment/Decrement Operation

Figure 2-26 shows the ODELAY line changing from tap 0 to tap 1 in response to INC and CE commands. Clearly, when the data value at tap 0 is different from the data value at tap 1, the output must change state. However, when the data values at tap 0 and tap 1 are the same (e.g., both 0 or both 1), then the transition from tap 0 to tap 1 causes no glitch or disruption on the output. This concept can be better comprehended by imagining the transmitter data signal passing through the ODELAY tap chain. If tap 0 and tap 1 are both near the center of the transmitted signal, the data at tap 0 will be no different than the data at tap 1. In this case, the transition from tap 0 to tap 1 causes no change to the output. To ensure that this is the case, the increment/decrement operation of ODELAY is designed to be glitchless.

The user can therefore dynamically adjust the ODELAY tap setting in real-time while live user data is passing through the ODELAYE2 primitive. The adjustments do not disrupt the live user data as long as the current delay line value is near the centre of the transmitted data signal.

The glitchless behavior also applies when an ODELAYE2 primitive is used in the path of a clock signal. Adjusting the tap setting does not cause a glitch or disruption on the output.

ODELAY VHDL and Verilog Instantiation Template

VHDL and Verilog instantiation templates are available in the Libraries Guide for all primitives and submodules.

In VHDL, each template has a component declaration section and an architecture section. Each part of the template should be inserted within the VHDL design file. The port map of the architecture section should include the design signals names.

Advanced SelectIO Logic Resources

Introduction

The I/O functionality in 7 series FPGAs is described in [Chapter 1](#) through [Chapter 3](#) of this user guide.

- [Chapter 1](#) covers the electrical characteristics of input receivers and output drivers, and their compliance with many industry standards.
- [Chapter 2](#) describes the register structures dedicated for sending and receiving SDR or DDR data.
- This chapter covers additional resources:
 - Input serial-to-parallel converters (ISERDESE2) and output parallel-to-serial converters (OSERDESE2) support very fast I/O data rates, and allow the internal logic to run up to 8 times slower than the I/O.
 - The Bitflip submodule can re-align data to word boundaries, detected with the help of a training pattern.

Input Serial-to-Parallel Logic Resources (ISERDESE2)

The ISERDESE2 in 7 series FPGAs is a dedicated serial-to-parallel converter with specific clocking and logic features designed to facilitate the implementation of high-speed source-synchronous applications. The ISERDESE2 avoids the additional timing complexities encountered when designing deserializers in the FPGA fabric.

ISERDESE2 features include:

- Dedicated deserializer/serial-to-parallel converter

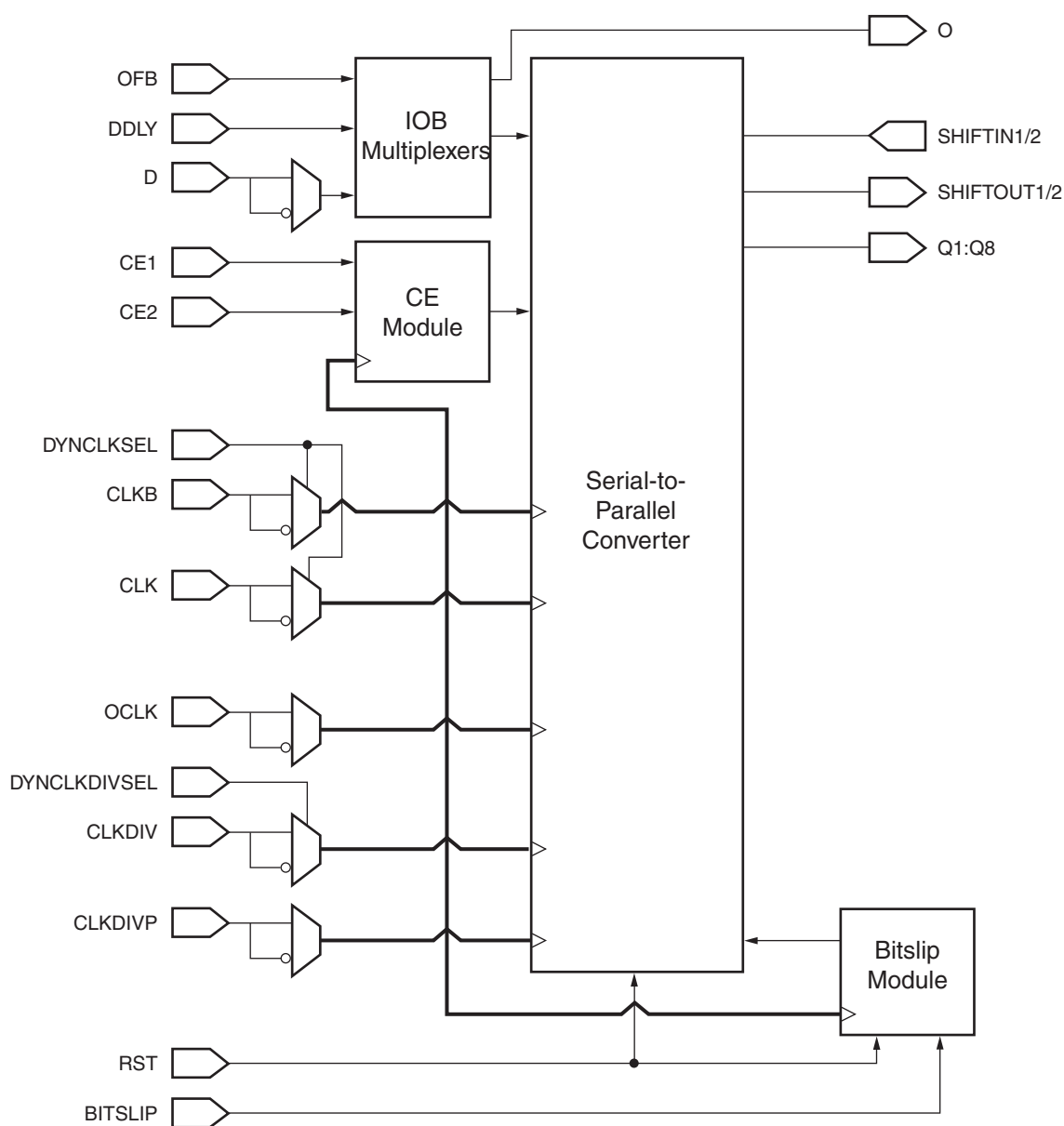
The ISERDESE2 deserializer enables high-speed data transfer without requiring the FPGA fabric to match the input data frequency. This converter supports both single data rate (SDR) and double data rate (DDR) modes. In SDR mode, the serial-to-parallel converter creates a 2-, 3-, 4-, 5-, 6-, 7-, or 8-bit wide parallel word. In DDR mode, the serial-to-parallel converter creates a 4-, 6-, 8-bit wide parallel word mode when using one ISERDESE2, and 10- or 14-bit-wide parallel word when using two cascaded ISERDESE2.
- Bitflip submodule

The Bitflip submodule allows designers to reorder the sequence of the parallel data stream going into the FPGA fabric. This can be used for training source-synchronous interfaces that include a training pattern.
- Dedicated support for strobe-based memory interfaces

ISERDESE2 contains dedicated circuitry (including the OCLK input pin) to handle the strobe-to-FPGA clock domain crossover entirely within the ISERDESE2 block. This allows for higher performance and a simplified implementation.

- Dedicated support for networking interfaces
- Dedicated support for DDR3 interfaces
- Dedicated support for QDR interfaces
- Dedicated support for asynchronous interfaces

Figure 3-1 shows the block diagram of the ISERDESE2, highlighting all the major components and features of the block including the optional inverters.

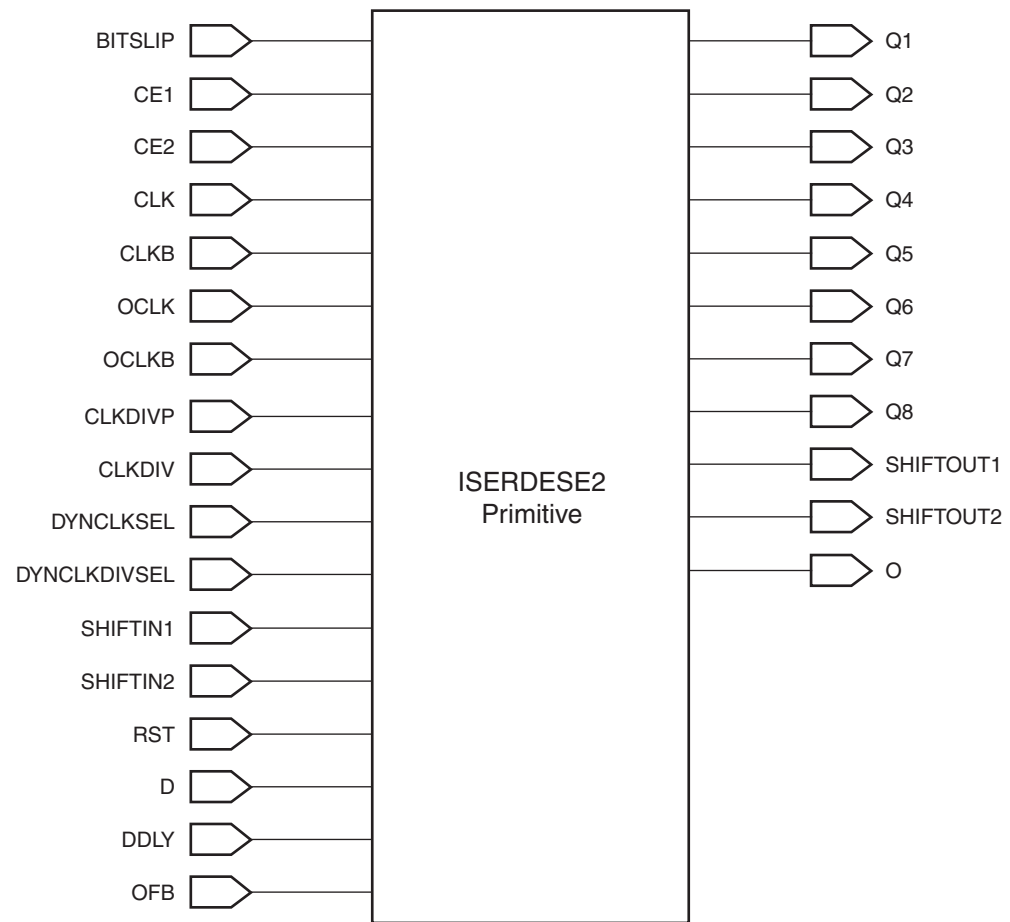


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Figure 3-1: ISERDESE2 Block Diagram

ISERDESE2 Primitive (ISERDESE2)

The ISERDESE2 primitive in 7 series devices (shown in [Figure 3-2](#)) is ISERDESE2.



UG471_c3_02_090810

Figure 3-2: ISERDESE2 Primitive

[Table 3-1](#) lists the available ports in the ISERDESE2 primitive.

Table 3-1: ISERDESE2 Port List and Definitions

Port Name	Type	Width	Description
Q1 – Q8	Output	1 (each)	Registered outputs. See Registered Outputs – Q1 to Q8 .
O	Output	1	Combinatorial output. See Combinatorial Output – O .
SHIFTOUT1	Output	1	Carry out for data width expansion. Connect to SHIFTIN1 of slave IOB. See ISERDESE2 Width Expansion .
SHIFTOUT2	Output	1	Carry out for data width expansion. Connect to SHIFTIN2 of slave IOB. See ISERDESE2 Width Expansion .
D	Input	1	Serial input data from IOB. See Serial Input Data from IOB - D .
DDLY	Input	1	Serial input data from IDELAYE2. See Serial Input Data from IDELAYE2 - DDLY .

Table 3-1: ISERDESE2 Port List and Definitions (Cont'd)

Port Name	Type	Width	Description
CLK	Input	1	High-speed clock input. Clocks serial input data stream. See High-Speed Clock Input - CLK .
CLKB	Input	1	Second High speed clock input only for MEMORY_QDR mode. Always connect to inverted CLK unless in MEMORY_QDR mode. See MEMORY_QDR Interface Type .
CE1, CE2	Input	1 (each)	Clock enable inputs. See Clock Enable Inputs - CE1 and CE2 .
RST	Input	1	Active High reset. See Reset Input - RST .
CLKDIV	Input	1	Divided clock input. Clocks delay element, deserialized data, Bitflip submodule, and CE unit. See Divided Clock Input - CLKDIV .
CLKDIVP	Input	1	Only supported via the MIG tool. Sourced by PHASER_IN divided CLK in MEMORY_DDR3 mode. All other modes connect to ground.
OCLK	Input	1	High-speed clock input for memory applications. See High-Speed Clock for Strobe-Based Memory Interfaces and Oversampling Mode - OCLK . (This clock resource is shared with the OSERDESE2 CLK pin.)
OCLKB	Input	1	Inverted high-speed clock input. (This clock resource is shared with the OSERDESE2 CLKB pin.)
BITSLIP	Input	1	Invokes the Bitflip operation. See Bitflip Operation - BITSLIP .
SHIFTIN1	Input	1	Carry input for data width expansion. Connect to SHIFTOUT1 of master IOB. See ISERDESE2 Width Expansion .
SHIFTIN2	Input	1	Carry input for data width expansion. Connect to SHIFTOUT2 of master IOB. See ISERDESE2 Width Expansion .
OFB	Input	1	Feedback Path from the OLOGICE2 or OLOGICE3 and OSERDESE2 output. See ISERDESE2 Feedback from OSERDESE2 .
DYNCLKDIVSEL	Input	1	Dynamically select CLKDIV inversion. See Dynamic Clock Inversions .
DYNCLKSEL	Input	1	Dynamically select CLK and CLKB inversion. See Dynamic Clock Inversions .

ISERDESE2 Ports

Registered Outputs – Q1 to Q8

The output ports Q1 to Q8 are the registered outputs of the ISERDESE2 module. One ISERDESE2 block can support up to eight bits (i.e., a 1:8 deserialization). Bit widths greater than eight (up to 14) can be supported in DDR mode only. See [ISERDESE2 Width Expansion](#). The first data bit received appears on the highest order Q output.

The bit ordering at the input of an OSERDESE2 is the opposite of the bit ordering at the output of an ISERDESE2 block, as shown in [Figure 3-3](#). For example, the least significant bit A of the word FEDCBA is placed at the D1 input of an OSERDESE2, but the same bit A emerges from the ISERDESE2 block at the Q8 output. In other words, D1 is the least significant input to the OSERDESE2, while Q8 is the least significant output of the ISERDESE2 block. When width expansion is used, D1 of the transmitter OSERDESE2 is the least significant input, while Q8 of the receiver ISERDESE2 block is the least significant output.

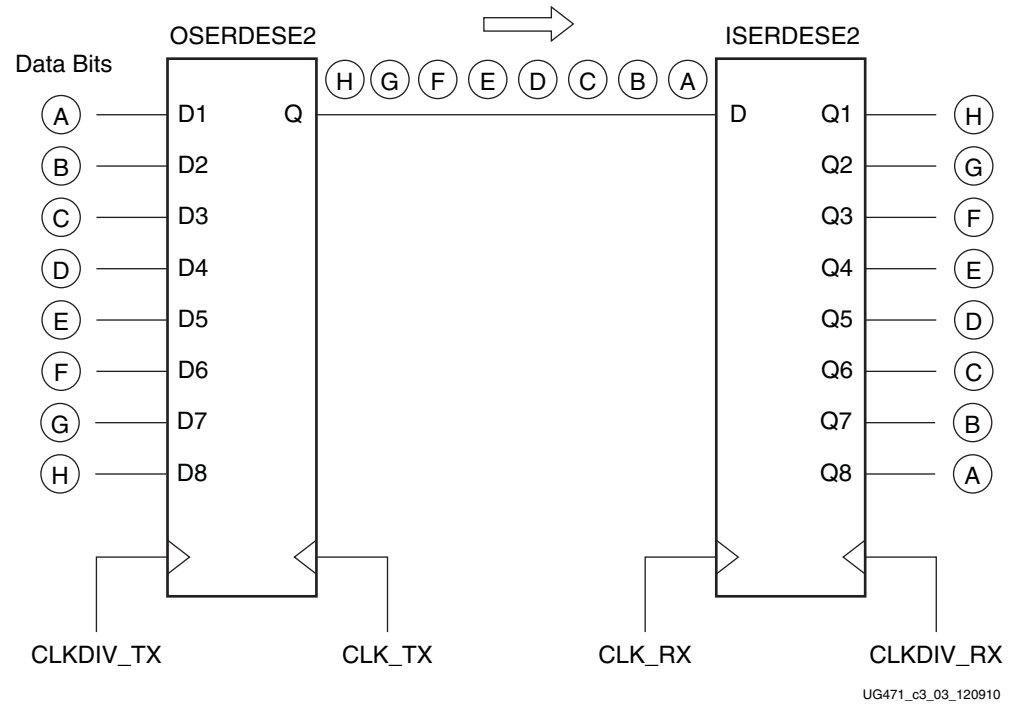


Figure 3-3: Bit Ordering on Q1-Q8 Outputs of ISERDESE2 Ports

Combinatorial Output – O

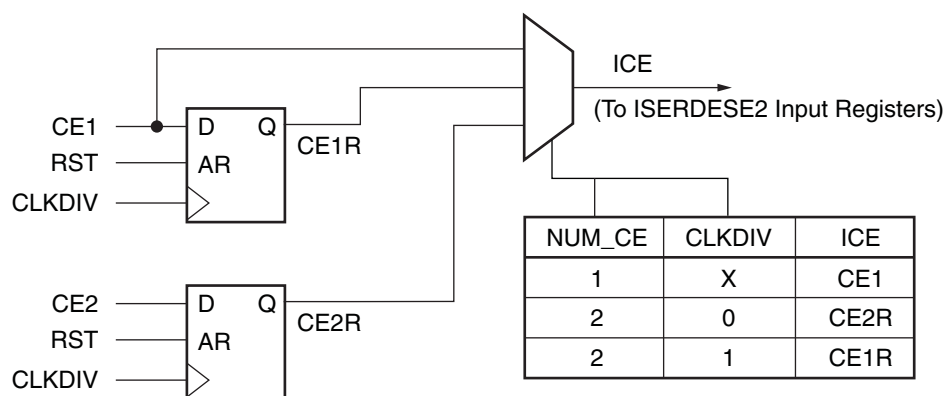
The combinatorial output port (O) is an unregistered output of the ISERDESE2 module. This output can come directly from the data input (D), or from the data input (DDLTY) via the IDELAYE2.

Bitslip Operation - BITSLLIP

The BITSLLIP pin performs a Bitslip operation synchronous to CLKDIV when asserted (active High). Subsequently, the data seen on the Q1 to Q8 output ports will shift, as in a barrel-shifter operation, one position every time Bitslip is invoked (DDR operation is different from SDR). See [BITSLLIP Submodule](#) for more details.

Clock Enable Inputs - CE1 and CE2

Each ISERDESE2 block contains an input clock enable module ([Figure 3-4](#)).



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Figure 3-4: Input Clock Enable Module

When NUM_CE = 1, the CE2 input is not used, and the CE1 input is an active High clock enable connected directly to the input registers in the ISERDESE2. When NUM_CE = 2, the CE1 and CE2 inputs are both used, with CE1 enabling the ISERDESE2 for $\frac{1}{2}$ of a CLKDIV cycle, and CE2 enabling the ISERDESE2 for the other $\frac{1}{2}$. The internal clock enable signal ICE shown in Figure 3-4 is derived from the CE1 and CE2 inputs. ICE drives the clock enable inputs of registers FF0, FF1, FF2, and FF3 shown in Figure 3-5, page 152. The remaining registers in Figure 3-5, page 152 do not have clock enable inputs.

The clock enable module functions as a 2:1 serial-to-parallel converter, clocked by CLKDIV. The clock enable module is needed specifically for bidirectional memory interfaces when ISERDESE2 is configured for 1:4 deserialization in DDR mode. When the attribute NUM_CE = 2, the clock enable module is enabled and both CE1 and CE2 ports are available. When NUM_CE = 1, only CE1 is available and functions as a regular clock enable.

High-Speed Clock Input - CLK

The high-speed clock input (CLK) is used to clock in the input serial data stream.

High-Speed Clock Input - CLKB

The high-speed secondary clock input (CLKB) is used to clock in the input serial data stream. In any mode other than MEMORY_QDR, connect CLKB to an inverted version of CLK. In MEMORY_QDR mode CLKB should be connected to a unique, phase shifted clock. See [ISERDESE2 Clocking Methods](#).

Divided Clock Input - CLKDIV

The divided clock input (CLKDIV) is typically a divided version of CLK (depending on the width of the implemented deserialization). It drives the output of the serial-to-parallel converter, the Bitflip submodule, and the CE module.

Serial Input Data from IOB - D

The serial input data port (D) is the serial (high-speed) data input port of the ISERDESE2. This port works in conjunction only with the 7 series FPGA IOB resource. See [Using D and DDLY in the ISERDESE2](#).

Serial Input Data from IDELAYE2 - DDLY

The serial input data port (DDLY) is the serial (high-speed) data input port of the ISERDESE2. This port works in conjunction only with the 7 series FPGA IDELAYE2 resource. See [Using D and DDLY in the ISERDESE2](#).

Serial Input Data from OSERDESE2 - OFB

The serial input data port (OFB) is the serial (high-speed) data input port of the ISERDESE2. This port works in conjunction only with the 7 series FPGA OSERDESE2 port OFB. See [ISERDESE2 Feedback from OSERDESE2](#).

High-Speed Clock for Strobe-Based Memory Interfaces and Oversampling Mode - OCLK

The OCLK clock input synchronizes data transfer in strobe-based memory interfaces. The OCLK clock is only unused when INTERFACE_TYPE is set to NETWORKING.

The OCLK clock input can be used to transfer strobe-based memory data onto a free-running clock domain. OCLK is a free-running FPGA clock at the same frequency as the strobe on the CLK input. The domain transfer from CLK to OCLK is shown in the [Figure 3-5](#) block diagram. The timing of the domain transfer is set by the user by adjusting the delay of the strobe signal to the CLK input (e.g., using IDELAY). Examples of setting the timing of this domain transfer for MEMORY_DDR3 and MEMORY_QDR modes are given in the Memory Interface Generator (MIG). When INTERFACE_TYPE is NETWORKING, this port is unused.

Reset Input - RST

When asserted, the reset input causes the outputs of most data flip-flops in the CLK and CLKDIV domains to be driven Low asynchronously. The exceptions are the first four flip-flops in the input structure whose value after RESET is selectable via attributes on the component. When deasserted synchronously with CLKDIV, internal logic re-times this deassertion to the first rising edge of CLK. Every ISERDESE2 in a multiple bit input structure should therefore be driven by the same reset signal, asserted, and deasserted synchronously to CLKDIV to ensure that all ISERDESE2 elements come out of reset in synchronization. The reset signal should only be deasserted when it is known that CLK and CLKDIV are stable and present, and should be a minimum of two CLKDIV pulses wide. After deassertion of reset, the output is not valid until after two CLKDIV cycles.

ISERDESE2 Attributes

Table 3-2 summarizes all the applicable ISERDESE2 attributes. A detailed description of each attribute follows the table. For more information on applying these attributes in UCF, VHDL, or Verilog code, refer to the Xilinx ISE Software Manual.

Table 3-2: ISERDESE2 Attributes

Attribute Name	Description	Value	Default Value
DATA_RATE	Enables incoming data stream to be processed as SDR or DDR data. See DATA_RATE Attribute .	String: SDR or DDR	DDR
DATA_WIDTH	Defines the width of the serial-to-parallel converter. The legal value depends on the DATA_RATE attribute (SDR or DDR). See DATA_WIDTH Attribute .	Integer: 2, 3, 4, 5, 6, 7, 8, 10 or 14. If DATA_RATE = DDR, value is limited to 4, 6, 8, 10, or 14. If DATA_RATE = SDR, value is limited to 2, 3, 4, 5, 6, 7, or 8.	4
DYN_CLKDIV_INV_EN	Enables DYNCLKDIVSEL inversion when TRUE and disables HDL inversions on CLKDIV pin. See Dynamic Clock Inversions .	Boolean: TRUE or FALSE	FALSE
DYN_CLK_INV_EN	Enables DYNCLKSEL inversion when TRUE and disables HDL inversions on CLK and CLKB pins. See Dynamic Clock Inversions .	Boolean: TRUE or FALSE	FALSE
INTERFACE_TYPE	Chooses the ISERDESE2 use model. See INTERFACE_TYPE Attribute .	String: MEMORY, MEMORY_DDR3, MEMORY_QDR, OVERSAMPLE, or NETWORKING	MEMORY
NUM_CE	Defines the number of clock enables. See NUM_CE Attribute .	Integer: 1 or 2	2
OFB_USED	Enables the path from the OLOGICE2/3, OSERDESE2 OFB pin to the ISERDESE2 OFB pin. Disables the use of the D input pin.	Boolean: TRUE or FALSE	FALSE
SERDES_MODE	Defines whether the ISERDESE2 module is a master or slave when using width expansion. See SERDES_MODE Attribute .	String: MASTER or SLAVE	MASTER
INIT_Q1	Sets the initial value for the first sample register.	Binary: 0 or 1	0
INIT_Q2	Sets the initial value for the second sample register.	Binary: 0 or 1	0
INIT_Q3	Sets the initial value for the third sample register.	Binary: 0 or 1	0
INIT_Q4	Sets the initial value for the fourth sample register.	Binary: 0 or 1	0
SRVAL_Q1	Sets the value after reset of the first sample register.	Binary: 0 or 1	1

Table 3-2: ISERDESE2 Attributes (Cont'd)

Attribute Name	Description	Value	Default Value
SRVAL_Q2	Sets the value after reset of the second sample register.	Binary: 0 or 1	1
SRVAL_Q3	Sets the value after reset of the third sample register.	Binary: 0 or 1	1
SRVAL_Q4	Sets the value after reset of the fourth sample register.	Binary: 0 or 1	1
IOBDELAY	Sets whether an input delay applies to registered and/or non-registered outputs. See Using D and DDLY in the ISERDESE2 .	NONE, IBUF, IFD, or BOTH	NONE

DATA_RATE Attribute

The DATA_RATE attribute defines whether the incoming data stream is processed as single data rate (SDR) or double data rate (DDR). The allowed values for this attribute are SDR and DDR. The default value is DDR.

DATA_WIDTH Attribute

The DATA_WIDTH attribute defines the parallel data output width of the serial-to-parallel converter. The possible values for this attribute depend on the INTERFACE_TYPE and DATA_RATE attributes. See [Table 3-3](#) for supported data widths.

Table 3-3: Supported Data Widths

INTERFACE_TYPE	DATA_RATE	Supported Data Widths
NETWORKING	SDR	2, 3, 4, 5, 6, 7, 8
	DDR	4, 6, 8, 10, 14
MEMORY MEMORY_DDR3 MEMORY_QDR	SDR	None
	DDR	4

When the DATA_WIDTH is set to widths larger than eight, a pair of ISERDESE2 must be configured into a master-slave configuration. See [ISERDESE2 Width Expansion](#). Width expansion is not allowed in memory mode.

INTERFACE_TYPE Attribute

The INTERFACE_TYPE attribute determines whether the ISERDESE2 is configured in memory or networking mode. The allowed values for this attribute are MEMORY, MEMORY_DDR3, MEMORY_QDR, OVERSAMPLE, or NETWORKING. The default mode is MEMORY.

[Figure 3-5](#) illustrates the ISERDESE2 internal connections when in MEMORY mode.

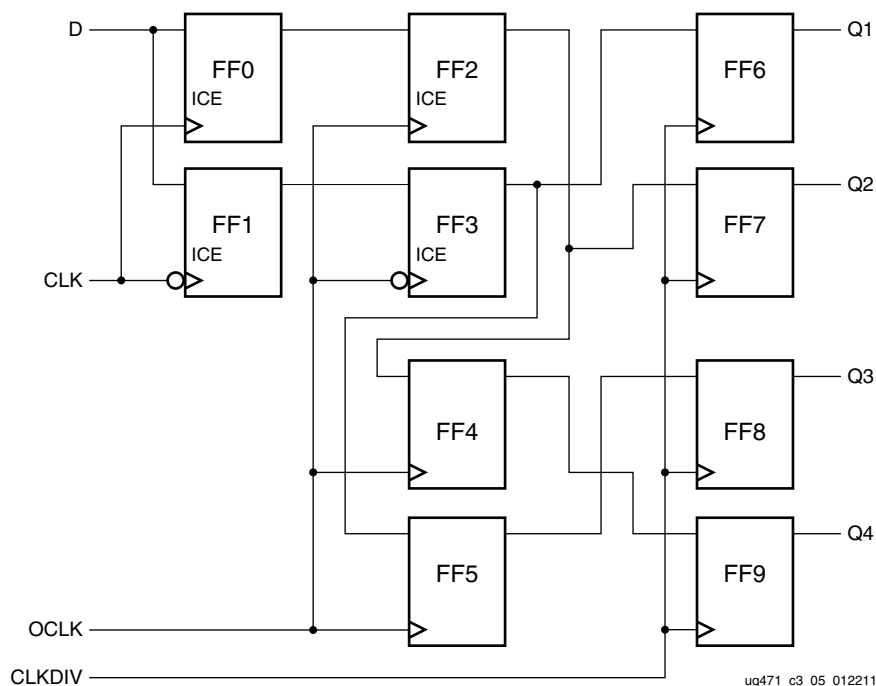


Figure 3-5: Internal Connections of ISERDESE2 When in MEMORY Mode

NUM_CE Attribute

The NUM_CE attribute defines the number of clock enables (CE1 and CE2) used. The possible values are 1 and 2 (default = 2).

SERDES_MODE Attribute

The SERDES_MODE attribute defines whether the ISERDESE2 module is a master or slave when using width expansion. The possible values are MASTER and SLAVE. The default value is MASTER. See [ISERDESE2 Width Expansion](#).

ISERDESE2 Clocking Methods

NETWORKING Interface Type

The phase relationship of CLK and CLKDIV is important in the serial-to-parallel conversion process. CLK and CLKDIV are (ideally) phase-aligned within a tolerance. There are several clocking arrangements within the FPGA to help the design meet the phase relationship requirements of CLK and CLKDIV.

The CLK and CLKDIV inputs must be nominally phase-aligned. For example, if CLK and CLKDIV in [Figure 3-6](#) were inverted by the designer at the ISERDESE2 inputs, then although the clocking arrangement is an allowed BUFIO/BUFR configuration, the clocks would still be out of phase. This also prohibits using DYNCLKINVSEL and DYNCLKDIVINVSEL.

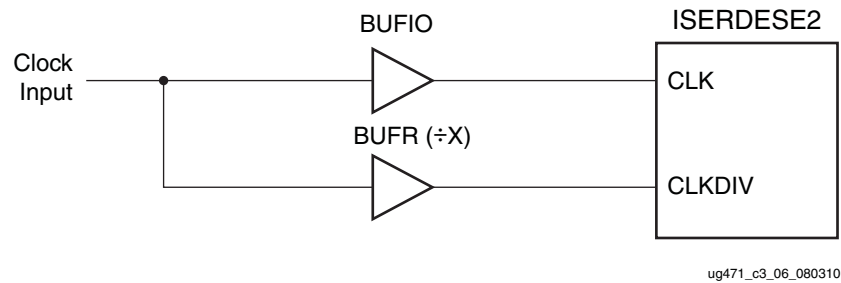


Figure 3-6: Clocking Arrangement Using BUFIO and BUFR

The only valid clocking arrangements for the ISERDESE2 block using the networking interface type are:

- CLK driven by BUFIO, CLKDIV driven by BUFR
- CLK driven by MMCM or PLL, CLKDIV driven by CLKOUT[0:6] of same MMCM or PLL

When using a MMCM to drive the CLK and CLKDIV of the ISERDESE2, the buffer types supplying the ISERDESE2 can not be mixed. For example, if CLK is driven by a BUFG, then CLKDIV must be driven by a BUFG as well. Alternatively, the MMCM can drive the ISERDESE2 through a BUFIO and BUFR.

MEMORY Interface Type

The only valid clocking arrangements for the ISERDESE2 block using the memory interface type are:

- CLK driven by BUFIO, OCLK driven by BUFIO, and CLKDIV driven by BUFR
- CLK driven by MMCM or PLL, OCLK driven by MMCM or PLL, and CLKDIV driven by CLKOUT[0:6] of same MMCM or PLL
- CLK driven by BUFG, OCLK driven by a BUFG, CLKDIV driven by a different BUFG

The OCLK and CLKDIV inputs must be nominally phase-aligned. No phase relationship between CLK and OCLK is expected. Calibration must be performed for reliable data transfer from CLK to OCLK domain. [High-Speed Clock for Strobe-Based Memory Interfaces and Oversampling Mode - OCLK](#) gives further information about transferring data between CLK and OCLK.

MEMORY_QDR Interface Type

The MEMORY_QDR mode has a complex clocking structure as a result of the QDR memory requirements. This INTERFACE_TYPE attribute setting is only supported when using the MIG tool.

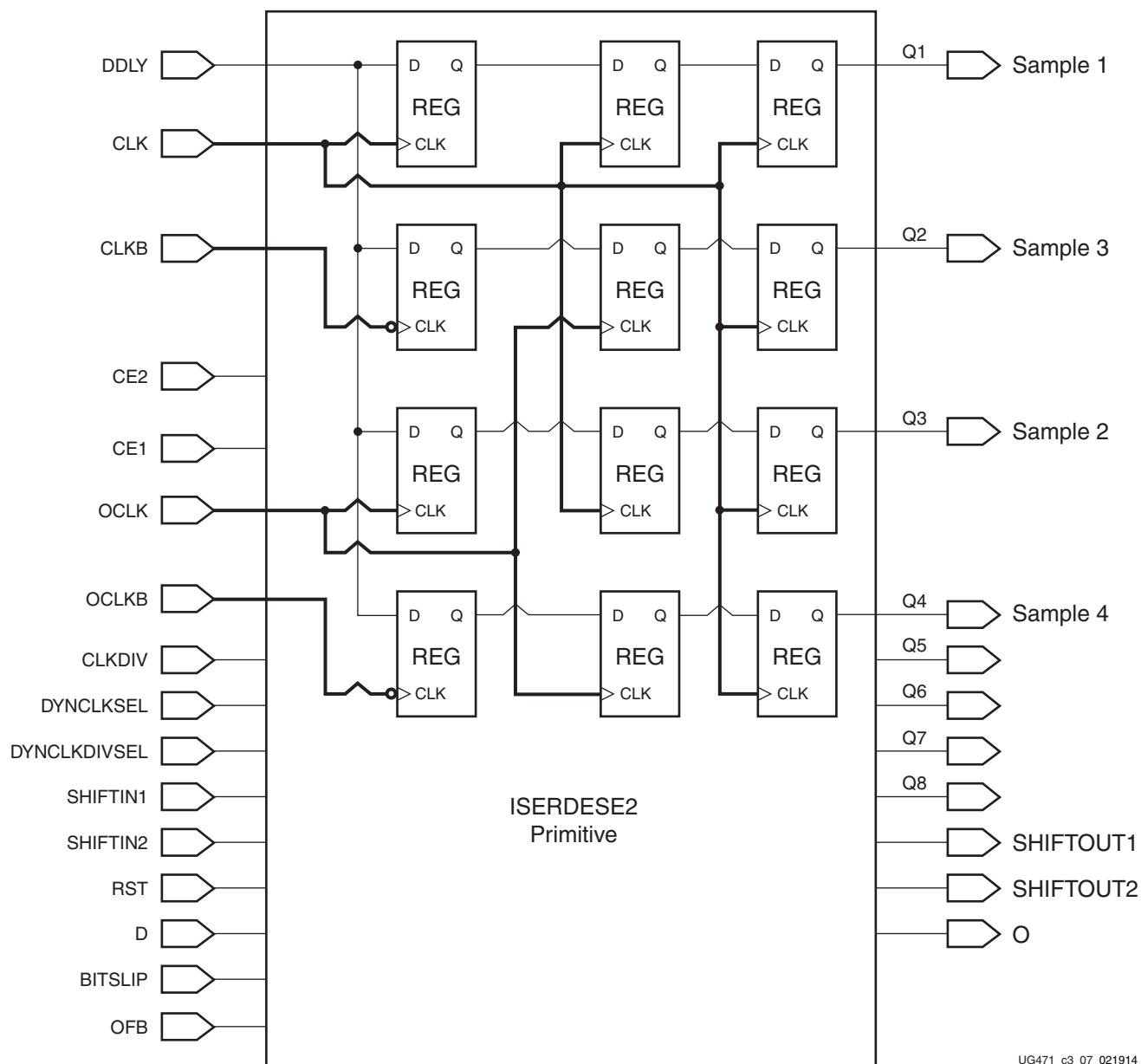
OVERSAMPLE Interface Type

The OVERSAMPLE mode is used to capture two phases DDR data. [Figure 3-7](#) shows a more detailed logical representation of the ISERDESE2 and how data is captured on both the rising and falling edge of CLK and OCLK. As shown in [Figure 3-7](#), there must be a 90° offset phase relationship between CLK and OCLK as the data is captured on both CLK and OCLK but is clocked out of the ISERDESE2 on the CLK domain. CLKDIV is not used in this mode. The only valid clocking arrangements for the OVERSAMPLE interface type are:

- CLK and CLKB are driven by a BUFIO. OCLK and OCLKB are driven by a BUFIO that is phase shifted by 90°. The two BUFIOs are driven from a single MMCM.

- CLK and CLKB are driven by a BUFG. OCLK and OCLKB are driven by a BUFG that is phase shifted by 90°. The BUFGs are driven from a single MMCM. In either case, the effective clocking is:

- CLK: 0°
- OCLK: 90°
- CLKB: 180°
- OCLKB: 270°



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Figure 3-7: Logical View of ISERDESE2 Primitive in Oversample Mode

MEMORY_DDR3 Interface Type

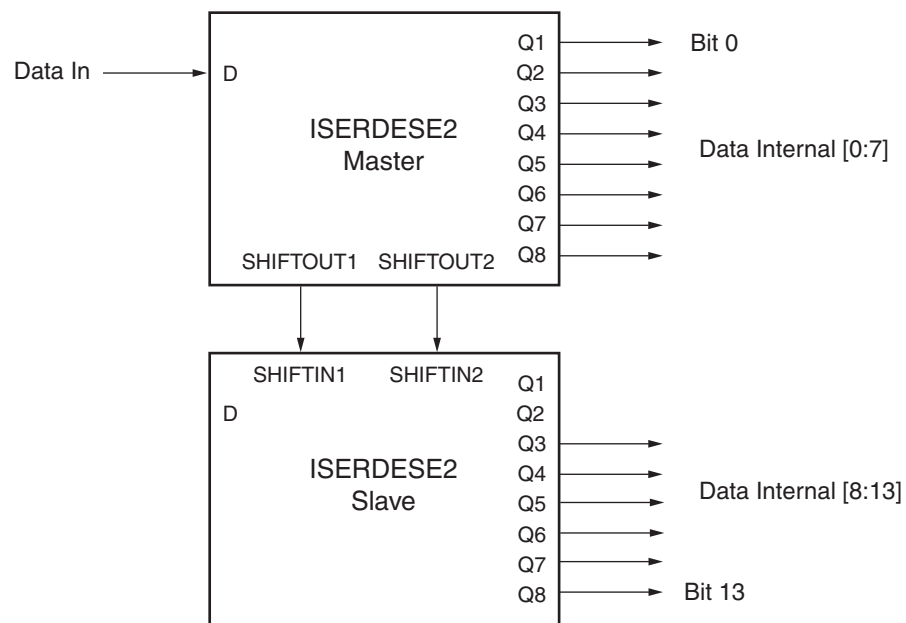
The MEMORY_DDR3 mode has a complex clocking structure as a result of the DDR3 memory requirements. This INTERFACE_TYPE attribute setting is only supported when using the MIG tool.

ISERDESE2 Width Expansion

Two ISERDESE2 modules can be used to build a serial-to-parallel converter larger than 1:8. In every I/O tile there are two ISERDESE2 modules; one master and one slave. By connecting the SHIFTOUT ports of the master ISERDESE2 to the SHIFTIN ports of the slave ISERDESE2 the serial-to-parallel converter can be expanded to up to 1:10 and 1:14 (DDR mode only).

Figure 3-8 illustrates a block diagram of a cascaded DDR serial-to-parallel converter using the master and slave ISERDESE2 modules. In the case of a 1:10 SERDES, slave ports Q3–Q4 are used for the last two bits of the parallel interface.

For a differential input, the master ISERDESE2 must be on the positive (_P pin) side of the differential input pair. When the input is not differential, the input buffer associated with the slave ISERDESE2 is not available, and so cascading cannot be used.



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Figure 3-8: Block Diagram of ISERDESE2 Width Expansion

Guidelines for Expanding the Serial-to-Parallel Converter Bit Width

1. Both ISERDESE2 modules must be adjacent master and slave pairs. Both ISERDESE2 modules must be in NETWORKING mode because width expansion is not available in MEMORY mode.
2. Set the SERDES_MODE attribute for the master ISERDESE2 to MASTER and the slave ISERDESE2 to SLAVE. See [SERDES_MODE Attribute](#).
3. The user must connect the SHIFTIN ports of the SLAVE to the SHIFTOUT ports of the MASTER.

4. The SLAVE uses the ports Q3 to Q8 as outputs.
5. DATA_WIDTH applies to both MASTER and SLAVE in [Figure 3-8](#).

ISERDESE2 Latencies

When the ISERDESE2 interface type is MEMORY, the latency through the OCLK stage is one CLKDIV cycle. However, the total latency through the ISERDESE2 depends on the phase relationship between the CLK and the OCLK clock inputs. When the ISERDESE2 interface type is NETWORKING, the latency is two CLKDIV cycles. See [Figure 3-12, page 160](#) for a visualization of latency in networking mode. The extra CLKDIV cycle of latency in networking mode (compared to memory mode) is due to the Bitflip submodule.

The latency in MEMORY_QDR and MEMORY_DDR3 is two CLKDIV cycles.

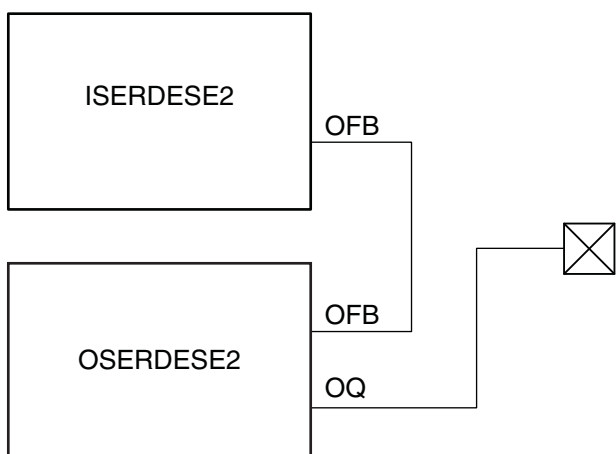
Dynamic Clock Inversions

The dynamic clock inversion pins DYNCLKSEL and DYNCLKDIVSEL when used in conjunction with DYN_CLK_SEL_EN and DYN_CLKDIV_SEL_EN respectively can enable the user to dynamically switch the polarity of the respective clock source. This operation causes the clock going into ISERDESE2 to switch asynchronously and will likely cause the ISERDESE2 to produce erroneous data until the ISERDESE2 is reset. This operation can only be supported in MEMORY_QDR and MEMORY_DDR3 mode.

ISERDESE2 Feedback from OSERDESE2

The OFB port in the ISERDESE2 and OSERDESE2 can be used to feed the data transmitted on the OSERDESE2 back to the ISERDESE2 ([Figure 3-9](#)). This feature is enabled when the attribute OFB_USED = TRUE. The OSERDESE2 and ISERDESE2 must have the same DATA_RATE and DATA_WIDTH setting for the feedback to give the correct data. When using the ISERDESE2 and OSERDESE2 in width expansion mode only, connect the master OSERDESE2 to the master ISERDESE2.

By using the ISERDESE2 as a feedback port, it can not be used as an input for external data.



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Figure 3-9: ISERDESE2 and OSERDESE2 Connected via the OFB Port

Using D and DDLY in the ISERDESE2

The D and DDLY pins are dedicated inputs to the ISERDESE2. The D input is a direct connection to the IOB. The DDLY pin is a direct connection to the IDELAYE2. This allows the user to either have a delayed or non-delayed version of the input to the registered (Q1-Q8) or combinatorial path (O) output. The attribute IOBDELAY determines the input applied to the ISERDESE2. Table 3-4 shows the result of each setting of the IOBDELAY value when both D and DDLY are connected.

Table 3-4: IOBDELAY Attribute Value on the Associated IOBDELAY Block⁽¹⁾

IOBDELAY Value	Combinatorial Output (O)	Registered Output (Q1-Q8)
NONE	D	D
IBUF	DDLY	D
IFD	D	DDLY
BOTH	DDLY	DDLY

Notes:

1. When both D and DDLY are connected to ISERDESE2.

ISERDESE2 Timing Model and Parameters

Table 3-5 describes the function and control signals of the ISERDESE2 switching characteristics in the 7 series FPGA data sheets.

Table 3-5: ISERDESE2 Switching Characteristics

Symbol	Description
Setup/Hold for Control Lines	
$T_{ISCK_BITS\text{LIP}} / T_{ISCKC_BITS\text{LIP}}$	BITSLIP pin Setup/Hold with respect to CLKDIV
$T_{ISCK_CE} / T_{ISCKC_CE}$	CE pin Setup/Hold with respect to CLKDIV (for CE1)
$T_{ISCK_CE2} / T_{ISCKC_CE2}$	CE pin Setup/Hold with respect to CLKDIV (for CE2)
Setup/Hold for Data Lines	
$T_{ISDCK_D} / T_{ISCKD_D}$	D pin Setup/Hold with respect to CLK
	D pin Setup/Hold with respect to CLK
	D pin Setup/Hold with respect to CLK
$T_{ISDCK_DDR} / T_{ISCKD_DDR}$	D pin Setup/Hold with respect to CLK at DDR mode
	D pin Setup/Hold with respect to CLK at DDR mode
	D pin Setup/Hold with respect to CLK at DDR mode
Sequential Delay	
T_{ISCKO_Q}	CLKDIV to Out at Q pins

Timing Characteristics

Figure 3-10 illustrates an ISERDESE2 timing diagram for the input data to the ISERDESE2. The timing parameter names change for different modes (SDR/DDR). However, the

names do not change when a different bus input width, including when two ISERDESE2 are cascaded together to form 10 or 14 bits. In DDR mode, the data input (D) switches at every CLK edge (rising and falling).

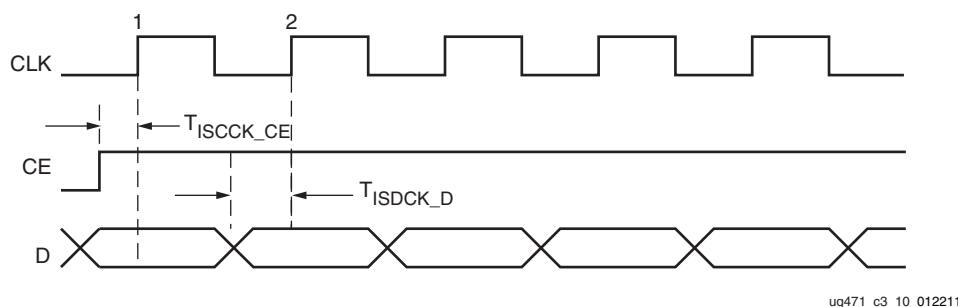


Figure 3-10: ISERDESE2 Input Data Timing Diagram

Clock Event 1

- At time T_{ISCCK_CE} , before Clock Event 1, the clock enable signal becomes valid-High and the ISERDESE2 can sample data.

Clock Event 2

- At time T_{ISDCK_D} , before Clock Event 2, the input data pin (D) becomes valid and is sampled at the next positive clock edge.

ISERDESE2 VHDL and Verilog Instantiation Template

VHDL and Verilog instantiation templates are available in the Libraries Guide for all primitives and submodules.

In VHDL, each template has a component declaration section and an architecture section.

Each part of the template should be inserted within the VHDL design file. The port map of the architecture section should include the design signal names.

BITSLIP Submodule

All ISERDESE2 blocks in 7 series devices contain a Bitflip submodule. This submodule is used for word-alignment purposes in source-synchronous networking-type applications. Bitflip reorders the parallel data in the ISERDESE2 block, allowing every combination of a repeating serial pattern received by the deserializer to be presented to the FPGA fabric. This repeating serial pattern is typically called a training pattern (training patterns are supported by many networking and telecommunications standards). In some interfaces, this can be a slow forwarded clock, which can be considered to be a repeating bit pattern.

Bitflip Operation

By asserting the Bitflip pin of the ISERDESE2 block, the incoming serial data stream is reordered at the parallel side. This operation is repeated until the required training pattern is seen at the ISERDESE2 outputs. The tables in Figure 3-11 illustrate the effects of a Bitflip operation in SDR and DDR mode. (Bit 8 of an input ISERDESE2 is the first bit received.) For illustrative purposes the data width is eight. The Bitflip operation is synchronous to CLKDIV. In SDR mode, every Bitflip operation causes the output pattern to shift left by one. In DDR mode, every Bitflip operation causes the output pattern to alternate between

a shift right by one and shift left by three. In this example, on the eighth Bitslip operation, the output pattern reverts to the initial pattern. This assumes that serial data is an eight bit repeating pattern.

Although the repeating pattern seems to show that bitslip is a barrel shifting operation, this is not the case. A bitslip operation adds one bit to the input data stream and loses the *n*th bit in the input data stream. This causes the operation on repetitive patterns to appear like a barrel shifter operation.

Bitslip Operation in SDR Mode		Bitslip Operation in DDR Mode	
Bitslip Operations Executed	Output Pattern (8:1)	Bitslip Operations Executed	Output Pattern (8:1)
Initial	10010011	Initial	00100111
1	00100111	1	10010011
2	01001110	2	10011100
3	10011100	3	01001110
4	00111001	4	01110010
5	01110010	5	00111001
6	11100100	6	11001001
7	11001001	7	11100100

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Figure 3-11: Bitslip Operation Examples

Guidelines for Using the Bitslip Submodule

In NETWORKING mode the Bitslip submodule is available. In all other modes, the module is not available.

To invoke a Bitslip operation, the BITSLIP port must be asserted High for one CLKDIV cycle. Bitslip cannot be asserted for two consecutive CLKDIV cycles; Bitslip must be deasserted for at least one CLKDIV cycle between two Bitslip assertions. In both SDR and DDR mode, the total latency from when the ISERDESE2 captures the asserted Bitslip input to when the “bit-slipped” ISERDESE2 outputs Q1–Q8 are sampled into the FPGA logic by CLKDIV is two CLKDIV cycles. From an applications perspective, a single Bitslip command must be issued for one CLKDIV cycle only. The user logic should wait for at least two CLKDIV cycles in SDR mode or three CLKDIV cycles in DDR mode, before analyzing the received data pattern and potentially issuing another Bitslip command. If the ISERDESE2 is reset, the Bitslip logic is also reset and returns back to its initial state.

Bitslip Timing Model and Parameters

This section discusses the timing models associated with the Bitslip controller in a 1:4 DDR configuration. Data (D) is a repeating, 4-bit training pattern ABCD. ABCD could appear at the parallel outputs Q1–Q4 of the ISERDESE2 in four possible ways: ABCD, BCDA, CDAB, and DABC. Only one of these four alignments of the parallel word makes sense to the user's downstream logic that reads the data from the Q1–Q4 outputs of the ISERDESE2. In this case, ABCD is assumed to be the word alignment that makes sense. Asserting Bitslip allows the user to see all possible configurations of the input data and then choose the required alignment (ABCD). Figure 3-12 shows the timing of a Bitslip operation and the corresponding re-alignments of the ISERDESE2 parallel outputs Q1–Q4.

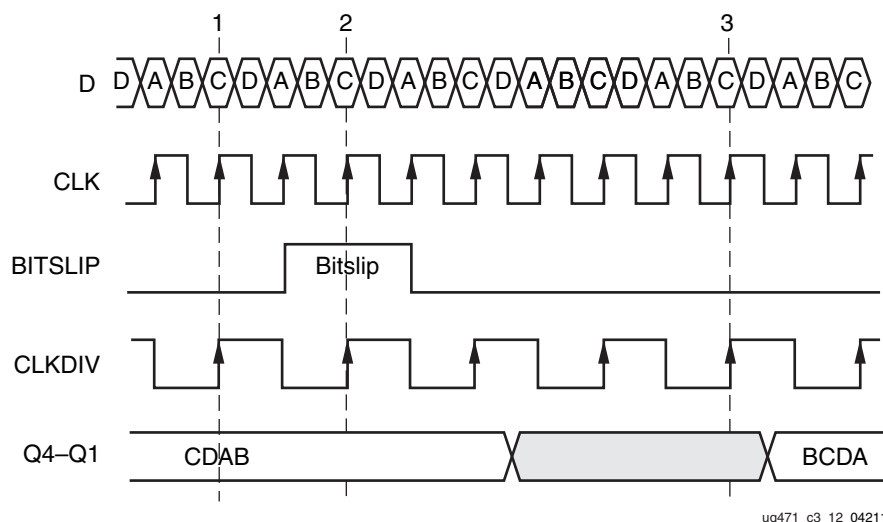


Figure 3-12: DDR Bitslip Functional Diagram

Clock Event 1

The entire first word CDAB has been sampled into the input side registers of the ISERDESE2. The Bitslip pin is not asserted; the word propagates through the ISERDESE2 without any realignment.

Clock Event 2

The Bitslip pin is asserted, which causes the Bitslip controller to shift all bits internally by one bit to the right. Bitslip is held High for one (only one) CLKDIV cycle.

Clock Event 3

Three CLKDIV cycles after asserting Bitslip, the Bitslip operation is completed and the new shifted data is available on the output as BCDA.

After Clock Event 3

Bitslip can be usefully asserted up to two more times as the ISERDESE2 is configured for 1:4. After the second shift (three positions left as this DDR), the (required) output ABCD is available on Q4-Q1. After a third shift (one position right), the output DABC is available on Q4-Q1. After a fourth shift (three positions left), the original output CDAB is available on Q4-Q1, and Bitslip has finished cycling through all four input combinations.

Output Parallel-to-Serial Logic Resources (OSERDESE2)

The OSERDESE2 in 7 series devices is a dedicated parallel-to-serial converter with specific clocking and logic resources designed to facilitate the implementation of high-speed source-synchronous interfaces. Every OSERDESE2 module includes a dedicated serializer for data and 3-state control. Both data and 3-state serializers can be configured in SDR and DDR mode. Data serialization can be up to 8:1 (10:1 and 14:1 if using [OSERDESE2 Width Expansion](#)). 3-state serialization can be up to 14:1. There is a dedicated DDR3 mode to support high-speed memory applications.

[Figure 3-13](#) shows a block diagram of the OSERDESE2, highlighting all the major components and features of the block.

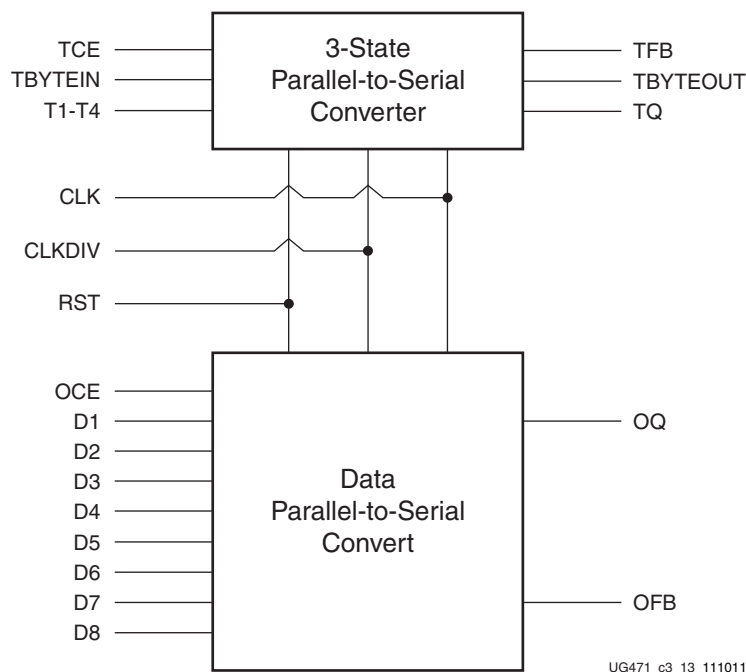


Figure 3-13: OSERDESE2 Block Diagram

Data Parallel-to-Serial Converter

The data parallel-to-serial converter in one OSERDESE2 blocks receives two to eight bits of parallel data from the fabric (14 bits if using [OSERDESE2 Width Expansion](#)), serializes the data, and presents it to the IOB via the OQ outputs. Parallel data is serialized from lowest order data input pin to highest (i.e., data on the D1 input pin is the first bit transmitted at the OQ pins). The data parallel-to-serial converter is available in two modes: single-data rate (SDR) and double-data rate (DDR).

The OSERDESE2 uses two clocks, CLK and CLKDIV, for data rate conversion. CLK is the high-speed serial clock, CLKDIV is the divided parallel clock. CLK and CLKDIV must be phase aligned. See [OSERDESE2 Clocking Methods](#).

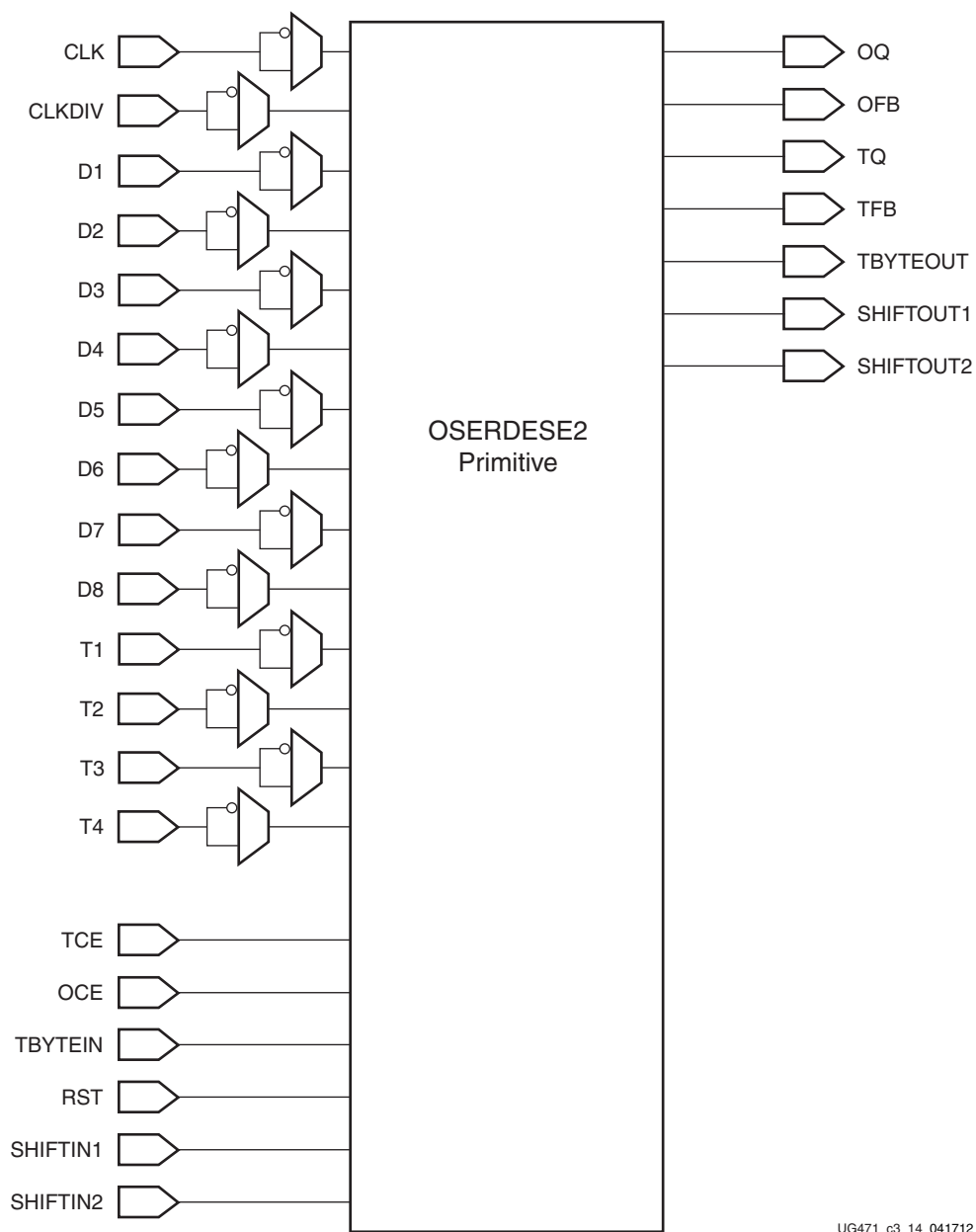
Prior to use, a reset must be applied to the OSERDESE2. The OSERDESE2 contains an internal counter that controls dataflow. Failure to synchronize the reset deassertion with the CLKDIV will produce an unexpected output.

3-State Parallel-to-Serial Conversion

In addition to parallel-to-serial conversion of data, an OSERDESE2 module also contains a parallel-to-serial converter for 3-state control of the IOB. Unlike data conversion, the 3-state converter can only serialize up to four bits of parallel 3-state signals. The 3-state converter cannot be cascaded.

OSERDESE2 Primitive

The OSERDESE2 primitive is shown in Figure 3-14.



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Figure 3-14: OSERDESE2 Primitive

OSERDESE2 Ports

Table 3-6 lists the available ports in the OSERDESE2 primitive.

Table 3-6: OSERDESE2 Port List and Definitions

Port Name	Type	Width	Description
OQ	Output	1	Data path output to IOB only. See Data Path Output - OQ .
OFB	Output	1	Data path output feedback to ISERDESE2 or connection to ODELAYE2. See Output Feedback .
TQ	Output	1	3-state control output to IOB. See 3-state Control Output - TQ .
TFB	Output		3-state control output to fabric. See 3-state Control Output - TFB .
SHIFTOUT1	Output	1	Carry output for data width expansion. Connect to SHIFTIN1 of master OSERDESE2. See OSERDESE2 Width Expansion .
SHIFTOUT2	Output	1	Carry output for data width expansion. Connect to SHIFTIN2 of master OSERDESE2. See OSERDESE2 Width Expansion .
CLK	Input	1	High-speed clock input. See High-Speed Clock Input - CLK .
CLKDIV	Input	1	Divided clock input. Clocks delay element, deserialized data, and CE unit. See Divided Clock Input - CLKDIV .
D1 to D8	Input	1 (each)	Parallel data inputs. See Parallel Data Inputs - D1 to D8 .
TCE	Input	1	3-state clock enable. See 3-state Signal Clock Enable - TCE .
OCE	Input	1	Output data clock enable. See Output Data Clock Enable - OCE .
TBYTEIN	Input	1	Byte group 3-state input.
TBYTEOUT	Output	1	Byte group 3-state output.
RST	Input	1	Active High reset.
SHIFTIN1	Input	1	Carry input for data width expansion. Connect to SHIFTOUT1 of slave OSERDESE2. See OSERDESE2 Width Expansion .
SHIFTIN2	Input	1	Carry input for data width expansion. Connect to SHIFTOUT2 of slave OSERDESE2. See OSERDESE2 Width Expansion .
T1 to T4	Input	1 (each)	Parallel 3-state inputs. See Parallel 3-state Inputs - T1 to T4 .

Data Path Output - OQ

The OQ port is the data output port of the OSERDESE2 module. Data at the input port D1 will appear first at OQ. This port connects the output of the data parallel-to-serial converter to the data input of the IOB. This port can not drive the ODELAYE2; the OFB pin must be used.

Output Feedback from OSERDESE2 - OFB

The output feedback port (OFB) is the serial (high-speed) data output port of the OSERDESE2 for use with the ODELAYE2 primitive, or the OFB port can be used to send out serial data to the ISERDESE2. See [Output Feedback](#).

3-state Control Output - TQ

This port is the 3-state control output of the OSERDESE2 module. When used, this port connects the output of the 3-state parallel-to-serial converter to the control/3-state input of the IOB.

3-state Control Output - TFB

This port is the 3-state control output of the OSERDESE2 module sent to fabric if required by the user. It indicates that the OSERDESE2 is 3-stated.

High-Speed Clock Input - CLK

This high speed clock input drives the serial side of the parallel-to-serial converters.

Divided Clock Input - CLKDIV

This divided high-speed clock input drives the parallel side of the parallel-to-serial converters. This clock is the divided version of the clock connected to the CLK port.

Parallel Data Inputs - D1 to D8

All incoming parallel data enters the OSERDESE2 module through ports D1 to D8. These ports are connected to the FPGA fabric, and can be configured from two to eight bits (i.e., a 8:1 serialization). Bit widths greater than eight (10 and 14) can be supported by using a second OSERDESE2 in SLAVE mode. See [OSERDESE2 Width Expansion](#). Refer to [Figure 3-3, page 147](#) for bit ordering at the inputs and output of the OSERDESE2 along with the corresponding bit order of the ISERDESE2.

Reset Input - RST

When asserted, the reset input causes the outputs of all data flip-flops in the CLK and CLKDIV domains to be driven low asynchronously. When deasserted synchronously with CLKDIV, internal logic re-times this deassertion to the first rising edge of CLK. Every OSERDESE2 in a multiple bit output structure should therefore be driven by the same reset signal, asserted asynchronously, and deasserted synchronously to CLKDIV to ensure that all OSERDESE2 elements come out of reset in synchronization. The reset signal should only be deasserted when it is known that CLK and CLKDIV are stable and present.

Output Data Clock Enable - OCE

OCE is an active High clock enable for the data path.

3-state Signal Clock Enable - TCE

TCE is an active High clock enable for the 3-state control path.

Parallel 3-state Inputs - T1 to T4

All parallel 3-state signals enter the OSERDESE2 module through ports T1 to T4. The ports are connected to the FPGA fabric. They can be configured as one, two, or four bits, or bypassed. The behavior of these ports is controlled by the DATA_RATE_TQ and TRISTATE_WIDTH attributes.

OSERDESE2 Attributes

The [Table 3-7](#) lists and describes the various attributes that are available for the OSERDESE2 primitive. The table includes the default values.

Table 3-7: OSERDESE2 Attribute Summary

Attribute	Description	Value	Default Value
DATA_RATE_OQ	Defines whether data (OQ) changes at every clock edge or every positive clock edge with respect to CLK.	String: SDR or DDR	DDR
DATA_RATE_TQ	Defines whether the 3-state (TQ) changes at every clock edge, every positive clock edge with respect to clock, or is set to buffer configuration.	String: BUF, SDR, or DDR	DDR
DATA_WIDTH	Defines the parallel-to-serial data converter width. This value also depends on the DATA_RATE_OQ value.	Integer: 2, 3, 4, 5, 6, 7, 8, 10, or 14 In SDR mode, 2, 3, 4, 5, 6, 7, and 8 are valid. In DDR mode, 2, 4, 6, 8, 10, and 14 are valid.	4
SERDES_MODE	Defines whether the OSERDESE2 module is a master or slave when using width expansion.	String: MASTER or SLAVE	MASTER
TRISTATE_WIDTH	Defines the parallel to serial 3-state converter width.	Integer: 1 or 4 See OSERDESE2 Attributes (Table 3-8) for valid combinations	4
TBYTE_CTL	Only for use via the MIG tool. Set to FALSE.	FALSE, TRUE	FALSE
TBYTE_SRC	Only for use via the MIG tool. Set to FALSE.	FALSE, TRUE	FALSE

DATA_RATE_OQ Attribute

The DATA_RATE_OQ attribute defines whether data is processed as single data rate (SDR) or double data rate (DDR). The allowed values for this attribute are SDR and DDR. The default value is DDR.

DATA_RATE_TQ Attribute

The DATA_RATE_TQ attribute defines whether 3-state control is to be processed as single data rate (SDR) or double data rate (DDR). The allowed values for this attribute are SDR, DDR, or BUF. The default value is DDR. In SDR and DDR modes, the four T inputs are used, and their behavior can be configured with the TRISTATE_WIDTH attribute. In BUF mode, the SDR and DDR mode registers are bypassed, and therefore, the T1 input should be used. The signal applied to the T1 input is asynchronous to all other signals because it simply passes through the OSERDESE2.

DATA_WIDTH Attribute

The DATA_WIDTH attribute defines the parallel data input width of the parallel-to-serial converter. The possible values for this attribute depend on the DATA_RATE_OQ attribute. When DATA_RATE_OQ is set to SDR, the possible values for the DATA_WIDTH attribute are 2, 3, 4, 5, 6, 7, and 8. When DATA_RATE_OQ is set to DDR, the possible values for the DATA_WIDTH attribute are 4, 6, 8, 10, and 14.

When the DATA_WIDTH is set to widths larger than eight, a pair of OSERDESE2 must be configured into a master-slave configuration. See [OSERDESE2 Width Expansion](#).

SERDES_MODE Attribute

The SERDES_MODE attribute defines whether the OSERDESE2 module is a master or slave when using width expansion. The possible values are MASTER and SLAVE. The default value is MASTER. See [OSERDESE2 Width Expansion](#).

TRISTATE_WIDTH Attribute

The TRISTATE_WIDTH attribute defines the parallel 3-state input width of the 3-state control parallel-to-serial converter. The possible values for this attribute depend on the DATA_RATE_TQ attribute. When DATA_RATE_TQ is set to SDR or BUF, the TRISTATE_WIDTH attribute can only be set to 1. When DATA_RATE_TQ is set to DDR, the possible values for the TRISTATE_WIDTH attribute are 1 and 4.

TRISTATE_WIDTH cannot be set to widths larger than 4. When a DATA_WIDTH is larger than four, set the TRISTATE_WIDTH to 1.

[Table 3-8](#) shows the valid setting and combinations of using the OSERDESE2.

Table 3-8: OSERDESE2 Attribute Combinations

INTERFACE_TYPE	DATA_RATE_OQ	DATA_RATE_TQ	DATA_WIDTH	TRISTATE_WIDTH
DEFAULT	SDR	SDR	1, 2, 3, 4, 5, 6, 7, 8	1
	DDR	DDR	4	4
		SDR	2, 6, 8, 10, 14	1

OSERDESE2 Clocking Methods

The phase relationship of CLK and CLKDIV is important in the parallel-to-serial conversion process. CLK and CLKDIV are (ideally) phase-aligned within a tolerance.

There are several clocking arrangements within the FPGA to help the design meet the phase relationship requirements of CLK and CLKDIV. The only valid clocking arrangements for the OSERDESE2 are:

- CLK driven by BUFIO, CLKDIV driven by BUFR
- CLK and CLKDIV driven by CLKOUT[0:6] of the same MMCM or PLL

When using a MMCM to drive the CLK and CLKDIV of the OSERDESE2 the buffer types supplying the OSERDESE2 can not be mixed. For example, if CLK is driven by a BUFG, CLKDIV must be driven by a BUFG as well.

OSERDESE2 Width Expansion

The OSERDESE2 modules be used to build a parallel-to-serial converter larger than 8:1. In every I/O tile there are two OSERDESE2 modules; one master and one slave. By connecting the SHIFTIN ports of the master OSERDESE2 to the SHIFTOUT ports of the slave OSERDESE2, the parallel-to-serial converter can be expanded to up to 10:1 and 14:1 (DDR mode only). For a differential output, the master OSERDESE2 must be on the positive (_P pin) side of the differential output pair. When the output is not differential, the output buffer associated with the slave OSERDESE2 is not available and width expansion cannot be used.

When using complementary single-ended standards (e.g., DIFF_HSTL and DIFF_SSTL), width expansion might not be used. This is because both OLOGICE2/3 blocks in an I/O tile are used by the complementary single-ended standards to transmit the two complementary signals, leaving no OLOGICE2/3 blocks available for width expansion purposes.

Figure 3-15 illustrates a block diagram of a 10:1 DDR parallel-to-serial converter using the master and slave OSERDESE2 modules. Ports D3–D4 are used for the last two bits of the parallel interface on the slave OSERDESE2 in this case.

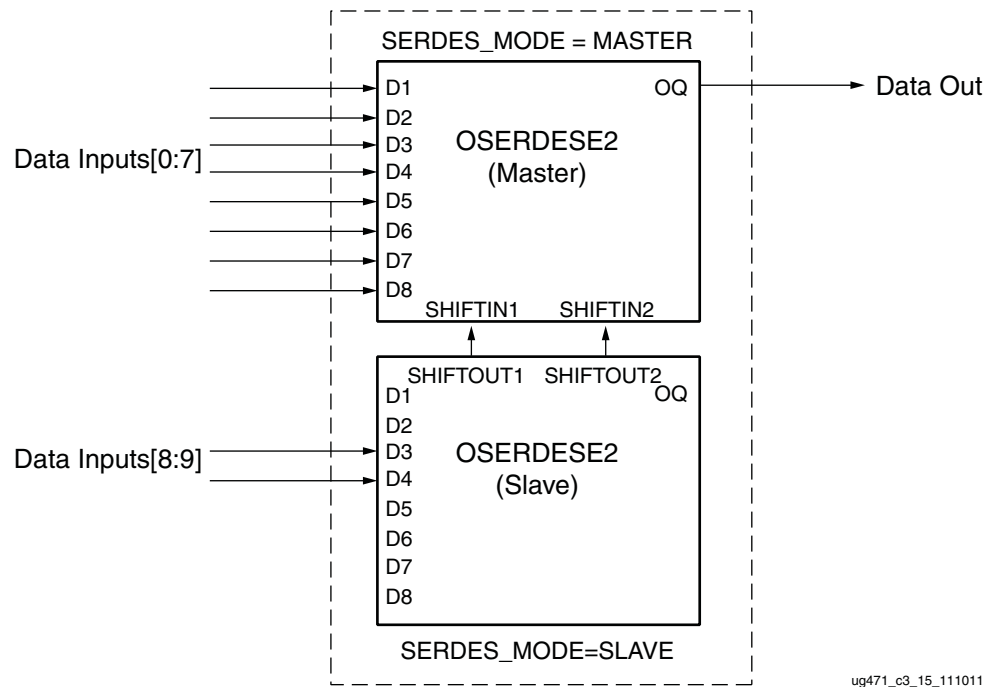


Figure 3-15: Block Diagram of OSERDESE2 Width Expansion

Table 3-9 lists the data width availability for SDR and DDR mode.

Table 3-9: OSERDESE2 SDR/DDR Data Width Availability

SDR Data Widths	2, 3, 4, 5, 6, 7, 8
DDR Data Widths	4, 6, 8, 10, 14

Guidelines for Expanding the Parallel-to-Serial Converter Bit Width

1. Both the OSERDESE2 modules must be adjacent master and slave pairs.
2. Set the SERDES_MODE attribute for the master OSERDESE2 to MASTER and the slave OSERDESE2 to SLAVE. See [SERDES_MODE Attribute](#).
3. The user must connect the SHIFTIN ports of the MASTER to the SHIFTOUT ports of the SLAVE.
4. The SLAVE only uses the ports D3 to D8 as an input.
5. DATA_WIDTH for Master and Slave are equal. See [DATA_WIDTH Attribute](#).
6. The attribute INTERFACE_TYPE is set to DEFAULT.

The slave inputs used for data widths requiring width expansion are listed in [Table 3-10](#).

Table 3-10: Slave Inputs Used for Data Width Expansion

Data Width	Slave Inputs Used
10	D3–D4
14	D3–D8

Output Feedback

The OSERDESE2 pin OFB has two functions:

- As a feedback path to the ISERDESE2 OFB pin. See [ISERDESE2 Feedback from OSERDESE2](#).
- As a connection to the ODELAYE2. The output of the OSERDESE2 can be routed through the OFB pin and then delayed through the ODELAYE2.

OSERDESE2 Latencies

DEFAULT Interface Type Latencies

The input to output latencies of OSERDESE2 blocks depend on the DATA_RATE and DATA_WIDTH attributes. Latency is defined as a period of time between the following two events: (a) when the rising edge of CLKDIV clocks the data at inputs D1–D8 into the OSERDESE2, and (b) when the first bit of the serial stream appears at OQ. [Table 3-11](#) summarizes the various OSERDESE2 latency values.

Table 3-11: OSERDESE2 Latencies

DATA_RATE	DATA_WIDTH	Latency
SDR	2:1	1 CLK cycle
	3:1	2 CLK cycles
	4:1	3 CLK cycles
	5:1	4 CLK cycles
	6:1	5 CLK cycles
	7:1	6 CLK cycles
	8:1	7 CLK cycles
DDR	4:1	2 CLK cycles
	6:1	3 CLK cycles
	8:1	4 CLK cycles
	10:1	5 CLK cycles
	14:1	5 CLK cycles

Note: In Table 3-11, the CLK and CLKDIV clock edges are normally not phase aligned. When the edges of both clocks are phase aligned, the latency can vary by one cycle.

OSERDESE2 Timing Model and Parameters

This section discusses all timing models associated with the OSERDESE2 primitive. Table 3-12 describes the function and control signals of the OSERDESE2 switching characteristics in the 7 series FPGA data sheets.

Table 3-12: OSERDESE2 Switching Characteristics

Symbol	Description
Setup/Hold	
T_{OSDCK_D}/T_{OSCKD_D}	D input Setup/Hold with respect to CLKDIV
T_{OSDCK_T}/T_{OSCKD_T}	T input Setup/Hold with respect to CLK
T_{OSDCK_T}/T_{OSCKD_T}	T input Setup/Hold with respect to CLKDIV
$T_{OSCKK_OCE}/T_{OSCKC_OCE}$	OCE input Setup/Hold with respect to CLK
$T_{OSCKK_TCE}/T_{OSCKC_TCE}$	TCE input Setup/Hold with respect to CLK
Sequential Delays	
T_{OSCKO_OQ}	Clock to Out from CLK to OQ
T_{OSCKO_TQ}	Clock to Out from CLK to TQ
Combinatorial	
T_{OSCO_OQ}	Asynchronous Reset to OQ
T_{OSCO_TQ}	Asynchronous Reset to TQ

Timing Characteristics of 2:1 SDR Serialization

In Figure 3-16, the timing of a 2:1 SDR data serialization is illustrated.

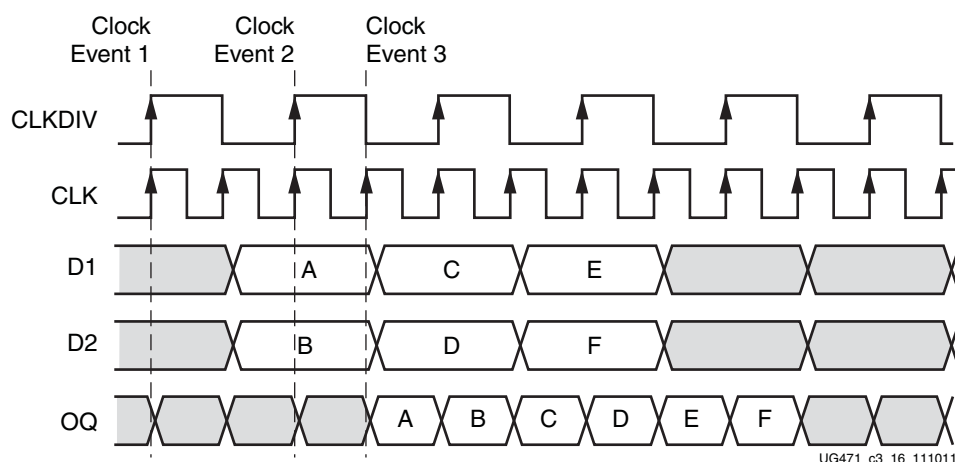


Figure 3-16: OSERDESE2 Data Flow and Latency in 2:1 SDR Mode

Clock Event 1

On the rising edge of CLKDIV, the word *AB* is driven from the FPGA logic to the D1 and D2 inputs of the OSERDESE2 (after some propagation delay).

Clock Event 2

On the rising edge of CLKDIV, the word *AB* is sampled into the OSERDESE2 from the D1 and D2 inputs.

Clock Event 3

The data bit *A* appears at OQ one CLK cycle after *AB* is sampled into the OSERDESE2. This latency is consistent with the Table 3-11 listing of a 2:1 SDR mode OSERDESE2 latency of one CLK cycle.

Timing Characteristics of 8:1 DDR Serialization

Figure 3-17 illustrates the timing of an 8:1 DDR data serialization. All eight of the bits are connected to D1–D8 of the master OSERDESE2 in contrast to previous generations where cascading was required.

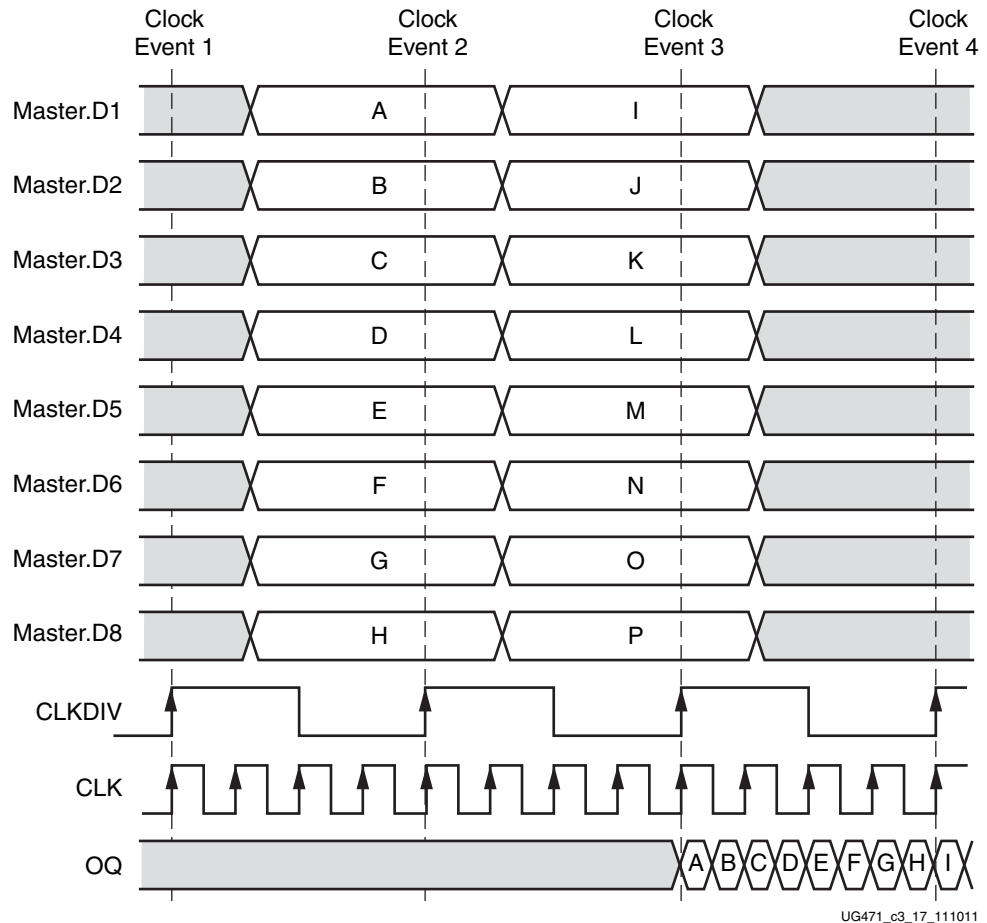


Figure 3-17: OSERDESE2 Data Flow and Latency in 8:1 DDR Mode

Clock Event 1

On the rising edge of CLKDIV, the word *ABCDEFGH* is driven from the FPGA logic to the D1–D8 inputs of the OSERDESE2.

Clock Event 2

On the rising edge of CLKDIV, the word *ABCDEFGH* is sampled into the OSERDESE2 from the D1–D8.

Clock Event 3

The data bit *A* appears at OQ four CLK cycles after *ABCDEFGH* is sampled into the OSERDESE2. This latency is consistent with the Table 3-11 listing of a 8:1 DDR mode OSERDESE2 latency of four CLK cycles.

The second word *IJKLMNOP* is sampled into the OSERDESE2 from the D1–D8.

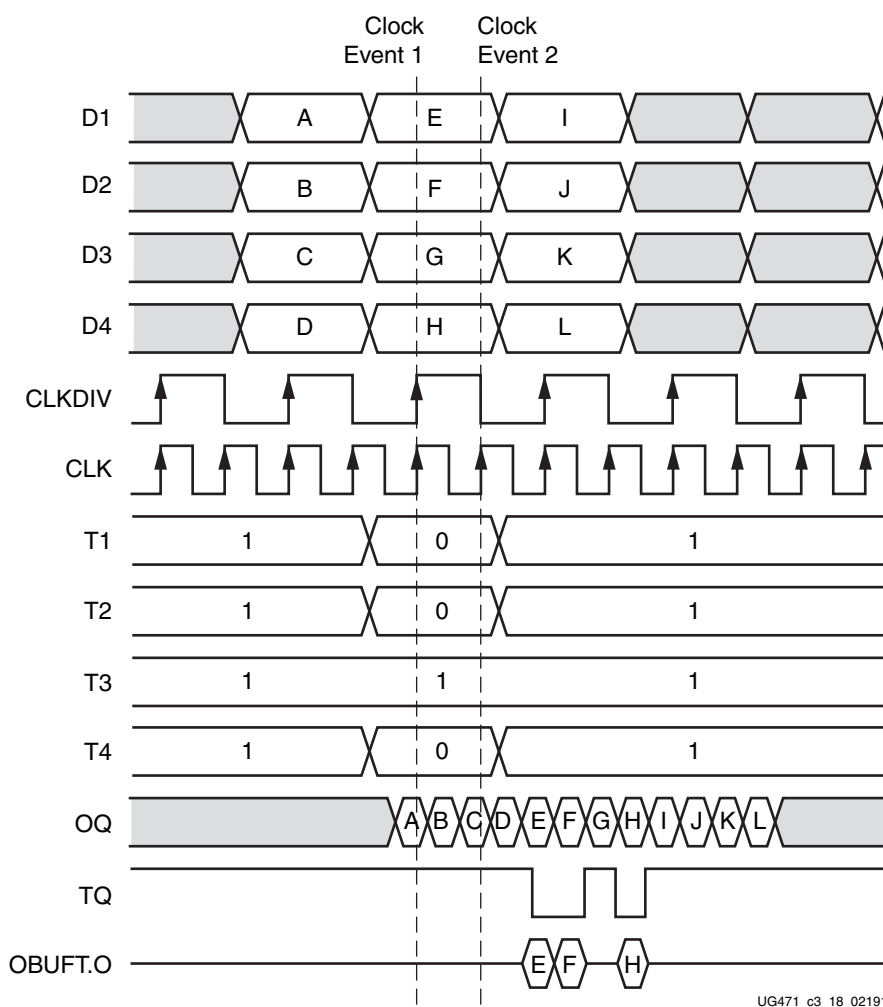
Clock Event 4

Between Clock Events 3 and 4, the entire word *ABCDEFGH* is transmitted serially on OQ, a total of eight bits transmitted in one CLKDIV cycle.

The data bit *I* appears at OQ four CLK cycles after *IJKLMNOP* is sampled into the OSERDESE2. This latency is consistent with the Table 3-11 listing of a 8:1 DDR mode OSERDESE2 latency of four CLK cycles.

Timing Characteristics of 4:1 DDR 3-State Controller Serialization

The operation of a 3-state controller is illustrated in Figure 3-18. The example is a 4:1 DDR case shown in a bidirectional system where the IOB must be frequently 3-stated.



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Figure 3-18: OSERDESE2 Data Flow and Latency in 4:1 DDR Mode

Clock Event 1

T1, T2, and T4 are driven Low to release the 3-state condition. The serialization paths of T1-T4 and D1-D4 in the OSERDESE2 are identical (including latency), such that the bits *EFGH* are always aligned with the 0010 presented at the T1-T4 pins during Clock Event 1.

Clock Event 2

The data bit *E* appears at OQ one CLK cycle after *EFGH* is sampled into the OSERDESE2. This latency is consistent with the [Table 3-11](#) listing of a 4:1 DDR mode OSERDESE2 latency of one CLK cycle.

The 3-state bit 0 at T1 during Clock Event 1 appears at TQ one CLK cycle after 0010 is sampled into the OSERDESE2 3-state block. This latency is consistent with the [Table 3-11](#) listing of a 4:1 DDR mode OSERDESE2 latency of one CLK cycle.

OSERDESE2 VHDL and Verilog Instantiation Templates

The Libraries Guide includes instantiation templates of the OSERDESE2 module in VHDL and Verilog.

IO_FIFO Overview

7 series devices have shallow IN_FIFOs and OUT_FIFOs (called IO_FIFOs collectively) located in each of the I/O banks. Although these IO_FIFOs are specifically designed for memory applications, they are available as general resources. For general use, all inputs and outputs are routed via interconnect. The most common use of IO_FIFOs is to interface with external components as an extension of IOLOGIC (e.g., ISERDES or IDDR and OSERDES or ODDR). Because of their general interconnect capability, IO_FIFOs can also serve as additional fabric FIFO resources.

Each I/O bank contains four IO_FIFOs with one IO_FIFO per byte group. A byte group is defined as 12 I/Os within a bank. The IO_FIFOs are physically aligned to an I/O byte group. This alignment yields the best performance when IO_FIFOs are used to interface to IOI components such as the input and output SERDES elements, which is their most common use. However, regardless of their location, IO_FIFOs can also interface to resources in the FPGA fabric and other I/O banks (see [UG475, 7 Series FPGAs Packaging and Pinout Specifications](#) for byte group pin arrangements). This section focuses on the use of IO_FIFOs to interface with IOI components.

For external data flowing into the FPGA, an IN_FIFO can connect to the ILOGIC (e.g., ISERDESE2, IDDR, or IBUF) to receive incoming data and pass it on to the fabric. For data flowing out of the FPGA, an OUT_FIFO can connect to the OLOGIC (e.g., OSERDESE2, ODDR, or OBUF) to pass data from the fabric and send it through to the IOB.

An IN_FIFO receives 4-bit data from an ILOGIC block while the fabric side reads either 4- or 8-bit data out of the array. An OUT_FIFO receives 4- or 8-bit data from the fabric while an OLOGIC block reads 4-bit data out of the array.

Each IO_FIFO has a 768-bit storage array and can be arranged as twelve groups of 4-bit data or ten groups of 8-bit data. An IO_FIFO is nine entries deep, including an input and output register. Typical IO_FIFO uses are as a buffer for a parallel I/O interface crossing between two frequency domains (e.g., the BUFR domain to/from the BUFG or BUFGH domain) or as a 2:1 serializer/deserializer to decouple the PHY from the fabric to relax fabric performance requirements.

IO_FIFOs are shallower versions of regular FIFOs and have similar functionality. The primary purpose of IO_FIFOs is to support I/O data transfer functions. They are not intended to replace built-in FIFOs or LUT-based FIFOs. IO_FIFOs support standard flag logic, clocks, and control signals. IO_FIFOs can operate in two modes, 4 x 4 mode (1:1) or 4 x 8/8 x 4 mode (1:2/2:1).

The IO_FIFOs have an input register, a 7-entry deep FIFO core, and an output register (see Figure 3-19). The input and output registers are an integral part of the IO_FIFO and provide the eighth storage location for the full IO_FIFO. The registers, FIFO core, and control signals are treated as a single atomic unit.

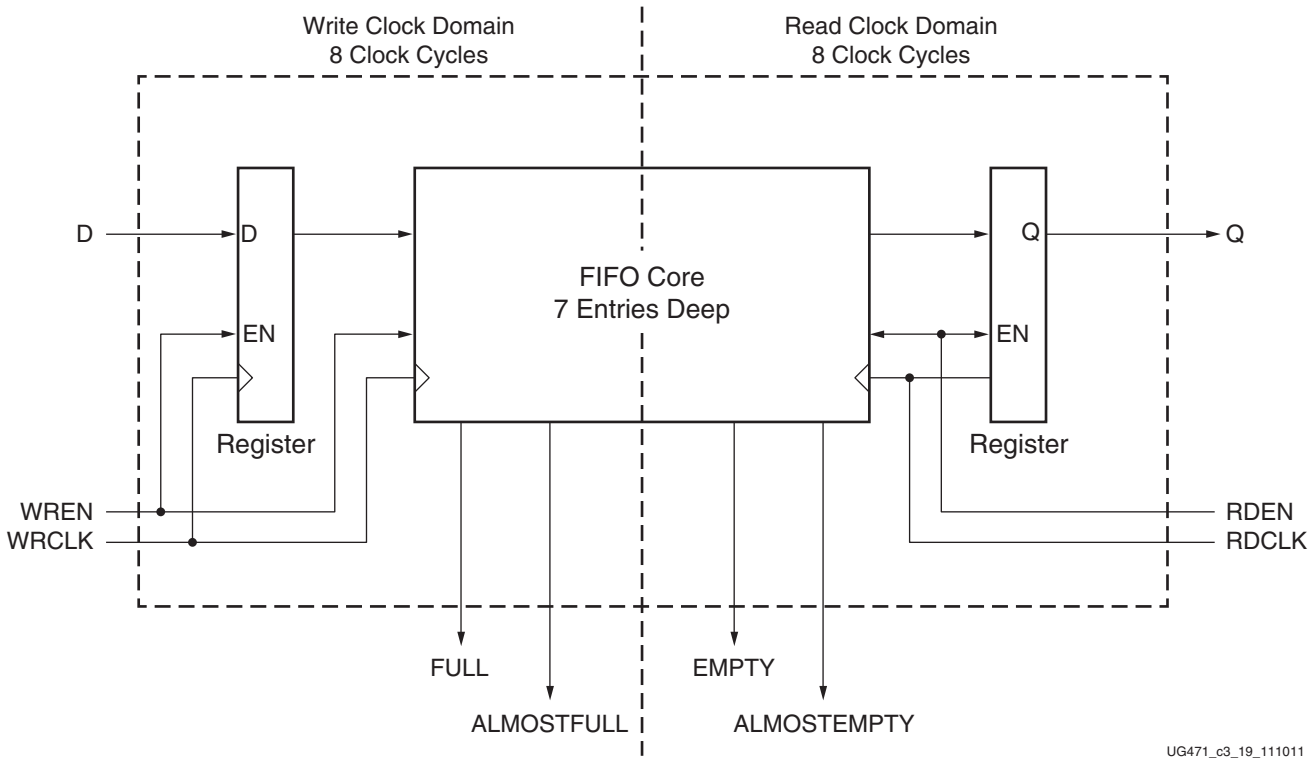


Figure 3-19: IO_FIFO Architecture: Top-Level View

IN_FIFO

The IN_FIFO is physically aligned to an I/O byte group for optimized performance. The 8-entry deep IN_FIFO supports data transfer using two modes of operation:

- 4 x 4 mode – This mode configures the FIFO to have 12 4-bit wide data inputs (D) and 12 4-bit wide data outputs (Q). The D0[3:0] – D9[3:0] ports map to the Q0[3:0] – Q9[3:0] ports. D5[7:4] and D6[7:4] are the two extra data input ports D10[3:0] and D11[3:0] and map to the Q5[7:4] and Q6[7:4] extra output ports Q10[3:0] and Q11[3:0]. The other Qn[7:4] ports are not used. Table 3-13 shows the 4 x 4 mode mapping in detail.

Table 3-13: IN_FIFO Input to Output Data Mapping in 4 x 4 Mode

Mapping	Not Used
D0[3:0] → Q0[3:0]	Q0[7:4]
D1[3:0] → Q1[3:0]	Q1[7:4]
D2[3:0] → Q2[3:0]	Q2[7:4]
D3[3:0] → Q3[3:0]	Q3[7:4]
D4[3:0] → Q4[3:0]	Q4[7:4]

Table 3-13: IN_FIFO Input to Output Data Mapping in 4 x 4 Mode (Cont'd)

Mapping	Not Used
D5[3:0] → Q5[3:0]	
D6[3:0] → Q6[3:0]	
D7[3:0] → Q7[3:0]	Q7[7:4]
D8[3:0] → Q8[3:0]	Q8[7:4]
D9[3:0] → Q9[3:0]	Q9[7:4]
D10[3:0] is D5[7:4] → Q5[7:4]	
D11[3:0] is D6[7:4] → Q6[7:4]	

- 4 x 8 mode – This mode configures the FIFO to have 10 4-bit wide data inputs (D) and 10 8-bit wide data outputs (Q). In 4 x 8 mode, the 4-bit input data is demultiplexed to form the 8-bit output data width. 4 x 8 mode is generally used when the output clock frequency is greater than one half the input clock frequency and thus output data is twice the width of the input data. Table 3-14 shows the 4 x 8 mode mapping in detail.

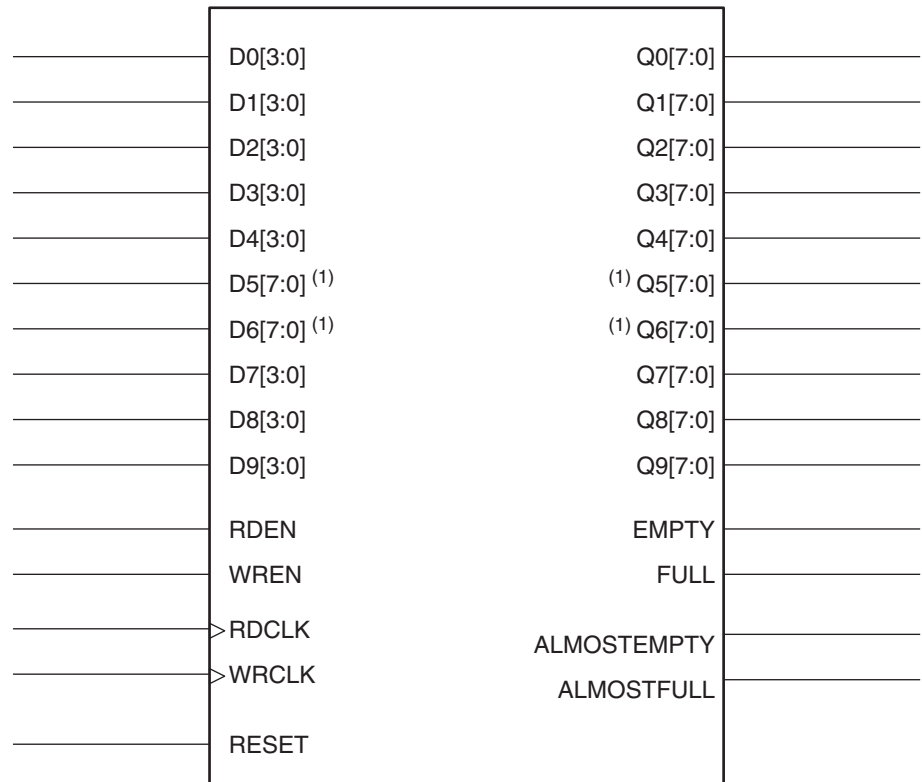
Table 3-14: IN_FIFO Input to Output Data Mapping in 4 x 8 Mode

Mapping	Not Used
D0[3:0] → Q0[7:0]	
D1[3:0] → Q1[7:0]	
D2[3:0] → Q2[7:0]	
D3[3:0] → Q3[7:0]	
D4[3:0] → Q4[7:0]	
D5[3:0] → Q5[7:0]	
D6[3:0] → Q6[7:0]	
D7[3:0] → Q7[7:0]	
D8[3:0] → Q8[7:0]	
D9[3:0] → Q9[7:0]	

Both modes support the FULL, EMPTY, ALMOSTFULL, and ALMOSTEMPTY flags.

IN_FIFO Primitive

The IN_FIFO primitive is shown in Figure 3-20.



Notes:
 1. Extra input ports D10 (D5[7:4]) and D11 (D6[7:4]) and output ports Q10 (Q5[7:4]) and Q11 (Q6[7:4]) in 4 x 4 mode.

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Figure 3-20: IN_FIFO Primitive

Table 3-15 lists the available ports in the IN_FIFO primitive.

Table 3-15: IN_FIFO Ports

Port Name	Input/output	Description
RDCLK	I	Read clock. Connect to BUFR, BUFG, or BUFH.
WRCLK	I	Write clock. Connect to BUFR, BUFG, or BUFH.
RESET	I	Reset, active High. Clears all counters, pointers, and data.
D0[3:0] – D9[3:0]	I	Ten 4-bit data in ports in 4 x 8 mode. Twelve 4-bit data in ports in 4 x 4 mode. Connect to ILOGIC if used for external interfaces.
D5[7:4], D6[7:4]	I	Supplemental data in ports D10 and D11. Used only in 4 x 4 mode. Data on the ports appears on corresponding output ports Q5[7:4] and Q6[7:4].
RDEN	I	Read enable.
WREN	I	Write enable.
Q0[7:0] – Q9[7:0]	O	Ten 8-bit data out buses in 4 x 8 mode, or ten 4-bit data out buses in 4 x 4 mode. Connect to fabric if used for external interfaces.

Table 3-15: IN_FIFO Ports (Cont'd)

Port Name	Input/output	Description
Q5[7:4], Q6[7:4]	O	Supplemental data out ports Q10 and Q11. Used only in 4x4 mode. Data on these ports is sourced from corresponding input ports D5[7:4] and D6[7:4].
EMPTY	O	Empty flag. Synchronized to RDCLK.
FULL	O	Full flag. Synchronized to WRCLK.
ALMOSTEMPTY ⁽¹⁾	O	Programmable level empty flag. Synchronized to RDCLK.
ALMOSTFULL ⁽¹⁾	O	Programmable level full flag. Synchronized to WRCLK.

Notes:

1. The corresponding attribute can be set to a value of 1 or 2 (see [Table 3-19, page 181](#)). Accordingly, at least one or two reads or writes occur after the flag asserts. Due to the asynchronous nature of the FIFO, there can be one or two additional reads or writes increasing the total reads or writes to three or four.

OUT_FIFO

The OUT_FIFO is co-located with the IN_FIFO and is also physically aligned to an I/O byte group for optimized performance. The 8-entry deep OUT_FIFO supports data transfer using two modes of operation:

- 4 x 4 mode – This mode configures the FIFO to have 12 4-bit wide data inputs (D) and 12 4-bit wide data outputs (Q). The D0[3:0] – D9[3:0] ports map to the Q0[3:0] – Q9[3:0] ports. D5[7:4] and D6[7:4] are the two extra data input ports that serve as D10 and D11 and map to the Q5[7:4] and Q6[7:4] output ports. The other D[7:4] ports are not used. [Table 3-16](#) shows the 4 x 4 mode mapping in detail.

Table 3-16: OUT_FIFO Input to Output Data Mapping in 4 x 4 Mode

Mapping	Not Used
D0[3:0] → Q0[3:0]	Q0[7:4]
D1[3:0] → Q1[3:0]	Q1[7:4]
D2[3:0] → Q2[3:0]	Q2[7:4]
D3[3:0] → Q3[3:0]	Q3[7:4]
D4[3:0] → Q4[3:0]	Q4[7:4]
D5[3:0] → Q4[3:0]	
D6[3:0] → Q6[3:0]	
D7[3:0] → Q7[3:0]	Q7[7:4]
D8[3:0] → Q8[3:0]	Q8[7:4]
D9[3:0] → Q9[3:0]	Q9[7:4]
D10[7:4] is D5[7:4] → Q5[7:4]	
D11[7:4] is D6[7:4] → Q6[7:4]	

- 8 x 4 mode – This mode configures the FIFO to have 10 8-bit wide data inputs (D) and 10 4-bit wide data outputs (Q). In 8 x 4 mode, a 2:1 multiplexer in the output datapath serializes the 8-bit input data to the 4-bit output data width. 4 x 8 mode is generally

used when the output clock frequency is twice the input clock frequency and thus output data is half the width of the input data. Table 3-17 shows the 8 x 4 mode mapping in detail.

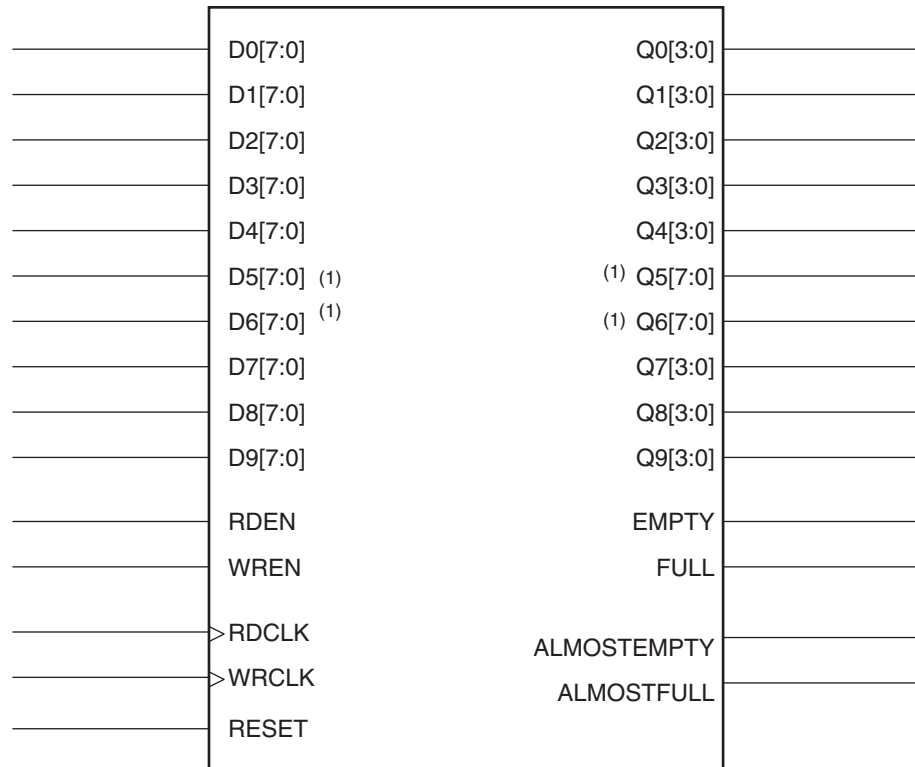
Table 3-17: OUT_FIFO Input to Output Data Mapping

Mapping	Not Used
D0[7:0] → Q0[3:0]	
D1[7:0] → Q1[3:0]	
D2[7:0] → Q2[3:0]	
D3[7:0] → Q3[3:0]	
D4[7:0] → Q4[3:0]	
D5[7:0] → Q5[3:0]	
D6[7:0] → Q6[3:0]	
D7[7:0] → Q7[3:0]	
D8[7:0] → Q8[3:0]	
D9[7:0] → Q9[3:0]	

Both modes support the FULL, EMPTY, ALMOSTFULL, and ALMOSTEMPTY flags.

OUT_FIFO Primitive

The OUT_FIFO primitive is shown in Figure 3-21.



Notes:

1. Extra input ports D10 (D5[7:4]) and D11 (D6[7:4]) and output ports Q10 (Q5[7:4]) and Q11 (Q6[7:4]) in 4 x 4 mode.

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Figure 3-21: OUT_FIFO Primitive

Table 3-18 lists the available ports in the OUT_FIFO primitive.

Table 3-18: OUT_FIFO Ports

Port Name	Input/output	Description
RDCLK	I	Read clock. Connect to BUFR, BUFG, or BUFH.
WRCLK	I	Write clock. Connect to BUFR, BUFG, or BUFH.
RESET	I	Reset, active High. Clears counters, pointers, and data.
D0[7:0] – D9[7:0]	I	Ten 8-bit data in ports in 8 x 4 mode. Twelve 4-bit data in ports in 4 x 4 mode. Connect to fabric if used for external interfaces.
D5[7:4], D6[7:4]	I	Supplemental data in ports D10 and D11. Used only in 4 x 4 mode. Data on the ports appears on corresponding output ports Q5[7:4] and Q6[7:4].
RDEN	I	Read enable.
WREN	I	Write enable.
Q0[3:0] – Q9[3:0]	O	Ten 4-bit data output buses. Connect to OLOGIC if used for external interfaces.

Table 3-18: OUT_FIFO Ports (Cont'd)

Port Name	Input/output	Description
Q5[7:4], Q6[7:4]	O	Supplemental data out ports Q10 and Q11. Used only in 4 x 4 mode. Data on these ports is sourced from the corresponding input ports D5[7:4] and D6[7:4].
EMPTY	O	Empty flag. Synchronized to RDCLK.
FULL	O	Full flag. Synchronized to WRCLK.
ALMOSTEMPTY ⁽¹⁾	O	Programmable level empty flag. Synchronized to RDCLK.
ALMOSTFULL ⁽¹⁾	O	Programmable level full flag. Synchronized to WRCLK.

Notes:

1. The corresponding attribute can be set to a value of 1 or 2 (see [Table 3-19, page 181](#)). Accordingly, at least one or two reads or writes occur after the flag asserts. Due to the asynchronous nature of the FIFO, there can be one or two additional reads or writes increasing the total reads or writes to three or four.

Resetting the IO_FIFO

An IO_FIFO has a single asynchronous reset that is internally resynchronized with both the read and write clock domains. To ensure proper reset, RESET must be asserted High for at least four cycles of either RDCLK or WRCLK, whichever is slower, before writing to the IO_FIFO. The RDEN and WREN must be held Low while RESET is asserted.

The IO_FIFOs should be held in reset until both the write and read clocks are present and stable. Similarly, if the read or write clocks are not available until after configuration, the IO_FIFO must be reset as described above after valid clocks are asserted.

EMPTY and FULL Flags

The FULL flag, when asserted High, signals that the FIFO core and the input register are both full. The state of the output register is ignored.

The EMPTY flag indicates the status of the data in the output register. When the EMPTY flag is asserted High, the data in the output register is not valid.

ALMOST EMPTY and ALMOST FULL Flags

The ALMOSTEMPTY and ALMOSTFULL flags provide an early indication that the IO_FIFO is approaching its limits. The flags can be configured to assert one or two cycles prior to the IO_FIFO reaching a full or empty state. A value of 1 indicates that there is only one word remaining to read or write. A value of 2 indicates that two words are remaining to read or write.

Due to the asynchronous nature of the IO_FIFO and internal synchronization, the flags might be overly pessimistic. During a read operation, there might be more data stored than is indicated by an ALMOSTEMPTY flag output of 1 or 2. During a write operation there can be more space available to write than indicated by an ALMOSTFULL flag output of 1 or 2.

The ALMOSTEMPTY and ALMOSTFULL flags do not necessarily overlap with the FULL and EMPTY flags. It is possible to have ALMOSTEMPTY assert and deassert before EMPTY asserts. This will occur if WRCLK is more than two times faster than RDCLK.

[Table 3-19](#) summarizes all the applicable IO_FIFO attributes.

Table 3-19: IO_FIFO Attributes

Attribute	Value	Default Value	Description
ARRAY_MODE (IN_FIFO)	String: ARRAY_MODE_4_X_8 ARRAY_MODE_4_X_4	ARRAY_MODE_4_X_8	Defines 4 input bits and 4 or 8 output bits per port.
ARRAY_MODE (OUT_FIFO)	String: ARRAY_MODE_8_X_4 ARRAY_MODE_4_X_4	ARRAY_MODE_8_X_4	Defines 4 or 8 input bits, and 4 output bits per port.
ALMOST_EMPTY_VALUE	Integer: 1 or 2	1	See ALMOST EMPTY and ALMOST FULL Flags , page 180.
ALMOST_FULL_VALUE	Integer: 1 or 2	1	See ALMOST EMPTY and ALMOST FULL Flags , page 180.
OUTPUT_DISABLE	Boolean: TRUE or FALSE	FALSE	OUT_FIFO: Output disable drives the Qx outputs High when RD_EN is low.

Termination Options for SSO Noise Analysis

The PlanAhead™ software has the ability to perform simultaneous switching noise (SSN) analysis for each design, taking into account the actual I/O standards and options assigned to the I/O pins in the target device and package. For details on how to use this feature and perform the SSN analysis, see the “Using Noise Analysis Predictors” section of [UG632: PlanAhead User Guide](#).

For each output pin, there is the option to specify whether or not termination is present on the board. The off-chip termination field automatically populates with the default terminations for each I/O standard, if one exists.

[Table A-1](#) lists all of the default terminations for each of the I/O standards supported by the 7 series FPGAs when using the SSN predictor tool within the PlanAhead™ software. For each I/O pin in the design, the user can specify whether to use these terminations, or to have no termination.

Table A-1: Default Terminations for SSN Noise Analysis by I/O Standard

IO Standard ⁽¹⁾	Default Termination
HSTL_I	Far V_{TT} 50Ω
HSTL_I_12	Far V_{TT} 50Ω
HSTL_I_18	Far V_{TT} 50Ω
HSTL_I_DCI	Far V_{TT} 50Ω
HSTL_I_DCI_18	Far V_{TT} 50Ω
HSTL_II	Near V_{TT} 50Ω & Far V_{TT} 50Ω
HSTL_II_18	Near V_{TT} 50Ω & Far V_{TT} 50Ω
HSTL_II_DCI	Far V_{TT} 50Ω
HSTL_II_DCI_18	Far V_{TT} 50Ω
HSTL_II_T_DCI	Far V_{TT} 50Ω
HSTL_II_T_DCI_18	Far V_{TT} 50Ω
HSUL_12	None
HSUL_12_DCI	None
LVC MOS (all voltages) LVTTL (2 mA, 4 mA, 6 mA, and 8 mA drive)	None

Table A-1: Default Terminations for SSN Noise Analysis by I/O Standard (Cont'd)

IO Standard ⁽¹⁾	Default Termination
LVC MOS (all voltages) LVTTL (12 mA, 16 mA, and 24 mA drive)	Far V_{TT} 50 Ω
MOBILE_DDR	None
SSTL12	Far V_{TT} 50 Ω
SSTL12_DCI	Far V_{TT} 50 Ω
SSTL12_T_DCI	Far V_{TT} 50 Ω
SSTL135	Far V_{TT} 50 Ω
SSTL135_DCI	Far V_{TT} 50 Ω
SSTL135_R	Far V_{TT} 50 Ω
SSTL135_T_DCI	Far V_{TT} 50 Ω
SSTL15	Far V_{TT} 50 Ω
SSTL15_DCI	Far V_{TT} 50 Ω
SSTL15_R	Far V_{TT} 50 Ω
SSTL15_T_DCI	Far V_{TT} 50 Ω
SSTL18_I	Far V_{TT} 50 Ω
SSTL18_I_DCI	Far V_{TT} 50 Ω
SSTL18_II	Near V_{TT} 50 Ω & Far V_{TT} 50 Ω
SSTL18_II_DCI	Far V_{TT} 50 Ω
SSTL18_II_T_DCI	Far V_{TT} 50 Ω
BLVDS_25	Near Series 165 Ω , Near Differential 140 Ω , and Far Differential 100 Ω
HSLVDCI_15	None
HSLVDCI_18	None
LVDCI_15	None
LVDCI_18	None
LVDCI_DV2_15	None
LVDCI_DV2_18	None
LVDS	Far Differential 100 Ω
LVDS_25	Far Differential 100 Ω
MINI_LVDS_25	Far Differential 100 Ω
PCI33_3	None
PPDS_25	Far Differential 100 Ω
RSDS_25	Far Differential 100 Ω

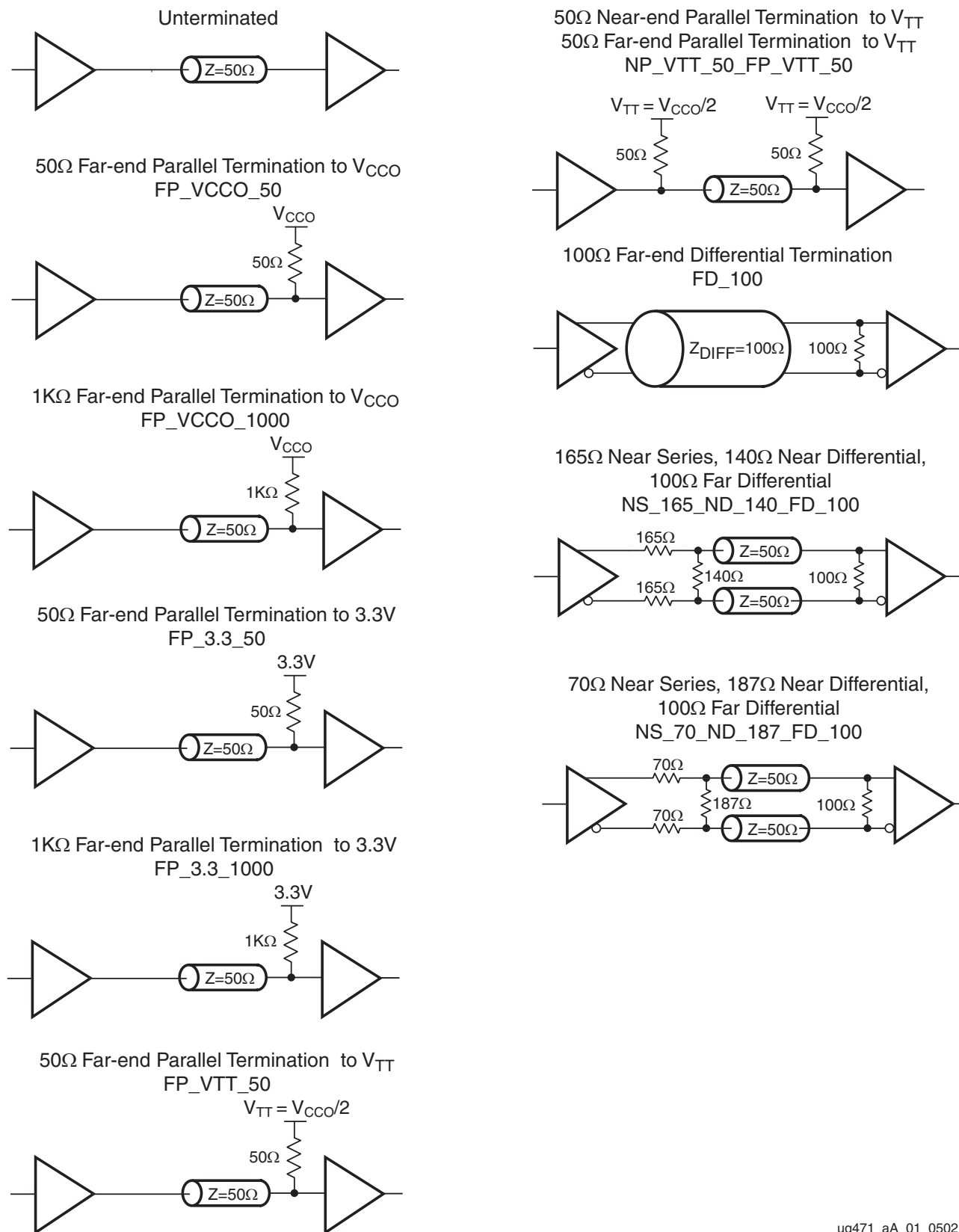
Table A-1: Default Terminations for SSN Noise Analysis by I/O Standard (Cont'd)

IO Standard ⁽¹⁾	Default Termination
TMD5_33	Far 3.3V 50Ω

Notes:

1. All differential versions of the HSTL, SSTL, HSUL, and MOBILE_DDR standards (e.g., DIFF_SSTL135) have the same termination as the single-ended versions.

Figure A-1 illustrates each of these terminations.



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Figure A-1: Default Terminations

